

Clinical Practice Guidelines for the Screening, Diagnosis, and Outpatient Management of Type 2 Diabetes Mellitus among Filipino Adults:

Part 2a - Guidelines for the Management of Type 2 Diabetes Mellitus



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Disclaimer and Contact Information

This clinical practice guideline is primarily intended for physicians, especially primary care providers, general practitioners, general medicine or family and community medicine specialists rather than subspecialists. Like most other guidelines it is intended to provide specific evidence-based guidance but is not intended to substitute for sound clinical judgment when faced with individual decisions, especially in consideration of a patient or client's values, needs and preferences. Although adherence to the recommendations is encouraged by the stakeholder organizations and by the Department of Health (DOH), there are many clinical situations when the recommendations may not always apply and thus, clinicians must always exercise their best judgment in individualized decision-making always of course, upholding the patient's welfare and best interests.

The contents of the recommendations could also potentially be used by decision makers for purposes other than primarily intended, which is patient care; some of these examples may include resource allocation and the prioritization of goods or services for health. As such, the recommendations may not always be the best.

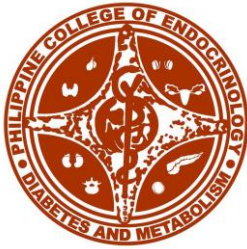
This CPG is intended to cover only the most common type of diabetes mellitus, that is Type 2 diabetes mellitus (T2DM) as well as the most common setting where they are seen, and this in the outpatient clinics. It does not provide recommendations for in-patient management, as well as for other forms of diabetes such as Type 1 DM or gestational diabetes. It is also not intended to be comprehensive and focuses on clinical questions where there may be variability in practice or where controversies may exist in the use of tests and treatments.

The specific considerations are always in the context of the Philippines and whenever possible, issues of equity were pointedly included in the discussions of the recommendations.

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Department of Health



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Philippine Society of Nephrology (PSN)



Philippine Heart Association (PHA)



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Executive Summary

Diabetes mellitus is a growing public health challenge in the Philippines, with increasing prevalence and delayed diagnosis leading to complications. This Clinical Practice Guideline (CPG) on the Screening and Diagnosis of Type 2 Diabetes Mellitus (T2DM) provides evidence-based recommendations for early detection and timely intervention. It primarily targets adults aged 18 years and older in outpatient settings, guiding general practitioners, specialists, and policymakers. This guideline does not cover Type 1 diabetes, gestational diabetes, or inpatient management. This CPG addresses three key areas: (1) screening for pre-diabetes and T2DM, (2) diagnosis of T2DM, and (3) screening for macrovascular and microvascular complications.

This CPG was developed following a rigorous methodology involving systematic review of existing guidelines using the ADAPTE framework and/or de novo systematic reviews. The GRADE approach was used to assess the quality of evidence and determine the strength of the recommendations. This guideline was developed through a multi-sectoral approach, ensuring representation from medical societies, primary care providers, endocrinologists, nephrologists, cardiologists, policymakers, and patient advocacy groups. The Philippine College of Endocrinology, Diabetes and Metabolism (PCEDM) led the initiative in collaboration with the Department of Health (DOH), the Philippine College of Physicians (PCP), Philippine Heart Association (PHA), Philippine Society of Nephrology (PSN), Philippine Academy of Family Physicians (PAFP), and other key stakeholders. Patients were actively involved in the consensus panel, clinical outcomes prioritization, and focus group discussions to ensure that their perspectives were considered. This guideline was independently developed by the PCEDM and collaborating organizations, with no direct influence from pharmaceutical companies or commercial entities.

This CPG recommends practical strategies for implementation across various healthcare settings which includes the following: (1) dissemination through professional societies, DOH, and academic institutions, (2) promoting the use of the screening algorithms in primary care, (3) integration of the recommendations into DOH's Universal Health Care framework and PhilHealth reimbursement policies, and (4) public awareness campaigns to promote early screening and diabetes prevention. Furthermore, addressing the identified limitations such as the availability of some tests (e.g., HbA1c in certain regions), need for cost-effectiveness analysis of screening interventions in the Philippine setting, and variability in access to diabetes care is needed.

Chapter 1. Background

Introduction

The prevalence of diabetes mellitus has been steadily rising since the late 1990s, when it was around 4%. By 2008, it had increased to 4.8% and reached approximately 8.0% by 2018.¹ More alarming than these numbers is the high prevalence of pre-diabetes (impaired fasting glucose, FBS 100-125 mg/dL) at 26% in 2018. When considering both types of dysglycemia, the prevalence is approximately one-third of the adult population 20 years and older.

Compounding this rising prevalence of diabetes mellitus is the pervasive problem of poor glycemic control. Between 1998-2008, the DiabCare Philippines survey estimated only about 15% of patients in outpatient clinics in the Philippines had an HbA1c level below 7.0%.² In this same survey, which examined a cohort of patients with established diabetes (mean duration: 9.0±6.0 years), complications were already common: 45% had neuropathy, 16% had microalbuminuria, and 20.1% had non-proliferative retinopathy. Severe late eye complications included photocoagulation in 7.4% of patients and proliferative diabetic retinopathy in 8.3%. Finally, 10.7% reported a history of angina pectoris while 4.7% had experienced a stroke.

Even among newly diagnosed cases, complications within one year are also already prevalent, indicating poor screening and late identification of both diabetes and its complications. A 2009 study in Manila found that among persons newly diagnosed with Type 2 diabetes mellitus (T2DM), 20% already had peripheral neuropathy, 42% had proteinuria, and 12% had diabetic retinopathy.³

Several initiatives can help improve diabetes care by addressing both systemic challenges and needs of individual patients. One key initiative is the development of clinical (practice) guidelines, which forms part of an evidence-based toolkit that has the potential to improve both the process of care and patient outcomes.⁴

It is therefore in this spirit of quality improvement in the way that care is delivered, and of improving the lives of persons with diabetes that this clinical practice guideline has been conceptualized and developed. The first multi-sectoral guideline on diabetes mellitus in the Philippines, UNITE for Diabetes CPG for the Diagnosis and Management of Diabetes, was published in 2013. Since its development, it has been integrated into the Department of Health (DOH) Primary Care National Formulary and has been used to inform decisions and priorities at various levels of healthcare, such as the DOH medicine access programs.

An update to the 2013 CPG is essential due to significant developments in diabetes care, including several new medications, expanded indications for existing treatments, and evolving public health policies locally and internationally. Advocacy efforts from public health organizations including the World Health Organization (WHO) have had a direct impact on the care of persons with diabetes, such as warnings on alcohol, cigarette smoking and vape, and artificial sweeteners.

Scope and Purpose of the CPG

This updated CPG intends to support the implementation of Universal Health Care (UHC) by providing evidence-based recommendations for improving diabetes care delivery. It focuses on the management of adult patients with T2DM—the most prevalent type of diabetes.

This CPG primarily covers clinical questions that are relevant for the outpatient management of adults with T2DM and does not include recommendations for pediatric patients or inpatient settings. The key topics include:

Part 1:

1. Screening for Pre-diabetes and T2DM
2. Diagnosis of T2DM
3. Screening for Complications

Part 2:

4. Prevention of T2DM and Pre-diabetes
5. Non-pharmacologic Management of T2DM
6. Pharmacologic Treatment of T2DM
7. Cardiovascular Risk Reduction in T2DM
8. Vaccination and Adjunctive Therapies

For purposes of this CPG, adults are defined as individuals 18 years and older, including older age groups. However, this CPG does not cover pregnant women with diabetes as the recommendations for this subgroup are typically different in many aspects, from screening, diagnosis, to pharmacologic treatment. Outpatient settings include hospital outpatient departments or doctors' clinics, various diabetes clinics (hospital-based, mall-based, or stand-alone), workplace clinics, and local health centers where many more persons with diabetes are being seen.

End users and settings

The primary intended users of this CPG are physicians who are in general practice, general medicine or family medicine specialists, Family Medicine and Internal Medicine residents-in-training and even medical students (as part of preparation for general practice). The recommendations can also be used by subspecialists who are managing persons with T2DM, including but not limited to endocrinologists, nephrologists, neurologists, or cardiologists.

Although the guidelines are intended for clinical use, some of the content may also be used to guide administrators, policymakers, and public/private health sector leaders for the procurement of medications, planning of health programs and priorities for persons with diabetes, or formulating diabetes screening strategies for at-risk populations.

Stakeholder Involvement

The organizations involved in developing this clinical practice guideline were intentionally identified to adequately represent the interests of persons with diabetes or at risk for diabetes. These organizations represent primary care physicians (general practice, family medicine, general medicine, or physicians in occupational health settings) as well as subspecialists managing patients with complex cases or complications. The representatives of these organizations were involved in the guideline development through these mechanisms: (1) serving as members of the Steering Committee and/or Consensus Panel; (2) developing, reviewing, and approving clinical questions as well as identifying and prioritizing clinical outcomes; and (3) reviewing, revising, and disseminating the draft guidelines through the organizations.

The involvement of lay persons with diabetes was three-fold: (1) representation in the consensus voting panel with active involvement during the discussions; (2) determination of outcomes and importance of clinical outcomes through focus group discussions with members of a lay diabetes club; (3) prioritization of the herbal products or dietary supplements which will be included in the recommendations through direct interaction with the members of an online diabetes group. These processes were intentionally done so that the voice of persons with diabetes is well heard, ensuring that their perspectives are reflected in the guideline.

Chapter 2. Guideline Development Methodology

Overview

This clinical practice guideline was developed following the prescribed methodology of the DOH.⁵ Following international recommendations, this guideline development followed 4 main phases or steps: (1) guideline preparation, (2) evidence synthesis, (3) evidence to decision (EtD), and (4) dissemination and evaluation. This manuscript will cover the first 3 phases and outline plans for dissemination and evaluation.

Phase 1 involves guideline preparation with three main steps: (1) organization of the guideline development teams, (2) formulation of clinical questions, and (3) search and retrieval of literature, appraisal of validity and applicability of literature by the evidence review experts (technical working group). *Phase 2* involves synthesizing and assessing the certainty of the body of evidence using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach. In this phase, draft recommendations are prepared and an initial strength of recommendation is identified. *Phase 3* involves EtD, where the evidence and the draft recommendations are presented by the EREs followed by a discussion and deliberation by the expert panel. The final recommendation, along with the corresponding strength of the recommendation, is decided by the expert panel by consensus building. Finally, *Phase 4* will involve presentations of the CPG in various meetings of medical societies, publication, providing online access to the guideline and its tools, and evaluation of its impact on the way that we care for people with diabetes.

Phase 1: Guideline Preparation

Organization of the Guideline Development Teams

The Guideline Development Teams are composed of the Steering Committee (SC), Technical Working Group/Evidence Review Experts (EREs), the Consensus Panel (CP), and the external Conflict of Interest Review Committee (COIRC).

Steering Committee (SC)

The Philippine College of Endocrinology, Diabetes and Metabolism (PCEDM) is the lead organization for this Clinical Practice Guideline (CPG), having volunteered to develop it in collaboration with the Department of Health (DOH). The Steering Committee is headed by the PCEDM Officers and includes representatives from the Philippine College of Physicians (PCP), the Philippine Heart Association (PHA), and the Philippine Society of Nephrology (PSN). These organizations represent specialists in internal medicine, cardiology, and nephrology, all of whom are frequently consulted in the management of diabetes and its complications.

Evidence Review Experts (EREs)

The organization of the evidence review experts (EREs) is unique because each team was organized from the training core of the various endocrinology fellowship programs of the PCEDM. They were joined by technical experts or methodologists with specific training on clinical epidemiology, guideline development, health technology assessment, and research methods, all of whom were also PCEDM members. The representatives from the training institutions underwent three training workshops to improve their skills and knowledge on guideline development: (1) guideline development using GRADE approach, the ADAPTE process, and the

AGREE tool for appraising CPGs; (2) formulating clinical questions; (3) assessing risk of bias for various clinical questions with focus on using diagnostic tests, systematic reviews with meta-analyses. These workshops were conducted by the Asia Pacific Center on Evidence-Based Medicine and the National Institutes of Health. Convened by the PCEDM, the EREs were tasked with reviewing existing CPGs, summarizing evidence, and drafting evidence-based recommendations.⁵

Consensus Panel (CP)

The members of the consensus panel were identified and invited by the SC as key stakeholders who are either expert in the field of diabetes, target CPG users, advocates for diabetes prevention and management, and persons with diabetes. Major diabetes organizations were represented, such as the PCEDM, Philippine Center for Diabetes Education Foundation, Inc. (PCDEFI), and Diabetes Philippines, alongside non-voting representatives from the Department of Health (DOH) and PhilHealth. Other organizations included the Philippine College of Physicians (PCP), Philippine Heart Association (PHA), Philippine Society of Nephrology (PSN), Philippine Academy of Family Physicians (PAFP), Philippine College of Occupational Medicine (PCOM), and the Philippine Society of General Internal Medicine (PSGIM).

Lay representation came from the Philippine Alliance of Patient Organizations (PAPO), the Chinese General Hospital and Medical Center Diabetes Club, and the St. Luke's Medical Center-Quezon City Diabetes Club, with at least one person with diabetes attending each consensus panel meeting.

The SC, EREs, and CP collaborated to develop and finalize clinical questions and prioritize critical outcomes, ensuring that stakeholders' values, including those of lay participants, were incorporated into the PICO questions. The consensus panel also reviewed evidence summaries, voted on draft recommendations, and finalized the CPG recommendations. A technical facilitator moderated the meetings to guide the consensus process.

Conflict of Interest Review Committee (COIRC)

Finally, COIRC reviewed possible conflicts of interest of all the participants involved in the development of this guideline. They are external to the various organizations and were contracted by the SC to provide an unbiased evaluation of the possible financial/material and intellectual or various types of conflicts of interest of the CP, EREs and the SC.

However, the COIRC was not engaged early enough in the project but evaluated the conflicts of interest while the remaining questions on pharmacotherapy and vaccinations have not yet been presented to the expert panel. Despite this oversight, this CPG incorporated several mitigating strategies to try to address the conflicts of interest.

Strategies to address conflicts of interest

Before the recommendations were developed:

1. Each of the members of the various panels (SC, EREs, CP) accomplished disclosure of conflicts of interest forms at the beginning of the project to individually evaluate any potential or actual conflicts which may introduce biases during the development of the recommendations. The standard form provided by the DOH was used and the declarations

extended back up to 4 years prior to the start of their involvement in this project. They were also asked to submit their curriculum vitae for evaluation by the COIRC.

2. Rather than an individual output of the guideline developer or ERE, the recommendations were the collective work of the group of 3 to 4 persons comprising the group. The output of each training institution was then presented to the bigger group of EREs comprising the technical working group of the PCEDM so that any biases that could have been introduced into the recommendation still also be addressed.
 - a. Those who are noted to have COI were assigned to the topics where they have direct or potential conflicts of interest.
 - b. Each of the EREs, most of whom have some form of primary or secondary COI, were paired up with other members of the team who have no or minimal COI.
 - c. Each of the EREs presented their work to their team to minimize COI as the other members of the team can check the work and ensure that there are no biases that are introduced. Subsequently, the statements were again presented in the larger group of EREs for comments and refinement to ensure that the process or methodology of guideline development was followed, including checking for any introduction of biases.
2. The oversight committee then gave their recommendations as to whether or not members of the CPG development process need to be removed. These were treated as recommendations rather than direct actions.
3. Those who can divest the sources of their conflicts of interest were asked to do so and these were noted.

During the CPG Development:

4. The clinical questions and recommendations were sent to the expert panel before each consensus meeting so that those who had major conflicts of interest (especially the advisory boards of pharmaceutical companies or speakers for products or who owned stocks in hospitals) could identify their conflicts and request their organization to provide a replacement who does not have COI.
5. At the beginning of each meeting, and with each recommendation that was going to be discussed, a declaration of conflicts of interest was called by the facilitator. If anyone in the consensus panel had significant conflicts, they were not allowed to vote. Since the topics were known ahead of time, the organizations that were represented were allowed to change their representative to the voting panel in case the regular representative had conflicts of interest.
6. External review of conflicts of interest– while this was not done early on because of the idea that declaration and management of conflicts is sufficient, this external review was done prior to the discussions on pharmacologic therapy and vaccination.

Generally, these were the strategies used in this CPG to manage conflicts of interest:

1. **Disclosure:** public disclosure of all potential & actual COIs before the start of the project and before each panel meeting.
2. **Divestment:** resignation from any positions that may cause COI or divestment of financial interests; this was identified as a strategy and the technical facilitator did not lecture on any of the topics that were tackled in the guidelines.

3. **Recusal:** several of the members of the expert panel recused themselves from participation in both the discussions and voting after declaring major conflict of interest. Those who participated in discussions always stated their COI and always recused themselves from voting.
4. **Monitoring the output** (draft recommendations) of the EREs by the steering committee, the technical facilitator and the other EREs within their team and the bigger group to ensure that biases are not introduced;
5. **Modification of the plan for development of the recommendations:** e.g. when one of the EREs within the team are assigned to a question where they have COI, then their assignments were revised by the team.
6. **Change of personnel:** Some of the members of the SC were replaced due to significant COI.

On [Annex A](#), are some of the tools used by the group and the COIRC for addressing conflicts of interest.

Development of clinical questions

The clinical questions framework was based on the UNITE for Diabetes Practice Guidelines (2013). While many original questions were retained, the significant time gap necessitated the development of additional, relevant questions. These were formulated by the SC, EREs, and technical or content experts in diabetes, then circulated to the CP for review and revision. The complete list of clinical questions can be found in [Annex B](#).

Selection and Rating of Critical Outcomes

Important clinical outcomes were likewise identified and prioritized, and these were also reviewed by the expert panel including (lay) persons with diabetes plus persons with diabetes other than the representatives in the expert panel for greater inclusivity. This was done through focus group discussions with members of a lay diabetes club in one of the endocrine training hospitals in Manila.

Although the outcomes were identified mainly by the SC, technical experts on diabetes, EREs and the expert (consensus) panel, persons with diabetes were asked for additional outcomes and their relative importance. These were rated using the GRADE categories of importance. The clinical outcomes were rated numerically on a 1-to-9 scale following the GRADE categories, where a score of 7-9 is critical; 4 -6 important; and 1- 3, of limited importance. According to GRADE, ranking outcomes by their relative importance can help to focus attention on those outcomes that are considered most important and help to resolve or clarify disagreements.⁶

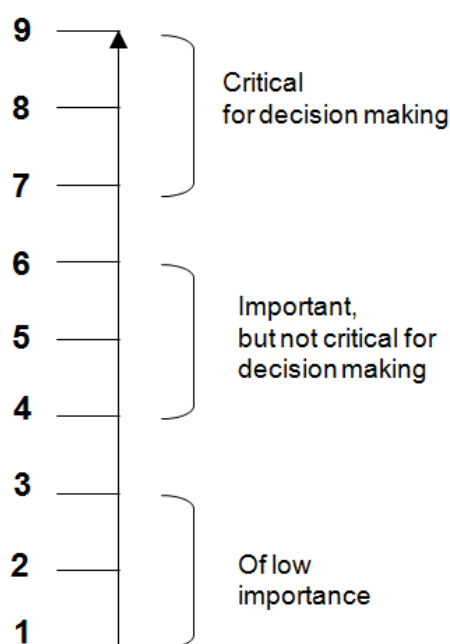


Figure 2. Interpretation of the numerical ranking for critical outcomes.⁶

[Table 2](#) lists the outcomes rated as critical for formulating the guideline recommendations. Since only typically only 7 outcomes per intervention/test are suggested, questions on therapeutics involves only outcomes #2 to #9. For diagnostic test questions, only outcomes #1 to #4 were assessed. Safety in pregnancy (#8) was only included whenever applicable for questions related to pharmacologic interventions.

Table 2. Interpretation of the numerical ranking for critical outcomes.

Outcomes	Score	Importance
1. Diagnostic accuracy (for tests)	9	Critical
2. Glycemic control	9	Critical
3. Microvascular complications	9	Critical
4. Macrovascular complications*	9	Critical
5. Hypoglycemia	9	Critical
6. ADRs especially Hepatotoxicity or Nephrotoxicity	8	Critical
7. Safety in Pregnancy	8	Critical
8. Quality of Life	7	Critical
9. Weight (loss/gain)	7	Critical

For questions that involve the cardiovascular outcomes of drugs (#4), macrovascular complications are further broken down into the following outcomes:

Table 3. Interpretation of the numerical ranking for critical outcomes.

Outcomes	Score	Importance
1. Total mortality	9	Critical
2. Cardiovascular death	9	Critical
3. Fatal and non-fatal myocardial infarction	9	Critical
4. Stroke or cerebrovascular disease	9	Critical
5. Major adverse cardiovascular events (MACE)	7	Critical
6. Coronary revascularization	6	Important

Funding and Independence Statement

This Clinical Practice Guideline (CPG) development project was funded by the Philippine College of Endocrinology, Diabetes, and Metabolism (PCEDM), which served as the lead organization in collaboration with the Department of Health (DOH). The development process was overseen by a Steering Committee led by PCEDM Officers, with representation from the Philippine College of Physicians (PCP), the Philippine Heart Association (PHA), and the Philippine Society of Nephrology (PSN). These organizations collectively contributed their expertise in internal medicine, cardiology, and nephrology—fields integral to the management of diabetes and its complications.

While PCEDM provided financial support and administrative coordination, it did not have exclusive authority over the development of the guideline. The content was determined through a consensus-driven process involving all participating organizations, ensuring a balanced, multidisciplinary, and evidence-based approach. The funding body had no influence on the recommendations, which were formulated based on the best available evidence and expert consensus.

To ensure transparency and safeguard the integrity of the guideline development process, all members of the development group—including the Steering Committee, Oversight Committee, Consensus Panel, and Technical Working Group—were required to complete a Declaration of Conflict of Interest (COI) form prior to their participation. The form required disclosure of any financial or intellectual conflicts of interest within the past four years.

An independent COI Oversight Committee reviewed all declarations, classified the nature and severity of each conflict, and advised the Steering Committee on the appropriate level of participation. The following classifications and management strategies were used:

1. **Allowed / Acceptable** – No relevant financial or intellectual conflicts of interest were identified. Full participation, including voting, was permitted.
2. **Manageable B (Broadcast)** – Typically applies to intellectual conflicts of interest (e.g., affiliations with relevant organizations, positions of leadership, authorship of related publications). Members were allowed to vote but were required to declare these conflicts during en banc meetings and relevant discussions.
3. **Manageable C** – Applied to individuals with both intellectual and limited financial conflicts of interest. These members could contribute to discussions as resource persons but were not allowed to vote on recommendations. Examples include representatives from implementing government agencies or the funding organization. Specific terms of participation were determined by the Oversight Committee, based on the clinical question.
4. **Disqualified** – Applied when an individual had significant financial and intellectual conflicts that could compromise objectivity. These individuals were not permitted to participate in the guideline development process.

This rigorous process for managing conflicts of interest reinforces the editorial independence of the guideline and ensures that the recommendations made in this document are unbiased, evidence-informed, and aligned with the best interests of Filipino patients and healthcare providers.

Management of Conflicts of Interest (COI) in the Steering Committee

The management of conflicts of interest (COI) for the steering committee was guided by principles of transparency, accountability, and integrity. To ensure full disclosure, all COIs of steering committee members were publicly broadcasted. This process allowed stakeholders to be informed of any potential competing interests, fostering trust in the committee's deliberations.

Notably, the overall chair had no conflicts of interest, reinforcing the impartiality of the group's oversight and coordination. Beyond disclosure, the committee implemented additional safeguards to uphold credibility. Any decisions made by the steering committee underwent further vetting to assess the potential perception of bias. This ensured that any concerns related to conflicts were identified and mitigated appropriately. However, it was important to note that the steering committee did not directly formulate recommendations. Instead, the final recommendations were developed by the consensus panel—an independent, multi-stakeholder body composed of experts representing diverse perspectives and disciplines. The consensus panel's decisions were grounded in a rigorous evaluation of the best available evidence and guided by explicit value judgments, ensuring a balanced and methodologically sound approach.

Justification for COI Management Approach

1. **Ensuring Transparency and Public Trust** – Public disclosure of COIs promoted openness, allowing stakeholders to assess any potential influences on decision-making. This aligned with international best practices for governance in evidence-based policymaking, as recommended by organizations such as the WHO, GRADE Working Group, and ICMJE.
2. **Strengthening Decision Integrity** – The absence of COI in the overall chair reinforced neutrality, ensuring that the committee's processes remained free from undue influence. Additionally, the independent vetting of decisions served as a safeguard against perceived bias.
3. **Alignment with Best Practices in Guideline Development** – The separation of roles between the steering committee and the consensus panel adhered to global standards for managing conflicts in clinical and policy guidelines. The presence of a diverse, multi-stakeholder consensus panel with a multiplicity of viewpoints further minimized the risk of any single interest disproportionately influencing the recommendations.
4. **Commitment to Rigorous, Evidence-Based Decision-Making** – The consensus panel formulated recommendations based on systematic evidence synthesis, explicitly weighing benefits, harms, feasibility, and acceptability. This structured approach mitigated the potential impact of conflicts of interest by ensuring that recommendations were primarily driven by empirical data and established value judgments.
5. **Enhancing Credibility and Adoption** – By implementing COI management strategies that exceeded standard disclosure requirements, the process enhanced the legitimacy of the final recommendations. This, in turn, increased stakeholder confidence and facilitated broader adoption of the recommendations in practice.

By employing a multi-layered approach to COI management—including disclosure, vetting, independent decision-making, and diverse stakeholder representation—the committee upheld the highest standards of integrity, ensuring that the recommendations were both scientifically robust and ethically sound.

Steering Committee

No.	Name	Area of Expertise	Affiliation	Sex	Summary of disclosures or other relevant interest
1	Ma. Encarnita C. Blanco-Limpin, MD, FPCP, FPSCCM, FPSSM	Internal Medicine, Pulmonary Medicine, Critical Care Medicine	Philippine Heart Center	F	Acceptable (A) No relevant financial or intellectual conflicts of interest. Full participation allowed.
2	Jude Erric Cinco MD, FPCP, FPCC, FPSCCM, FAsCC, FESC, FACC	Cardiology and Critical Care Medicine	Philippine Heart Association; The Medical City	M	Manageable B Received honoraria as a speaker, module creator, or key opinion leader for Abbott, AstraZeneca, Bayer, Boehringer Ingelheim, Biomedis, Corbridge, Delex, Getz, JustRight, LRI Therapharma, Novartis, Philcare Pharma, Pfizer, and Sanofi (2018–present).
3	Roberto C. Tanchanco, MD, MBA, FPCP, FPSN, FISN	Internal Medicine, Nephrology	Ateneo de Manila University School of Medicine and Public Health ; Philippine Society of Nephrology	M	Manageable B Received honoraria for speaking engagements from AstraZeneca (2021–present) and MSD (prior to 2020).
4	Marichel D. Pile-Coronel, MD, FPCP, FPSN	Internal Medicine- Adult Nephrology	Philippine Society of Nephrology	F	Manageable B Received honoraria for speaking engagements related to Dapagliflozin (Forxiga) from AstraZeneca (2021–2022), and for Vildagliptin (Galvus) and Sacubitril/Valsartan (Entresto) from Novartis (2019–2022).
5	Carolyn Narvacan-Montano, MD, FPCP, FPCEDM	Endocrinology, Diabetes and Metabolism	Makati Medical Center	F	Manageable B Served as a speaker for Sanofi Aventis, Novo Nordisk, Boehringer Ingelheim, and Gx Philippines.
6	Daveric Pagsisihan, MD, FPCP, FPCEDM	Endocrinology, Diabetes and Metabolism	De La Salle Medical and Health Sciences Institute	M	Manageable B Served on the advisory board of Eli Lilly (2018–2019). Sub-investigator in clinical trials sponsored by Sanofi Aventis (2017–2020). Received travel grants from Corbridge, Natrapharm, LRI Therapharma, and Servier Philippines (2017–present). Holds shared stocks in Gentry Medical Center, Dasmariñas City Medical Center, and Asia Medic Family Hospital. Receives income as a member of a pharmacy corporation (2022–present).

Technical Working Group

Question	Specific Assignments	Decision/Management	Assessments/Remarks
Prevention Q4. Should we recommend non-pharmacologic methods (LSM) for preventing diabetes among those with risk factors for diabetes or pre-diabetes?			
Q 4.1 Should we recommend dietary interventions?	Daphne Gayle Galang	A - Allowed – participation with no constraints	
	Ma. Cecilia G. Gonzales	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
	Maria Jocelyn C. Isidro	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
	Aimee Andag-Silva	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Q 4.2 Should we recommend physical activity and exercise?	Aimee Andag-Silva	A - Allowed – participation with no constraints	
	Daphne Gayle Galang	A - Allowed – participation with no constraints	
	Maria Jocelyn C. Isidro	A - Allowed – participation with no constraints	
Q 4.3 Should we recommend limitation of alcohol intake?	Aimee Andag-Silva	A - Allowed – participation with no constraints	
	Daphne Gayle Galang	A - Allowed – participation with no constraints	

Q 4.4. Should we recommend sleep hygiene	Aimee Andag-Silva	A - Allowed – participation with no constraints	
	Daphne Gayle Galang	A - Allowed – participation with no constraints	
Q 4.5. Smoking cessation	Ma. Cecilia G. Gonzales	A - Allowed – participation with no constraints	
Prevention Q5. Should we recommend pharmacologic therapy (metformin, pioglitazone, SGLT-2 inhibitors, acarbose, orlistat, DPP4-inh, GLP-1 RAs) among adults with pre-diabetes?			
Metformin and Pioglitazone Acarbose and Orlistat SGLT-2 inhib and GLP-1/ GIP agonists	Aimee Andag-Silva	D- Disallowed – participation is not allowed (Metformin Pioglitazone; SGLT-2 inhib and GLP-1/ GIP agonists)	Merck Serono Global (Glucophage XR); pioglitazone LRI/ Therapharma; Astrazeneca (Forxiga, Xigduo, Q-tern); Novo Nordisk (Ryzodeg, Ozempic, Rybelsius)
	Aimee Andag-Silva	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
	Olivia P. Roderos	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
	Antonio L. Faltado Jr.	D- Disallowed – participation is not allowed (SGLT-2 inhib and GLP-1/ GIP agonists)	SGLT2i speaker Astrazeneca (Forxiga, Xigduo, Q-tern); Investor (120,000 PhP) at Zure Rx Pharmacy (2022-present)
Mgt Q6.0 Among persons with T2DM, should we recommend non-pharmacologic methods (LSM)?			
Q 6.1 Should we recommend dietary interventions for glucose control, weight management and overall cardiovascular health? Done	Rebecca Lim-Alba	D- Disallowed – participation is not allowed	Novo Nordisk Has stocks (50,000 PhP) in Promedica Pharma (food supplements and nutraceuticals), Exact Check Lab, Accuserve Lab, Intercon Lab (2016-present)

			Proprietor (30,000 PhP) of Accuserve plus, Mediscount Pharmacy (2017-present)
	Mark David Francisco	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Q 6.2 Should we recommend physical activity and exercise for glucose control, weight management and overall cardiovascular health? What are the specific recommendations? [type, duration, frequency, targets]	Rebecca Lim-Alba	D- Disallowed – participation is not allowed	
	Mark David Francisco	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Q 6.3 Should we recommend limitation of alcohol intake? Done	Lora Tan Tin-hay	A - Allowed – participation with no constraints	
Q 6.4. Should we recommend non-nutritive sweeteners (aspartame, sucralose, stevia) for glycemic control and weight loss? Done	Lora Tan Tin-hay	A - Allowed – participation with no constraints	
Q 6.5 Should we recommend KD and intermittent fasting for weight loss and glycemic control?	Daphne Gayle Galang	A - Allowed – participation with no constraints	
Mgt Q7. Among persons with T2DM, should we recommend glycemia-targeted or diabetes-specific formulas			
Q 7.1 for weight management?	Lora Tan Tin-Hay	D- Disallowed – participation is not allowed	LRI-Therapharma (2018-2022); GlucoBest
	Mark David Francisco	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	

Q7.2 for glucose control?	Lora Tan Tin-Hay	D- Disallowed – participation is not allowed	LRI-Therapharma (2018-2022); GlucoBest
	Mark David Francisco	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Mgt Q8. Among persons with T2DM, should we recommend bariatric surgery for weight loss and glycemic control? What are the specific indications for bariatric surgery?	Rebecca Lim-Alba	D- Disallowed – participation is not allowed	Novo Nordisk
	Mark David Francisco	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Mgt Q 9. Among adults with T2 DM, should we recommend metformin as the preferred first line antidiabetic agent?			
Mgt Q 9.1. Among adults with T2 DM, should we recommend metformin as the preferred first line antidiabetic agent as monotherapy? [specify cut-off HbA1c or FBS]	Anna Elvira S. Arcellana	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	Janumet
	Roy Raoul Felipe	D- Disallowed – participation is not allowed	Med Choice
Mgt Q 9.2. Among adults with T2 DM, should we recommend metformin as the preferred first line antidiabetic agent as the base drug for combination therapy?	Anna Elvira S. Arcellana	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	Janumet
	Roy Raoul Felipe	D- Disallowed – participation is not allowed	Med Choice
Q 10.1 Among adults with T2DM, should we recommend initial oral	Gabriel V. Jasul Jr.	D- Disallowed – participation is not allowed	Novo Nordisk

combination therapy for an HbA1c > 9%? HbA1c > 10% ? FBS > 250mg/dL?	Anna Angelica Macalalad-Josue	D- Disallowed – participation is not allowed	Degludec
Q10.2 Among adults with T2DM, should we recommend initial injectable therapy combined with oral agents/ s for an HbA1c > 9%? HbA1c > 10% ? FBS > 250mg/dL?			
10.2.1 Basal Insulin	Gabriel V. Jasul Jr.	D- Disallowed – participation is not allowed	Novo Nordisk
	Anna Angelica Macalalad-Josue	D- Disallowed – participation is not allowed	Degludec
10.2.2 GLP-1 mimetics or agonists	Gabriel V. Jasul Jr.	D- Disallowed – participation is not allowed	Novo Nordisk
	Anna Angelica Macalalad-Josue	D- Disallowed – participation is not allowed	Degludec
Q 11.1 Among adults with T2DM on metformin, should we recommend SU/DPP4-inh/SGLT-2 inhibitors/TZD/alpha-glucosidase inhibitors after failure to achieve glycemic targets?	Karla Kristine S. Fernando	D- Disallowed – participation is not allowed	TReceived travel meeting sponsorship (100,000 PhP/meeting) from Novo Nordisk, Natrapharm (2019) Received honoraria as part of speakers' bureau (10,000 to 25,000 PhP/lecture) from Boehringer, Astra, Novo Nordisk, Sanofi, MSD (2018-2022)
	Harold Henrison C. Chiu	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
	Sahra O. Paragas	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Q 11.2 Among adults with T2DM on 2-3 oral agents and with HbA1c above target, should we recommend	Karla Kristine S. Fernando	D- Disallowed – participation is not allowed	TReceived travel meeting sponsorship (100,000 PhP/meeting) from Novo Nordisk, Natrapharm (2019)

injectable therapy (basal insulin or GLP-1 receptor agonists)?			Received honoraria as part of speakers' bureau (10,000 to 25,000 PhP/lecture) from Boehringer, Astra, Novo Nordisk, Sanofi, MSD (2018-2022)
	Anna Elvira Arcellana	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	Janumet
	Katherine Anne C. Banal	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	Received honoraria for speaking engagements for FGDs from Sanofi, Servier, Astrazeneca (2018-2021)
Mgt Q12. Among adults with T2DM and specific disease conditions, should we recommend specific oral antidiabetic agents or injectables?			
12.1 Cirrhosis/chronic liver disease 12.2 Chronic kidney disease, ESRD on dialysis 12.4 NAFLD 12.5 TB [with recommendations from TB consensus] 12.6 Obese 12.7 Elderly & those with risks for hypoglycemia 12.8 Prior stroke or CVD	Ma. Cecille Anonuevo-Cruz	D- Disallowed – participation is not allowed	Receives honoraria as Advisory Board Member (50,000 PhP) from Zuellig-Lilly, now ZP Therapeutics (2019-present) Received research grant as primary investigator (279,145.75 PhP) from Servier Philippines (2019) Received sponsorship for conventions (120,000 PhP) from Servier Philippines, Astrazeneca, MSD, Boehringer Ingelheim, LRI-Therapharma (2018-2021) Received honoraria as member of speakers' bureau (600,000 PhP) from Servier Philippines, Novo Nordisk, Zuellig-Lilly (now ZP Therapeutics) MSD, Astrazeneca, Getz Pharma, LRI-Therapharma (2018-2022) Part of group investment (100,000

			PhP) in ZureRx, Tanauan, Batangas (2020)
	Mark David Francisco	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Mgt Q14. Among persons with T2DM, should we recommend the following herbal medications and supplements?			
14.1 Chromium	Mia Fojas	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
14.2 Ampalaya [Momordica]	Mia Fojas	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
14.3 Serpentina	Mia Fojas	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
14.4 Vitamin B complex	Mia Fojas	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
14.5 Insulin Plant	Mia Fojas	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Mgt Q15. Among adults with T2DM, should we recommend the following tests for monitoring glycemic control?			

Q15.1 FBS, HbA1c	Elaine C. Cunanan	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	Has hospital stocks (450.000 PhP) in Great Valley Medical Center (still under construction) (2021-present)
Q15.2 1,5-anhydroglucitol test (Glycomark)	Elaine C. Cunanan	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Q15.3 Fructosamine	Elaine C. Cunanan	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Q15.4 SMBG	Elaine C. Cunanan	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Q15.5 CGM	Elaine C. Cunanan	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Mgt Q16. Among adults diagnosed with diabetes, should we recommend the following glycemic targets?			
Q 16.1 FBS < 100 mg/dL (or 110)	Elaine C. Cunanan	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Q 16.2 HbA1c < 7.0%, < 6.5%	Erick Mendoza	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	

Q 16.3 SMBG pre-breakfast < 110 mg/dL (< 100 mg/dL), 1 hr-post-prandial < 140 mg/dL	Aurora G. Macaballug	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	Has hospital stocks (500,000 PhP) in Las Pinas Doctors Hospital, Medical Center Muntinlupa (2010-present)
Q17.1 Among persons with T2DM, should we recommend ASA or clopidogrel for primary prevention of CV events?	Jeremy Jones F. Robles	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Q17.2 Among persons with T2DM with ASCVD or previous acute coronary syndrome, should we recommend ASA and/ clopidogrel for secondary prevention of CV events?	Waynila Lim-Cuizon	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Q18. Among persons with Type 2 DM and hypertension, should we recommend low dose combination drugs for the control of hypertension? [RAS blocker + CCB, RAS blocker + HCTZ] [From Philippine CPG on Hypertension]	Cecilia A. Jimeno	D- Disallowed – participation is not allowed	Sanofi - Irbesartan
Q19.1 Among persons with Type 2 DM and dyslipidemia, should we recommend statins for primary prevention of CV events? [from Phil CPG ON Dyslipidemia]	Cecilia A. Jimeno	D- Disallowed – participation is not allowed	Sanofi - generic STATIN; Member of the technical review board of the UP NIH FDA and Board Member of the Philippine Lipid and Atherosclerosis Society
Q 19.2 Among persons with Type 2 DM and dyslipidemia, should we recommend fibrates for primary prevention of CV events? [from Phil CPG ON Dyslipidemia]	Cecilia A. Jimeno	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Q20.1 Among adults with T2DM and established atherosclerotic cardiovascular disease (ASCVD), should we recommend SGLT-2 inhibitors?	Waynila Lim-Cuizon	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	

Q20.2 Among adults with T2DM and established atherosclerotic cardiovascular disease (ASCVD), should we recommend GLP-1 Receptor agonists in combination with metformin? [from Philippine consensus recommendations]	Celito A. Tamban	A - Allowed – participation with no constraints	
Vac Q21.1 Among adults with T2DM, should we give influenza vaccine yearly for the prevention of respiratory tract infections?	Mia Fojas	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	
Vac Q21.2 Among adults with T2DM, should we give pneumococcal vaccination for the prevention of hospitalization from bacterial pneumonia?	Mia Fojas	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);	

Expert Panel or Consensus Panel

Name	COI Declaration	COI Assessment
Dr. Maria Rosario Sylvia Uy	Nothing to disclose	A - Allowed – no constraints
Dr. Charisse Banaag	Nothing to disclose	A - Allowed – no constraints
Dr. Bernice Ong-Dela Cruz	Nothing to disclose	A - Allowed – no constraints
Dr. Rowena Roselle P. Blanco-Santos	Nothing to disclose	A - Allowed – no constraints
Ms. Agnes Angelo Aquino	Nothing to disclose	A - Allowed – no constraints
Ms. Zenaida Velasco	NDAP, expert lectures on nutrition	For questions on diet and nutrition - B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); All other questions: A - Allowed – no constraints
Dr. Gerald B. Natanauan	Nothing to disclose	A - Allowed – no constraints
Dr. Cynthia H. Manabat	Receives honorarium as Advisory Board member of Boehringer Ingelheim, Novo Nordisk, Sanofi, Zuellig-Lilly (2018–2022) Receives fees for equipment/travel to meetings from Boehringer Ingelheim, Cathay YSS, Corbridge, Kaufmann, LRI-Therapharma, Natrapharm, Novo Nordisk, Sanofi, Servier, Westmont, Zuellig-Lilly (2018–2022) Receives honorarium as Lecturer for Boehringer Ingelheim, Cathay YSS, Corbridge, Kaufmann, LRI-Therapharma, Natrapharm, Novo Nordisk, Sanofi, Servier, Westmont, Zuellig-Lilly (2018–2022) Secretary of the Board, Philippine Center for Diabetes Education Foundation (NGO) (2018–present)	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (e.g., DPP4, SGLT2, GLP1RA)
Ms. Ma. Regina Rosario 'Gina' Javier	Board of Trustee, RAINBOW Camp Foundation Philippines, Member of Philippine Alliance of Patients Organizations	B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);
Dr. Waynila Lim-Cuizon	Received honoraria for RTDs on Gliclazide (SU) MR, Gemigliptin (DPP4) from GX Pharma, Med Choice (2020-2021)	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)

Dr. Monica Therese Cating-Cabral	Travel meetings paid by LRI-Therapharma (March 2019) Received honoraria for speaking engagements from AstraZeneca, Johnson & Johnson, Novo Nordisk, Sanofi, Medtronic (November 2019 – October 2022) Board of Trustee, Philippine College of Endocrinology, Diabetes and Metabolism a. Holds stocks in Makati Medical Center b. Holds stocks in Rinon Care Pharma, Inc.	Manageable B for non-pharmacologic intervention questions D – disallowed for pharmacologic, diagnostic, and screening questions
Dr. Walid Amil	Receives honorarium as Advisory Board member of Boehringer Ingelheim Receives honorarium as Speaker for LRI-Therapharma, Servier, AstraZeneca, Novartis, and Boehringer Ingelheim (2018–2022) Board Member, Hospital de Zamboanga, Zamboanga City (2018–present) Treasurer, Philippine Heart Association (2020–present) Holds stocks in hospitals and a diagnostic laboratory	Manageable B for non-pharmacologic intervention questions D – disallowed for pharmacologic/treatment, diagnostic, and screening questions
Dr. Rebecca Lim-Alba	Received honoraria for lectures/RTDs from Abbott, Astra, Boehringer Ingelheim, LRI, Menarini, MSD, Novo Nordisk, and Westmont (2018–2022) Has stocks in Promedica Pharma, Exact Check Lab, Accuserve Lab, and Intercon Lab (2016–present) Proprietor of Accuserve Plus and Mediscount Pharmacy (2017–present) Vice Chair, Filipino-Chinese Charitable Foundation Receives honoraria for lectures on SGLT2 inhibitors, novel insulin, and GLP-1 agonists from Astra, Boehringer Ingelheim, Novo Nordisk, Menarini, and MSD (2017–present)	Manageable B for non-pharmacologic intervention questions D – disallowed for pharmacologic/treatment, diagnostic, and screening questions
Dr. Jerome R. Barrera	Received honoraria as Advisory Board member of Eli Lilly (2019 only) Received airfare, accommodation, and registration for conventions from Sanofi, Natrapharm, Cathay, Getz Pharma, Scigen, Corbridge, Kauffman, AstraZeneca, GX Pharma (2018–2022) Received honoraria for CME lectures from GX Pharma, Sanofi Aventis, Eli Lilly, Novo Nordisk, Getz Pharma, Multicare, Scigen Pharma, Cathay, Novartis, Servier, Boehringer, AstraZeneca, Merck, Abbott, Nestlé (2018–2022) Received stock fees from Healthwise Laboratory and Rite Drug Pharmacy (2021–2022)	Manageable B for non-pharmacologic intervention questions D – disallowed for pharmacologic/treatment, diagnostic, and screening questions

	Board of Trustee, Philippine College of Endocrinology, Diabetes and Metabolism	
Dr. Ma. Theresa M. Chua-Agcaoili	Sponsored in conventions by Boehringer Ingelheim Receives honoraria for lectures on SGLT-2 and GLP-1 round table discussions from AstraZeneca, Boehringer Ingelheim, and Novo Nordisk Member of the Medical Advisory Council for Ayala Healthcare	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Elizabeth Angelica Lapid-Roasa	Receives honoraria for module development from Boehringer Ingelheim (2017–present) and AstraZeneca	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Katherine Anne C. Banal	Received honoraria for speaking engagements for FGDs from Sanofi, Servier, Astrazeneca (2018-2021)	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Richard Henry Tiongco	Receives honoraria for speaking engagements from Boehringer-Ingelheim and Astrazeneca (2020-2022) Director II, Philippine Heart Association (non-stock, non-profit organization)	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions

		D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Erick Mendoza	Vice President, Philippine Thyroid Association (2022–present) Received honoraria for lectures/RTDs from Sanofi (2018–2022), Novo Nordisk (2020–2022), AstraZeneca (2019–2021), Boehringer Ingelheim (2021–2022), and GX International (2021–2022) Speaker for Lantus, Toujeo, Apidra, Soliqua (2018–2022); Tresiba, Ryzodeg (2020–2022); Forxiga, Xigduo (2019–2021); Jardiance and Jardiance Duo (2021–2022); Zemiglo (2021–2022)	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Endrik H. Sy	Government employee with private clinic connected to a pharmacy and diagnostic laboratory	For screening and diagnostic questions - D-Disallowed – participation is not allowed For other questions, B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);
Dr. Michael L. Villa	Receives honoraria for speaking engagements from Novo Nordisk, Unilab, Servier (2019–2022), AstraZeneca, and Abbott	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Jeremy Jones F. Robles	Speaker AZ, Abbott	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Franklin Guillano	Founding members of Tagum Global Medical Center in Tagum City. It is a member of ACE	For screening and diagnostic questions - D-Disallowed – participation is not allowed

	GROUP of Hospitals. Part ownership of 3 dialysis centers in Davao region under the Nephro group of companies.	Manageable B for non-pharmacologic intervention questions For other questions, B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions);
Dr. Reynaldo Rosales	Receives honoraria as Medical Advisory Board member of Novo Nordisk (1 year) Receives honoraria for speaking engagements from Novo Nordisk, Boehringer Ingelheim, Sanofi, and Nestlé Health Science Board of Director, Diabetes Philippines	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Grace K. de los Santos	Receives honoraria as adviser and for speaking engagements from AstraZeneca, Boehringer Ingelheim, Corbridge, LRI-Therapharma, Menarini, Merck, Novartis, Novo Nordisk, OEP, and Sanofi (2019–present)	For screening and diagnostic questions B - Broadcast - Manageable with minor constraints – there are COIs: participation is allowed with minor constraints (e.g., – COIs declared aloud during panel discussions); Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)
Dr. Aurora G. Macaballug	Received travel sponsorship from LRI/Therapharma, Natrapharm, and Corbridge (2018–2019) Received honoraria for speaking engagements from AstraZeneca, MSD, Novo Nordisk, Zuellig/Eli Lilly, Boehringer Ingelheim, LRI, Natrapharm, and Sandoz (2018, 2019, 2020, 2022) Has hospital stocks in Las Piñas Doctors Hospital and Medical Center Muntinlupa (2010–present) Advisory Board Member for dulaglutide, Zuellig Secretary, Philippine College of Endocrinology, Diabetes and Metabolism Head of Advocacy, Philippine College of Endocrinology, Diabetes and Metabolism	For screening and diagnostic questions - D-Disallowed – participation is not allowed Manageable B for non-pharmacologic intervention questions D – Disallowed for questions on pharmacologic therapy (DPP4, SGLT2, GLP1RA)

Phase 2: Evidence synthesis

The technical experts and the EREs, working with the SC decided on two main frameworks for guideline development. The first is to use ADAPTE in order to make use of the evidence in existing and updated guidelines that have similar clinical questions to inform the current guideline development, and the second is to use GRADE for many of the de-novo clinical questions.

ADAPTE framework

A systematic literature search for existing CPGs was done using MEDLINE through PubMed (<https://pubmed.ncbi.nlm.nih.gov/>). The initial search strategy and the results are summarized in [Table 4](#). A team of 4 EREs then refined the search in order to have an optimal output.

Table 4. Literature search on MEDLINE.

Search number	Query	Search Details	Results
3	(#2) AND (("2017"[Date - Publication] : "3000"[Date - Publication]))	"guideline"[Publication Type] AND ("diabetes mellitus, type 2"[MeSH Terms] OR "diabetes mellitus, type 2"[MeSH Terms] OR "diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "diabetes mellitus type 2"[All Fields] OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields])) AND 2017/01/01:3000/12/31[Date - Publication]	78
2	("guideline"[Publication Type]) AND (#1)	"guideline"[Publication Type] AND ("diabetes mellitus, type 2"[MeSH Terms] OR "diabetes mellitus, type 2"[MeSH Terms] OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "diabetes mellitus type 2"[All Fields] OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields]))	361
1	((("Type 2 diabetes mellitus"[MeSH Terms]) OR (diabetes mellitus, type 2[MeSH Terms])) OR (diabetes mellitus, Type 2)) OR (Type 2 diabetes mellitus)	"diabetes mellitus, type 2"[MeSH Terms] OR "diabetes mellitus, type 2"[MeSH Terms] OR "diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "diabetes mellitus type 2"[All Fields] OR "diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields]	163,540

Additional guidelines were identified from cross-referencing and consultation with the rest of the EREs, which yielded a total of 86 reports after deduplication.

Eligibility criteria for guidelines

The team of four EREs then applied the following eligibility criteria to identify CPGs eligible for inclusion and further appraisal using AGREE ([Table 5](#)). From the initial 86 guidelines, the list was then narrowed to 26 with the following distribution: (a) general guidelines (n=13), (b) specific guidelines on pharmacologic therapy (n=9), and (c) guidelines specifically tackling complications (n=4).

Table 5. Eligibility criteria for guidelines included for adaptation.

Inclusion criteria	Exclusion criteria
1. Guideline must be about diabetes in the outpatient setting 2. General guidelines covering the entire scope of diabetes with focus on those which include the management of Type 2 diabetes in an outpatient setting 3. Published (in text or online) since the details of the review must be available 4. Written in English or with English translation 5. Published in the last five years (2017-onwards) to ensure that evidence base is current. In case that the guideline has an update, then both the original guideline and the update will be retrieved and reviewed. 6. Only evidence-based guidelines will be included (guideline must include a report on systematic literature searches and explicit links between individual recommendations and their supporting evidence) 7. Only national and/or international guidelines will be included (see exclusion #2)	1. For duplicate guidelines (e.g., update or revision of previous guidelines), reviewers will only consider the most recent guidelines 2. Guidelines commissioned by or published by HMO's will not be included since the intent and the use of these guidelines is different from the intended users of this guideline 3. Guidelines for special situations which may be unique to the western population will not be included e.g., care of institutionalized patients, homeless, nursing homes, etc. 4. Guidelines written by a single author not on behalf of an organization; in order to be valid and comprehensive, a guideline ideally requires multidisciplinary input. 5. Guidelines published without references – as the panel needs to know whether a thorough literature review was conducted and whether current evidence was used in the preparation of the recommendations

Appraisal of guideline quality

The EREs were divided into teams according to their training institutions, and the 26 guidelines then underwent appraisal using the [AGREE II tool](#). While there are no specific guidelines for the “cut-off” or passing scores in the AGREE instrument, the EREs decided on the following criteria:

1. Both of the reviewers have to have at least a final score of 5 and answer YES (or yes with Modification) that they will use the guideline, and for the
2. Rigour criteria to have at least a score of 70% and higher

After applying AGREE, the final list of guidelines was narrowed to 15 as follows: (a) general guidelines (n=8); (2) specific guidelines on pharmacologic therapy (n=4); and (3) guidelines specifically tackling complications (n=3). The final list of the 15 included guidelines is listed in [Annex C](#). These guidelines were used as the source of the evidence base.

De novo systematic reviews

Whenever the listed CPGs were not able to provide direct evidence or address the same guideline question, de novo systematic reviews and meta-analysis were performed. Aside from cross-referencing from the international guidelines, systematic literature search was done for online databases including MEDLINE via PubMed (<https://pubmed.ncbi.nlm.nih.gov/>), Google Scholar, as well as COCHRANE database (<https://www.cochranelibrary.com/central> and <https://www.cochranelibrary.com/cdsr/reviews>). Local databases such as HERDIN and the PCEDM list of researches were also reviewed for studies relevant to the clinical question. A separate search strategy was done to obtain cost-effectiveness studies or similar studies on the treatment of Type diabetes.

Systematic literature search

Keywords for the systematic literature search were derived from the PICO (Population, Intervention/Exposure, Comparison, Outcomes) framework for each clinical question, utilizing both MeSH terms and free-text searches. When necessary, EREs contacted authors of relevant studies, including local research, to obtain copies or clarify study details for appraisal. Search strategies for each clinical question are detailed in the subsequent chapter.

For diagnostic questions, the included studies comprised cross-sectional (simple or criterion-referenced), cohort, case-control studies, and diagnostic systematic reviews and meta-analyses. For therapy-related questions, the review incorporated observational studies (cohorts and case-control), randomized controlled trials (RCTs), systematic reviews, meta-analyses, and network meta-analyses.

Quality assessment of studies included in the review

Individual studies, including RCTs, diagnostic accuracy studies, and studies on harm, were evaluated using appraisal guides tailored to each study type.⁷ The QUADAS tool⁸ was applied for assessing the quality of diagnostic accuracy studies, while the Newcastle-Ottawa Scale⁹ was used for evaluating non-randomized clinical and observational studies.

Evidence synthesis using GRADE approach

The primary framework for developing this CPG is **GRADE**, a widely used tool for both grading the quality of evidence and formulating recommendations. GRADE offers a transparent, systematic approach for summarizing evidence and making clinical practice recommendations.¹⁰⁻¹²

The process of guideline development begins with formulating clinical questions in the PICO format, which includes the population that the question applies to, the two or more alternatives (the comparison), and the outcomes that matter most to those faced with the decision.¹³ The best estimates of treatment effects are determined for each outcome, ideally through systematic reviews and meta-analyses.¹⁴ The EREs then rate the certainty of evidence for each outcome. An overall GRADE quality (or certainty) rating is given to the body of evidence, taking into account the lowest quality of evidence from outcomes critical to decision-making. The quality of evidence reflects the extent to which our confidence in an estimate of the effect is adequate to support a particular recommendation.¹⁴

GRADE classifies evidence into four levels of certainty ([Table 6](#)): very low, low, moderate, and high. For intervention or pharmacologic treatment questions, evidence from randomized controlled trials starts as high quality, while evidence from observational studies starts as low quality due to potential residual confounding.

Table 6. GRADE Categories of Certainty of Evidence.

Certainty of Evidence	Interpretation
Very low ⊕○○○	The true effect is probably markedly different from the estimated effect [the estimate of the effect is very uncertain].
Low ⊕⊕○○	The true effect might be markedly different from the estimated effect [further research is very likely to have an important impact].
Moderate ⊕⊕⊕○	The true effect is probably close to the estimated effect [further research is likely to have an important impact].
High ⊕⊕⊕⊕	The authors have a lot of confidence that the true effect is similar to the estimated effect [further research is unlikely to change our confidence in the estimate of the effect].

After determining the baseline certainty of evidence (e.g. high for data from RCTs), the certainty can be rated up or down based on various domains. Certainty can be rated down for the following reasons: serious risk of bias (as obtained from appraisal of individual studies), imprecision, inconsistency, indirectness, and publication bias. Conversely, certainty can be upgraded in the presence of a large magnitude of effect, dose-response gradient, or when confounding is deemed to have no effect on the results.

As GRADE is used to rate the body of evidence at the *outcome level* rather than the study level, EREs must make a judgement about whether the risk of bias in the individual studies is sufficiently large that their confidence in the estimated treatment effect is lower. Imprecision is rated down if the 95% confidence interval (CI) suggests that clinical decisions might differ depending on whether the true effect lies at the upper or lower CI.¹⁵ Similarly, certainty may be downgraded if evidence comes from only a few small studies or if there are few events.¹⁶ Inconsistency across the studies involves evaluating the similarity of point estimates and the overlap of their confidence intervals, usually through statistical tests for heterogeneity (e.g., the I^2 and chi-squared test).¹⁷ Indirectness involves comparing the included studies with the population, intervention, or outcomes of interest for the guideline question. Certainty can be rated down if the populations or patients studied are different from those for whom the recommendations apply (e.g., no Filipinos or even Asians are included in the studies, or when the interventions studied are different than the real outcomes, the outcome in the studies is a surrogate for a different outcome).¹⁸ For publication bias, inferences are made about the missing evidence. Visual methods such as the funnel plot in meta-analyses and various statistical methods are helpful in detecting publication bias. In assessing publication bias, consider the following: what is the study design (experimental vs. observational), study size (small studies vs. large studies), lag bias (early publication of positive results), search strategy (was it comprehensive?) and asymmetry in the funnel plot.¹⁹

Phase 3: Evidence-to-Decision

Generating recommendations

GRADEpro online software (<https://www.gradepro.org>) online was used to generate the Summary of Findings (SoF) tables for each outcome. The EREs present the SoF tables to the expert panel, summarizing the results for each of the relevant outcomes and giving an over-all certainty of evidence. An Evidence to Decision (EtD) framework was used to process the evidence to develop recommendations, which takes into consideration several domains that contribute to the strength of the recommendation ([Table 7](#)).

Determining the direction strength of recommendations

During the consensus meetings, the wording of the recommendations as well as the strength of each recommendation was determined by the panel by considering the domains that are given in [Table 7](#). A strong recommendation means that the panel is “*confident that the desirable effects of adherence to a recommendation outweighs the undesirable effects*,” while a weak recommendation means that the “*desirable effects of adherence to a recommendation probably outweigh the undesirable effect*” but the panel cannot say so with confidence.

Table 7. Domains that contribute to the strength of a recommendation.

Domain	Comment
Balance between desirable and undesirable outcomes (trade-offs) e.g. prevention of complications of diabetes versus adverse effects of drugs, taking into account: - Best estimates of the magnitude of effects on desirable & undesirable outcomes - Importance of outcomes (estimated typical values and preferences).	The larger the differences between desirable and undesirable outcomes, the more likely a strong recommendation is warranted. The smaller the net benefit and the lower the certainty for that benefit, the more likely is a weak recommendation warranted.
Confidence in the magnitude of estimates of effect of the interventions on important outcomes (overall quality of evidence for outcomes)	The higher the quality of evidence, the more likely a strong recommendation is warranted.
Confidence in values and preferences, and their variability	The greater the variability in values and references, or uncertainty about typical values and preferences, the more likely a weak recommendation is made.
Resource use.	The higher the cost of an intervention (the more resources consumed), the less likely a strong recommendation is warranted.

Source: Schünemann H, Brożek J, Guyatt G, and OxmanA (eds). GRADE Handbook. October 2013.

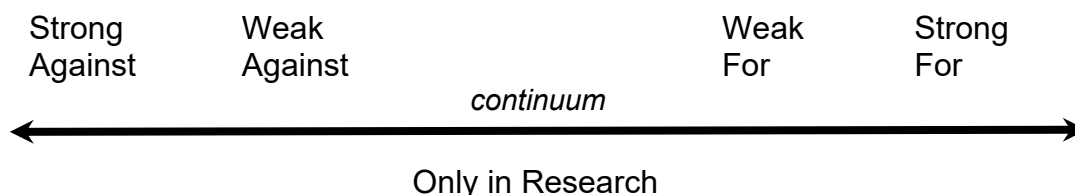


Figure 3. Strength of recommendation: a continuum divided into categories.

Consensus process

The recommendation for each question and the strength of each recommendation were determined through voting. A consensus decision was reached if 75% of all CP members agreed. In case that consensus was not reached in the first voting, questions and discussion were encouraged. Afterwards, another round of voting was done and if still, no consensus is reached after voting, then the recommendation is tabled for discussion at the next meeting. Evidence-based recommendations were based on input arrived at by consensus in the en banc discussions. The recommendations presented by the evidence-based reviewers were moved forward and revised depending on the decision of the consensus panel during voting.

In cases where direct evidence is lacking and the consensus panel deems it necessary that a statement be articulated regarding the intervention in question, a good practice statement is included. A good practice statement implies that there is general agreement that the intervention will do more good than harm.

Table 8. Summary of Recommendations.

No.	Recommendations	Overall Certainty of Evidence	Strength
Q1. Should we recommend non-pharmacologic interventions — including weight loss, dietary interventions, physical activity and exercise, proper sleep hygiene, smoking cessation, and avoidance of excessive alcohol intake — among adults at risk for diabetes or with pre-diabetes?			
1	Recommendation 1. We recommend a minimum weight loss of at least 5 to 7% body weight through diet and increased physical activity to delay or prevent diabetes among adults who are at risk for diabetes or with pre-diabetes who are overweight and/or obese. In patients with BMI of less than 23 kg/m ² , weight maintenance is advised.	High	Strong
2	Recommendation 2. We recommend an individualized approach to diet, focusing on the reduction in total caloric intake, rather than a specified macronutrient distribution to achieve weight loss goals among adults who are at risk for diabetes or with pre-diabetes. In general, a diet rich in vegetables and fiber is encouraged, along with a reduced intake of refined carbohydrates, sugars, alcohol and fats.	High	Strong
3	Recommendation 3. We recommend at least 150 mins of moderate intensity exercise to be distributed at least three times a week among adults who are at risk for diabetes or with pre-diabetes.	High	Strong
4	Recommendation 4. We recommend seven to eight hours of sleep at night among adults who are at risk for diabetes or with pre-diabetes	Low	Strong
5	Recommendation 5. We recommend avoidance of excess* alcohol intake among adults who are at risk for diabetes or with pre-diabetes <i>*Excess alcohol intake refers to binge drinking, defined as consumption of more than 63 grams of alcohol per day. A dose–response relationship was observed, with the lowest risk of diabetes seen at alcohol consumption of approximately 10 to 14 grams per day (equivalent to about one standard drink, where one drink contains 14 grams of alcohol).</i>	Low	Strong
6	Recommendation 6. We recommend smoking cessation with active measures to prevent weight gain among adults who are at risk for diabetes or with pre-diabetes	Low	Strong
Q2. Should we recommend pharmacologic therapy (metformin, pioglitazone, SGLT-2 inhibitors, acarbose, orlistat, GLP-1 RAs) among adults with pre-diabetes?			
7	Recommendation 7. We recommend pharmacologic therapy among adults with prediabetes who do not respond to lifestyle modification	Moderate	Strong

No.	Recommendations	Overall Certainty of Evidence	Strength
8	Recommendation 8. We recommend metformin at a dose of 0.5 to 1.7 grams/ day among adults with pre-diabetes who do not respond to lifestyle change* *The following patients who are proven to benefit the most from long term metformin treatment: women with prior history of gestational diabetes, those with fasting glucose > 110 mg/dL, those with HbA1c 6 to 6.4%, adults less than 60 years old, and those with obesity especially BMI > 35 kg/m2	Moderate	Strong
9	Recommendation 9. We recommend assessing the response by monitoring 75-gram OGTT or HbA1c every 6 months among adults with prediabetes.	Moderate	Strong
10	Recommendation 10. We suggest using other pharmacologic agents among patients who cannot tolerate or who have contraindications to metformin; or have co-existing co-morbidities of pre-diabetes that benefit from specific agents as tabulated (Table 9).	Low	Weak
Q3. Should we recommend dietary interventions among adults with Type 2 DM?			
11	Recommendation 11. We recommend an individualized medical nutrition therapy to achieve better glycemic control, weight management, and overall cardiovascular health among adults with Type 2 DM	Low	Strong
12	Recommendation 12. We recommend referral to a dietitian/nutritionist/diabetes educator who is specially trained and knowledgeable on diabetes nutrition to provide individualized nutritional counselling among adults with Type 2 DM	Low	Strong
Q4. Should we recommend physical activity and exercise for glucose control, weight management and overall cardiovascular health among adults with Type 2 DM?			
13	Recommendation 13. We recommend increased physical activity*, moderate to vigorous aerobic, resistance, or combined aerobic and resistance exercises** of at least 150 minutes per week for glucose control, weight management, and over-all cardiovascular health. * physical activity, defined as any bodily movement produced by skeletal muscles that results in energy expenditure (e.g., walking, occupational, household, or recreational activities) ** exercise, defined as a planned, structured, and repetitive subset of physical activity	Low	Strong
Q5. Should we recommend limitation of alcohol intake among adults with Type 2 DM?			

No.	Recommendations	Overall Certainty of Evidence	Strength
14	Recommendation 14. We suggest against the intake of alcohol among patients with Type 2 DM.	Very Low	Weak
Q6. Should we recommend the use of non-nutritive sweeteners among adults with Type 2 DM?			
15	Recommendation 15. We recommend against the long term (more than 6 months) high intake of non-nutritive sweeteners among adults with T2DM due to lack of long-term safety data and suggestion of possible link to increased mortality, stroke, cardiovascular events, and cancer seen with artificially sweetened beverages and other highly processed foods.	Very Low	Strong
16	Recommendation 16. We suggest allowing the use of non-nutritive sweeteners (NNS) as a short-term (less than 6 months) alternative for patients with T2DM accustomed to sweet beverages, provided that the consumption is below acceptable daily intake, and there is no compensatory increased intake of additional calories and carbohydrates from other food sources.	Very Low	Weak
Q7. Should we use diabetes-specific nutrition formulas among adults with type 2 DM?			
17	Recommendation 17. We suggest that diabetes-specific nutrition formulas (DSNF) be used as meal or snack replacement for adults with type 2 diabetes mellitus to minimize post-prandial glucose excursions and improve overall glycemic control in the context of achieving their optimal daily caloric requirements.	Low	Weak
Q8. Should we recommend referral for bariatric surgery to achieve sustained glycemic control and weight loss among obese adults with type 2 DM?			
18	Recommendation 18. We suggest referral for bariatric surgery among adults with type 2 diabetes and obesity w/ BMI ≥ 37.5 kg/m ² and those with BMI ≥ 32.5 -37.4 kg/m ² who do not achieve desired glycemic control and weight loss with medical therapy.	Low	Weak
Q9. Should we recommend HbA1c to monitor glycemic control among nonpregnant adults with type 2 DM?			
19	Recommendation 19. We recommend the use of laboratory-measured HbA1c to monitor glycemic control among nonpregnant adults with T2DM	High	Strong
20	Recommendation 20. We recommend Hba1c testing every 3 months until HbA1c is stable on unchanging therapy, then every 6 months once the HbA1c level and blood glucose lowering therapy are stable.	Moderate	Strong
Q10. Should we use fasting plasma glucose (FPG) to monitor glycemic control among nonpregnant adults with type 2 DM?			

No.	Recommendations	Overall Certainty of Evidence	Strength
21	Recommendation 21. We recommend the use of FPG in conjunction with HbA1c and/or SMBG to monitor glycemic control among nonpregnant adults with T2DM	Moderate	Strong
22	Recommendation 22. We suggest against the routine use of FPG alone to monitor glycemic control among nonpregnant adults with T2DM	Moderate	Weak
23	Recommendation 23. We suggest FPG monitoring of response to therapy, determination of the risk for nocturnal hypoglycemia, and investigation of discordant levels of blood glucose monitoring and HbA1c among nonpregnant adults with T2DM	Low	Weak
Q11. Should we recommend the use of 1,5-anhydroglucitol (1,5-AG) to monitor glycemic control among nonpregnant adults with type 2 DM?			
24	Recommendation 24. We suggest against the use of 1,5-AG to monitor glycemic control among nonpregnant adults with T2DM.	Very Low	Weak
Q12. Should we use fructosamine to monitor glycemic control among adults with type 2 DM			
25	Recommendation 25. We suggest against the use of fructosamine to monitor glycemic control among adults with type 2 DM.	Very Low	Weak
Q13. Should we do SMBG to monitor glycemic control among nonpregnant adults with type 2 DM?			
26	Recommendation 26. We recommend self-monitoring of blood glucose (SMBG) for individuals on insulin therapy among nonpregnant adults with type 2 DM.	Moderate	Strong
27	Recommendation 27. We suggest self-monitoring of blood glucose (SMBG) among nonpregnant adults with T2DM even if they are not insulin if they: • have poor glycemic control (HbA1c >8%) • are at increased risk of hypoglycemic episodes • have an acute illness • will be started on corticosteroids • have HbA1c results that do not accurately reflect glycemia • are planning to become pregnant	Low	Weak
28	Recommendation 28. We suggest self-monitoring of blood glucose (SMBG) at least once a day before breakfast	Low	Weak

No.	Recommendations	Overall Certainty of Evidence	Strength
	among nonpregnant adults with T2DM who are on once-daily insulin		
29	Recommendation 29. We suggest self-monitoring of blood glucose (SMBG) at least twice a day before breakfast and before dinner among nonpregnant adults with T2DM who are on twice-daily insulin	Low	Weak
30	Recommendation 30. We suggest that the frequency of SMBG be individualized depending on the type of hyperglycemic agents, level of glycemic control, and risk of hypoglycemia among nonpregnant adults with T2DM not on insulin	Low	Weak
Q14. Should we use a continuous glucose monitor (CGM) to monitor glycemic control among nonpregnant adults with type 2 DM?			
31	Recommendation 31. We suggest CGM among nonpregnant adults with T2DM on insulin therapy in the following situations: • There is suspected nocturnal hypoglycemia or hypoglycemia unawareness. • There is suspected dawn phenomenon (early morning hyperglycemia due to the diurnal increase in cortisol or growth hormone) or Somogyi phenomenon (early morning hyperglycemia due to a rebound effect from late-night hypoglycemia). • The HbA1c levels are still not at target despite being on multiple daily insulin injections. • There is discordant SMBG and HbA1c results.	Low	Weak
Q15. Should we use aspirin among adults with type 2 DM for primary prevention of cardiovascular events?			
32	Recommendation 32. We suggest against the use of aspirin for primary prevention of cardiovascular events among patients with type 2 diabetes mellitus.	Very Low	Weak
Q16. Should we use aspirin among adults with type 2 DM for secondary prevention of cardiovascular events?			
33	Recommendation 33. We suggest the use of aspirin for secondary prevention of cardiovascular events among persons with type 2 diabetes mellitus.	Low	Weak
Q17. Should we use clopidogrel for the secondary prevention of cardiovascular events among adults with type 2 DM?			
34	Recommendation 34. We suggest the use of clopidogrel as an alternative in the setting of aspirin allergy among adults with type 2 DM and established cardiovascular disease.	Low	Weak

No.	Recommendations	Overall Certainty of Evidence	Strength
Q18. Should we recommend low dose combination drugs for the control of hypertension among adults with type 2 DM and hypertension?			
35	Recommendation 35. We recommend the use of low-dose combinations of a RAAS blocker (ACEi or ARB) with a CCB or thiazide/thiazide-like diuretic, preferably using a single-pill combination (SPC) to reach a blood pressure target of < 130/80 mm Hg among adults with type 2 DM. Free tablet combinations may also be given if SPCs are not available.	Moderate	Strong
36	Recommendation 36. We recommend against the combination of ACEi and ARB among adults with type 2 DM due to a higher risk of hyperkalemia and renal failure.	Moderate	Strong
Q19. Should we recommend statins for primary prevention of cardiovascular (CV) events among adults with type 2 DM?			
37	Recommendation 37. We recommend the use of moderate intensity statins for the primary prevention of cardiovascular events among adults with type 2 DM, even without evidence of atherosclerosis (ASCVD).	Moderate	Strong
Q20. Should we use sodium-glucose cotransporter-2 (SGLT2) inhibitors among patients with type 2 DM with established atherosclerotic cardiovascular disease (ASCVD)?			
38	Recommendation 38. We recommend the use of SGLT2 inhibitors among adults with type 2 DM and established cardiovascular disease or elevated cardiovascular risk.	Moderate	Strong
Q21. Should we use GLP-1 Receptor agonists (GLP-1 RA) among adults with type 2 DM and established atherosclerotic cardiovascular disease (ASCVD)?			
39	Recommendation 39. We recommend the use of GLP-1 RA with proven CV benefit* among adults with type 2 DM and established ASCVD. *GLP-1 RA with proven CV benefits includes Liraglutide, Semaglutide Subcutaneously, Dulaglutide, Albiglutide and Efpeglenatide. *In the Philippines, the available GLP-1 RA with CV benefits includes Liraglutide, Semaglutide Subcutaneously and Dulaglutide.	Moderate	Strong

Table 9: PHARMACOTHERAPY CHOICES for DM Prevention

Drug:	Benefits/ Strengths	Side Effects
Metformin	Longest follow-up with assurance of safety for prevention of T2 DM Cost-effective treatment; cost-saving in post GDM pts	Gastro-intestinal side effects; contraindicated in eGFR < 30 ml/min
Acarbose	Lowers risk of diabetes	GI side effects
Pioglitazone	Strongest rate on prevention of DM (60%) Drug of choice for MetS with MASH Reduces MI and recurrent stroke in patients with insulin resistance with prior history of stroke or TIA	Weight gain Edema Worsening of pre-existing CHF NYH III-IV Increased fracture risk
Orlistat	Reduces body weight	GI side effects
GLP-1 RA based incretin therapy	Strongest class to reduce weight Reduction in CVD, CKD in obese prediabetes	GI side effects Prohibitive cost Lacks long term safety data in preDM
SGLT-2 inhibitors	Reduces body weight Reduces hard points including mortality in pre-DM patients with CHF Reduces hard endpoints including mortality in patients with CKD	Genito-urinary infections

Chapter 3. Evidence and Recommendations

A. Prevention among those at Risk for Diabetes or with Prediabetes

Guideline Question 1. Should we recommend non-pharmacologic interventions — including weight loss, dietary interventions, physical activity and exercise, proper sleep hygiene, smoking cessation, and avoidance of excessive alcohol intake — among adults at risk for diabetes or with pre-diabetes?

Recommendation 1.

We recommend a minimum weight loss of at least 5 to 7% body weight through diet and increased physical activity to delay or prevent diabetes among adults who are at risk for diabetes or with pre-diabetes who are overweight and/or obese. In patients with BMI of less than 23 kg/m², weight maintenance is advised.

Certainty of Evidence:	High
Strength of recommendation:	Strong

Recommendation 2. We recommend an individualized approach to diet, focusing on the reduction in total caloric intake, rather than a specified macronutrient distribution to achieve weight loss goals among adults who are at risk for diabetes or with pre-diabetes. In general, a diet rich in vegetables and fiber is encouraged, along with a reduced intake of refined carbohydrates, sugars, alcohol and fats.

Certainty of Evidence:	High
Strength of recommendation:	Strong

Recommendation 3. We recommend at least 150 minutes of moderate intensity exercise to be distributed at least three times a week among adults who are at risk for diabetes or with pre-diabetes.

Certainty of Evidence:	High
Strength of recommendation:	Strong

Recommendation 4. We recommend seven to eight hours of sleep at night among adults who are at risk for diabetes or with pre-diabetes.

Certainty of Evidence:	Low
Strength of recommendation:	Strong

Recommendation 5. We recommend avoidance of excess* alcohol intake among adults who are at risk for diabetes or with pre-diabetes.

**Excess alcohol intake refers to binge drinking, defined as consumption of more than 63 grams of alcohol per day. A dose-response relationship was observed, with the lowest risk of diabetes seen at alcohol consumption of*

approximately 10 to 14 grams per day (equivalent to about one standard drink, where one drink contains 14 grams of alcohol).

Certainty of Evidence:	Low
Strength of recommendation:	Strong

Recommendation 6. We recommend smoking cessation with active measures to prevent weight gain among adults who are at risk for diabetes or with pre-diabetes.

Certainty of Evidence:	Low
Strength of recommendation:	Strong

Background

There is strong evidence to support the positive effect of lifestyle intervention in the prevention of Type 2 Diabetes. Preventive measures through non-pharmacologic interventions are being studied, fine-tuned, reviewed, discussed and being implemented globally and within regional sectors aside from the creation of population, country, and culture-specific health care policies. This is in recognition of the heavy burden of diabetes in the healthcare system if it is allowed to grow continuously.

Summary of the Evidence Combined Lifestyle Intervention

A meta-analysis of seven high quality randomized controlled trials showed that lifestyle intervention through diet and exercise reduced the risk of diabetes by 47% [RR= 0.53 (95% CI 0.41, 0.67)].¹ These lifestyle intervention studies combine the use of weight reduction for overweight and obese people, improved diet, and increased physical activity.

In the Da Qing Study which had the longest follow-up and involved Asians, it was found that diet, exercise, and diet plus exercise interventions were associated with a 31% ($P < 0.03$), 46% ($P < 0.0005$), and 42% ($P < 0.005$) decrease in the incidence of diabetes over a 6-year period, respectively.² After 30-years of follow up, the Da Qing Study showed that the combined intervention group had 26% decrease in cardiovascular events (HR 0.74, 95% CI 0.59–0.92; $p=0.0060$), 35% decrease in microvascular complications (HR 0.65, 0.45, 0.95; $p=0.025$), 33% in cardiovascular deaths (HR 0.67, 0.48, 0.94; $p=0.022$), and 26% in all-cause mortality (HR 0.74, 0.61, 0.89; $p=0.0015$), leading to an increase of 4.82 years in median survival and a mean increase of 1.44 years in life expectancy in the intervention group compared with control.³ The diet recommendations in the trial included eating more vegetables, decreasing alcohol intake and reducing simple sugars in the diet. The exercise recommendations involved increasing leisure time physical activity (e.g. 30 minutes walking per day). In the same trial, participants with a body mass index (BMI) of more than 25 were encouraged to lose weight until a BMI of less than 23 is reached.²

The benefits of weight reduction among patients with pre-diabetes was also emphasized by the Diabetes Prevention Program (DPP) trial. The landmark study found that achieving at least 7% weight loss through intensive lifestyle intervention reduced diabetes progression by 58%, with 1 case of diabetes prevented per 7 persons treated for 3 years.³ In this trial, intensive lifestyle intervention consisted of an initial structured 16-session core curriculum on nutrition, physical

activity and behavioral self-management, followed by close contact with lifestyle coaches to provide support and ensure adherence. Weight reduction was achieved through a healthy, low calorie, low fat diet and moderate intensity physical activity (similar in intensity to brisk walking) more than or equal to 150 minutes per week.

Dietary intervention

Many studies exploring the role of dietary intervention in the prevention of diabetes focused at the reduction of total caloric intake rather than a specific food group or diet type.^{4,5,6} There is no evidence to support an ideal percentage of calories from carbohydrate, protein and fat.¹ It is best that patient preferences and culture will be considered to ensure the long-term implementation of dietary intervention. Specific dietary intervention in the trials exploring lifestyle change in the prevention of diabetes include the consumption of a diet rich in vegetables, reducing intake of refined carbohydrates, sugars, alcohol, fatty acids and increasing dietary fiber.^{4,5,6}

Other dietary patterns have been studied in the prevention of diabetes. The use of Mediterranean diet in diabetes prevention is supported by strong evidence. In a meta-analysis by Esposito et al, adherence to a Mediterranean diet significantly reduced new onset diabetes by 19% to 23%.^{7,8} This diet uses olive oil as the main source of dietary fat, rich in whole grains, fish, fruits, vegetables and has a reduced carbohydrate content.⁷

When it comes to specific foods, some observational studies showed consumption of greater than 2 servings of whole grains per day had a 43% reduced risk of incident type 2 diabetes compared with women who consumed no whole grain at all.⁹ Some observational studies also found that increased intake of total dairy products and cheese is associated with a decreased risk of type 2 diabetes.¹⁰

In the Philippines, the Food and Nutrition Research Institute is promoting the use of the “pinggang pinoy” but there are no studies published to date to show its impact on the prevention of diabetes among those at risk for DM.

The most important strategy is to do the nutrition counseling as part of a health policy with proper supervision, individualized diet, and with the frequency of counseling determined by patient's progress/ adherence. Table 4.1 lists the lifestyle intervention strategies in the various clinical trials on diabetes prevention.

Physical activity

In the Da Qing study where exercise showed a 46% decrease in the risk of diabetes, the recommended duration of exercise depends on the intensity of the activity, which is 30 mins of mild intensity exercise such as slow walking, housecleaning, 20 mins of moderate intensity exercise like brisk walking, cycling, ballroom dancing, 10 mins of strenuous activities like slow running, climbing stairs, playing volleyball or tennis and 5 mins of very strenuous activity such as jumping rope, playing basketball and swimming.² In other studies, a simple activity such as 30-40 mins of walking daily has been prescribed to participants in the lifestyle intervention arm.^{5,6}

In the DPP, the goal of physical activity was 700 kcal/week, approximated by at least 150 min of moderate intensity physical activity similar in intensity to brisk walking, to be distributed throughout the week with a minimum frequency of thrice a week. This was chosen due to feasibility, effectiveness as supported by previous studies, and long-term maintenance. Other activities that were given as equivalent to brisk walking include aerobic dance, bicycle riding, skating,

swimming. Lifestyle activities such as taking the stairs instead of the elevator, stretching, household chores, gardening, were not counted in the 150-min goal. ⁴

Table 10. Specific Counselling Strategies

STUDY	Control group	Intensive Lifestyle intervention
Da Qing ²	brochures given, no formal personal/ group counseling	% macro-nutrient distribution with reduced calories with food exchange list through <u>individual counseling</u> followed by <u>weekly small group counseling</u> for 1 month, then <u>monthly counseling</u> for 3 months, then <u>quarterly for 6 years</u> . Same frequency of counselling to monitor physical activity.
Finland Diabetes Prevention Program ⁶	oral and written information_(a two-page leaflet)_about diet and exercise at base line and at subsequent <u>annual</u> visits; no individualized program and formal counselling	reduction in weight of 5 %; fat intake <30 %, SFA < 10%, increase in fiber, whole-grain products, vegetables, fruits, low-fat milk and meat products, MUFA. Moderate exercise > 30 minutes per day. <u>7 individual nutrition counselling sessions in 1st year then quarterly for 3 yrs. Supervised exercise programs in training circuit</u>
U.S. Diabetes Prevention Program ⁴	reading materials given with counselling 1x/ year	<u>16- session formal curriculum on exercise, nutrition, behavior modification to achieve 7% weight loss given one-on-one over 24 weeks followed by monthly individual + group sessions for 2.8 years</u>
Indian Diabetes Prevention Program ⁵	standard health care advice, no individualization or formal counselling and no telephone reminders for adherence	advice on healthy diet and regular physical activity reduction in total calories, refined carbohydrates and fats, avoidance of sugar, and inclusion of fibre-rich foods. Counselling given <u>every 2 weeks for first month, then through telephone monthly to encourage adherence, and face-to-face sessions every 6 months for 2.5 years.</u>

Other lifestyle modifications (LSM):

Adequate sleep

In 2019, an Asia Pacific Consensus panel which included representation for the Philippines convened and reviewed the evidence of other lifestyle interventions in the prevention of diabetes among Asians, with subsequent recommendations published in 2021.¹¹ One of the more unique items discussed was the link between inadequate sleep and diabetes. Adequate sleep duration of at least 7 hours at night but no more than 8 hours is encouraged to decrease risk for diabetes. The association of both long and sleep duration with a significantly higher risk of diabetes was studied in a meta-analysis of 10 prospective studies, and a more recent meta-analysis of 17 cohort studies.^{12,13} Potential mechanisms linking diabetes and sleep restriction include decreased insulin sensitivity and glucose tolerance, increased inflammatory markers. Potential mechanisms linking diabetes and longer hours of sleep are more speculative and remain to be explored. Prolonged sleep was associated with other confounders like depression, undiagnosed medical disease, poor physical health, and poor physical activity. ^{12,13} The latest version of the ADA 2024 Standards of

Care now has this similar statement on the duration of sleep, albeit as part of healthy lifestyle for patients with overt diabetes.²⁸

Effect of alcohol intake

The effect of alcohol intake on the incidence of Type 2 diabetes remains controversial due to inconsistent results across studies. Cohort studies have shown associations between moderate alcohol consumption and a decreased risk of diabetes and coronary heart disease.¹⁴ Moderate alcohol consumption is significantly associated with increased insulin-mediated glucose uptake due to enhanced insulin sensitivity.¹⁵ Another hypothesis is the anti-inflammatory effects of alcohol. Alcohol alters the expressions of inflammatory proteins implicated in metabolic processes including adiponectin and interleukin 1b and a possible stimulatory effect of alcohol on HDL synthesis. But the generalizability of these findings is limited by short follow up periods and small sample sizes in the previous studies.

A meta-analysis of Knott et al. published in *Diabetes Care* 2015 studied the dose response relationship between average daily alcohol consumption and the incident risk of type 2 diabetes. In the 1.9M participants included levels of alcohol consumption above 63 g/day showed a consistent increased risk of developing Type 2 Diabetes. In levels below 63 g/day, there was a noted decreased risk of type 2 diabetes with peak reduction at 10-14 g/day, however this was limited to non-Asians and in women only. The dose response relationship between average daily alcohol consumption and incident type 2 diabetes mellitus did not show benefit with moderate alcohol consumption among Asians and continued higher risk for T2DM at all levels of alcohol consumption.¹⁶ (See Grade Evidence Table). Hence, the metabolic benefit of alcohol drinking cannot be applied to the Filipino population in general, particularly since the latest 2021 expanded national nutrition and health survey showed that 54.7% of adult Filipinos aged 20 to 59 are binge drinkers.¹⁷

Moreover, the hypothesized theory on improvement of insulin sensitivity associated with moderate alcohol drinking in the Western studies may not be totally applicable to Asians. Recent studies have suggested that impaired β -cell function has a greater effect on the development of diabetes than does insulin resistance in Asian people.¹⁸ In the study of Yoo, et al.,¹⁹ heavy alcohol consumption was associated significantly with reduced insulin secretion, but not insulin sensitivity, in Koreans. It was identified that the mechanism behind chronic alcohol consumption and the development of diabetes was through pancreatic B cell dysfunctions and not insulin resistance.¹⁹

Smoking Cessation

Smoking cessation should be encouraged. The risk of diabetes and smoking has been established in many studies.²⁰⁻²⁴ For example, a large meta-analysis of 88 studies found that smokers had a 37% increased risk of developing T2DM compared with nonsmokers [pooled RR 1.37 (CI 1.33- 1.42)].²¹ Although many environmental and genetic factors are known to be the cause of diabetes, the hypothesis that smoking increases the risk for incident T2DM is plausible due to the general inflammation caused by smoking, along with increased insulin resistance and greater abdominal obesity in smokers.²⁴

A common effect of smoking cessation is weight gain. A recent Chinese cohort study suggested that recent smoking quitters who gain substantial weight (>2.0 kg) after quitting were at an increased risk for type 2 diabetes than sustained smokers.²⁵ Data from a Korean adult population found that smoking cessation with no subsequent weight gain is associated with a reduced risk of developing diabetes.²⁶ Compared with current smokers, the adjusted hazard ratios for the risk

of type 2 diabetes was lower at 0.86 (95% CI=0.80, 0.93) in recent quitters with no weight gain; but the differences were not significant in those who gained weight after cessation with hazard ratios of 0.94 (95% CI=0.87, 1.03) in those with 0.1–5 kg weight gain, 0.93 (95% CI=0.73, 1.19) in those with 5.1–10 kg weight gain, and 1.49 (95% CI=0.84, 2.62) in those with a weight gain of >10 kg.

Weight gain may lead to the disruption of metabolism, such as decreased glucose tolerance and insulin resistance which may amplify or prolong deterioration of glucose metabolism caused by smoking cessation.²⁷ These findings suggest that weight management after smoking cessation has an impact on the prevention of type 2 diabetes.

Certainty of Evidence

Physical activity, nutrition

For the intensive lifestyle intervention with physical activity, nutrition alone or in combination, the systematic review and meta-analysis of 7 randomized controlled trials are rated as high quality of evidence given the design and the long duration of follow-up. Likewise, the studies included Asian cohorts which makes them very applicable.

Sleep duration

For other lifestyle intervention measures, the evidence for short sleep duration is based on nine observational studies with moderate certainty. The intervention relied on self-reported sleep duration rather than more objective measures like actigraphy and polysomnography. Sleep duration was assessed only once, so participants might have changed their sleep patterns during follow-up.¹² The measurement of diabetes outcomes was not standardized across all studies, showing substantial heterogeneity in the shortest sleep duration category ($I^2 = 57.1\%$, $P = 0.017$). There is low certainty regarding the link between long sleep duration and diabetes based on seven observational studies, due to substantial heterogeneity in the second longest sleep duration ($I^2 = 59.0\%$, $P = 0.023$) and the longest sleep duration ($I^2 = 75.8\%$, $P < 0.001$). Additionally, only 2% of the population studied was Asian (from Japan), with the rest from the US, Australia, Germany, and Finland. The GRADE for the other meta-analysis¹³, which adjusted for confounding factors, analyzed weight effects per trial, and included studies examining both long and short sleep durations within the same population cohort with consistent follow-up and assessments, has higher certainty of evidence.

Alcohol intake

For the role of reducing alcohol intake in the prevention of diabetes, the quality of evidence is moderate based on 38 observational studies with possible reporting bias due to self-reported alcohol consumption, presence of confounders, and considerable heterogeneity I^2 of 75%. Analyses hinted at the possibility that existing studies overestimated the degree to which the risk of T2DM is reduced among moderate consumers of alcohol; and noted funnel plot asymmetry among female data points.

Smoking

The evidence linking smoking to an increased risk of type 2 diabetes is graded with high certainty.²¹ It has been downgraded due to significant heterogeneity and potential publication bias, but on the other hand was upgraded because of the robust dose-response relationship between

smoking intensity and diabetes risk. While a strong link between smoking and type 2 diabetes was identified, the evidence supporting smoking cessation as a preventive measure for diabetes is of low certainty. This is based on ten observational studies, which also noted significant publication bias, reporting bias and the possibility of reverse causality.

Other considerations

As mentioned, other countries such as the USA and the UK consider diabetes prevention programmes as a cost effective measure that is why their health care policy supports lifestyle programs that should be implemented as a public health measure. Currently, intervention programs for healthy lifestyle including specific nutrition counselling are not covered by both private and public health insurance

Even if the link of alcohol to type 2 DM shows low quality of evidence, we should not promote alcohol intake because in our local setting and based on the latest 2021 expanded national nutrition and health survey, 54.7% of adult Filipinos aged 20 to 59 are binge drinkers with excessive alcohol intake.

B. Prevention (Pharmacologic Therapy)

Guideline Question 2. Should we recommend pharmacologic therapy (metformin, pioglitazone, acarbose, orlistat, SGLT-2 inhibitors, GLP-1 RAs) among adults with pre-diabetes?

Recommendation 7. We recommend pharmacologic therapy among those at risk for diabetes (pre-diabetes) who do not respond to lifestyle modification among adults with prediabetes

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 8. We recommend metformin at a dose of 0.5 to 1.7 grams/day among adults with pre-diabetes who do not respond to lifestyle change*

*The following patients who are proven to benefit the most from long term metformin treatment: women with prior history of gestational diabetes, those with fasting glucose > 110 mg/dL, those with HbA1c 6 to 6.4%, adults less than 60 years old, and those with obesity especially BMI > 35 kg/m²

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 9. We recommend assessing the response by monitoring 75-gram OGTT or HbA1c every 6 months among adults with prediabetes.

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 10. We suggest using other pharmacologic agents (Table 5.1) among patients who cannot tolerate or who have contraindications to metformin; or have co-existing co-morbidities of pre-diabetes that benefit from specific agents as tabulated.

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Consensus statements

The expert panel and Chair of the technical review committee decided unanimously to present the statements, recommendations and summary of evidence highlighting pharmacotherapy in general with emphasis on metformin as separate recommendation since it is standard of care globally. So, even if similar rigorous search for other agents were done- these were presented as table of medications rather than specific questions and recommendations per drug. This is a way to simplify the guidelines for all.

Table 11: PHARMACOTHERAPY CHOICES for DM Prevention

Drug	Benefits/ Strengths	Side Effects	Cost/ day (PhP) as of 2025
Metformin	Longest follow-up with assurance of safety for prevention of T2 DM Cost-effective treatment; cost-saving in post GDM pts	Gastro-intestinal side effects; contraindicated in eGFR < 30 ml/min	4 to 70
Acarbose	Lowers risk of diabetes	GI side effects	22 to 110
Pioglitazone	Strongest rate on prevention of DM (60%) Drug of choice for MetS with MASH Reduces MI and recurrent stroke in patients with insulin resistance with prior history of stroke or TIA	Weight gain Edema Worsening of pre-existing CHF NYH III-IV Increased fracture risk	13 to 105
Orlistat	Reduces body weight	GI side effects	105 to 148
GLP-1 RA based incretin therapy	Strongest class to reduce weight Reduction in CVD, CKD in obese prediabetes	GI side effects Prohibitive cost Lacks long term safety data in preDM	188 to 230
SGLT-2 inhibitors	Reduces body weight Reduces hard points including mortality in pre-DM patients with CHF Reduces hard endpoints including mortality in patients with CKD	Genito-urinary infections	33 to 56

Background

Diabetes causes significant burden on the healthcare system if it is allowed to progressively rise in the population. There is strong evidence to support the positive effect of lifestyle intervention in the prevention of Type 2 Diabetes, however it is difficult to sustain lifestyle changes, and if this strategy fails to lower glucose levels after 6 to 12 months, then it is reasonable to start pharmacotherapy. There are certain situations and conditions where pharmacologic therapy should be started together with lifestyle intervention. Several medications are now available and have shown benefits beyond diabetes prevention and this section will tackle these therapies.

Summary of the Evidence

Metformin

Among the clinical trials, it is the U.S. Diabetes Prevention Program (DPP) and its extension – the DPP Outcomes Study (DPPOS), which has the longest duration of follow up of over 22 years using metformin in patients with pre-diabetes.²⁻³ In the first 3 years of treatment, the relative risk reduction was 31% with a number needed to treat of only 13 patients for 3 years to prevent 1 case of DM.² At the 15 year follow-up, the efficacy of metformin was superior over and was not statistically different from the effect of intensive lifestyle changes at 18% and this degree of reduction in the development of overt type 2 diabetes remained significant up to the 22-year analysis.²⁻⁵ Treatment with metformin has the strongest impact among women with prediabetes who had history of gestational diabetes with a 50% reduction in the incidence of DM after 2.8 years of follow-up, and a number needed to treat of only 6 for 2.8 years to prevent 1 case of diabetes.⁶⁻⁷ Metformin is cost-effective in the prevention of diabetes versus placebo, and for women with prediabetes post GDM, it is considered cost-saving.⁸⁻¹⁰ The other subsets of patients who benefit the most from metformin in prediabetes are adults younger than 60 years old, those with obesity, those with FPG of 110 mg/dL or 6 mmol/L and up, and those with HBA1c at 6% to 6.4%.^{2,7} The safety and effect of metformin on mortality was comparable to lifestyle intervention and placebo in the 21-year follow-up of the DPP/ DPPOS.¹¹ Aside from the DPP/ DPPOS landmark study, a meta-analysis of four randomized controlled trials including Asian population in the prevention of diabetes with metformin showed the relative risk ratio of 0.76 [Risk ratio 0.65 to 0.84 at 95% CI] with reduction in conversion to overt diabetes at 24% which was statistically significant.¹² From the Cochrane Library, there was a systematic review that looked at metformin and its effects on prevention of type DM and its associated complications in persons at risk for DM with 20 RCT's concluding that metformin compared with placebo or diet and exercise reduced or delayed the risk of T2DM in people at increased risk for the development of T2DM by 50% [HR:0.50 95% CI (0.38, 0.65)] (moderate-quality evidence). However, metformin compared to intensive diet and exercise did not reduce or delay the risk of T2DM (moderate-quality evidence). Data on patient-important outcomes such as mortality, macrovascular and microvascular diabetic complications and health-related quality of life were sparse or missing.¹³

Acarbose

Acarbose is the only α -glucosidase inhibitor available in the Philippines. Using the search terms “prediabetes”, “Acarbose”, “impaired glucose tolerance”, “diabetes prevention”, “meta-analysis” and “randomized controlled trials”, PubMed and Open Athens search retrieved 2 relevant articles: 1 randomized controlled trial and 1 meta-analysis.

The Study to Prevent Non-Insulin-Dependent Diabetes Mellitus (STOP-NIDDM) trial by Chiasson et al reported that Acarbose was associated with a 25% reduction in the incidence of diabetes over a follow-up period of 3.3 years in a large cohort of 1429 high-risk individuals with impaired

glucose tolerance (IGT), a mean age of 54.5 years and BMI of 31.¹⁴ Efficacy of Acarbose was independent of age, sex, or BMI. Chiasson et al, in a related article, emphasized that if acarbose is discontinued, its beneficial protective effect of is expected to wear off over time, similar with metformin and lifestyle modification.¹⁵

The meta-analysis by Hu et al involved 8 randomized controlled trials in which participants were allocated to either Acarbose 100 mg TID or to control consisting of placebo and/or detailed advice on diet and exercise. IGT patients who took Acarbose have 59% less chance of progressing to Type 2 diabetes mellitus (Please see Annexes 1.A and 2.A).¹⁶ As for the 5 RCTs which studied reversal to normoglycemia, IGT patients who took Acarbose for 3 years have a 66% chance of reverting to normal glucose tolerance as compared to placebo and/or detailed advice on diet and exercise (Please see Appendices).¹⁶

Because of its wide use and experience in Asia, the 2019 Asia-Pacific consensus statements for the prevention of diabetes, where the Philippines participated, recommend both metformin and acarbose as options for the prevention of type 2 diabetes.¹⁷

Pioglitazone

Pioglitazone for the prevention of diabetes among those with impaired glucose level has been reviewed by the Cochrane Library.¹⁸ There were 6 trials comparing pioglitazone to placebo showing a 60% reduction [RR 0.40, 95% CI 0.17, 0.95] in the progression to overt diabetes.¹⁸ This effect was consistent regardless of the criteria used for pre-diabetes, whether isolated IFG or IGT or the elevation of both, and regardless of age, gender, and ethnicity. There was also no difference in the occurrence of myocardial infarction, congestive heart failure, and mortality. When compared with other drugs (metformin, acarbose, repaglinide)- there were no significant difference.¹⁸ Using Medline PubMed, the search terms “pioglitazone”, “meta-analysis”, “insulin resistance” published between 2000 to 2024 yielded 12 results. After excluding the analysis on polycystic ovary syndrome-related infertility and amenorrhea, and type 2 diabetes, four analyses on insulin resistance, pre-diabetes and its co-morbidities became relevant for citation. In a 2019 systematic review and meta-analysis of 29 RCT's examining the effect of various medicines on metabolic dysfunction-associated steatotic liver disease (formerly termed and published as non-alcoholic fatty liver disease), it is only pioglitazone in 6 RCT's that was able to reduce liver enzyme tests, improve histology, and reverse mild fibrosis in trials of patients with metabolic syndrome including pre-diabetes.¹⁹ In a 2022 meta-analysis of 15 randomized controlled trials on pioglitazone in the treatment of metabolic dysfunction associated steatohepatitis, liver enzymes, fibrosis, glycemic parameters and HOMA insulin resistance, the overall efficacy rate of pioglitazone was 78% [CI: 1.78 (1.31–2.43), P = .0003].²⁰ Insulin resistance has been closely linked with the development of ischemic stroke or transient ischemic attack (TIA), and this is the basis for the landmark multi-center randomized controlled trial of using pioglitazone in the secondary prevention of ischemic stroke and transient ischemic attack or the IRIS trial. In a population of 3876 non-diabetic adults with recent ischemic stroke or TIA, and documented insulin resistance using HOMA-IR score > 3, the use of pioglitazone for 4.8 years reduced stroke or myocardial infarction by 24% (HR 0.76, 95% CI 0.62 to 0.93, p = 0.007, but weight gain, edema and fractures were more frequent.²⁰ Development of diabetes was reduced by 52% (HR 0.48, 95% CI 0.33-0.69, p < 0.0001) with the effect driven primarily by those with FPG > 100 mg/dL and A1c > 5.7%.^{21, 22} A meta-analysis of IRIS and eight other pioglitazone trials including Asian studies with a total of 12, 026 participants showed that pioglitazone therapy was associated with a lower risk of MACE in patients with pre-diabetes or insulin resistance (RR 0.77, 95% CI 0.64 to 0.93), and among those with diabetes (RR 0.83, 95% CI 0.72 to 0.97). Risks of heart failure (RR 1.32;

CI 1.14 to 1.54), bone fracture (RR 1.52, 95% CI 1.17 to 1.99), edema (RR, 1.63; CI 1.52 to 1.75) and weight gain (RR 1.60; CI 1.50 to 1.72) increased in pioglitazone group.²³

Weight-reducing agents

Orlistat

Using the search terms “prediabetes”, “Orlistat”, “diabetes prevention”, “impaired glucose tolerance”, “randomized controlled trial” and “meta-analysis” in Medline PubMed and Open Athens, a randomized controlled trial was retrieved.

The XENical in the Prevention of Diabetes in Obese Subjects (XENDOS) study by Torgerson et al is a 4-year randomized, double-blind, placebo-controlled, prospective study involving a total of 3,305 individuals of age 30-60 years with BMI > 30 kg/m² and were either with normal or with impaired glucose tolerance. The individuals were either assigned to take Orlistat 120 mg TID or placebo TID on top of lifestyle modification. At the end of the study, the mean total body weight loss compared to baseline was 5.8 kg for the intervention group and 3.0 kg for the placebo group and the cumulative incidence of diabetes was 9.0% with placebo and 6.2% with Orlistat, corresponding to a risk reduction of 37.3% (P = 0.0032). (Please see Annexes 1, 2 and 3).²⁴

Glucagon-like Peptide 1 Receptor Agonists

Glucagon-like peptide-1 receptor agonist is an agent that has been recommended by the ADA as treatment for both obesity and type 2 diabetes mellitus²⁵. They also have been found beneficial on cardiovascular outcome trials in type 2 diabetes.²⁶⁻²⁹

We performed a comprehensive systematic search of related literature from Medline and CENTRAL. Freehand search using Google was also done and the reference list of all relevant studies were also checked to look for other sources of information. The search was conducted using the following search terms: ("glucagon-like peptide-1 receptor agonist"[Title/Abstract] OR "liraglutide"[Title/Abstract] OR "dulaglutide"[Title/Abstract] OR "exenatide"[Title/Abstract] OR "lixisenatide"[Title/Abstract] OR "semaglutide"[Title/Abstract]) AND ("prediabetes"[Title/Abstract] OR "prediabetic state"[MeSH Terms]) AND (("randomized controlled trial"[Publication Type] OR "controlled clinical trial"[Publication Type] OR "randomized"[Title/Abstract] OR "placebo"[Title/Abstract] OR "drug therapy"[MeSH Subheading] OR "randomly"[Title/Abstract] OR "trial"[Title/Abstract] OR "groups"[Title/Abstract]) NOT ("animals"[MeSH Terms] NOT "humans"[MeSH Terms]))

Population	Prediabetes patients
Intervention/Exposure	GLP1-RA (Exenatide, Lixisenatide, Liraglutide, Dulaglutide, Semaglutide)
Comparison	Placebo
Outcomes	Prevention of diabetes
Methodological filter	randomized controlled trials (RCT), observational clinical studies, systematic review and meta-analysis available

A total of 105 articles were found but only twenty-five articles fulfilled the criteria and were analyzed. Among the twenty-five articles was a recently published meta-analysis that included the 8 articles found in our search.

The recently published meta-analysis (11 June 2023) looked at the effects of treatment with glucagon-like peptide-1 receptor agonist on prediabetes with overweight and obesity.³⁰ The meta-analysis included studies using the following criteria: (1) the subjects were clinically diagnosed with prediabetes who had received a diagnosis of impaired glucose tolerance or impaired fasting glucose according to the ADA guidelines, (2) subjects were treated with GLP1-RA (3) minimum BMI > 25 kg/m² and (4) the study design was randomized controlled trial. A total of eight eligible studies³¹⁻³⁸ involving 2530 patients were included in this meta-analysis. The study duration of included articles ranged from 14 to 160 weeks. A total of six studies used liraglutide, one study used exenatide and one used semaglutide (see appendix). They found that results from five studies showed greater proportion of patients in the GLP1-RA group that regressed from prediabetes to normoglycemia than individuals in the placebo group (OR=4.56, 95% CI: 3.58, 5.8, P=0.004, I² = 68.6%) whereas results from the other 3 studies showed that lesser proportion of patients in the GLP1-RAs group progressed to diabetes than individuals in the placebo group (OR=0.31, 95% CI: 0.12, 0.81, P=0.017, I²=60.7%). There is also greater fasting glucose reduction in the GLP1-RAs group compared to placebo group (mean difference (MD)= -0.41 mmol/L, 95% CI: -0.58, -0.25, P<0.00001) in the five studies. Results from the three studies showed that GLP1-RA could significantly reduce the weight (MD=-4.01kg, 95% CI: -4.74, -3.28, P<0.00001, I² = 74.5%), BMI (MD= -1.70 kg/m², 95% CI:-1.94, -1.46, P<0.00001, I²=0%), waist circumference (MD=-3.47cm, 95% CI: -4.10, -2.84, P<0.00001, I² = 0%). GLP1-RAs could also significantly reduce the systolic blood pressure (MD=-2.85mmHg, 95% CI: -3.98, -1.71 P<0.00001, I² = 0%) however, there was no significant effect on diastolic blood pressure (MD = -0.41mmHg, 95% CI: -1.18, 0.36, P=0.295, I² =0%). Patients on the GLP1-RA group had higher incidence of adverse events (OR=2.38, 95% CI: 2.08, 2.73, P<0.00001) majority were gastrointestinal events (e.g., nausea, vomiting, diarrhea, constipation, abdominal pain, gastroesophageal reflux and dyspepsia) and were considered mild to moderate in severity. There is no statistical significance in serious adverse events between the two groups (OR=1.18, 95% CI: 0.98, 1.43, P=0.09). There were no cost effectiveness studies found with the use of GLP1-RA among prediabetic patients. In the Philippines, the price of a GLP1-RA ranges from Php 5,633/month (Saxenda), 6,260/month (Trulicity), Ozempic Php 6900/month.

Sodium Glucose Ligand Transporter-2 inhibitors

Sodium-glucose cotransporter 2 inhibitors (SGLT2i) were initially developed as agents for lowering blood glucose levels. They exhibit insulin-independent effects, hence, a low risk of hypoglycemia. Several trials have confirmed the weight loss effects of SGLT2 inhibitors.³⁹⁻⁴² Furthermore, cardiovascular and renal outcome trials have demonstrated their benefits for patients, both with and without type 2 diabetes mellitus.⁴³⁻⁴⁶

We performed a comprehensive systematic search of related literature from Medline and CENTRAL. Freehand search using Google was also done and the reference list of all relevant studies were also checked to look for other sources of information. The search was conducted using the following search terms: ("sodium glucose cotransporter 2 inhibitors"[Title/Abstract] OR "dapagliflozin"[Title/Abstract] OR "empagliflozin"[Title/Abstract] OR "canagliflozin"[Title/Abstract] OR "ertugliflozin"[Title/Abstract]) AND ("prediabetes"[Title/Abstract] OR "prediabetic state"[MeSH Terms]) AND (("randomized controlled trial"[Publication Type] OR "controlled clinical trial"[Publication Type] OR "randomized"[Title/Abstract] OR "placebo"[Title/Abstract] OR "drug therapy"[MeSH Subheading] OR "randomly"[Title/Abstract] OR "trial"[Title/Abstract] OR "groups"[Title/Abstract]) NOT ("animals"[MeSH Terms] NOT "humans"[MeSH Terms])).

Population	Prediabetes patients
Intervention/Exposure	SGLT2i (Dapagliflozin, Empagliflozin, Canagliflozin, Ertugliflozin)
Comparison	Placebo
Outcomes	Prevention of diabetes
Methodological filter	randomized controlled trials (RCT), systematic review and meta-analysis available

The search strategy yielded a total of 41 articles. After screening titles and abstracts, four randomized controlled trials and a recently published meta-analysis were reviewed.

The meta-analysis of Mori et al published in December 2022 included four randomized controlled trials based on the following eligibility criteria: 1) the population should be patients with prediabetes either as total sample or subgroup, 2) compared sglt2i with placebo, 3) new-onset diabetes was reported as one of the samples and 4) randomized controlled trials. The four RCTs were large cardiovascular and renal outcome trials that included a total of 5,655/ participants with prediabetes.^{43,46,47} Two trials compared dapagliflozin with placebo whereas the other 2 compared empagliflozin with placebo. The median follow up ranged from 16 months to 28.8 months. All of the studies were judged as low risk of bias using the Cochrane Risk of Bias 2 tool.

The incidence of new-onset diabetes was 157 (9.0%) in DAPA-HF, 54 (8.2%) in DAPA-CKD, 151 (11.9%) in EMPEROR-Reduced and 257 (13%) in EMPEROR-Preserved. Using random-effects, they found out that SGLT2i were associated with a lower risk of developing new-onset diabetes among people with prediabetes (RR, 0.79; 95% CI, 0.68-0.93). The results were consistent even when using the raw number of events in each of the 4 trials (RR, 0.80; 95% CI, 0.69-0.93) (see appendix 4.) No statistical difference between dapagliflozin and empagliflozin after random effects stratified across SGLT2i. However, summary estimates showed lower risk with dapagliflozin (RR, 0.68; 95% CI, 0.52-0.89) compared with empagliflozin (RR, 0.87; 95% CI, 0.72-1.04).

There were no reported hypoglycemia events among people without diabetes in the DAPA-CKD and DAPA-HF. In the EMPEROR-Preserved, there were 10 (0.7%) hypoglycemic events among participants without diabetes in the intervention group and 12 (0.8%) in the placebo group. In the EMPEROR-Reduced, hypoglycemic events occurred in 7 (0.7%) participants without diabetes in the treatment group and 6 (0.6%) in the placebo group. Overall, the frequency of severe hypoglycemia was not increased in the SGLT2i group compared to the placebo group. Using GRADEpro, these results were rated as high level of certainty. The cost effectiveness of using SGLT2i in diabetes prevention among people with prediabetes has not been elucidated. As of July 27, 2023, the cost of dapagliflozin is Php 44.00/tab while empagliflozin costs Php 55.25/tab.

Certainty of Evidence

The meta-analysis of several pharmacologic agents in prevention of diabetes had moderate certainty of evidence due to significant heterogeneity in terms of population and classes of drugs and duration of study. The meta-analysis on metformin and SGLT-2 inhibitors in preventing diabetes have high certainty of evidence while the studies on acarbose, pioglitazone and orlistat have moderate certainty of evidence due to allocation bias, high attrition and low Asian inclusion.

The studies on GLP-1 RA on prevention of diabetes related to weight loss have low certainty due to the allocation bias, high attrition rates and compared to other drugs, there is limited long term safety data at this time. The certainty for important clinical endpoints such as reduction in steatosis, stroke, CVD events were low to moderate for pioglitazone, but high certainty for SGLT-2 inhibitors- particularly CHF and CKD. Majority of the studies included few Asian patients and were done in settings very different from the Philippine health care system.

Other considerations

As mentioned, other countries such as the USA and the UK consider diabetes prevention programs as cost effective. Metformin use for the prevention of diabetes is considered cost-effective versus placebo in high-risk groups. The cost of other medications will need to be discussed with patients.

In the Philippines, there has been a deluge of misinformation on social media versus metformin so this warrants further education and explanation when prescribing it to the patients who benefit from its use.

Research Gaps and Recommendations:

Cost effectiveness studies of other drugs other than metformin should be done. The cost effectivity of pioglitazone and SGLT-2 inhibitors in preventing other important co-morbidities in patients with insulin resistance/ metabolic syndrome and pre-diabetes should be looked at.

Dual agonist GLP-1 RA and GIP, GLP-1 RA plus amylin; triple incretin agonist GLP-1 RA plus GIP and glucagon data in pre-diabetes and prevention of diabetes are lacking; these drugs are NOT yet available in the Philippines.

C. Nonpharmacologic management Among Patients with Diabetes Mellitus

Guideline Question 3. Should we recommend dietary interventions among adults with Type 2 DM?

Recommendation 11. We recommend an individualized medical nutrition therapy to achieve better glycemic control, weight management, and overall cardiovascular health.

Certainty of Evidence: Low
Strength of recommendation: Strong

Recommendation 12. We recommend referral to a dietitian/nutritionist/diabetes educator who is specially trained and knowledgeable on diabetes nutrition to provide individualized nutritional counselling.

Certainty of Evidence: Low
Strength of recommendation: Strong

Background

Nutrition therapy is an integral part in the holistic management of diabetes. It is one of the core aspect of diabetes self-management education and support (DSME) rendered by a diabetes-specific healthcare team. It aims to educate individuals to adopt a healthy eating pattern, which is a fundamental aspect in the management of metabolic diseases. Medical nutrition therapy (MNT) is defined as a “nutrition-based treatment provided by a dietitian/nutritionist according to the Center for Disease Control and Prevention. It includes a nutrition diagnosis with counselling services through an in-depth, individualized nutritional evaluation and recommendations. The different types of diet are defined as follows:

1. Low carbohydrate (LC) diet <25% carbohydrates of total energy intake,
2. Moderate carbohydrate diet w/ 25-45% CHO of total energy intake w/ 10-20% protein
3. High protein diet w/ >20% protein
4. Low fat diet <30% fat of total energy intake high intake of grains & cereals, 10-15% protein
5. Low glycemic index diet
6. Vegetarian diet
7. Mediterranean dietary pattern w/ fruits vegetables, olive oil, legumes, cereals, fish, moderate intake of red wine during meals
8. Paleolithic diet (emphasizes on unprocessed foods, excluding milk, cereals, breads)
9. Control diet w/ minimal intervention.

Unfortunately, most of the studies included different lifestyle changes with health teaching, exercise, and general dietary advice without any specified dietary intervention. The articles did not directly answer the clinical question with regards to dietary intervention alone.

Summary of the Evidence

Characteristics of included studies

Thirty-nine (39) studies were included w/ 30 on Type 2 DM and 9 studies on hypertension. The meta-analysis included studies that utilized educational/ counselling programs w/ multiple

component lifestyle intervention strategies and other studies combined with group educational programs w/ diet and exercise.

Efficacy outcomes

Educational/counseling programs and food replacement programs in primary care were able to reduce HbA1c on patients with T2DM (mean difference (MD): -0.37, 95% CI: -0.57 to -0.17, I^2 : 85.8%, among 7,437 patients, 27 studies; MD: 0.54, 95% CI: 0.75 to 0.32, 440 patients, 2 studies, respectively). Educational and counseling programs likely reduce DBP in patients with hypertension (MD: -1.79, 95% CI: -3.46, -0.12, 2840 patients, 9 studies, moderate quality of the evidence), but the effect in SBP was unclear due to risk of bias and imprecision, SBP (MD: 2.17, 95%CI 6.81, 2.46, 4518 patients, 9 studies, I^2 : 88.8%, low quality of the evidence. (Fig 3) Unfortunately, most of the studies included various types of nutritional therapy strategies (i.e., educational/counseling programs and food replacement programs) together with different lifestyle changes through health teaching, exercise, and general dietary advices without any specified dietary intervention. The counselling interventions were delivered by any trained physician, nurse, dietitian, or psychologists. The articles therefore did not directly answer the clinical question with regards to dietary intervention alone.

Another meta analysis was done by Juri K, et al² looking at RCTs comparing the efficacy of dietary education compared with general intervention on diabetes control. Primary outcome was HbA1c level at different follow up periods and other outcome variables assessed were body weight, BMI, blood pressure, and postprandial glucose. 33 RCTs were included in the metaanalysis, 5 provided only dietary intervention, 9 provided diet and exercise therapy, while the remaining 19 provided diet, exercise, and psychosocial therapy. The effect size between dietary education and general interventions was 0.42 ($n = 5639$, MD = 0.42; 95% CI 0.53 to 0.31) and was significantly different ($Z = 7.73$, $p < 0.001$). When subgroup analysis was done based on period of education, intervention method, and intervention content, the mean differences in 4–6-month application, individual education, and diet-exercise-psychosocial intervention were 0.51, ($n = 2742$, 95% CI 0.71 to 0.32), 0.63 ($n = 627$, 95% CI 1.00 to 0.26), and 0.51 ($n = 3244$, 95% CI 0.71 to 0.32), respectively. Dietary education intervention provided for at least 3 months was effective in reducing HbA1c.

There is uncertainty regarding the optimal proportion of macronutrient distribution for carbohydrates, fats, and proteins to be recommended to achieve sustained glycemic control. There are meta-analyses which showed different dietary approaches like a low-carbohydrate, low glycemic index, high protein, vegetarian, and Mediterranean diet that were effective in reducing HbA1c but results were conflicting. But there are clinical evidences to show that a low carbohydrate diet is more effective in achieving HbA1c reduction. In a network meta-analysis by Schwingshackl et al, the authors investigated comparative efficacy of different dietary approaches on glycemic control with glycosylated hemoglobin as the outcome. The use of any dietary approach resulted in small to moderate reduction of HbA1c with low certainty of evidence. The different types of diet are defined as follows:

1. Low carbohydrate (LC) diet <25% carbohydrates of total energy intake
2. Moderate carbohydrate diet w/ 25-45% CHO of total energy intake w/ 10-20% protein
3. High protein diet w/ >20% protein
4. Low fat diet <30% fat of total energy intake high intake intake of grains & cereals, 10-15% protein
5. Low glycemic index diet
6. Vegetarian diet

7. Mediterranean dietary pattern w/ fruits vegetables, olive oil, legumes, cereals, fish, moderate intake of red wine during meals
8. Paleolithic diet (emphasizes on unprocessed foods, excluding milk, cereals, breads)
9. Control diet w/ minimal intervention.

Fifty-six trials met eligibility criteria and were published between 1978-2016 w/ a total population of 4,937 type 2 diabetics. Study duration ranged between 3 to 48 months and BMI between 25 (Asians) and 43 kg/m². Of the 56 trials, 21 were considered to be low risk, 7 at high risk and 28 as moderate /unclear risk of bias. The studies used heterogeneous definitions for the different intervention diets and dietary protocols varied among the trials. NMA simultaneously analyzes direct comparisons of interventions within trials and indirect comparisons across trials based on a common comparator. The authors ranked the 9 dietary approaches according to the SUCRA (Surface Under the Cumulative Ranking curve) value which represents the mean probability to be the best with a higher value directly correlated with better performance. All dietary approaches were more effective in reducing HbA1c (-0.82 to -0.47% reduction) and fasting glucose (-1.61 to -1 mmol/l) compared to a control diet. Low-carbohydrate diet (MD -0.82; 95% CI -1.11, -0.53) [84%] ranked first in diet approaches, followed by Mediterranean diet (MD -0.79; 95% CI -1.07, -0.52) [80%] and Palaeolithic diet (MD -0.87; 95% CI -1.56, -0.18) [76%] based on surface under the cumulative ranking (SUCRA) See Annex 2 for the network estimates of different approaches.

Medical nutrition therapy assisted by a dietitian specially trained on diabetes management was evaluated in a metaanalysis done by Razaz J., et al (health effects of MNT2019). This included 11 studies which examined the effect on HbA1c, weight, blood pressure, and cholesterol. Eight studies that reported HbA1c as an outcome were combined using a random-effects model which included 1,032 participants (intervention = 522 and control 510) It demonstrated that MNT intervention significantly reduced HbA1c levels (WMD:-0.43%, 95% CI:-0.69, -0.17) in the intervention group compared to the control routine care.(Fig 1)

In 6 studies that reported weight as an outcome variable, there were a total of 556 participants in the intervention group and 534 in the control group. Pooled results showed a significant reduction in weight in the MNT group compared with the control group (WMD: -1.54 kg, 95%CI:-2.44, -0.64). However, there was significant heterogeneity among the studies ($p=0.001$, $I^2 = 77\%$).

Eight studies including 748 participants reported cholesterol as an outcome measure using a random effects model. It demonstrated a significant reduction in cholesterol following MNT intervention (WMD: -4.06 mg/dl, 95% CI: -7.31, -0.81). There was no significant heterogeneity between studies ($p=0.29$, $I^2 = 16$). Four studies reported LDL as outcome variable, with the MNT group showing significant reduction in LDL level (WMD: -4.43 mg/dl, 95% CI:-13.66, 4.80) compared with the control group. There was no significant heterogeneity between studies ($p=0.12$, $I^2 = 89$) For TG, the combined effect size was -8.82 mg/dl(95%CI -20.1, 2.45) but there was significant heterogeneity ($p=0.03$, $I^2 = 57\%$).

In geographically challenged areas where dietitians/diabetes educators are not available, community barangay nutrition scholars may adeptly provide nutritional counselling. An easy-to-understand visual tool, “The Filipino Plate Method,” was developed by the Department of Science and Technology-Food and Nutrition Research Institute (DOST-FNRI) to educate patients on healthy eating habits. This was based on the Idaho Plate Method or “The Healthy Diabetes Plate” wherein a dinner plate serves as a pie chart to show the proportions of food groups and appropriate food choices. This is a simple model replacing the more complex traditional exchange-based method for meal planning.

Table 12. Summary of Findings and Certainty of evidence

Outcomes	Basis	Effect Estimate	95% Confidence Interval	Interpretation	Certainty of Evidence
HbA1c reduction(1)	27 RCTs N= 7,437	-0.37	-0.57, -0.17	Beneficial	Moderate
HbA1c reduction (2)	33 RCTs N=5,639	-0.42	-0.53, -0.31	Beneficial	Low
HbA1c (4)	8 RCTs N = 1032	-0.43	-0.69, -0.17	Beneficial	Low
weight reduction (4)	6 RCTs N= 1,090	-1.54 kg	-2.44, -0.64	Beneficial	Low
DBP reduction (1)	9 RCTs 2,840 pts	-1.79	-3.46, -0.12	Beneficial	Low
SBP reduction (4)	7 RCTs	-7.1	-13.03, -2.77	Beneficial	Low
Cholesterol	6 RCTs	-4.06 mg/dl	-7.31, -0.81	Beneficial	Low
LDL chole	4 RCTs	-4.43	-13.66, 4.8	No significant benefit	Low
HDL chole	6 RCTs	-0.40 mg/dl	-3.2, 2.4	No significant benefit	Low
TG	6 RCTs	-8.82 mg/dl	-20.1, 2.45	No significant benefit	Low

Guideline Question 4. Should we recommend physical activity and exercise for glucose control, weight management and overall cardiovascular health among adults with Type 2 DM?

Recommendation 13. We recommend increased physical activity, moderate to vigorous aerobic, resistance, or combined exercises of at least 150 minutes per week for glucose control, weight management, and over-all cardiovascular health.

* **physical activity**, defined as any bodily movement produced by skeletal muscles that results in energy expenditure (e.g., walking, occupational, household, or recreational activities)

** **exercise**, defined as a planned, structured, and repetitive subset of physical activity

Certainty of Evidence: Low

Strength of recommendation: Strong

Background

Physical activity is defined as any skeletal muscle movement which expends energy beyond resting levels, such as walking or gardening. Exercise, on the other hand, is a more structured subset of physical activity, performed regularly with the goal of improving or maintaining physical fitness.

Acute effects of exercise include enhanced glucose metabolism through increased glucose disposal and improved insulin sensitivity. It also increases the aerobic threshold and concentration of mitochondria in skeletal muscles. Evidence suggests that combined resistance and aerobic training produces the most pronounced improvements in glycemic control with enhancement of sensitivity of muscle cells to insulin and increases activity of oxidative enzymes.

There are different types of exercises (Table 1). Aerobic exercise involves continuous, rhythmic movements of large muscle groups, such as brisk walking, bicycling, swimming, and jogging. Resistance exercise utilizes weights, resistance bands, or one's own body weight like push-ups to increase muscle strength. Flexibility exercises, on the other hand, enhance the ability to move through fuller range of motion like for the lower back and hamstring stretching. Most studies on exercises involve a combination of exercises.

Table 1. Different exercise modalities, their definitions, and examples.

Type of exercise ^{1,2}	Definition	Examples
1. Aerobic	Rhythmic, repeated, and continuous movement of the same large muscle groups for at least 10 minutes at a time	
1.1. Moderate intensity	Target of 64%-76% of person's maximum heart rate*	Biking, brisk walking, dancing, swimming, water aerobics
1.2. Vigorous intensity	Target >76% of person's maximum heart rate*	Brisk walking up an incline, jogging
2. Resistance training	Activities of short duration with use of free weights, machines, resistance bands to increase muscle strength and endurance	Exercise with free weights, resistance bands, one's own body weight like push-ups

3. Flexibility training	Activities that enhance the ability to move through fuller ranges of motion like lower back or hamstring stretching	Yoga, tai chi
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*Maximum heart rate = 220 – Age; based on ADA 2023 Standards of Care¹ and Diabetes Canada 2018 CPG²

Summary of the Evidence

Characteristics of included studies

Evidence on the effect of exercise and physical activity in adults with T2DM was derived from 32 RCTs. The most common interventions were aerobic exercise and resistance training, with some studies incorporating combined modalities. The primary outcomes of interest were glycosylated hemoglobin (HbA1c) and weight reduction.

Efficacy outcomes

Evidence from multiple RCTs demonstrates that structured exercise produce clinically meaningful reductions in HbA1c and waist circumference in adults with type 2 diabetes, especially those exceeding 150 minutes per week. Effects on body weight are modest, and there is insufficient direct evidence on long-term vascular outcomes.

A meta-analysis by Shah et al. in 2021 (26 RCTs; interventions lasting ≥8 weeks) evaluated the effect of exercise on glycemic control in 2,456 participants with type 2 diabetes.³ Exercise sessions averaged 45.2 minutes in duration, 3.25 days/week, over a mean follow-up of 21.9 weeks. A variety of exercise interventions were used in the studies, but aerobic training (12 studies) and resistance training (14 studies) were the most common. A separate meta-analysis by Umpierre et al. in 2011 reviewed 47 RCTs (n=8,358 patients) comparing structured exercise training or physical activity advice versus no exercise in adults with T2DM.⁴

Glycemic control, waist circumference, and weight reduction

Based on the Shah 2021 meta-analysis, the exercise group showed a mean HbA1c reduction of –0.34% (95% CI –0.50 to –0.19%; p<0.0001; n=2,456) and a significant average decrease in waist circumference of –2.63 cm (95% CI –4.54 to –0.73; p=0.007) among 1,138 individuals. However, the average degree of weight reduction was not statistically significant at –0.73 kg (95% CI –1.62 to +0.16; p=0.11). The authors suggested that greater weight loss may require more vigorous or longer-term interventions. Overall, exercise was considered a safe adjunct to medical therapy to optimize glycemic control and reduce central adiposity. The majority (75%) of studies had issues in allocation concealment, blinding of participants and therapists, and blinding of outcome assessors.

Structured exercise training was associated with more pronounced HbA1c reduction compared with physical activity advice and physical activity advice in conjunction with dietary intervention. Structured exercise resulted in a mean HbA1c reduction of –0.67% (95% CI –0.84 to –0.49; I² = 91.3%).⁴ When analyzed by exercise type, aerobic training (18 studies; 848 participants) achieved a reduction of –0.73% (95% CI –1.06 to –0.40), resistance training (4 studies; 261 participants) yielded –0.57% (95% CI –1.14 to –0.01), and combined aerobic plus resistance training (7 studies; 404 participants) resulted in –0.51% (95% CI –0.79 to –0.23). In terms of exercise duration, interventions exceeding 150 minutes per week (18 observations; 826 participants) produced a greater HbA1c reduction of –0.89% (95% CI –1.26 to –0.51) compared with those lasting 150 minutes or less per week (12 observations; 687 participants), which achieved –0.36% (95% CI –0.50 to –0.23). When compared with physical activity advice, advice

alone (14 studies; 712 participants) was associated with a reduction of -0.43% (95% CI -0.59 to -0.28), and advice combined with dietary counseling (12 studies; 6,313 participants) resulted in a reduction of -0.58% (95% CI -0.74 to -0.43). Overall, structured exercise achieved a more pronounced reduction in HbA1c than either physical activity advice alone or advice combined with dietary intervention.

Microvascular and macrovascular complications

No meta-analyses were identified that directly evaluated the effect of exercise on the incidence of microvascular or macrovascular complications. However, a Cochrane review⁵ found favorable changes in triglyceride levels with a mean reduction of -0.25 mmol/L (95% CI -0.48 to -0.02 ; $p < 0.05$; 5 studies). However, cholesterol levels (total, HDL, LDL) as well as systolic blood pressure changes were not significantly changed.

Safety outcomes

There was no safety outcomes reported.

Certainty of Evidence

Outcomes	Basis	Effect Estimate	95% Confidence Interval	Interpretation	Certainty of Evidence
HbA1c reduction ³	26 RCTs	-0.34	-0.5, -0.19	Benefit	Moderate
HbA1c reduction ⁴	23 RCTs (N=1533)	-0.67	-0.84, -0.49	Benefit	Low
HbA1c reduction ⁵	13 RCTs (N= 361)	-0.62	-0.91, -0.33	Benefit	Moderate
Weight Reduction ³	11 RCTs	-0.73	-1.62, 0.16	Tendency towards benefit	Moderate
Weight Reduction ⁵	10 RCTs (N =248)	-0.04	-3.83, 3.76	No benefit	Moderate
Waist circumference	10 RCTs (N= 1,138)	-2.63	-4.54, -0.73	Benefit	Moderate
Triglycerides	5 RCTs (N= 139)	-0.25	-0.48, -0.02	Benefit	Moderate
Total cholesterol	5 RCTs (N=139)	-0.11	-0.41, 0.18	No benefit	Moderate
LDL cholesterol	3 (N= 73)	0.12	-0.29, 0.53	No benefit	Moderate
HDL cholesterol	5 (N=139)	-0.02	-0.10, 0.06	No benefit	Moderate
Systolic BP	4 (n= 127)	-4.16	-9.46, 1.14	No benefit	Moderate
Diastolic BP	3 (n= 78)	-0.13	-3.7, 3.45	Tendency towards benefit	Moderate

Other Considerations:

No studies found regarding patient preferences and values.

Guideline Question 5. Should we recommend limitation of alcohol intake among adults with Type 2 DM?

Recommendation 14. We suggest against the intake of alcohol among adults with type 2 DM.

Certainty of Evidence: Very Low
Strength of recommendation: Weak

Background

The intake of alcoholic beverage is so ingrained into the Filipino culture that surveys such The Philippine Nutrition Facts and Figures 2015 by the Food and Nutrition Research Institute published in December 2016, stated that 4 out of 10 (44.9%) of Filipino adults were drinking alcoholic beverages at that time; five (54.9%) out of ten of these consuming alcoholic drinks engaged in binge drinking.¹

The distribution of adults by alcohol consumption, age group and sex, as published in this monograph, as well as the prevalence of binge drinking in adults by age group and sex, are shown below.

Table 1. Percent distribution of adults, 20.0 years and over, by alcohol consumption status, age group and sex: Philippines, 2015

Age Group/ Sex	n	Alcohol Consumption Status		
		Never	Current	Former
		% (95% CI)	% (95% CI)	% (95% CI)
All	21,871	39.4 (38.0-40.8)	44.9 (43.7-46.0)	15.8 (14.8-16.7)
20.0-29.9	4,814	34.0 (31.9-36.1)	50.9 (49.0-52.8)	15.1 (13.8-16.5)
30.0-39.9	4,229	36.0 (34.1-37.9)	49.5 (47.7-51.3)	14.5 (13.0-15.9)
40.0-49.9	4,636	38.8 (36.8-40.8)	46.6 (44.7-48.4)	14.6 (13.3-15.9)
50.0-59.9	4,025	40.9 (38.9-42.9)	41.8 (40.0-43.7)	17.3 (15.9-18.7)
60.0-69.9	2,491	50.8 (48.4-53.3)	32.2 (30.1-34.3)	16.9 (15.2-18.7)
≥ 70.0	1,606	57.6 (54.7-60.4)	20.3 (18.2-22.4)	22.1 (19.6-24.6)
Male	10,118	18.7 (17.3-20.1)	69.1 (67.7-70.6)	12.2 (11.3-13.1)
20.0-29.9	2,396	19.1 (16.8-21.3)	73.2 (70.8-75.5)	7.8 (6.4-9.1)
30.0-39.9	1,976	15.0 (13.1-16.9)	76.3 (74.1-78.4)	8.7 (7.3-10.1)
40.0-49.9	2,148	17.1 (15.1-19.2)	72.1 (69.7-74.5)	10.8 (9.3-12.2)
50.0-59.9	1,877	17.9 (15.8-20.0)	66.3 (63.7-68.9)	15.8 (13.8-17.8)
60.0-69.9	1,112	25.6 (22.5-28.7)	54.7 (51.5-57.8)	19.7 (17.0-22.5)
≥ 70.0	609	31.0 (26.9-35.1)	36.7 (32.6-40.7)	32.4 (27.9-36.8)
Female	11,753	56.9 (55.2-58.6)	24.3 (23.1-25.5)	18.8 (17.5-20.0)
20.0-29.9	2,418	47.3 (44.7-50.0)	31.0 (28.7-33.3)	21.6 (19.6-23.7)
30.0-39.9	2,323	53.4 (50.8-55.9)	27.4 (25.2-29.6)	19.2 (17.2-21.3)
40.0-49.9	2,488	57.5 (54.9-60.1)	24.6 (22.5-26.7)	17.9 (16.0-19.8)
50.0-59.9	2,148	61.0 (58.4-63.6)	20.4 (18.4-22.5)	18.6 (16.6-20.6)
60.0-69.9	1,379	71.7 (68.9-74.4)	13.7 (11.7-15.7)	14.6 (12.6-16.7)
≥ 70.0	997	75.5 (72.4-78.6)	9.3 (7.4-11.3)	15.2 (12.6-17.8)

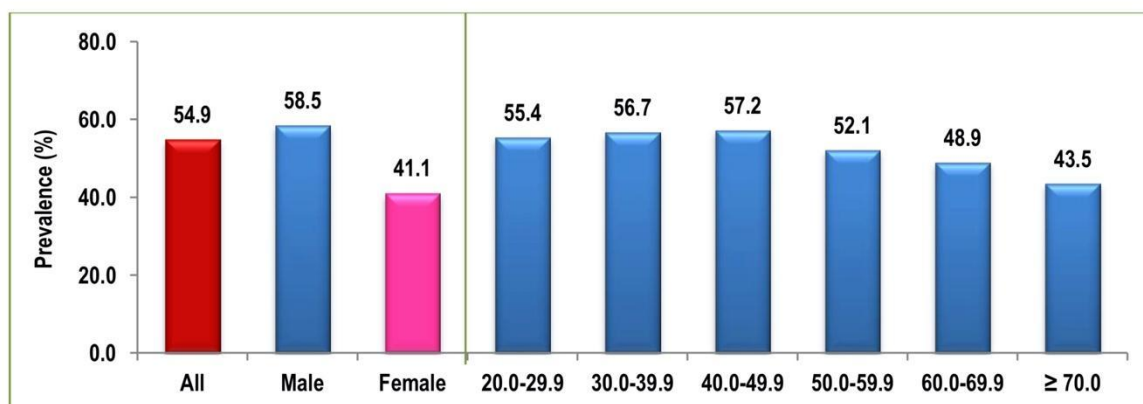


Figure xx. Prevalence of binge drinking among currently drinking adults (those who reported drinking alcoholic beverages in the past 30 days) 20.0 years old and over, by sex and age group” Philippines, 2015

On August 17, 2021, the Department of Health released the result of a mobile phone health survey conducted in January 2021. It was stated therein that 4 out of 10 Filipinos (40.1%) had drunk alcoholic beverages in the 30 days prior to the survey. One out of every three of those adult Filipinos who drank (33.1%) also stated that they had consumed six or more alcoholic drinks in a single session; this constituted binge drinking.²

The definition of light, moderate and heavy drinking varies per country. However, according to the World Health Organization, a daily intake of <1 drink, 2-3 drinks and >4 drinks corresponds to light, moderate and heavy drinking, respectively. 3 Binge drinking of alcohol refers to drinking four or more standard drinks in a row for females and five or more standard drinks in a row for males.⁴

The criteria for one alcoholic drink also varies per country. However, in the Philippines, the measurement of standard drinks, according to the Department of Science and Technology-Food and Nutrition Research Institute.

Summary of the Evidence

Efficacy Outcomes:

Alcohol Intake and Glycemic Control

A cohort study involving 38,564 patients was done by A.T. Ahmeed et al, which involved the assessment of glycemic control via HbA1c within a year of a self-reported survey on the number of drinks these patients took per day. According to this study, using multivariate analysis, those who took 2-2.9 drinks per day had the lowest HbA1c, with increasing HbA1c levels for heavier drinkers. Those who took smaller amounts of alcohol (<0.1 drinks/day) had the highest HbA1c, followed by lifetime abstainers. 5

On the contrary, randomized trials ranging in duration from 3 months to 2 years in adults with type 2 diabetes showed that moderate alcohol intake, specifically from wine may lead to insignificant reductions in fasting plasma glucose and HbA1c as compared to non-alcoholic beverages. In the study by Gepner Y, et al, this reduction was significant only in patients who were slow ethanol metabolizers. 6,7,8 These RCTs generally have issues though, with binding and attrition bias for most of the outcomes including FPG, HbA1c and BP levels.

Alcohol Intake and Lipid Profile

The above mentioned randomized controlled trials showed that with regards to lipid levels, overall, there was no significant differences in levels of LDL cholesterol, but significant improvement in triglyceride levels and HDL levels, the latter especially among those who took red wine.

On the other hand, according to a meta-analysis by S.A. Naame, et al of randomized controlled trials that included those of R.Marfella, et al and Y. Gepner, et al cited above, moderate intake of red wine did not significantly affect body mass index (BMI) nor total cholesterol and LDL-cholesterol levels, but had favorable effects on HDL-cholesterol. However, there were significant heterogeneity for the analyses for BMI and for cholesterol levels.⁹

ALCOHOL AND BP

The randomized controlled trials by Gepner, Marfella and Shai showed that moderate intake of alcohol did not significantly impact blood pressure (MD -0.50, -2.12, 1.11).^{6,7,8}

On the other hand, according to a meta-analysis by S.A. Naame, et al of randomized controlled trials that included those of R.Marfella, et al and Y. Gepner, et al cited above, had favorable effects on systolic (MD -1.33; 95% CI -1.81, -0.85) and diastolic blood pressures (MD = -1.10; 95% CI -1.42, -0.79). ⁹

Alcohol and CV outcomes: [total patients 11,140]

A post hoc analysis of the Action in Diabetes and Vascular Disease: Preterax and Diamcron Modified-Release Controlled Evaluation (ADVANCE) trial¹⁰ examined the relationship of alcohol use in the patients enrolled in this trial. This was an observational analysis based on self-reporting of alcohol use at the baseline visit, at the 24-month follow-up visit and at the 5-year final study visit. There were a total of 11,140 patients and they reported that patients who took alcoholic beverages had fewer cardiovascular events, less microvascular complications and less all-cause mortality, as noted in the figure below. These benefits were primarily seen in those who drank moderate amounts of alcohol.

However, moderate alcohol use did not show any significant relationship with cardiovascular and microvascular events, as well as all-cause mortality, when examined separately from heavy alcohol intake. On the other hand, high alcohol intake, defined as more than 21 standard drinks weekly for men and more than 14 standard drinks weekly for women, was inversely related to cardiovascular events and all-cause mortality, with a similar trend for microvascular complications.

Safety Outcomes: Alcohol And Hypoglycemia

With regards to the possibility of hypoglycemia after the intake of alcoholic beverages, JA Hirst, et al. ¹¹ did a systematic review and meta-analysis of randomized controlled trials looking at the effects of light to moderate alcohol intake on glycemic levels of diabetic patients. Here they also analyzed the incidence of hypoglycemia after light to moderate alcohol intake. Short-term trials (cross-over design lasting less than or equal to 2 days) of blood glucose concentrations immediately after and within 24 hours of people who drank alcohol amounting to 16-80 gm or approximately 2.5 units daily versus those had no alcohol intake did not reveal any increase in the incidence of hypoglycemia in the alcohol arm. Medium-term trials (4-104 weeks) of people who drank alcohol 11-18 gm or 1.5 units daily versus the comparator arm also did not show any significant effect of alcohol on acute hypoglycemic events. This meta-analysis, which included

the randomized trials of Marfella, Gepner and Shai mentioned above, also did not show any significant effect of light to moderate alcohol intake on glycemic control.

However, other data have shown that the effects of alcohol on glycemia and cardiovascular health may follow a U-shaped or J-shaped curve, wherein low to moderate doses of alcohol may be beneficial, but excessive or heavy drinking can cause deterioration in blood glucose levels in the long term coupled with increased risk of hypoglycemia in the short term.

Certainty of Evidence

Summary of Evidence of the Efficacy and Safety (Hypoglycemia) Outcomes of Moderate Alcohol Intake among patients with type 2 diabetes mellitus:

Outcomes	Basis	Effect Estimate	95% Confidence Interval	Interpretation	Certainty of Evidence	NNT
CV events	1 cohort (n=10,727)	HR 0.82	0.71, 0.95	As good as or better	Very Low	142.9
Microvascular Complications	1 cohort (n=10,727)	HR 0.83	0.72, 0.97	As good as or better	Very Low	38.5
Glycemic Control (HbA1c)	4 RCT (n=362)	MD 0.02	-0.18, 0.15	equivalent	Low	N/A
Weight (kilogram)	2 RCT (n=247)	MD 0.18	-0.58, 0.95	equivalent	Low	N/A
Hypoglycemia	2 RCT (n=105)	RR 2.72	0.15, 50.79	Harm	Low	26.3
Triglyceride (mg/dl)	3 RCT (n=362)	MD -8.74	-11.95, -5.53	Benefit	Moderate	N/A
High Density Lipoprotein (mg/dl)	2 RCT (n=362)	MD	0.72, 1.09	Benefit	Moderate	N/A
Low Density Lipoprotein (mg/dl)	2 RCT (n=247)	MD 6.31	-15.8, 3.18	Inconclusive	Low	N/A
Systolic Blood Pressure	4 RCT (n=386)	MD 0.1	-0.07, 0.26	Equivalent	Low	N/A
Diastolic Blood Pressure	4 RCT (n=386)	MD -0.57	-2.03, 0.88	Inconclusive	Low	N/A

Other Considerations

ALCOHOL USE IN THE PHILIPPINES

Although alcohol taken in light quantities may confer some benefit, these benefits may not outweigh the potential risks even at low levels of alcohol intake. More so for moderate to heavy drinking, which is associated with detrimental health outcomes. Given that, as stated

above, many Filipinos drink alcohol, and a significant percentage of these drinkers engage in binge drinking, we do not recommend the initiation of alcohol intake for individuals who are currently not drinking alcoholic beverages. To avoid possible harm, those who choose to drink are advised to limit their consumption of alcohol to light-moderate levels only. They are likewise advised to be aware of, and take the necessary precautions, against the possible risks for hypoglycemia.

Recommendations from other guidelines:^{20,21,22,23}

Generally, guidelines state that the diabetic population can follow the recommendations for the non-diabetic population with regards to alcohol intake. For women who choose to drink, they should limit themselves to 14 grams to less than 20 grams of alcohol per day. For men, this should be limited to 28 grams to less than 30 grams of alcohol per day. The Japanese did not distinguish between men and women but placed an overall limit of 25 grams of alcohol per day.

For individuals with diabetes who choose to drink alcohol, it is recommended that they be advised regarding the risk of moderate alcohol consumption and subsequent acute and delayed hypoglycemia, especially for those taking insulin or insulin-secretagogues. They should also be advised that acute alcohol intake may produce hypoglycemia unawareness. Steps, such as more frequent blood glucose monitoring, should be initiated.

Research Gaps and Recommendations:

There is a limited number of randomized-controlled trials examining the effects of alcohol on human health. Moreover, many confounding variables such as gender, ethnicity, genetics and other environmental and behavioral factors make the interpretation of the impact of alcohol on human health difficult.

Guideline Question 6. Should we recommend the use of non-nutritive sweeteners among adults with type 2 DM?

Recommendation 15. We recommend against the long term (more than 6 months) high intake of non-nutritive sweeteners among adults with type 2 DM due to lack of long-term safety data and suggestion of possible link to increased mortality, stroke, cardiovascular events, and cancer seen with artificially sweetened beverages and other highly processed foods.

Certainty of Evidence: Very low
Strength of recommendation: Strong

Recommendation 16. We suggest allowing the use of non-nutritive sweeteners (NNS) as a short-term (less than 6 months) alternative for patients with T2DM accustomed to sweet beverages, provided that the consumption is below acceptable daily intake, and there is no compensatory increased intake of additional calories and carbohydrates from other food sources.

Certainty of Evidence: Very low
Strength of recommendation: Weak

Background

Non-nutritive sweeteners (NNS) are low calorie alternatives to sugar commonly used to sweeten beverages and foods. They can be derived from natural sources or synthetic (artificial). Examples of artificial sweeteners include aspartame (Equal), sucralose (Splenda), saccharin (Sweet and low), while examples of natural sweeteners include steviol glycosides (Stevia) and monk fruit (luo han guo) extracts. NNS are popular due to their potential benefits, including reduced caloric intake to support weight management, improved blood sugar control for individuals with or at risk of diabetes, dental health benefits, and compatibility with various dietary preferences.

Summary of the Evidence

Characteristics of included studies

Glycemic control

In a systematic review and meta-analysis conducted by Nichol et al.¹, referenced in the American Diabetes Association guideline, the researchers investigated the impact of NNS on blood glucose levels. The analysis included 29 randomized controlled trials (RCTs) involving 741 participants aged 9 to 69 years (mean age 38 years). Participants' ages ranged from 9 to 69 years old, with an average age of 38 years. Of these, 406 participants had normal weight, 71 were overweight, 20 were obese, and 244 did not report body mass index (BMI). In terms of health status, 69 participants had type 2 diabetes, 526 had no chronic conditions, and 146 did not specify health status.

The interventions involved oral consumption of NNS after overnight fasting, including aspartame (n = 280), saccharin (n = 214), stevia (n = 105), and sucralose (n = 142). Blood glucose measurement methods varied: four studies used the glucose oxidase method, one used the

hexokinase method, and the remainder used glucometers; three studies did not report the method used.

Weight loss

A meta-analysis by Miller et al. in 2014² evaluated the effects of non-nutritive sweeteners (NNS) on body weight and composition, pooling data from 15 randomized controlled trials (RCTs) and 9 prospective cohort studies involving a total of 1,951 participants. Most studies were conducted in adults, with four trials including children. Study durations ranged from 3 to 78 weeks, and participant mean ages ranged from 4 to 65 years. Baseline body mass index (BMI) varied widely from 16.8 (in young children aged 4–11 years, corresponding to a 0.03 z-score for age) to 37.7 kg/m², with a median BMI of 29.1 kg/m² across studies. Eight trials were conducted exclusively in overweight or obese populations. The interventions compared NNS with regular-calorie sweeteners, and outcomes assessed included body weight, BMI, and body composition (including fat mass).

Safety outcomes

Safety data were primarily derived from the systematic review and meta-analysis by Rios-Leyvraz and Montez for the WHO (2022).³ This review included RCTs and prospective studies in generally healthy adults (≥ 18 years), children (< 18 years), and pregnant women, with a minimum follow-up duration of one year. The exposure of interest was any non-sugar sweetener (NSS), excluding sugar alcohols and natural caloric sweeteners, consumed either alone or in combination, within acceptable daily intake limits. The safety outcomes assessed included all-cause mortality, cardiovascular disease, stroke, coronary heart disease, chronic kidney disease, and cancer incidence.

Efficacy outcomes

Glycemic control

Non-nutritive sweeteners (NNS) are not shown to impact glycemic control of persons with diabetes.^{1,2} NNS consumption did not result in an increase in blood glucose levels when compared with baseline measurements.¹ In fact, glucose concentrations tended to decline over time following ingestion. Estimated changes relative to baseline were as follows:

- 1–29 minutes: +0.048 mmol/L (95% CI: –0.037 to 0.133);
- 30–59 minutes: +0.006 mmol/L (95% CI: –0.177 to 0.189);
- 60–89 minutes: –0.113 mmol/L (95% CI: –0.248 to 0.022);
- 90–119 minutes: –0.041 mmol/L (95% CI: –0.194 to 0.112);
- 120–149 minutes: –0.245 mmol/L (95% CI: –0.335 to –0.156);
- 150–179 minutes: –0.108 mmol/L (95% CI: –0.170 to –0.045); and
- 180–210 minutes: –0.359 mmol/L (95% CI: –0.434 to –0.283).

This absence of glycemic impact makes NNSs potentially useful for people with diabetes or those on weight loss regimes. As some persons with diabetes prefer sugar sweetened beverages and experience craving for sweet drinks, NNS can be an alternative, provided there is no compensatory increased intake of additional calories and carbohydrates from other food sources.

Weight loss

Since NNS contains little to no calories, their use as an adjunct to multidisciplinary weight loss programs may contribute to weight management in diabetes patients with obesity. Short-term studies have found modest effects on weight loss when substituted for sugar sweetened beverages. In the meta-analysis of Miller et al containing 15 RCTs, the use of non-nutritive sweeteners versus control groups is associated with a modest but statistically significant weight

loss of -0.8 kg (95% CI -1.17 to -0.43).² Non-nutritive sweeteners (NNS) may be safely used to sweeten beverages, provided that recommended acceptable daily intake limits are followed.²

Safety outcomes

While short term NNS consumption may contribute to weight management and glycemic control, there is insufficient evidence to recommend for or against the long term consumption of NNS. Some observational studies suggest an association with cardiovascular events, stroke and even cancer among patients with long term NSS use, particularly high intake of soft drinks and ultra-processed food that contain NNS.^{3–7} The WHO's systematic review and meta-analysis found that that higher consumption of beverages containing non-sugar sweeteners (NSS) was linked to increased risks for several health outcomes.³

RCTs have found no significant link between the consumption of NNS and the risk of diabetes, based on fasting blood sugar (FBS) and HbA1c levels. The mean differences were -0.01 (95% CI: -0.05 to 0.04) for FBS and 0.02 (95% CI: -0.03 to 0.07) for HbA1c. Conversely, prospective studies have indicated a 23% increased risk of diabetes (HR 1.23, 95% CI: 1.14 to 1.32) with the intake of NNS-containing beverages and a 34% higher risk (HR 1.34, 95% CI: 1.21 to 1.48) when consuming tabletop NNS. These differences may be attributed to the varying nature of real-world studies and RCTs. This underscores the importance of considering both study types to fully understand the impact of non-nutritive sweeteners on diabetes risk.

Mortality and cardiovascular outcomes from prospective cohorts revealed a 12% higher risk of all-cause mortality (95% CI: 1.05 to 1.19), a 19% increased risk of cardiovascular mortality (95% CI: 1.07 to 1.32), and a 32% increased risk of composite cardiovascular events including stroke, myocardial infarction, and vascular death (95% CI: 1.17 to 1.50). A significant 19% increase in any type of stroke was also observed (HR: 1.19; 95% CI: 1.09 to 1.29). No significant associations were found between NSS intake and chronic kidney disease (MD: 8.80; 95% CI: -14.65 to 32.25) or cancer risk (HR: 1.02; 95% CI: 0.95 to 1.09).

The proposed biological mechanisms linking non-nutritive sweeteners (NNS) to metabolic diseases involve the activation of sweet taste receptors located outside the mouth, such as those in the gastrointestinal tract and pancreas. When these receptors are stimulated by NNS, they may trigger a hyperinsulinemic response to glucose, potentially leading to increased fat accumulation and insulin resistance. Additionally, interactions with the gut microbiota, particularly with the sweeteners saccharin, sucralose, and steviol glycosides, could also play a role in the alteration of metabolic functions. Research indicates that these sweeteners may alter glucose metabolism and insulin sensitivity, though the precise health effects remain uncertain.⁸

Certainty of Evidence

Glycemic impact

The certainty of evidence for NNS varied by outcome. For glycemic impact, certainty was low due to small, unrepresentative samples, inconsistent measurement methods, and publication bias. For weight loss, certainty was low because of uncontrolled confounding factors and indirectness from including non-target populations. For safety outcomes, certainty ranged from very low to moderate: moderate for diabetes outcomes (FBS and HbA1c) and all-cause mortality, but low for cardiovascular mortality, cardiovascular events, stroke, and chronic kidney disease, and very low for coronary heart disease and cancer incidence. Downgrades were mainly due to risks of bias, small sample sizes, observational study designs, heterogeneity, and wide confidence intervals crossing the null value.

Other Considerations

Avoiding non-nutritive sweeteners (NSS) may lead to reduced purchasing of both NSS products and foods or beverages containing them. As NSS provide no caloric or nutritional value, avoiding them generally requires no dietary substitutions. This decision may result in modest cost savings, with potentially meaningful impacts on household budgets—particularly for individuals from lower socioeconomic groups, who tend to spend a greater proportion of their income on food and beverages.⁹

Adherence to recommendations discouraging NNS use may be low, especially among individuals who regularly consume sugar-sweetened beverages. Reducing both added sugars and NNS can result in a noticeable decrease in the overall sweetness of the diet, which may be difficult for some to accept. To improve acceptability, public information campaigns that raise awareness about the potential health risks associated with NNS use and the benefits of reducing overall dietary sweetness.

Avoiding NNS may be feasible by leveraging product awareness initiatives, government regulations on food labeling, revising product marketing plans, and incorporating these recommendations in current healthy diet education programs.

Guideline Question 7. Should we use diabetes-specific nutrition formulas among adults with type 2 DM?

Recommendation 17. We suggest that diabetes-specific nutrition formulas (DSNF) be used as meal or snack replacement for persons with type 2 diabetes mellitus to minimize post-prandial glucose excursions and improve overall glycemic control in the context of achieving their optimal daily caloric requirements.

Certainty of Evidence: Low

Strength of recommendation: Weak

Background

Diabetes-specific nutrition formulas are formulas available in powder or liquid form that have as their ingredients consist of macro- and micro-nutrients that are designed to help address malnutrition as well as glycemic control. They have low glycemic indices or have slowly-digestible carbohydrates. They may also have lower carbohydrate content, with some carbohydrates being replaced by monounsaturated fatty acids (MUFA) or polyunsaturated fatty acids, as compared to other standard nutrition formulas (STDF).

These formulas are available in calorie-controlled portions that can be used for the following indications:

1. To provide meals or snack replacements in an iso- or hypo-caloric manner
2. To provide additional calories for the undernourished
3. To serve as aid for very-low-calorie diets
4. To provide enteral nutrition in tube-fed patients.

DSNFs are available in both powdered and premixed liquid forms, and depending on the preparation, will provide a specific number of calories. These may be utilized to completely replace a meal, such as breakfast, lunch or dinner, or a snack, in which case no other food is eaten. Partial meal replacement may also be done, wherein part of the calories and nutrients for a meal is replaced by DSNF. Care must be taken not to provide an excess amount of nutrients and calories per meal or snack.

Summary of the Evidence

Characteristics of included studies

We found 12 RCTs, 5 meta-analyses and reviews and 1 observational study. The RCT's looked at the glycemic outcomes of patients with type 2 diabetes mellitus in terms of postprandial glucose excursions, glycemic variability/excursions, fasting plasma glucose and glycosylated hemoglobin after being given DSNF as compared to standard formulas, high-carbohydrate liquid diet, or regular food. Other outcomes measured were impact on lipid profile (3 studies [4,7, 14]), blood pressure [3] and composite of death from CV causes, nonfatal MI, nonfatal stroke or hospitalization for angina [15]. DSNFs were provided either as meal or snack replacements or as a sole source of nutrition especially for tube-fed patients. Trial durations ranged from a few days to 9.6 years.

The meta-analysis and systematic reviews included RCTs that compared DSNF versus standard formula. Outcomes of interest included glycemic control as outlined above. 4 of the reviews also looked at lipid profiles [14, 16, 17, 18], and 1 review [14] also looked at blood pressure control.

Efficacy outcomes

Overall, the data has shown that there is improvement in glycemic control, in terms of decreased postprandial glucose excursions, less glycemic variability, and improvements in fasting plasma glucose and glycosylated hemoglobin, in patients with diabetes who are given DSNF as meal replacement compared to standard formula or regular meals/snacks. In the meta-analysis conducted by Elia et al (16), DSNF decreased peak glucose by 1.58mmol/L (95%CI [0.88, 2.32]) compared to standard formula. Further, the meta-analysis by Ojo (17) showed a mean difference of -1.15mmol/L (95% CI, [-2.07, -0.23]) in fasting blood glucose and a mean difference of -0.67% (95%CI, [-1.14, -0.21]) in glycated hemoglobin.

There is conflicting data on the impact on lipid profiles, with some analysis showing no significant differences in total cholesterol, LDL and triglyceride levels between DSNF and standard formula (16). However, a different meta-analysis showed positive impact on lipid profiles such as lower triglyceride and higher HDL levels (13). The latter were associated with inter-study homogeneity. In terms of clinical endpoints, the LOOK AHEAD trial did not show any reduction in adverse cardiovascular outcomes in patients provided DSNF as meal replacements in the long term, leading to study termination after a median of 9.6 years.

Safety outcomes

The meta-analysis of Sanz-Paris et al (18) looked at a total of 9 RCTs that reported on adverse events and noted that there was no overall increase in adverse events with the use of DSNF (0.95 [0.87,1.05] p=0.007).

Certainty of Evidence

With regards to the outcome of fasting blood glucose, there is low certainty of evidence due to serious inconsistencies and imprecision. The other outcomes of glycosylated hemoglobin, LDL cholesterol, triglycerides and HDL cholesterol had low levels of certainty due serious risks of bias and serious inconsistencies. The outcome of total cholesterol has a moderate level of certainty despite a high risk for bias. Overall level of certainty was deemed to be low.

Table 7.1. Efficacy and Safety Outcomes of Moderate Alcohol Intake among patients with type 2 diabetes mellitus

Outcomes	Basis	Effect Estimate	95% Confidence Interval	Interpretation	Certainty of Evidence
CV events	1 cohort (n=10,727)	HR 0.82	0.71, 0.95	As good as or better	Very Low
Microvascular Complications	1 cohort (n=10,727)	HR 0.83	0.72, 0.97	As good as or better	Very Low
Glycemic Control (HbA1c)	4 RCT (n=362)	MD 0.02	-0.18, 0.15	equivalent	Low
Weight (kilogram)	2 RCT (n=247)	MD 0.18	-0.58, 0.95	equivalent	Low
Hypoglycemia	2 RCT (n=105)	RR 2.72	0.15, 50.79	Harm	Low
Triglyceride (mg/dl)	3 RCT (n=362)	MD -8.74	-11.95, -5.53	Benefit	Moderate
High Density Lipoprotein (mg/dl)	2 RCT (n=362)	MD	0.72, 1.09	Benefit	Moderate
Low Density Lipoprotein (mg/dl)	2 RCT (n=247)	MD 6.31	-15.8, 3.18	Inconclusive	Low
Systolic Blood Pressure	4 RCT (n=386)	MD 0.1	-0.07, 0.26	Equivalent	Low
Diastolic Blood Pressure	4 RCT (n=386)	MD -0.57	-2.03, 0.88	Inconclusive	Low

Other Considerations

Recommendations from other groups:

The American Diabetes Association (ADA) Standards of Care in Diabetes – 2024 stated that partial or total meal replacements may be effective tools to aid individuals in attaining glycemic control, to improve nutrient quality and to promote weight loss by reducing calorie intake.

The following is a list of some of the DSNF available in the Philippines with a comparison of some of their properties.

	Glucobest (chocolate) Per 100 gm	Glucobest (vanilla) Per 100 gm	Nutren Diabpro Per 100 gm	Glucerna (Powder) per 100 gm	Diabetasol per (capuccino) 100 gm	Diabetamil per 100 gm
Energy (KCal)	439	440	455	437	430	400
Fat (gm)	17	16.3	19.5	16.7	11.8	11
	MUFA 9.8 gm PUFA 4.8 gm	MUFA 9.5 gm PUFA 4.8 gm	MUFA 10.2 gm PUFA 3.2 gm	MUFA 7.77 PUFA 3.76	MUFA 6 PUFA 4	No mention of MUFA/PUFA in nutritional composition
Total Carbohydrates (gm)	44.8	49	43.7	57.6	65	64
Protein (gm)	21.4 (whey)	20 (whey)	22.8 (whey and milk protein: cassein)	19.5 (soy and milk protein; cassein)	16 (whey and cassein)	16 (whey and milk)
Carbohydrate/ Sugar	Dextrin Isomaltulose Sugars 7.5 gm	Dextrin Isomaltulose Sugars 7.5 gm	Dextrin Isomaltulose Sugars 9.5 gm	Maltitol Sugars 10.77 gm	Maltodextrin Isomaltulose Sugars 9.9 gm	Maltodextrin
Artificial sweetener	Acesulfame K Sucralose	Acesulfame K Sucralose	Acesulfame K	(None)	Sucralose Sorbitol	Sucralose
Lactose	Negligible	Negligible	Negligible	Negligible	Negligible	Contains milk (skim milk powder and full cream milk powder)

Guideline Question 8. Should we recommend referral for bariatric surgery to achieve sustained glycemic control and weight loss among obese adults with type 2 DM?

Recommendation 18. We suggest referral for bariatric surgery among adults with type 2 diabetes and obesity w/ BMI ≥ 37.5 kg/m² and those with BMI ≥ 32.5 -37.4 kg/m² who do not achieve desired glycemic control and weight loss with medical therapy.

Certainty of Evidence: Low

Strength of recommendation: Weak

Background

Surgical procedures for the treatment of obesity are interchangeably referred to as bariatric, metabolic, or weight loss surgery. In adults with T2DM and BMI ≥ 40 kg/m² (> 37.5 kg/m² for Asians), or those with BMI 35-39.9 kg/m² (32.5-37.4 kg/m² for Asians) who have not achieved glycemic targets and weight loss w/ non-surgical methods, bariatric surgery may be offered as a treatment option for durable glycemic control and weight management.¹

For Asian populations, the BMI cut-offs for considering bariatric surgery are reduced by 2.5 kg/m² based on epidemiologic evidence² and expert consensus indicating that South Asians have higher visceral adiposity and risk of developing diabetes and cardiovascular diseases at a lower BMI. This threshold adjustment aligns with recommendations from international societies such as the American Diabetes Association (ADA) and the American Association of Clinical Endocrinology (AACE).

The most frequently performed metabolic/bariatric surgeries include:

- Vertical sleeve gastrectomy (VSG): removal of approximately 80% of the stomach, creating a thin sleeve of stomach pouch. This reduces ghrelin secretion (a hunger-stimulating hormone) and increases levels of GLP-1 and PYY, which promote satiety and improve glucose metabolism.
- Roux-en-Y gastric bypass (RYGB): creation of a small gastric pouch (<20 mL) anastomosed to the jejunum, bypassing part of the stomach and the proximal small bowel
- Biliopancreatic diversion (BPD): partial gastrectomy with a long intestinal bypass.

Summary of the Evidence

Characteristics of included studies

Evidence on the effects of bariatric surgery compared with medical and/or lifestyle management was derived from 19 randomized controlled trials (RCTs) enrolling a total of 674 participants. These comprised 11 primary randomized controlled trials³⁻¹³ and 8 follow up RCTs¹⁴⁻¹⁷. The trials enrolled patients with type 2 diabetes mellitus and obesity. Different bariatric surgery types were utilized:

- Roux-en-Y gastric bypass (RYGB): 7 RCTs
- Laparoscopic adjustable gastric banding (LAGB): 4 RCTs
- Biliopancreatic diversion (BPD): 1 RCT

Sleeve gastrectomy (SG): 1 RCT

Treatment effects were assessed at multiple follow-up intervals, most commonly at 12, 24, 60, and 120 months. The primary outcomes reported across trials were:

- sustained glycemic control, defined as achieving glycosylated hemoglobin (HbA1c) less than 7.0%;
- change in body weight in kilogram (kg), defined as the difference in weight from baseline to end of trial.
- incidence of macrovascular complications, including myocardial infarction, stroke, peripheral vascular disease, and ischemic heart disease; and
- incidence of microvascular complications, including nephropathy, retinopathy, and neuropathy.

Efficacy outcomes

Glycemic Control

Bariatric surgery may be associated with a significantly higher odds of achieving sustained glycemic control compared with medical and/or lifestyle management at all reported follow-up periods:

- ~ 12 months: OR 11.11 (95% CI 3.72, 33.19); 8 RCTs
- ~ 24 months: OR 15.06 (95% CI 4.15, 54.72); 4 RCTs
- 36-60 months: OR 5.76 (95% CI 2.59, 12.81); 4 RCTs
- ~120 months: OR 18.84 (95% CI 1.05, 337.71); 1 RCT.

The certainty of evidence was low due to high heterogeneity and imprecision (low event rate).

Change in Body Weight (Kilogram)

Based on 7 RCTs, bariatric surgery may produce a significantly greater reduction in body weight compared with medical and/or lifestyle management, with mean difference of -17.31 kg (95% CI -24.68 to -9.94 kg). The certainty of the evidence was low, downgraded for serious risk of bias (attrition bias) and serious inconsistency (heterogeneity across studies).

Microvascular Complications

Pooled risk ratios based on 3 RCTs showed no statistically significant benefit of surgery in reducing the incidence of microvascular complications (RR 0.30 [95% CI 0.07 to 1.27]). Certainty was very low, downgraded for serious risk of bias (attrition bias), inconsistency, and very serious imprecision (small sample size, few events).

Macrovascular Complications

Based on 6 RCTs, there may be no significant benefit of surgery for reducing macrovascular complications (RR 0.57 [95% CI 0.17 to 1.99]). Certainty was low, downgraded due to serious risk of bias (attrition bias) and serious imprecision (low sample size and few events).

Safety outcomes

Evidence on perioperative mortality and complications was obtained from a retrospective analysis of an Italian database, which included 13,871 bariatric surgical procedures performed between 1996 to 2006.¹⁸ Procedures included gastric bypass, biliopancreatic diversion, silicone gastric banding, vertical banded gastroplasty, and biliointestinal bypass. The 60-day mortality rate was 0.25%, with pulmonary embolism as the most common cause of death (38.2% of cases). Mortality was observed more frequently in the biliopancreatic diversion group than in other surgical groups. Retrospective data from 50 patients (of which 60% underwent laparoscopic adjustable gastric banding (LAGB) and 40% Roux-en-Y gastric bypass (RYGBP)) in a tertiary care private hospital in the Philippines (2002-2004) showed low complication rates with bariatric surgery and similar outcomes with other Asian and Western countries. There was no mortality, and early complications were: wound infection (2/50 or 4%), and 1/50 or 2% each

for pneumonia, dehydration, gastritis, and leakage. Late complications were: band slippage (2/20 or 10%), stomal stenosis (1/20 or 5%), and ventral hernia (1/5 or 20%).¹⁹

Certainty of Evidence

Overall certainty of evidence was then downgraded to low due to high heterogeneity and high risk of attrition bias.

Outcomes	Basis	Effect Estimate	95% Confidence Interval	Interpretation	Certainty of Evidence	NNT
Sustained Glycemic Control (HbA1c <7.0%)						
* By 12 months	8 RCTs	OR 11.11	3.72, 33.19	Benefit	Low	3
* By 24 months	4 RCTs	OR 15.06	4.15, 54.72	Benefit	Low	2
* By 36-60 months	4 RCTs	OR 5.76	2.59, 12.81	Benefit	Moderate	3
* By 120 months	1 RCT	OR 18.84	1.05, 337.71	Benefit	Low	3
Weight Reduction	7 RCTs	MD -17.31	-24.68, -9.94	Benefit	Low	
Microvascular Complications	3 RCTs	RR 0.30	0.07,1.27	Inconclusive	Low	8
Macrovascular Complications	6 RCTs	RR 0.57	0.17,1.99	Inconclusive	Low	65

NNT, number needed to treat; MD, mean difference; OR, odds ratio; RCT, randomized controlled trials; HbA1c, glycosylated hemoglobin

D. Monitoring and treatment targets

Guideline Question 9. Should we recommend HbA1c to monitor glycemic control among nonpregnant adults with type 2 DM?

Recommendation 19. We recommend the use of laboratory-measured HbA1c to monitor glycemic control among nonpregnant adults with T2DM

Certainty of Evidence:	High
Strength of recommendation:	Strong

Recommendation 20. We recommend Hba1c testing every 3 months until HbA1c is stable on unchanging therapy, then every 6 months once the HbA1c level and blood glucose lowering therapy are stable.

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Background

Glycated hemoglobin (A1C, hemoglobin A1C, HbA1c) is the most widely used laboratory test to monitor glycemic control. It reflects the average blood glucose level over approximately 3 months. It is the metric used to date in clinical trials demonstrating the benefits of improved glycemic control on microvascular and macrovascular outcomes. It is the primary endpoint for the demonstration of glycemia-lowering efficacy for new diabetes drugs. The performance of HbA1c is generally excellent for National Glycohemoglobin Standardization Program (NSGP)-certified assays. It should ideally be performed routinely in all patients with diabetes. The A1C-Derived Average Glucose (ADAG) study established a reliable regression equation that can be used to translate HbA1c results into an estimated average glucose value. The differences in HbA1c across racial/ethnic groups are so small (approximately 0.3 to 0.4 percentage-point difference in HbA1c) that it will not significantly modify the association between HbA1c and diabetes-related complications.

Summary of the Evidence

In various large randomized controlled trials, HbA1c control led to significant reductions in microvascular complications. The quality of the evidence is high with consistent estimate of effects. Most trials included all racial groups, eg the ADVANCE trial in 2008 included 20 countries from Asia, Australia, Europe, and North America together with 4 centers in the Philippines.

The frequency of HbA1c testing should depend on the clinical situation, the treatment and the clinician's judgment. HbA1c can be checked every 3 months among patients whose therapy has recently changed and/or in those who are not meeting glycemic goals. On the other hand, in patients who are meeting treatment goals and who have stable glycemic control, HbA1c may be determined every 6 months.

Point-of-care devices are used to measure HbA1c in clinics using a fingerstick sample. Although there is concern about the analytical performance of some of these devices, the results were comparable with laboratory-measured HbA1c in one study. Although point-of-care testing is not recommended for diagnosis of diabetes or prediabetes, it may play a valuable role in monitoring diabetes control in clinical settings as it can provide immediate availability of HbA1c

results at the time of the patient visit. Immediate patient feedback at clinician visits regarding HbA1c values may improve glycemic control by facilitating clinical decision-making and more efficient patient-provider communication.

When HbA1c values are unexpected or discordant with other information such as glucose measurements, the following factors should be considered: red cell turnover, hemoglobin variants, chronic kidney disease. Likewise, HbA1c does not provide a measure of glycemic variability or hypoglycemia. For patients prone to glycemic variability because of severe insulin deficiency, glycemic control is best evaluated by the combination of results from blood glucose monitoring and HbA1c.

In a study by Imai, the risk of developing CKD for patients with high adherence to HbA1c monitoring was significantly lower than for patients with low adherence (OR: 0.42, 95% CI 0.18 to 0.99). According to Driskell, testing HbA1c every 3 months was associated with a 3.8% reduction in HbA1c compared with a 1.5% increase observed with annual testing. Compared with annual monitoring, 3-monthly testing was associated with a halving of the proportion showing a significant rise in HbA1c (7–10 vs. 15–20%). Testing more frequently than 3 months provided no additional benefit over testing every 3 to 4 months. In a study by Duff, compared with four tests/year, having more than four tests/year were associated with lower likelihoods of achieving either target; two to three tests/year gave similar likelihoods to four tests/year.

Certainty of Evidence

GRADE Table Summary for HbA1c as a Glycemic Control Monitor

Outcome	Anticipated Effects	Certainty of Evidence (GRADE)	Remarks
Glycemic Control	HbA1c is the most widely used laboratory test to monitor glycemic control, reflecting average blood glucose over approximately 3 months. It is the standard metric used in clinical trials demonstrating benefits on microvascular and macrovascular outcomes.	High	The test is a primary endpoint for evaluating the efficacy of new diabetes medications. While there are small racial/ethnic differences, they do not significantly impact its association with diabetes-related complications.
Frequency of Testing	Testing frequency (twice a year or quarterly) is associated with improved glycemic control. Quarterly testing is recommended for patients with recently changed therapy or those not meeting glycemic goals.	Moderate	More frequent testing (3-monthly) is linked to a significant reduction in HbA1c compared to annual testing. Studies suggest 2-3 tests per year yield similar outcomes to 4 tests per year in some cases.

Feasibility and Implementation	NSGP-certified assays provide excellent performance. Point-of-care devices can provide immediate results.	High	While point-of-care tests are not recommended for diagnosis, they are valuable for monitoring, facilitating immediate patient feedback and improving clinical decision-making. The test is widely available and supported by major international guidelines.
Limitations	HbA1c does not measure glycemic variability or hypoglycemia. Its values can be affected by factors like red cell turnover, hemoglobin variants, and chronic kidney disease.	High	When HbA1c results are unexpected, clinicians should consider alternative methods and further investigation. For patients with severe insulin deficiency, a combination of blood glucose monitoring and HbA1c is recommended.

Research Gaps and Recommendations

1. **Effectiveness of Point-of-Care Testing:** While some studies show comparable results, further research is needed to more widely validate the analytical performance of different point-of-care HbA1c devices, especially across various clinical settings and patient demographics.
2. **Optimal Testing Frequency:** While some studies suggest frequencies, more definitive research is needed to establish the optimal testing frequency for different patient profiles (e.g., type 1 vs. type 2 diabetes, different comorbidities, or those using specific medications) to maximize clinical benefit and cost-effectiveness.
3. **Integration of HbA1c with Other Metrics:** Investigate the most effective strategies for combining HbA1c with other metrics like Continuous Glucose Monitoring (CGM) or fructosamine, especially in patient populations with high glycemic variability or conditions affecting red blood cell turnover.
4. **Addressing Confounding Factors:** Develop clearer, evidence-based guidelines for clinicians on how to interpret HbA1c results in the presence of confounding factors (e.g., chronic kidney disease, hemoglobinopathies) and when to transition to alternative markers.

Other Considerations

Conditions Associated with Falsely Elevated or Lowered HbA1c

Conditions associated with falsely increased HbA1c
Anemias associated with decreased red cell turnover (iron deficiency, vitamin B-12, folate deficiency)
Asplenia (increased erythrocyte lifespan)
Uremia (formation and detection of carbamyl-hemoglobin)
Severe hypertriglyceridemia (>1,750mg/dL)
Severe hyperbilirubinemia (>20mg/dL)
Chronic alcohol consumption (formation of acetaldehyde-HbA1 compound)
Chronic salicylate or opioid ingestion or lead poisoning (mechanism uncertain)
Conditions associated with falsely decreased HbA1c
Anemia from acute or chronic blood loss (eg hemolytic anemia)
Splenomegaly ((decreased erythrocyte lifespan)
Pregnancy (decreased erythrocyte lifespan)
- Falsely low HbA1c values through the 2nd trimester, but may rise during the 3rd trimester
Ribavirin and interferon-alpha (possibly due to hemolytic anemia)
Conditions that may falsely increase or decrease HbA1c
Red blood cell transfusion
- High glucose concentration in storage medium (false increase)
- Dilutional effect (false decrease)
Hemoglobin variants
- Depends on method and assay used
A1c generally reliable for heterozygous variants, but not homozygous variants
Vitamin C ingestion
- May increase HbA1c when measured by electrophoresis
- May decrease HbA1c when measured by chromatography due to competitive inhibition of glycosylation

Adapted from Radin MS. Pitfalls in hemoglobin A1c measurement: when results may be misleading. J Gen Intern Med. 2014 Feb;29(2):388-94.

Guideline Question 10. Should we use fasting plasma glucose (FPG) to monitor glycemic control among nonpregnant adults with type 2 DM?

Recommendation 21. We recommend the use of FPG in conjunction with HbA1c and/or SMBG to monitor glycemic control among nonpregnant adults with type 2 DM

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 22. We suggest against the routine use of FPG **alone** to monitor glycemic control among nonpregnant adults with type 2DM

Certainty of Evidence:	Moderate
Strength of recommendation:	Weak

Recommendation 23. We suggest FPG monitoring of response to therapy, determination of the risk for nocturnal hypoglycemia, and investigation of discordant levels of blood glucose monitoring and HbA1c among nonpregnant adults with T2DM

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Background

Fasting plasma glucose alone cannot be used as the basis of glycemic target. It has utility in conjunction with other parameters in monitoring glycemic variability and risk of hypoglycemia. It is not used in clinical trials as the primary metric to correlate with glycemic control and development of complications.

Summary of the Evidence

In a meta-analysis to assess the effects of reduction of FPG and PPG on cardiovascular outcomes in randomized controlled trials, the Mantel-Haenszel Odds Ratio (MH-OR) for cardiovascular events, all-cause and cardiovascular mortality in trials with a between-group FPG difference of 1mmol/l produced a significantly lower all-cause (MH-OR 0.90 [0.81, 0.99], $p=0.03$) and cardiovascular mortality (MH-OR 0.86 [0.76, 0.97], $p=0.012$), with no significant effect on the incidence of major cardiovascular events. Generally, the quality of the evidence is moderate since there are minor inconsistencies (the estimates of effects in the trials were generally consistent) and minor imprecision. An issue on indirectness is that in all the trials, HbA1c was still the primary parameter utilized, FPG was only secondary. However, fasting plasma glucose may be checked in certain situations such as immediate monitoring of response to therapy, determination of the risk for hypoglycemia and investigation of discordant levels of blood glucose monitoring and HbA1c. For monitoring in low resource settings where HbA1c is not available, fasting plasma glucose value of <100 mg/dL is correlated with intensified regimen achieving target HbA1c.

Certainty of Evidence

Table 10.1: Summary of Evidence: Effect of fasting plasma glucose reduction on major cardiovascular events, all-cause mortality, and cardiovascular mortality.

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	More intensive FPG reduction	Less intensive FPG reduction	Relative (95% CI)	Absolute (95% CI)		
Major cardiovascular events												
13	randomised trials	not serious	not serious	not serious	not serious	none			OR 0.950 (0.818 to 1.105)	1 fewer per 1,000	⊕⊕⊕ ○ Moderate	CRITICAL
All-cause mortality												
8	randomised trials	not serious	not serious	not serious	not serious	none			OR 0.904 (0.825 to 0.991)	1 fewer per 1,000	⊕⊕⊕ ○ Moderate	CRITICAL
Cardiovascular mortality												
8	randomised trials	not serious	not serious	not serious	not serious	none			OR 0.857 (0.760 to 0.967)	1 fewer per 1,000	⊕⊕⊕ ○ Moderate	CRITICAL

Monami M, Adalsteinsson JE, Desideri CM, Ragghianti B, Dicembrini I, Mannucci E. Fasting and post-prandial glucose and diabetic complication. A meta-analysis. *Nutr Metab Cardiovasc Dis*. 2013 Jul;23(7):591-8.

GRADE Table Summary for Fasting Plasma Glucose (FPG)

Outcome	Anticipated Effects	Certainty of Evidence (GRADE)	Remarks
Glycemic Control	FPG is a poor measure of overall glycemic control, with a weaker correlation to HbA1c than postprandial glucose (PPG). It can be used for immediate monitoring of therapy response, and a value of <100 mg/dL may be associated with an intensified regimen that achieves target HbA1c in low-resource settings.	Low	FPG is not used as a primary metric in clinical trials to correlate with long-term glycemic control or the development of complications. Its utility is in conjunction with other parameters to monitor glycemic variability.
Prognostic Significance	FPG is not a primary metric used in clinical trials to correlate with the development of complications.	Very Low	There is a high positive correlation between FPG and HbA1c for the diagnosis of diabetes, but this does not translate to its use as a prognostic tool for monitoring.
Feasibility and Implementation	FPG measurement is widely available, relatively inexpensive, and easy to perform.	High	FPG is used for diagnosis and as an adjunct to other monitoring tools but is not recommended in major guidelines for monitoring long-term glycemic control. Self-monitoring of blood glucose on multiple occasions is considered more practical.
Limitations	FPG alone cannot be used as a basis for glycemic targets. It does not provide a comprehensive picture of glycemic variability. The ability of beta cells to secrete insulin may be restricted at levels >110 mg/dL.	High	Clinicians should not rely on FPG alone for monitoring. It has limited utility beyond short-term assessments and specific situations.

Research Gaps and Recommendations

1. **Standardized FPG Targets for Long-Term Outcomes:** Conduct studies to establish if specific FPG target ranges, when achieved, are independently associated with improved long-term microvascular and macrovascular outcomes.
2. **Role of FPG in Hypoglycemia Risk:** Further investigate the precise role of FPG, in combination with other metrics, in predicting and preventing hypoglycemic events in patients on various diabetes medications.
3. **Utility of FPG in Resource-Limited Settings:** Research is needed to develop and validate clear protocols for using FPG for glycemic monitoring in low-resource settings

where HbA1c testing is unavailable. This could include a specific algorithm for adjusting therapy based on FPG trends.

Correlation with Postprandial Glucose (PPG): Conduct more studies to better understand the relative contribution of FPG to overall hyperglycemia in different patient populations and on different diets, and how it correlates with PPG. This would help refine its use as a supplementary monitoring tool.

Guideline Question 11. Should we recommend the use of 1,5-anhydroglucitol (1,5-AG) to monitor glycemic control among nonpregnant adults with type 2 DM?

Recommendation 24. We suggest against the use of 1,5-AG to monitor glycemic control among nonpregnant adults with type 2 DM.

Certainty of Evidence:	Very Low
Strength of recommendation:	Weak

Background

1,5-anhydroglucitol is one of the common polyols in humans. It is metabolically stable and is excreted in the urine when the renal threshold for glucosuria is exceeded. It is a short-term indicator of glycemic status over the past 48 hours to 2 weeks. Additionally, the measurement of 1,5-AG does not require fasting and the colorimetric assay has stable reagents and is easily automated and reproducible (1,2).

Summary of the Evidence

We didn't find any study that directly assessed the effect of 1,5-AG on diabetic complications or glycemic control. There were also no studies assessing its accuracy. In patients with HbA1c between 6.5 to 8%, the 1,5-AG has good correlation to HbA1c. In a study done among Asian Indians, 1,5-AG showed a significant negative correlation with HbA1c ($r = -0.7$, $p < 0.001$) (3). 1,5-AG is inversely associated with postprandial glucose. In those with HbA1c of $<7\%$, about a quarter had 1,5-AG that is <10 ug/mL which may denote postprandial glucose excursions. Studies by Won Jun Kim et al in 2013 showed correlation of low 1,5-AG in the development of diabetic retinopathy among diabetics with HbA1c $<8\%$. In another study by Ohira in 2012, it was shown that a change in 1,5-AG is significantly correlated with cardio-ankle vascular index reflecting arterial stiffness (4). Despite the advancement in the study of 1,5-AG, there is paucity of data about its translation into average glucose levels and its prognostic significance is not as clear. Moreover, there is no standardized cut-off level that has been established for 1,5-AG as a monitoring tool for diabetes. Also, 1,5-AG can be influenced by other factors such as dairy product, race, uric acid, triglycerides, liver disease, gastrectomy state, and cystic fibrosis (5). Significant renal disease with serum creatinine exceeding 2.0-3.0 mg/dl can reduce 1,5-AG (6).

In DM, glucose excretion increases and exceeds the capacity of SGLT2 and SGLT1 to reabsorb glucose. When this happens reabsorption of glucose happens via SGLT4 which is the site for 1,5-AG absorption. The competitive inhibition with glucose in this channel causes a reduction in the reabsorption of 1,5-AG. In patients given an SGLT2 inhibitor, there is inhibition of reabsorption of glucose through SGLT2 and glucose appears in the renal tubule downstream of SGLT2, serum 1,5-AG shows a low value and cannot be used as an index of glycemic control (6).

Certainty of Evidence

No direct studies were found.

GRADE Table Summary for 1,5-Anhydroglucitol

Outcome	Anticipated Effects	Certainty of Evidence (GRADE)	Remarks
Glycemic Control	Correlates well with HbA1c in patients with HbA1c between 6.5% and 8%. 1,5-AG is inversely associated with postprandial glucose excursions.	Low	While a negative correlation with HbA1c is noted, there's no standardized cut-off level for 1,5-AG as a diabetes monitoring tool. The sensitivity is low at high plasma glucose levels.
Prognostic Significance (Cardiovascular and Retinopathy)	Low 1,5-AG (< 6 µg/mL) is associated with an increased risk of coronary heart disease, heart failure, stroke, and death. Low 1,5-AG (< 8 µg/mL) is also correlated with the development of diabetic retinopathy.	Very Low	The ADA 2022 guidelines state that the prognostic significance of 1,5-AG is not as clear as that of CGM or HbA1c. There is no established correlation between its use and diabetic complications/outcomes.
Use in Special Populations	May be a reliable glycemic index in mild-to-moderate renal dysfunction. The use of SGLT2 inhibitors and severe renal disease (serum creatinine > 2.0-3.0 mg/dL) significantly reduces 1,5-AG levels.	Low	Other factors, such as race, uric acid, and certain medical conditions (liver disease, cystic fibrosis), can also influence 1,5-AG levels.
Feasibility and Implementation	Measurement does not require fasting. Colorimetric assays are stable, easily automated, and reproducible.	High	The test is not included in the Canadian, Malaysian, or Japanese Diabetes guidelines and is only cited in the ADA guidelines as a secondary measure.

Other Considerations

COST

Php1,200.00 to Php3,200.00

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY

We did not find any study that addressed patient's values or preferences.

Recommendations From Other Groups

None of the clinical guidelines reviewed mentioned 1,5-AG.

Research Gaps and Recommendations

1. **Standardized Cut-off Levels:** Further research is needed to establish standardized, evidence-based cut-off values for 1,5-AG to be used as a monitoring tool for diabetes and to define what constitutes a clinically significant change.
2. **Correlation with Average Glucose:** Additional studies should focus on translating 1,5-AG values into estimated average glucose (eAG) levels, similar to the established relationship for HbA1c.
3. **Prognostic Significance and Outcomes:** Large-scale, prospective studies are required to more clearly define the prognostic significance of 1,5-AG and to establish its correlation with long-term diabetic complications and outcomes, such as retinopathy, nephropathy, and cardiovascular disease.
4. **Influence of Confounding Factors:** More data is needed to understand the quantitative impact of various confounding factors, such as race, diet, and concurrent medications (beyond SGLT2 inhibitors), on 1,5-AG levels to improve its clinical interpretation.
5. **Role in Glycemic Variability:** Investigate the specific role of 1,5-AG as a marker for short-term glycemic variability and its utility in monitoring patients who may have well-controlled HbA1c but significant postprandial glucose excursions.

Guideline Question 12. Should we use fructosamine to monitor glycemic control among nonpregnant adults with type 2 DM?

Recommendation 25. We suggest against the use of fructosamine to monitor glycemic control among nonpregnant adults with type 2 DM.

Certainty of Evidence:	Very Low
Strength of recommendation:	Weak

Background

Fructosamine is formed from glycation reactions between glucose and plasma proteins. It reflects blood glucose levels over the previous 2 to 4 weeks.¹ It may be useful as an adjunct to HbA1c to provide information on short-term glycemic control, and/or for monitoring glycemic control when HbA1c is unreliable such as in patients with anemia, kidney disease and hemoglobinopathies.²

Summary of the Evidence

Characteristics of included studies

Indirect evidence for this recommendation was based on the Atherosclerosis Risk in Communities (ARIC) Study and AMORIS cohort studies. In ARIC a cross-sectional study of 1,600 community-based participants (227 with a history of diabetes and 1,323 without), the associations of diabetes-specific tertiles of fructosamine with prevalence of chronic kidney disease, albuminuria, and retinopathy was investigated.²

After adjustment for demographic, clinical, and lifestyle variables, participants in the highest fructosamine categories also had a greater risk of microvascular complications, including retinopathy and albuminuria (P trend = 0.003). Elevated baseline concentrations of fructosamine has also been shown to be associated with coronary heart disease, ischemic stroke, heart failure, and death, with patterns of association similar to those observed for HbA1c.³ Even after adjustment for HbA1c, fructosamine remained significantly associated with vascular outcomes and death.

The AMORIS cohort study from Sweden reported that fructosamine and HbA1c values tended to move in parallel over time, suggesting that both markers reflect similar aspects of glycemic control in individuals with type 1 and type 2 diabetes.⁴ A strong positive correlation between fructosamine and HbA1c has also been observed in studies involving Southeast Asian populations.⁵

Despite these correlations, no randomized controlled trials have evaluated whether interventions targeting fructosamine reduction translate into improved patient-important outcomes. The accuracy of fructosamine measurements can be affected by conditions associated with altered protein metabolism or protein loss as in nephrotic syndrome as well as diminished protein synthesis, thyroid disease and malnutrition.¹ Fructosamine has the advantage of not requiring a fasting sample.⁶ However, analytical challenges include lack of assay standardization, potential interference from hyperlipidemia, and limited test availability in many settings in the Philippines.⁷

The limited period of glycemia that they reflect limits their clinical utility. Similarly, the limited data that support their relationship with risk of complications makes them less useful than HbA1c. Moreover, treatment goals have not been established. While efforts are underway to develop a reference method for glycated albumin, neither fructosamine nor glycated albumin is standardized.

Certainty of Evidence

The evidence is of very low certainty due to the lack of randomized studies.

GRADE Table Summary for Fructosamine

Outcome	Anticipated Effects	Certainty of Evidence (GRADE)	Remarks
Glycemic Control	Reflects short-term blood glucose levels over 2-4 weeks. Fructosamine and HbA1c levels tend to parallel each other, indicating a similar reflection of glucose control.	Low	Useful as an adjunct to HbA1c or when HbA1c is unreliable (e.g., in patients with anemia, kidney disease, or hemoglobinopathies). Accuracy can be affected by conditions with altered protein metabolism.
Prognostic Significance (Cardiovascular and Microvascular)	Associated with an increased risk of diabetes, retinopathy, and albuminuria. Also associated with cardiovascular outcomes (coronary heart disease, ischemic stroke, heart failure, and death) similar to HbA1c.	Very Low	The American Diabetes Association (ADA) and Canadian Diabetes Association do not recommend its use due to a lack of clear established prognostic significance and correlation with HbA1c.
Feasibility and Implementation	Does not require a fasting sample for measurement.	Moderate	Not widely available in many parts of the country, which limits its use.

Research Gaps and Recommendations

1. **Standardized Cut-off Levels:** Establish standardized cut-off values for fructosamine that correlate with established glycemic control targets and a specific risk for microvascular and macrovascular complications.
2. **Long-Term Prognostic Studies:** Conduct large, long-term prospective studies to clearly establish the prognostic significance of fructosamine and its independent association with diabetic complications, especially when used in combination with or as an alternative to HbA1c.
3. **Accuracy in Diverse Populations:** Investigate the accuracy and reliability of fructosamine measurements in diverse patient populations with different confounding conditions (e.g., various chronic illnesses, different stages of kidney or liver disease, and malnutrition).
4. **Clinical Utility and Outcomes:** Research how the use of fructosamine to guide treatment decisions impacts patient outcomes, such as rates of hospitalization, hypoglycemia, and long-term complication development.

Other Considerations

Cost

We did not find any local economic evaluation studies.

Patient's Values And Preference, Equity, Acceptability, And Feasibility

We did not find any study that addressed patient's values or preferences

Guideline Question 13. Should we recommend self-monitoring of blood glucose (SMBG) to monitor glycemic control among adults with type 2 DM?

Recommendation 26. We recommend self-monitoring of blood glucose (SMBG) for individuals on insulin therapy among adults with type 2 DM.

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 27. We suggest self-monitoring of blood glucose (SMBG) among adults with type 2 DM who are not on insulin, if they:

- have poor glycemic control (HbA1c >8%)
- are at increased risk of hypoglycemic episodes
- have an acute illness
- will be started on corticosteroids
- have HbA1c results that do not accurately reflect glycemia
- are planning to become pregnant

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Recommendation 28. We suggest self-monitoring of blood glucose (SMBG) at least once a day before breakfast among adults with T2DM who are on once-daily insulin

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Recommendation 29. We suggest self-monitoring of blood glucose (SMBG) at least twice a day before breakfast and before dinner among adults with T2DM who are on twice-daily insulin

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Recommendation 30. We suggest that the frequency of SMBG be individualized depending on the type of anti-hyperglycemic agents, level of glycemic control, and risk of hypoglycemia among adults with T2DM who are not on insulin

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Background

Self-monitored blood glucose (SMBG) refers to finger prick testing of capillary blood glucose by the patient or caregiver at home or in the hospital. It can provide immediate feedback on the impact of behavioral (diet and physical activity) and pharmacological interventions on blood glucose levels (1,2). Mean blood glucose determined from SMBG values has a strong positive correlation with HbA1c in both the Diabetes Control and Complications Trial (DCCT) among patients with type 1 diabetes as well as UK Prospective Diabetes Study (UKPDS) among subjects with type 2 diabetes (3). Clinical practice guidelines universally recommend SMBG for the optimal management of patients on insulin therapy in consideration of safety in terms of detection and prevention of hypoglycemia), insulin dose adjustment, and flexibility with regard to dietary choices and physical activity.

Summary of the Evidence

Characteristics of included studies

We have identified two meta-analyses that investigated the effect of SMBG on glycemic efficacy (HbA1c reduction). We also found one meta-analysis that looked into the effect of SMBG frequency on hypoglycemia reduction. The rationale for the use of SMBG in insulin-treated type 2 diabetes is supported by data from large prospective observational trials. The UKPDS and the Kumamoto study showed that intensive insulin therapy guided by SMBG can reduce the risk of microvascular complications compared with conventional therapy in type 2 diabetes. In the Diabetes Outcomes in Veterans Study (DOVES), intensified SMBG improved glycemic control (0.3% reduction in HbA1c over 1 year) in a large cohort of stable, insulin-treated veterans with type 2 diabetes. Benefits were only seen in patients whose SMBG compliance exceeded 75% or those with baseline HbA1c >8% (4,5). Guidelines also generally suggest that SMBG may be beneficial in non-insulin-treated type 2 diabetes, especially when incorporated into an educational program. Recent meta-analyses have shown a statistically significant modest benefit with absolute HbA1c reductions ranging from 0.2% to 0.5% (see Table). It was more effective in patients with baseline HbA1c >8% (6).

SMBG appeared to be most effective when combined with an educational program that incorporates instructions for patients on dietary and lifestyle changes in response to SMBG values and for health-care providers on how to adjust medications based on SMBG readings. Patient instructions should include the following: purpose of SMBG, proper techniques required to use specific glucose meters, when to perform SMBG, how to record the results in an organized fashion, how to interpret and use the results to adjust treatment, diet, and other lifestyle factors (3).

In a qualitative analysis done in 2012, SMBG showed no effect on patient satisfaction, general well-being or general health-related quality of life (7). In a pragmatic, open-label randomized trial conducted in US primary care practices from 2014 to 2015, there were also no significant differences found in health-related quality of life measures (8).

A longitudinal observational data from the Fremantle Diabetes Study found that SMBG was not independently associated with improved survival, and there were inconsistent findings relating to the association of SMBG with cardiac death and retinopathy (9).

The guidelines generally recommend that SMBG be performed at least three times per day in patients with type 1 diabetes and that it should include both fasting and postprandial glucose measurements. In a large, non-randomized study of individuals with stable type 2 diabetes using insulin, testing at least 3 times a day was associated with improved glycemic control (10). More frequent testing, including preprandial and 2-hour postprandial PG and occasional overnight BG measurements, is often required to provide the information needed to reduce hypoglycemia risk, including unrecognized nocturnal hypoglycemia (11).

Insulin-treated patients with type 2 diabetes are a heterogeneous group. Insulin regimens can vary from a daily long-acting basal insulin injection to basal-bolus regimens. The pattern and frequency will depend on the particular insulin regimen and use of other specific glucose-lowering drugs, as well as the intrinsic risk of hypoglycemia.

For patients on a single injection of long-acting insulin, the minimum SMBG requirement should be one measurement per day. Morning fasting measurements are important for adjusting insulin doses to achieve target fasting blood glucose. Once fasting blood glucose is at target, postmeal measurement can help to determine if postmeal glucose control is adequate. For patients on premixed insulins, the minimum should be a fasting measurement plus one postmeal measure each day (using a staggered pattern by testing after different meals on different days). For patients on basal-bolus insulin regimens, the recommendations should mirror those for type 1 diabetes.

Everyday circumstances and other factors contributing to an increased risk of hypoglycemia (e.g., illness, exercise, etc.) should also be taken into consideration. Furthermore, if the HbA1c level remains elevated in spite of good fasting glucose control, more postmeal SMBG measurements may be required to determine the extent of postmeal glucose control.

In a meta-analysis of randomized controlled trials among non-insulin treated patients with type 2 DM, performing SMBG for 8 to 14 times per week exhibited a small but significant improvement in glycemic control at 6 and 12 months, together with a reduction in BMI. The reduction of HbA1c associated with SMBG is likely due to modification of lifestyle and pharmacological therapy triggered by the SMBG results. However, performing SMBG up to 7 times weekly did not show significant HbA1c reduction. Although the optimal frequency of weekly SMBG was determined, the appropriate timing of capillary glucose measurements was not established as the information was not made available in the majority of the studies. Further studies are needed to specify the optimal timing of testing (12).

The recommended frequency of monitoring BG may be individualized to each person's unique circumstances. Factors influencing this recommendation include type of diabetes, type of antihyperglycemic therapy, changes to antihyperglycemic therapy, adequacy of glycemic control, literacy and numeracy skills, propensity to hypoglycemia, awareness of hypoglycemia, occupational requirements and acute illness.

In order to ensure accuracy of SMBG, results should be compared with a laboratory measurement of FPG at least annually or when HbA1C does not match SMBG readings. A difference of <15% between SMBG and simultaneous venous fasting plasma glucose (after at least an 8-hour fast), is considered acceptable (3).

Certainty of Evidence

The risk of bias were low to moderate due high risk of performance bias, as patients themselves carried out the intervention tests.

Table 13.1: Evidence summary table of SMBG versus no SMBG among non-insulin treated patients with Type 2DM.

Certainty assessment							N _e of patients		Effect		Certainty	Importance
N _e of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	SMBG	no SMBG	Relative (95% CI)	Absolute (95% CI)		
mean HbA1c difference (assessed with: mean HbA1c difference)*												
14	randomised trials	serious	not serious	not serious	not serious	none	1752	1524	-	MD 0.3 % lower (0.4 lower to 0.2 lower)	⊕⊕⊕○ Moderate	CRITICAL
severe hypoglycemia (assessed with: incidence rate)#												
3	randomised trials	serious	not serious	not serious	not serious	none	1/853 (0.1%)	4/727 (0.6%)	RR 0.35 (0.07 to 1.77)	4 fewer per 1,000 (from 5 fewer to 4 more)	⊕⊕⊕○ Moderate	CRITICAL

CI: confidence interval; MD: mean difference; RR: risk ratio

*Chircop J, Sheffield D, Kotera Y. Systematic Review of Self-Monitoring of Blood Glucose in Patients With Type 2 Diabetes. Nurs Res. 2021 Nov-Dec 01;70(6):487-497.

#NICE Grade tables and metaanalysis results. Accessed <https://www.nice.org.uk/guidance/ng28/evidence/appendix-d-grade-tables-and-metaanalysis-results-pdf-78671532573>

Table 13.2: Evidence Summary of SMBG: 8-14 measurements /week compared to SMBG: none or less than one measurement /week for glycemic control among patients with non-insulin treated T2DM

Certainty assessment							№ of patients		Effect		Certain ty	Importan ce
№ of studi es	Study design	Risk of bias	Inconsi stency	Indire ctnes s	Impre cision	Other conside rations	SMBG: 8-14 measurements /week	SMBG: none or less than one measurement /week	Relative (95% CI)	Absolute (95% CI)		
HbA1c reduction (follow-up: mean 12 months; assessed with: mean difference)												
3	random ised trials	not serio us	not serious	not serio us	not serio us	none	619	629	-	MD 0.2 % lower (0.29 lower to 0.11 lower)	⊕⊕⊕ ○ Moder ate	CRITICA L
HbA1c reduction (follow-up: mean 6 months; assessed with: mean difference)												
7	random ised trials	not serio us	not serious	not serio us	not serio us	none	707	697	-	MD 0.46 % lower (0.54 lower to 0.39 lower)	⊕⊕⊕ ○ Moder ate	CRITICA L

Citation: Xu Y, Tan DHY, Lee JY. Evaluating the impact of self-monitoring of blood glucose frequencies on glucose control in patients with type 2 diabetes who do not use insulin: A systematic review and meta-analysis. Int J Clin Pract. 2019 Jul;73(7):e13357.

Table 13.3: Evidence Summary of SMBG: upto 7 measurements /week compared to SMBG: none or less than one measurement /week for glycemic control among patients with non-insulin treated T2DM

Certainty assessment							N _e of patients		Effect		Certainty	Importance
N _e of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	SMBG: upto 7 measurements/week	SMBG: one or less than one measurement/week	Relative (95% CI)	Absolute (95% CI)		

HbA1c reduction (follow-up: mean 12 months; assessed with: mean difference)

4	randomised trials	not serious	not serious	not serious	not serious	none	462	466	-	MD 0.07 % higher (0.05 lower to 0.2 higher)	⊕⊕⊕○ Moderate	CRITICAL
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HbA1c reduction (follow-up: mean 6 months; assessed with: mean difference)

3	randomised trials	not serious	not serious	not serious	not serious	none	402	404	-	MD 0.15 % lower (0.4 lower to 0.1 higher)	⊕⊕⊕○ Moderate	CRITICAL
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Citation: Xu Y, Tan DHY, Lee JY. Evaluating the impact of self-monitoring of blood glucose frequencies on glucose control in patients with type 2 diabetes who do not use insulin: A systematic review and meta-analysis. Int J Clin Pract. 2019 Jul;73(7):e13357.

Table 13.4: Evidence Summary of SMBG results applied to adjust diabetes medication compared to SMBG results not applied for glycemic control among non-insulin treated T2DM

Certainty assessment							N _e of patients		Effect		Certainty	Importance
N _e of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	SMBG results applied to adjust diabetes medication	SMBG results not applied	Relative (95% CI)	Absolute (95% CI)		

HbA1c reduction (follow-up: mean 12 months; assessed with: mean difference)

4	randomised trials	not serious	not serious	not serious	not serious	none	836	862	-	MD 0.23 % lower (0.31 lower to 0.15 lower)	⊕⊕⊕○ Moderate	CRITICAL
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Citation: Xu Y, Tan DHY, Lee JY. Evaluating the impact of self-monitoring of blood glucose frequencies on glucose control in patients with type 2 diabetes who do not use insulin: A systematic review and meta-analysis. Int J Clin Pract. 2019 Jul;73(7):e13357.

Other Considerations

Recommendations From Other Groups

Group or Agency	Recommendation	Strength of Recommendation/ Certainty/Quality of Evidence
ADA 2023	Yes No mention regarding frequency	Grade A Grade B Grade B
NICE 2022	"Do not routinely offer" except if on insulin, hypoglycemia-prone No mention regarding frequency	Not mentioned
CDA	Yes (Grade A level 1 T1DM) (Grade C, level 3 T2DM) Individualized frequency (Grade D consensus)	Not mentioned
IDF	Yes Mandatory for patients on insulin Useful for treatment adjustment and acute illness No mention regarding frequency	Not mentioned
SIGN	Yes for those on insulin and prone to hypoglycemia or planning to become pregnant	Not mentioned
Malaysian	Yes, recommended if on insulin, desired if on OAD - Individualized frequency	Grade C

Research Gaps and Recommendations:

- What is the optimal timing for self-monitoring of blood glucose testing in adults with type 2 diabetes?
- What is the cost-effectiveness of SMBG among non-insulin treated patients in the local setting?
- What is its impact on quality of life?

COST

Php1,200.00 to Php2500.00 glucose meter
Php700.00 to Php1,000.00 per cannister of 50 test strips
Php200.00 per box of 100 lancets
Prevailing costs as of September 13, 2022.

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY

We did not find any study that addressed patient's values or preferences.

Guideline Question 14. Should we use a continuous glucose monitor (CGM) to monitor glycemic control among nonpregnant adults with type 2 DM?

Recommendation 31. We suggest CGM among nonpregnant adults with type 2 DM on insulin therapy in the following situations:

- There is suspected nocturnal hypoglycemia or hypoglycemia unawareness.
- There is suspected dawn phenomenon (early morning hyperglycemia due to the diurnal increase in cortisol or growth hormone) or Somogyi phenomenon (early morning hyperglycemia due to a rebound effect from late-night hypoglycemia).
- The HbA1c levels are still not at target despite being on multiple daily insulin injections.
- There is discordant SMBG and HbA1c results.

Certainty of Evidence: Weak
Strength of recommendation: Low

Background

Continuous glucose monitoring (CGM) systems involve the use of subcutaneously inserted sensors with scanning devices to determine the glucose concentrations in interstitial fluid. There are two types of personal devices: real-time CGM (RTCGM) and flash or intermittently-scanned CGM (FGM). FGM is the one that is currently available locally.

FGM is a factory-calibrated sensor-based technology utilizing a small sensor skin patch lasting up to 14 days. On-demand sensor scanning provides patients with comprehensive glucose data, including current glucose levels updated every minute, historical glucose readings from the last 8 hours, and trend arrows. Also, ambulatory glucose profiles can be reviewed and shared with the physician.

Summary of the Evidence

Characteristics of included studies

We found two RCTs that compared the efficacy and safety of CGM and SMBG among patients on insulin.

Summary of Evidence

In a meta-analysis of 21 studies involving 2149 individuals with DM type 1 (DMT1), CGM significantly decreased HbA1c levels compared with SMBG (mean difference -0.23% [95% CI $-3.83, -1.08$], $p=0.0005$), with larger effects among those with higher baseline HbA1c of $>8\%$ (mean difference -0.43% [95% CI $-6.04, -3.30$], $p<0.00001$). CGM demonstrated a non-significant decrease in severe hypoglycemia events (RR 0.61 [95% CI 0.33, 1.15]; $p=0.13$). CGM demonstrated no significant reduction in DKA events (RR 1.06 [95% CI 0.49, 2.32]; $p=0.88$) compared with SMBG. Most studies had low risk of bias. The certainty of evidence was moderate (1).

Use of CGM is, therefore, strongly recommended for patients with DMT1 particularly if they have poor glycemic control (HbA1c >8%) despite being on SMBG.

A meta-analysis of 5 randomized controlled trials comparing CGM (both RTCGM and FGM) versus SMBG among patients with DMT2 showed a statistically significant pooled mean difference in HbA1c of -0.25 (-0.45, -0.06, $p=0.01$) favoring those on CGM. However, there was no significant difference in the rates of hypoglycemia. All 5 studies had small sample sizes and were of short duration (range 3 to 8 months) (2).

In a systematic review and meta-analysis comparing FGM with SMBG among insulin-treated patients with Type 1 and Type 2 DM, FGM was associated with a trend to greater HbA1c change ($p=0.09$) from baseline to the last available follow-up, but this difference was not statistically significant. There was no difference in the total daily insulin dose changes from baseline to the last follow-up ($p=0.77$). FGM was associated with a lower RR of discontinuation (0.42; 95% CI 0.25 to 0.71). There were not enough data to perform a meta-analysis for the following outcomes: difference in change in time in range on FGM versus SMBG; difference in change in time above 180 mg/dL on FGM versus SMBG; difference in change in time below 70 mg/dL on FGM versus SMBG; difference in change in frequency of hypoglycemic events on FGM versus SMBG. For patient-reported outcomes (eg level of satisfaction), conflicting findings were reported on the comparison between FGM and SMBG (3).

Patients with T2DM on insulin therapy who have recurrent or severe hypoglycemic events are likely to benefit most from FGM.

Certainty of Evidence

For the meta-analyses involving randomized controlled trials, the risk of bias were low. Over-all certainty of evidence was downgraded from “High” to “Moderate” because of indirectness.

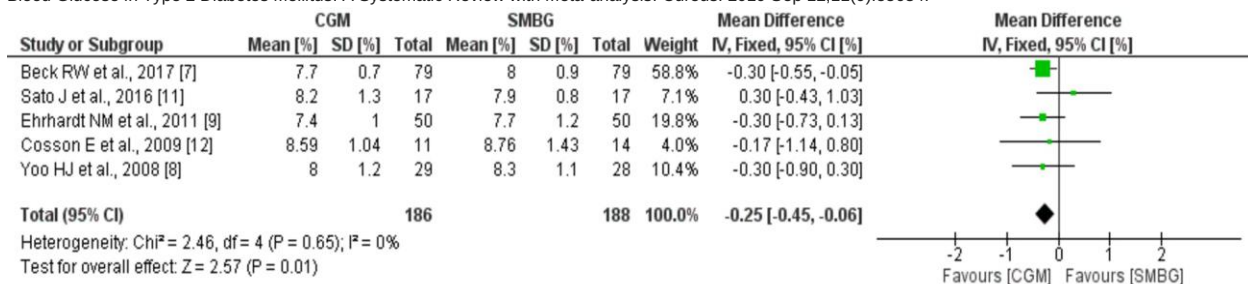
Table 14.1: Evidence Summary: CGM (both RTCGM and FGM) compared to SMBG for monitoring glycemic control among individuals with T2DM

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	CGM (both RTCGM and FGM)	SMBG	Relative (95% CI)	Absolute (95% CI)		

HbA1c reduction (follow-up: range 3 months to 8 months)

5	randomised trials	not serious	not serious	not serious	not serious	none	186	188	-	MD 0.25 % lower (0.45 lower to 0.06 lower)	⊕⊕○ ○ LOW	CRITICAL
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Citation: Janapala RN, Jayaraj JS, Fathima N, Kashif T, Usman N, Dasari A, Jahan N, Sachmechi I. Continuous Glucose Monitoring Versus Self-monitoring of Blood Glucose in Type 2 Diabetes Mellitus: A Systematic Review with Meta-analysis. Cureus. 2019 Sep 12;11(9):e5634.



Citation: Janapala RN, Jayaraj JS, Fathima N, Kashif T, Usman N, Dasari A, Jahan N, Sachmechi I. Continuous Glucose Monitoring Versus Self-monitoring of Blood Glucose in Type 2 Diabetes Mellitus: A Systematic Review with Meta-analysis. Cureus. 2019 Sep 12;11(9):e5634.

Table 14.2: Evidence Summary: Flash CGM compared to SMBG for glycemic control among patients with type 2 DM on insulin

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Flash CGM	SMBG	Relative (95% CI)	Absolute (95% CI)		

Insulin-treated T2DM: HbA1c reduction (follow-up: mean 17 weeks; assessed with: mean difference in HbA1c)

2	randomised trials	not serious	not serious	not serious	not serious	none	202	123	-	MD 0.22 % lower (0.72 lower to 0.28 higher)	⊕⊕○ ○ LOW	CRITICAL
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Citation: Castellana M, Parisi C, Di Molfetta S, Di Gioia L, Natalicchio A, Perrini S, Cignarelli A, Laviola L, Giorgino F. Efficacy and safety of flash glucose monitoring in patients with type 1 and type 2 diabetes: a systematic review and meta-analysis. BMJ Open Diabetes Res Care. 2020 Jun;8(1):e001092.

Other Considerations

Recommendations From Other Groups

Group or Agency	Recommendation	Strength of Recommendation/ Certainty/Quality of Evidence
ADA 2023	Continuous glucose monitoring is recommended for individuals with type 1 diabetes and for certain individuals with type 2 diabetes to improve glycemic control and reduce the risk of hypoglycemia.	Grade A
AACE 2023	CGM is recommended for individuals with type 1 diabetes and for those with type 2 diabetes who are on insulin therapy.	Grade A
NICE 2022	Continuous glucose monitoring should be offered to individuals with type 1 diabetes, particularly those who experience frequent episodes of hypoglycemia or have difficulty managing their blood glucose levels.	Grade A
Canadian Diabetes Association 2023	Continuous glucose monitoring (CGM) is recommended for individuals with diabetes, particularly those with type 1 diabetes and some with type 2 diabetes who are on intensive insulin therapy.	Grade A

Research Gaps and Recommendations:

- What is the effectiveness of CGM across different age groups and insulin regimens?
- What is the long-term effectiveness of CGM?
- What is the cost-effectiveness of CGM in the local setting?

COST

Php 5,500.00 (reader with sensor)

Php 3,000.00 (sensor to be changed every 2 weeks) (@FreeStyle Libre 1)

Barriers to use include its cost. Also, SMBG is still necessary since SMBG ensures the accuracy of the CGM device and serves as a back-up for situations such as when the technology fails. Healthcare professionals and patients need to have the proper education and skills to interpret the data effectively.

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY

We did not find any study that addressed patient's values or preferences.

E. Cardiovascular Risk Reduction

Guideline Question 15. Should we use aspirin among patients with type 2 DM for primary prevention of cardiovascular events?

Recommendation 32. We suggest against the use of aspirin for primary prevention of cardiovascular events among patients with type 2 DM.

Certainty of Evidence: Very Low

Strength of recommendation: Weak

Background

The diabetic population is at an increased risk of cardiovascular events. Beyond generalized and accelerated atherosclerosis, these patients have increased platelet turnover and reactivity, as well as higher thrombin generation. Therefore, anti-thrombotic agents, including aspirin and clopidogrel, have been thought to be suitable for reducing the risk of cardiovascular events in primary prevention.

Summary of the Evidence

Characteristics of included studies

In a recent systematic review, evidence for the use of aspirin in the primary prevention of cardiovascular events was derived from 10 randomized control trials.¹ Four studies exclusively enrolled diabetic patients (ASCEND, ETDRS, JPP, and POPADAD), while the other six conducted subgroup analyses from their main analyses (PPP, JPAD, ASPREE, WHS, Physician's Study, HOT). Only data pertaining to primary prevention was obtained and analyzed. The studies included 34,058 diabetic patients, with a mean age of 63.75 years, and 50.6% of participants were male. The median follow-up period was 5.73 years. One-third of the patients were overweight or obese, the proportion of hypertensive patients ranged from 26% to 100%, and statin use varied from 30% to 76%. All trials utilized a placebo as a control, except for the open-label studies JPPP, JPAD, and PPP, which compared aspirin with a no-aspirin or standard care arm. See Appendix A for a table of the characteristics of the included studies.

Efficacy outcomes

All-cause mortality was not significantly reduced with RR 0.96 (95% CI 0.90, 1.03; 7 RCT; 27,595 patients). Regarding MACE, there was an 8% risk reduction (RR 0.92, 95% CI 0.84-0.999; $I^2 = 0\%$; 8 RCT; 29,814 patients). The risk of cardiovascular mortality was not significantly decreased by aspirin (RR 0.92, 95% CI 0.82, 1.02; $I^2 = 26\%$; 6 RCTs, 25,538 patients) as well myocardial infarction (RR 0.89, 95% CI 0.74, 1.07; $I^2 = 51\%$; 8 RCTs, 27,097 patients) or stroke (RR 0.91, 95% CI 0.80, 1.03; $I^2 = 27\%$; 7 RCTs, 26,564 patients). See Appendix B.1 and B.2 for forest plots.

Safety outcomes

The risk of major bleeding was significantly increased (RR 1.30, 95% CI 1.10, 1.53; $I^2 = 16\%$; 2 RCTs, 18,019 patients) as well as major GI bleeding (RR 1.39, 95% CI 1.08, 1.80; $I^2 = 39\%$; 2 RCTs, 18,019 patients) — Fig. 3. The difference in intracranial bleeding risk did not achieve statistical significance (RR 1.31, 95% CI 0.90, 1.91; $I^2 = 21\%$; 2 RCTs, 18,019 patients).

Certainty of evidence

Table 15.1. Efficacy outcomes of aspirin on primary prevention among patients with type 2 diabetes mellitus

OUTCOMES	Basis	Effect Estimate	95% Confidence Interval	Interpretation	Certainty of Evidence	NNT
All Cause Mortality	7 RCTs	RR 0.96	0.90 – 1.03	As good as or better	Moderate	264
MACE	8 RCTs	RR 0.92	0.84 – 0.999	As good as or better	Low	178
CV mortality	6 RCTs	RR 0.92	0.82 – 1.02	As good as or better	Moderate	262
MI	8 RCTs	RR 0.89	0.74 – 1.07	As good as or better	Very Low	172
Stroke	7 RCTs	RR 0.91	0.80 – 1.03	As good as or better	Low	289

Table 15.2. Safety outcomes of aspirin on primary prevention among patients with type 2 diabetes mellitus

OUTCOMES	Basis	Effect Estimate	95% Confidence Interval	Interpretation	Certainty of Evidence	NNH
Major Bleeding	2 RCTs	RR 1.30	1.10 – 1.53	Harm	Moderate	121
Major GI Bleeding	2 RCTs	RR 1.39	1.08 – 1.80	Harm	Moderate	224
Intracranial Bleeding	2 RCTs	RR 1.31	0.90 – 1.91	As good as or worse	Moderate	597

Analysis on tables 1 and 2 show that most of the efficacy outcomes have very low to low certainty of evidence while all cause and CV mortality have moderate certainty of evidence, while safety outcomes (bleeding) have moderate certainty of evidence. Aspirin reduced the risk of MACE in 6 of each 1000 treated patients but increased the risk of major bleeding in 8 of each 1000 treated patients and 5 patients with major GI bleeding.

This systematic review suggests that the use of aspirin in primary prevention on diabetic patients is associated with a significant 8% risk reduction of MACE and 30% increase of major bleeding risk, without significantly reducing the risks of all-cause mortality, cardiovascular mortality, MI, stroke, coronary revascularization or amputation, individually. Previous systematic reviews regarding this subject suggested that aspirin was beneficial, even though also causing a significant increased risk of bleeding. The recent trials, particularly the ASCEND study, gave robustness to previous evidence of increased bleeding risk, allowing us to clarify this controversial but important subject regarding cardiovascular benefit.

Other considerations

Drug	Brand	Cost (range)	Cost per month (range)
Aspirin	Aspilet EC 80 MG	2.91 PHP/tab	87.3 PHP/month
	Asthromed 100 mg	2.00 PHP/tab	60.00 PHP/month
	Bayer Aspirin 100 mg	2.80 PHP/tab	84.00 PHP/month
	Bespirin EC 100 mg	2.54 PHP/tab	76.20 PHP/month
Clopidogrel	Clopidra 75 mg	29.00 PHP/tab	870.00 PHP/month
	Clopivaz 75 mg	28.00 PHP/tab	840.00 PHP/month
	Noklot 75 mg	25.89 PHP/tab	776.70 PHP/month
	Plogrel 75 mg	37.93 PHP/tab	1137.90 PHP/month

Research Gaps and Recommendations

In 2024, two randomized trials were published about primary prevention of cardiovascular disease with aspirin. A post-hoc analysis of patients included in the ACCORD Trial without cardiovascular disease at baseline showed that the use of aspirin was not associated with a decreased risk of cardiovascular disease or all-cause mortality.⁹ Another subgroup analysis of the ADAPTABLE Trial showed no added clinical benefit for higher doses of aspirin for primary prevention in patients with established ASCVD to mitigate the risk of future adverse cardiovascular events.¹⁰ Both studies have no significant impact on the recommendations included by previous studies.

Guideline Question 16. Should we use aspirin among patients with type 2 DM for secondary prevention of cardiovascular events?

Recommendation 33. We suggest the use of aspirin for secondary prevention of cardiovascular events among persons with type 2 DM.

Certainty of Evidence:	Low
Strength of recommendation:	Weak

Background

The use of aspirin for the secondary prevention of cardiovascular disease in people with type 2 diabetes has been shown to be beneficial. Most of the guidelines agree with this recommendation. Aspirin exerts its effect primarily by interfering with the biosynthesis of cyclic prostanoids.¹ This antithrombotic effect is attributed to the irreversible inhibition of platelet thromboxane A₂ synthesis. Major coronary events and stroke accounted for a large proportion of serious vascular events. The net benefit of long-term low-dose aspirin in the background of existing occlusive vascular disease for fatal and nonfatal events would be substantial despite the adverse effects.

Summary of the Evidence

Characteristics of included studies

Evidence was obtained from a systematic review of 16 RCTs comparing the use of aspirin to a control group (n = 17, 029, 43,000 person-years). Incidence of diabetes ranged from 4 to 23% in each trial. Further research did not yield additional randomized controlled trials relevant to aspirin as secondary prevention for patients with type 2 diabetes.²

Efficacy outcomes

Aspirin use reduced the risk of nonfatal myocardial infarction (RR 0.69; 95% CI 0.60, 0.80), coronary heart disease mortality (RR 0.87; 95% CI 0.78, 0.98), stroke (RR 0.81; 95% CI 0.71, 0.92), and any serious vascular event (RR 0.81; 95% CI 0.75, 0.87), compared to control. Any serious vascular events were defined as myocardial infarction, stroke, or vascular death. However, aspirin use did not significantly reduce vascular mortality compared to the control group (RR 0.91; 95% CI 0.82, 1.00). Due to indirectness, the quality of the evidence was rated as moderate.

Safety Outcomes

Aspirin use compared to placebo resulted in increased major extracranial bleeding with a rate ratio of 2.61; 95% CI 1.25, 5.76. However, it did not result in a statistically significant reduction in hemorrhagic strokes (RR 1.67; 95% CI 0.97, 2.90). The quality of evidence was low due to indirectness and imprecision.

Certainty of evidence

Table 16.1. Effect of Aspirin vs Placebo for Secondary Prevention of Cardiovascular Events for Patients with T2DM

Outcomes	No. of Studies (person-years)	Rate Ratio (95% CI)	Interpretation	Certainty of Evidence
Non Fatal Myocardial Infarction	16 RCTs (n = 86,000 person-years)	Rate ratio 0.69 (0.60 to 0.80)	Benefit	Moderate
Coronary Heart Disease Mortality	16 RCTs (n = 86,000 person-years)	Rate ratio 0.87 (0.78 to 0.98)	Benefit	Moderate
Stroke	16 RCTs (n = 86,000 person-years)	Rate ratio 0.81 (0.71 to 0.92)	Benefit	Moderate
Vascular Death	16 RCTs (n = 86,000 person-years)	Rate ratio 0.91 (0.82 to 1.00)	As good as or better	Moderate
Any serious vascular event	16 RCTs (n = 86,000 person-years)	Rate ratio 0.81 (0.75 to 0.87)	Benefit	Moderate
Hemorrhagic Stroke	16 RCTs (n = 86,000 person-years)	Rate ratio 1.67 (0.97 to 2.90)	As good as or worse	Low
Major Extracranial Bleeding	5 RCTs (n = 86,000 person-years)	Rate ratio 2.61 (1.25 to 5.76)	Harm	Low

Other considerations

COST

Drug	Brand	Cost	Cost per month
Aspirin	Aspilet EC 80 MG	2.91 PHP/tab	87.3 PHP/month
	Asthromed 100 mg	2.00 PHP/tab	60.00 PHP/month
	Bayer Aspirin 100 mg	2.80 PHP/tab	84.00 PHP/month
	Bespirin EC 100 mg	2.54 PHP/tab	76.20 PHP/month
Clopidogrel	Clopidra 75 mg	29.00 PHP/tab	870.00 PHP/month
	Clopivaz 75 mg	28.00 PHP/tab	840.00 PHP/month
	Noklot 75 mg	25.89 PHP/tab	776.70 PHP/month
	Plogrel 75 mg	37.93 PHP/tab	1137.90 PHP/month

Research Gaps

No recent data relevant to this study have been published beyond the presented evidence. We recommend a randomized controlled trial directly comparing aspirin and clopidogrel for the secondary prevention of cardiovascular events specifically involving diabetic patients.

Guideline Question 17. Should we use clopidogrel for the secondary prevention of cardiovascular events among patients with type 2 DM?

Recommendation 34. We suggest the use of clopidogrel as an alternative in the setting of aspirin allergy among adults with type 2 DM and established cardiovascular disease.

Certainty of Evidence: Very Low

Strength of recommendation: Weak

Background

Patients with diabetes mellitus have accelerated atherosclerosis with an increased risk for atherothrombotic cardiovascular complications. Clopidogrel is one of the three different classes of antiplatelet drugs, which are the cornerstones in the prevention and treatment of these prothrombotic states [1,2]. Clopidogrel is a second-generation thienopyridine, which is a nondirect irreversible antagonist of the P2Y₁₂ receptor. However, it is administered as an inactive prodrug and requires hepatic conversion via cytochrome (CYP) P450 enzymes. This results in a slow onset of action and interindividual variability in clopidogrel response. Third generation P2Y₁₂ inhibitors including prasugrel and ticagrelor were developed to address this issue, but real-world studies have not confirmed the benefits of these newer P2Y₁₂ inhibitors over clopidogrel in terms of efficacy and safety [3].

Current practice guidelines recommend the use of P2Y₁₂ inhibitors as effective alternatives to aspirin, although aspirin is still the default agent in most clinical scenarios. Guidelines recommend the use of Clopidogrel 75 mg/day for patients with atherosclerotic cardiovascular disease and documented aspirin allergy [4]. Clopidogrel has also been extensively studied as part of dual anti-platelet therapy (DAPT) with aspirin following acute coronary syndrome and ischemic stroke [3,4,5].

Summary of the Evidence

Characteristics of included studies:

Five randomized controlled trials (RCTs) compared aspirin with clopidogrel with a duration of follow-up between 3 and 36 months. Studies included CAPRIE in 1996 to the most recent HOST-EXAM, which was just published in 2021. Population included patients with prior coronary and/or cerebrovascular events. Most studies except for the CADET study in the UK included diabetic patients, with the HOST-EXAM having the advantage of an exclusively East Asian population, making it more applicable to our own patients. All the five RCTs used clopidogrel of the same dose of 75mg daily as intervention. The comparison used variable doses of Aspirin ranging from 75 to 325mg daily. See Table 1 for the summary of characteristics for the included studies.

Table 17.1. Characteristics of included studies

Study	Study design	Number of patients (diabetic)	Population (qualifying event)	Intervention/Treatment	Control/ Comparison	Outcomes	Follow-up (months)
CAPRIE 1996 multinational	RCT multicentre double blind	19185 (20.0%)	Patients with prior ischemic stroke, peripheral artery disease, myocardial infarction	Clopidogrel 75mg daily	Aspirin 325mg daily	MI, ischemic stroke, vascular death	36
ASCET 2012 Norway	RCT single centre double blind	1001 (19.9%)	Patients with coronary artery disease	Clopidogrel 75mg daily	Aspirin 160 mg daily	death, MI, ischemic stroke, unstable angina	24
HOST EXAM 2021 South Korea	RCT multicentre open label	5438 (34.2%)	Patients with CAD post-PCI	Clopidogrel 75mg daily	Aspirin 100mg daily	All cause mortality, MI, stroke, readmission for ACS	24
CADET 2004 UK	RCT multicentre double blind	184 (not specified)	Patients with recent MI	Clopidogrel 75mg daily	Aspirin 75 mg daily	Thrombotic variables and CRP	6
CHANCE 2013 China	RCT multicentre double blind	5170 (21.2%)	Patients with stroke or high risk TIA	Clopidogrel 75mg daily	Aspirin 75 mg daily	Thrombotic variables and CRP	3

Efficacy Outcomes

There was a significant reduction of MACE and myocardial infarction events with use of clopidogrel compared with aspirin with a pooled RR of 0.89 (95% CI 0.82, 0.96) and RR 0.80 (95% CI 0.65, 0.99), respectively. There were similar rates of all-cause mortality and stroke events between clopidogrel and aspirin with pooled RR of 1.05 (95% CI 0.88 - 1.26) and RR 0.67 (95% CI 0.38, 1.17)

Safety Outcomes

There were similar rates of major bleeding between aspirin and clopidogrel with a pooled RR of 0.78 (0.5, 1.2).

Table 17.2. Efficacy and safety outcomes of clopidogrel compared with aspirin

Outcomes	Number of RCTs	Participants	Effect Estimate (95% CI)	Interpretation	Certainty of Evidence	NNT/NNH
MACE	5	30,504	0.89 (0.82, 0.96)	Benefit (?)	High	103 (NNT)
All-cause death	4	25,808	1.05 (0.88 - 1.26)	Inconclusive	High	1793 (NNH)
MI	4	25,808	0.80 (0.65, 0.99)	Benefit (?)	High	177
Stroke	3	25,624	0.67 (0.38 (1.17)	Inconclusive	Moderate	234
Major Bleeding	4	30,320	0.78 (0.5, 1.2)	Inconclusive	Moderate	381 (NNH)

Certainty of evidence

The risk of bias for most studies included was low. Some concern over randomization was noted for HOST-EXAM, due to the open label design. Another concern regarding intervention deviation and overall risk was noted as well for ASCET.

Other considerations

Cost considerations

The cost of clopidogrel in the Philippines can be as high as 83.75 per tablet depending on the brand name. Many generic brands are available and come at a much cheaper price.

Research Gaps and Recommendations

We recommend a randomized controlled trial directly comparing aspirin and clopidogrel for the secondary prevention of cardiovascular events specifically involving diabetic Filipino patients.

Guideline Question 18. Should we recommend low dose combination drugs for the control of hypertension among adults with Type 2 DM and hypertension?

Recommendation 35. We recommend the use of low-dose combinations of a RAAS blocker (ACEi or ARB) with a CCB or thiazide/thiazide-like diuretic, preferably using a single-pill combination (SPC) to reach a blood pressure target of < 130/80 mm Hg among adults with type 2 DM. Free tablet combinations may also be given if SPCs are not available.

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Recommendation 36. We recommend against the combination of ACEi and ARB among adults with type 2 DM due to a higher risk of hyperkalemia and renal failure.

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Summary of the Evidence:

The recommendations and the summary of evidence was modified with permission from the original recommendations of the Philippine Practice Guidelines for the Management of Hypertension in the Philippines. A total of 13 guidelines were used and adapted, updating the original 12 from the 2020 Philippine CPG on Hypertension with the latest publications and adding one from the current search.

Persons with diabetes (PWD) are generally considered to have higher cardiovascular risks. Filipinos in particular whether newly diagnosed, or already with established diabetes are seen to be of higher cardiovascular risks as compared to the general population. The CANDI-Manila study among newly diagnosed adults with T2DM (mean age of 50 years) demonstrated a high prevalence of diabetes complications and CV risk factors. The electrocardiographic findings showed that 5% had myocardial infarcts or ischemic changes, and 6% had left ventricular hypertrophy. Hypertension was found in 42% of individuals with a mean BP of 144/88 mm Hg, and 80% of all subjects had LDL-C of at least 100 mg/dL, with another 38% with elevated triglyceride of at least 150 mg/dL.¹ Among persons with established T2DM in the Diabcare Philippines (2008), 95% had dyslipidemia and nearly 70% had hypertension. Around 20% already had some form of nephropathy, 35% retinopathy (8% severe), and 15% already had a stroke, myocardial infarction or had undergone cardiac bypass or angioplasty or had overt CAD.²

The use of combination therapy of a RAS blocker with a CCB or thiazide/thiazide-like diuretics is advocated by at least 4 of the reviewed guidelines, and acknowledged to have emerging evidence by the Canadian Diabetes Association.³⁻⁷ Some of the guidelines only recommend combination therapy for those with BP > 150/100 mm Hg (AACE) or $\geq$ 160/100 mm Hg (ADA). Starting persons with hypertension on initial combination therapy results in greater achievement of BP lowering at the shortest amount of time, and is generally preferred for higher CV risk patients including those with T2DM. Some of the advantages of initial combination therapy are the greater BP reduction versus monotherapy, there is no or just a small increase in hypotensive episodes with greater likelihood of achieving good BP control; better adherence to drug treatment and reduced therapeutic inertia.⁸

Low-dose combination therapy has been shown to be more effective than maximal dose monotherapy in the general population of persons with hypertension⁹. In a 2015 network meta-analysis of 27 studies with nearly 50,000 participants, it was seen that there was no benefit of any single antihypertensive class in the reduction of mortality in hypertensive persons with type 2 diabetes. Reduction of cardiovascular mortality was only observed among patients treated with ACE-inhibitors and CCB combination, which may be related to lower blood pressure levels.¹⁰ Combination therapy between an ARB and another drug class was not included in this network meta-analysis.

RAAS based therapies (ACE-inhibitors and Angiotensin-2 receptor blockers) are the drugs of choice as base medications for persons with diabetes who have hypertension, although CCBs and diuretics may also be given. These drugs are especially preferred among those with proteinuria, known CV disease, and additional CV risk factors such as dyslipidemia. Majority of the guidelines recommend a step-wise approach that begins with monotherapy and where treatment can be further escalated according to their recommended treatment algorithms^{7,11,12}. However, multiple drug therapy is generally required to achieve blood pressure targets among persons with diabetes (PWD). As much as 90% of persons with established T2DM and hypertension require three anti-hypertensive medication to achieve target.¹³ These drugs may be used as add-on therapy if BP targets are not reached: diuretics, calcium channel blockers (CCBs), beta blockers and peripheral alpha blockers. However, the combination of ARB and ACEI more than doubles the risk of renal failure and hyperkalemia and is therefore not recommended.¹⁴⁻¹⁵

The other clinical trial that supports the use of combination therapy but which was not included in the previous meta-analysis is the ADVANCE trial. The BP-lowering arm of the Action in Diabetes and Vascular Disease: Preterax and Diamicon MR Controlled Evaluation (ADVANCE) trial used the ACE-inhibitor perindopril combined with indapamide as a single-pill combination, resulting not only to decreased CV outcomes (death from CVD was reduced by 18%) but also with significant impact on microvascular outcomes including nephropathy and retinopathy. The six-year observational study called ADVANCE-ON still found significant reduction in risk of death although already attenuated. The achieved blood pressure in the intervention group was 136/73 mmHg.¹⁶⁻¹⁷ The Avoiding Cardiovascular Events Through Combination Therapy in Patients Living With Systolic Hypertension (ACCOMPLISH) trial although included in the meta-analysis is worthy of mention because similar to ADVANCE, it advocates SPC as initial therapy. It enrolled participants at high risk of cardiovascular events (60% with diabetes) and demonstrated a decrease in morbidity and mortality with the ACE inhibitor benazepril plus the dihydropyridine CCB amlodipine versus benazepril and the thiazide-like diuretic hydrochlorothiazide¹⁸⁻¹⁹.

While more trials are needed to support the use of single pill combinations for the initial therapy of hypertension among persons with diabetes, there is already enough data to support the use of initial combination therapy of ACE-inhibitors or ARB with CCB compared to monotherapy, or alternatively ACE-I/ARB with thiazide or thiazide-like diuretic as supported by the ADVANCE trial¹⁷.

While cardiovascular risk reduction (myocardial infarction, CV death) is already significant for BP < 140/90 mm Hg (with no additional benefit for < 120 mm Hg), a lower BP target has additional benefit for stroke reduction and decreased risk for nephropathy. However, there is also a recommendation not to lower the BP to less than 120/70 mm Hg due to increased risk for adverse cardiovascular outcomes.

Although proportional associations of BP lowering treatment for most CV outcomes studied were attenuated below a systolic BP level of 140mmHg, data indicate that further reduction below 130

mmHg is associated with a lower risk of stroke, retinopathy, and albuminuria, potentially leading to net benefits for many individuals at high risk for those outcomes⁷. In the past, the recommendation to lower systolic BP to <130 mmHg had been partly based on prospective cohort data; specifically, the Pittsburgh Epidemiology of Diabetes Complications Study²⁰ (in people with type 1 diabetes mellitus) and the UKPDS²¹ (in people with type 2 diabetes). These studies demonstrated a linear relationship between systolic BP levels and mortality, CAD, overt diabetic nephropathy and proliferative retinopathy^{20,22}. These associations were maintained even after adjustment for other confounding factors (such as lipid levels, age, sex and glycemic control). In these studies, direct relationships were seen between the magnitude of incremental BP reduction and reductions in risk of hypertension-related complications, over time.

This target was challenged in the ACCORD BP study arm which showed that a blood pressure of < 140 mm Hg did not differ to a BP < 120 in terms of CV risk reduction; however, the same study showed that there is still substantial stroke reduction with lower systolic BP²³. The meta-analyses of Bangalore et al also showed that while the other components of major adverse cardiac events were not improved, lowering BP <130 mmHg conferred additional protection against stroke^{7, 24-26}. Lowering diastolic BP to equal to or less than 80 mm Hg is also supported by the HOT trial where 1,500 persons with diabetes among 18,790 participants were included. In the over-all trial, there was no cardiovascular benefit with more intensive targets but in the subpopulation with diabetes, an intensive diastolic BP target of less than or equal to 80 mm Hg showed significantly reduced risk (51%) of CVD events²⁷.

However, there is also a recommendation not to lower the BP to less than 120/70 mm Hg due to increased risk of adverse cardiovascular outcomes.²⁸

GRADE Evidence Profile

Criteria	Points
Study design (randomized trials only)	+ 4
Risk of bias	0
Inconsistency	-1
Indirectness	0
Imprecision	0
Large effect	0
Dose response	0
All plausible residual confounding	0
TOTAL	+ 3 (moderate)

Source: Supplementary materials from <https://doi.org/10.1016/j.jdiacomp.2016.04.020> from Luciana R. Remonti, Sofia Dias, Cristiane B. Leitão, Caroline K. Kramer, Lucas P. Klassman, Nicky J. Welton, A.E. Ades, Jorge L. Gross, Classes of antihypertensive agents and mortality in hypertensive patients with type 2 diabetes—Network meta-analysis of randomized trials, *Journal of Diabetes and its Complications*, Volume 30, Issue 6, 2016. Pages 1192-1200, ISSN 1056-8727.

According to GRADE system, the quality of the evidence was considered moderate.

Certainty of the evidence

All studies were randomized, however in ten they could not define the method used for randomization and, therefore, its concealment. Eleven trials were not double blinded; however in all but 2 trials the outcome evaluators were blinded. From the 21 studies included in cardiovascular mortality analysis, 16 had all events adjudicated by an independent committee.

The other 5 trials do not describe if outcomes were adjudicated and, in 3 of these, the events of death were described in adverse event section. In 9 trials, the study describes clearly a standardized method for blood pressure measurement. Six trials describes that clinical assessments including blood pressure were conducted according to the study protocol. Only in one case, there is no information regarding blood pressure measurement technique.

Research Gaps:

1. Cost-effectiveness analysis of the use of combination therapy for the prevention of CV events among persons with diabetes mellitus and hypertension.
2. Compliance with the combination therapy: fixed dose versus separate tablets.

Guideline Question 19. Should we recommend statins for primary prevention of cardiovascular (CV) events among adults with type 2 DM?

Recommendation 37. We recommend the use of moderate intensity statins for the primary prevention of cardiovascular events among adults with type 2 DM, even without evidence of atherosclerosis (ASCVD).

Certainty of Evidence: Moderate
Strength of recommendation: Strong

Summary of the Evidence:

Evidence on the use of statins for primary prevention of cardiovascular outcomes were derived from 8 different clinical trials. In the Philippine guideline on dyslipidemia published in 2006, only 5 studies were included; this has been updated in the 2015 CPG on dyslipidemia to include 8 RCTs. In the current recommendation, no additional clinical trials have been added.

Of the eight studies that were included, 5 were sub-studies from a larger group of individuals who had no previous cardiovascular events (AFCAPS/TEXCAPS, ALLHAT-LLA, ASCOT-LLA, PROSPER, MEGA) while the other 3 were primarily studies done on individuals with diabetes (ASPEN, CARDS, HPS). Whenever a study involved a combination of patients for primary and secondary prevention, then only the data for primary prevention was obtained and analyzed e.g. ASPEN. Table 1 below summarizes the results of the review and the relevant outcomes. Statistically significant results are seen from the outcomes of fatal and nonfatal MI (N=27,810, RR 0.73 [0.64, 0.83]), stroke (N=27,810, RR 0.75 [0.63, 0.89]), acute major CV events (MACE) (N=16,095, RR=0.78 [0.70, 0.86]) and coronary revascularization (N=25,783, RR= 0.84 [0.73, 0.97]). A trend to benefit is seen for the outcomes of total mortality, CV death, and unstable angina.

The suggested dose for most patients with diabetes is a moderate intensity statin but the actual dose will depend on the treatment targets. Consistent with the original recommendations taken from the CPG for Dyslipidemia, the targets for persons with diabetes and dyslipidemia are an LDL-cholesterol of < 100 mg/dL for newly diagnosed, < 70 mg/dL if with one other CV risk factor (albuminuria, retinopathy, CKD), < 55 mg/dL for secondary prevention for those who have had any cardiovascular event like acute coronary syndrome or stroke..

Table 19.1. Summary Table of Outcomes for the Use of Statins versus Placebo among Persons with Diabetes

Outcome	Studies	Participants	Effect Estimate (RR, 95% CI)	NNT
Total Mortality	1	2837	0.73 (0.53-1.010)	ns
Fatal CHD/CV death	3	7544	0.98 (0.68, 1.41)	ns
Fatal & Nonfatal MI	4	27,810	0.73 (0.64, 0.83)	100
Unstable Angina	3	7274	0.77 (0.53, 1.10)	ns
Stroke	4	27,810	0.75 (0.63, 0.89)	200
Acute major CV events (MACE): composite	8	16,095	0.78 (0.70, 0.86)	45
Coronary revascularization	3	25,783	0.84 (0.73, 0.97)	200

Certainty of Evidence

The GRADE Summary of Findings table (Table 2) shows that generally, the quality of the evidence is moderate with the downgrade due to the question of directness. The studies were all done in

Caucasian populations (Table X) and Asians, and in particular Filipinos were not included in the samples that were included in these trials. Likewise, 5 out of the 8 studies were subgroup analysis of diabetic individuals from a larger group of individuals with no previous cardiovascular events.

Thus, the certainty of the evidence is only moderate for the use of statins for primary prevention. Table 19.2. GRADE Summary of Findings on Statins and CV Outcomes Among Persons with diabetes.

diabetes.

Quality assessment							No of patients		Effect		Quality	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Statins	Placebo	Relative (95% CI)	Absolute (95% CI)		
Total Mortality												
1	randomised trials	not serious	not serious	serious ¹	not serious	none	61/1428 (4.3%)	82/1409 (5.8%)	RR 0.73 (0.53 to 1.01)	NA	⊕⊕⊕○ MODERATE	CRITICAL
Fatal CHD/Cardiovascular death												
3	randomised trials	not serious	not serious	serious ²	not serious	none	60/3645 (1.6%)	58/3629 (1.6%)	RR 0.98 (0.68 to 1.41)	NA	⊕⊕⊕○ MODERATE	CRITICAL
Fatal and Non-fatal MI												
4	randomised trials	not serious	not serious	serious ³	not serious	none	378/13914 (2.7%)	518/13896 (3.7%)	RR 0.73 (0.64 to 0.83)	10 fewer per 1000 (from 6 fewer to 13 fewer)	⊕⊕⊕○ MODERATE	CRITICAL
Unstable angina												
3	randomised trials	not serious	not serious	serious ⁴	not serious	none	51/3645 (1.4%)	60/3629 (1.7%)	RR 0.77 (0.53 to 1.01)	NA	⊕⊕⊕○ MODERATE	IMPORTANT
CVD/Stroke												
4	randomised trials	not serious	not serious	serious ⁵	not serious	none	224/13914 (1.6%)	298/13896 (2.1%)	RR 0.75 (0.63 to 0.89)	5 fewer per 1000 (from 2 fewer to 8 fewer)	⊕⊕⊕○ MODERATE	CRITICAL
Acute Major CVD Events (composite)												
8	randomised trials	not serious	not serious	serious ⁵	not serious	strong association	597/8083 (7.4%)	766/8012 (9.6%)	RR 0.78 (0.7 to 0.86)	21 fewer per 1000 (from 13 fewer to 29 fewer)	⊕⊕⊕⊕ HIGH	IMPORTANT
Interventional Procedures												
3	randomised trials	not serious	not serious	serious ⁵	not serious	none	328/12908 (2.5%)	390/12875 (3.0%)	RR 0.84 (0.73 to 0.97)	5 fewer per 1000 (from 1 fewer to 8 fewer)	⊕⊕⊕○ MODERATE	IMPORTANT

Research Gaps

1. Cost -effectiveness studies on the use of statins for most adults with diabetes mellitus and high LDL-cholesterol.
2. Studies on the compliance and risk factors for non-compliance with statins among persons with diabetes.

Guideline Question 20. Should we use sodium-glucose cotransporter-2 (SGLT2) inhibitors among adults with type 2 DM with established atherosclerotic cardiovascular disease (ASCVD)?

Recommendation 38. We recommend the use of SGLT2 inhibitors among adults with type 2 DM and established cardiovascular disease or elevated cardiovascular risk.

Certainty of Evidence: Moderate
Strength of recommendation: Strong

Background

Sodium-glucose co-transporter-2 (SGLT2) inhibitors improve glycemic control by inhibiting the reabsorption of glucose and sodium from the proximal renal convoluted tubules. This results in not only glucosuria but also mild natriuresis that leads to reductions in plasma volume and blood pressure and improved vascular resistance, which may underlie its benefits in improving cardiovascular outcomes [1,2]. Guidelines currently recommend the use of SGLT2 inhibitors in patients with type 2 diabetes and cardiovascular disease (CVD) or at high/very high CV risk to reduce CV events and the risk of death [3]. In addition, trials have demonstrated that the positive CV effects of these medications extend to individuals without diabetes [4].

Summary of the Evidence

Characteristics of included studies

Four major RCTs with good methodology were included in the meta-analysis and involved 42,568 participants with a follow-up period between 3-6 years. Primary outcomes included the composite outcome of CV death and hospitalization for heart failure (HHF), HHF, and renal composite outcomes. Renal Composite 1 concluded 40% decrease in eGFR, end-stage kidney disease, or renal death; renal Composite 2 included renal death, renal dialysis/transplant, or doubling of serum creatinine from the baseline). Secondary outcomes were major adverse cardiovascular events (MACE), CV death, all-cause mortality, and change from the baseline in HbA1c.

Table 20.1 Characteristics of included studies

Study	Design	Number of diabetic patients/total number of patients (%)	Intervention/Treatment	Control/Comparat or	Outcomes	Median Follow-up Duration
VERTIS CV (2020)	RCT	8,246 (100%)	Ertugliflozin 5/15mg OD	Placebo	MACE (composite of CV death, nonfatal MI, or stroke	6.1 years
DECLARE-TIMI 58 (2019)	RCT	17,160 (100%)	Dapagliflozin 10mg OD	Placebo	MACE, composite of CV death or HHF	5.2 years
CANVAS Program (2018)	RCT	10,142 (100%)	Canagliflozin 100/300mg OD	Placebo	Composite of CV death, nonfatal MI or stroke	3.6 years

EMPA-REG OUTCOME (2017)	RCT	7,020 (100%)	Empagliflozin 10/25mg OD	Placebo	Composite of CV death, nonfatal MI, or stroke	4.6 years
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Efficacy outcomes

Among the 42,568 participants, the use of SGLT2i was associated with significantly reduced CV death/HHF (HR, 0.84; 95% CI, 0.73 to 0.98; $p = 0.02$; appendix 3.1), HHF (HR, 0.84; 95% CI, 0.73 to 0.98; $p = 0.02$; appendix 3.2), and renal composite outcomes (HR, 0.83; 95% CI, 0.74 to 0.92; $p = 0.0007$; appendix 3.3) in diabetic patients with CVD or at high CV risk. In terms of secondary outcomes, the effects of SGLT2i on MACE, CV death, and all-cause mortality were not statistically significant in diabetic patients with CVD or at high CV risk (appendices 3.4, 3.5, 3.6). However, the use of SGLT2i in these patients showed a statistically significant reduction in the HbA1c level from baseline (MD, -0.30; 95% CI, -0.36 to -0.23; $p < 0.00001$, appendix 3.7).

Outcome	Number of RCTs	Participants	Effect Estimate (95% CI)	Interpretation	Certainty of Evidence
CV death/HHF	4	42,568	0.90 (0.84-0.98)	Benefit	High
HHF	3	25,408	0.84 (0.73-0.98)	Benefit	High
All-cause mortality	4	42,568	0.94 (0.87-1.01)	Inconclusive	Moderate
MACE	4	42,568	0.96 (0.90-1.13)	Inconclusive	Moderate
Glycemic control (change in HbA1c)	2	18,388	0.3% (MD)	Benefit	High
Renal composite outcome 1*	3	34,322	0.78 (0.67-0.89)	Benefit	High
Renal composite outcome 2**	2	15,266	0.91 (0.76-1.09)	Inconclusive	Moderate

Certainty of evidence (Risk of bias)

The quality of the studies was assessed by the Cochrane Collaboration Risk of Bias (RoB 2) tool. All of the four included RCTs had low risk of bias.

Other considerations

In the Philippines, SGLT2 inhibitors are available at varying price points depending on the drug and dosage. Dapagliflozin 10 mg is priced at approximately Php 44.00, while empagliflozin costs around Php 53.75 for the 10 mg tablet and Php 55.25 for the 25 mg tablet. Canagliflozin is available at Php 53.75 for the 100 mg dose and Php 87.50 for the 300 mg dose. Luseogliflozin is also marketed locally in 2.5 mg and 5 mg formulations. Notably, the patent for dapagliflozin expired in 2023, allowing the entry of generic brands in the market, which has made the drug more accessible and affordable for patients.

Guideline Question 21. Should we use GLP-1 Receptor agonists (GLP-1 RA) among adults with Type 2 Diabetes Mellitus (T2DM) and established atherosclerotic cardiovascular disease (ASCVD)?

Recommendation 39. We recommend the use of GLP-1 RA with proven CV benefit* among adults with type 2 DM and established ASCVD.

**GLP-1 RA with proven CV benefits includes Liraglutide, Semaglutide Subcutaneously, Dulaglutide, Albiglutide and Efglenatide.*

**In the Philippines, the available GLP-1 RA with CV benefits includes Liraglutide, Semaglutide Subcutaneously and Dulaglutide.*

Certainty of Evidence:	Moderate
Strength of recommendation:	Strong

Background

GLP-1 Receptor agonists (GLP-1 RA) is a class of anti-diabetic drug that decrease HbA1c by stimulating glucose-dependent insulin secretion and by reducing glucagon secretion, gastric emptying, and appetite [1,2]. A significant reduction in the incidence of major cardiovascular events (MACE) has been observed in patients with type 2 diabetes mellitus (T2DM) treated with GLP 1RA, suggesting the possibility of cardioprotective actions for some molecules of the class [3,4]. Recently published meta-analysis showed that this class of drug reduced MACE as well as the risk of cardiovascular death, nonfatal stroke, hospitalization for heart failure and all-cause mortality [5].

Summary of the Evidence

Characteristics of included studies

We found 2 meta-analyses that showed cardiovascular outcomes among patients with T2DM treated with GLP-1RA. One meta-analysis included 8 CVOTs (about 60,080 patients of which 72.4% with established cardiovascular disease) [5]. The other meta-analysis which included 4 RCTs (about 43,456 patients) explored the effect of background treatment with metformin on the efficacy of GLP-1 receptor agonists (GLP-1 RAs) on cardiovascular outcomes in T2DM [6].

Efficacy outcomes

Meta-analysis done by Giugliano et al. showed that GLP-1 RA reduced MACE by 14% (HR=0.86, 95% CI 0.79 to 0.94, P=0.006) as well as reduction in the risk of cardiovascular death by 13% (P = 0.016), nonfatal stroke by 16% (P = 0.007), hospitalization for heart failure by 10% (P = 0.023) and all-cause mortality by 12% (P = 0.012) [5].

This cardiovascular benefit can be seen even if GLP-1 RA is given without Metformin. The meta-analysis published by Tsapas et al. explored the effect of background treatment with Metformin on the efficacy of GLP-1 RA on cardiovascular outcomes in T2DM. This study showed that overall, treatment with GLP-1 RA reduced MACE by 13% (HR 0.87, 95% CI 0.82 to 0.93, P= 0.06) both in patients on background Metformin (HR 0.91, 95% CI 0.85 to 0.97) and in patients not on Metformin (HR 0.80, 95% CI 0.72 to 0.90). A sensitivity analysis excluding a trial that reported results for patients treated at baseline with any oral antihyperglycemic agent rather than just with Metformin, verified the favorable effect of GLP-1 RAs on MACE, irrespective of baseline use of metformin (HR 0.85, 95% CI 0.78 to 0.93, p for interaction = 0.18). The study showed that the presence of Metformin at baseline did not affect the overall favorable effect of GLP-1 RAs both on cardiovascular and all-cause mortality [6].

Safety outcomes

There were no reported adverse outcomes on both meta-analysis included on this study.

Certainty of evidence

There is a reported low risk of bias on both meta-analysis included on this study.

Chapter 4. Application and Implementation

External review

The Clinical Practice Guideline (CPG) manuscript **underwent** a structured external review process. This process **involved** the engagement of external reviewers representing end-users, stakeholders, or technical experts who **performed** a comprehensive review of the draft manuscript. The external review **focused** on evaluating methodological rigor, local applicability, and general usability. The following review process **was implemented**:

1. **Methodological Review** – A clinical epidemiologist or guideline methodology expert **assessed** the methodological rigor and reporting quality of the CPG using the AGREE-II checklist (non-scored version).
2. **Content Expert Review** – A content expert / clinician **evaluated** the technical soundness and relevance of the recommendations using the AGREE-REX checklist, focusing on clinical validity and applicability.
3. **Non-Content Expert Review** – A clinician without specialized expertise in the topic area **provided** feedback to ensure the recommendations were clear, applicable, and user-friendly for broader audiences.

External reviewers also **provided** their feedback using a structured template that **addressed** the following aspects for each recommendation:

- General response to the guideline as a whole
- Specific comments or feedback on individual recommendations
- Assessment of completeness in terms of search and synthesis of the evidence base
- Evaluation of the acceptability of the recommendation statement, including justification for agreement or disagreement
- Suggestions for alternative recommendation statements, if applicable
- Comments on the rationale behind the recommendation, such as consensus issues, trade-offs between benefits and harms, cost considerations, patient preferences, availability, and feasibility
- Feedback on the implementability of the recommendations

Feedback from the external reviewers **was consolidated** and **deliberated upon** by the Steering Committee (SC) and Overall Technical Lead. A consensus-based approach **was followed** to address all comments and ensure relevant feedback **was incorporated** into the final manuscript. After resolving all identified issues, the Technical Writer **prepared** the final version of the manuscript, integrating approved revisions. This final draft **was submitted** to the DOH National Practice Guidelines Program for further review and approval.

Dissemination

The CPG was submitted to the Department of Health (DOH) National Practice Guideline Clearinghouse for review, assessment, and approval. The DOH, PCEDM, and other collaborating organizations will promote the use and adoption of these recommendations nationwide through publications, lectures, and various forms of communication targeting relevant stakeholders.

Dissemination efforts will include presentations at conferences and stakeholder meetings, submission for publication in both local and international journals, and availability of the guideline through the official platforms of the Philippine College of Endocrinology, Diabetes, and Metabolism and the Compendium of DOH-Approved Clinical Practice Guidelines (<https://doh.gov.ph/dpcb/doh-approved-cpg/>).

The DOH will distribute copies of the CPG to the Philippine Health Insurance Corporation, health maintenance organizations, and pharmaceutical partners. Additionally, the guideline will be presented during medical society conferences and annual conventions. Endorsed copies will also be sent to medical schools and libraries to support integration into training curricula, with the collaboration of faculty members and department heads from relevant medical specialties.

Implementation strategies

To facilitate the nationwide uptake of the newly developed guidelines for screening and diagnosis of diabetes and pre-diabetes, implementation tools will be developed either de novo or adapted from existing resources. Protocols, algorithms, and care pathways will be designed to be culturally relevant, ensuring accessibility and practicality across diverse local settings. The Philippine College of Endocrinology, Diabetes and Metabolism (PCEDM), in collaboration with the Department of Health (DOH), will employ a comprehensive implementation and monitoring strategy that includes capacity-building, system support, and performance tracking mechanisms.

Implementation tools will be developed de novo or adapted from existing resources. These include:

- Standardized protocols, algorithms, and care pathways tailored to the Philippine context, designed to be relevant across diverse local settings including primary care and community health stations.
- Educational resources, including training modules, presentation slides, and case studies.
- Digital platforms (e.g., interactive websites, mobile-accessible portals) for dissemination of tools, continuing education, and updates.

In partnership with government agencies, academic institutions, local governments, and professional societies, the PCEDM aims to promote widespread access and uptake of the CPG through medical societies, webinars, CME programs, and facility-based rollouts.

Availability of Implementation Tools

In accordance with the AGREE II checklist, the guideline development process considers the availability of practical implementation tools that support the application of recommendations in clinical practice. As part of this consideration, existing implementation-oriented guideline materials developed in other health systems, such as those from Canada (<https://guidelines.diabetes.ca/cpgrefguide>), were identified as reference examples that illustrate how clinical practice guidelines may be operationalized at the point of care.

These materials do not replace the full guideline document but are designed to translate recommendations into concise, actionable formats, including algorithms, tables, targets, and safety prompts. While direct transferability cannot be assumed, these tools demonstrate established approaches used internationally and may serve as models for potential adoption or adaptation to the Philippine context, taking into account local epidemiology, health system capacity, and resource availability.

The 2025 Canadian Clinical Practice Guidelines Quick Reference Guide provides an example of how guideline recommendations are accompanied by embedded implementation tools to facilitate screening, diagnosis, management, monitoring, and patient safety. The following section enumerates and describes these actual implementation tools as examples of approaches used in other countries.

Examples of Implementation Tools Used in Other Settings (Canadian Experience)

1. Screening and Diagnostic Algorithms

The 2025 Canadian Clinical Practice Guidelines Quick Reference Guide includes structured screening and diagnostic pathways that specify:

- Eligibility for screening based on age and risk category
- Recommended screening intervals for different risk strata
- Diagnostic thresholds and confirmatory testing strategies using A1C, fasting plasma glucose, OGTT, or random plasma glucose

These pathways function as clinical decision support tools that promote consistency in screening and diagnosis and may inform locally adapted algorithms.

2. Risk-Based Screening Tools

The document references the use of a validated diabetes risk calculator (e.g., CANRISK) to guide screening decisions. This reflects the incorporation of risk stratification tools within guideline implementation and provides an example of how risk-based screening approaches may be operationalized using validated instruments.

3. Treatment Targets as Monitoring and Audit Tools

The 2025 Canadian Clinical Practice Guidelines Quick Reference Guide presents explicit treatment targets, including:

- A1C targets tailored to patient characteristics (e.g., frailty, hypoglycemia risk, life expectancy)
- Fasting and post-prandial blood glucose targets
- Continuous glucose monitoring (CGM) metrics such as time in range, time below range, and glycemic variability

These targets function as monitoring and audit reference points, supporting evaluation of practice alignment with guideline recommendations.

4. Pharmacotherapy Pathways Linked to Comorbidities

Structured guidance is provided for pharmacologic management based on the presence of comorbid conditions such as atherosclerotic cardiovascular disease, chronic kidney disease, and heart failure. This represents a prescribing support tool that translates recommendations into clinically actionable treatment pathways.

5. Dose Adjustment Tables Based on Kidney Function

The 2025 Canadian Clinical Practice Guidelines Quick Reference Guide includes tabulated dose recommendations and restrictions for antihyperglycemic agents according to estimated glomerular filtration rate (eGFR). These tables function as medication safety tools that support appropriate dosing and monitoring in patients with impaired kidney function.

6. Safety and Sick-Day Management Guidance

In the 2025 Canadian Clinical Practice Guidelines Quick Reference Guide, practical guidance is provided on sick-day management, including hydration advice and temporary withholding of specific medication classes during acute illness or dehydration risk. These elements serve as patient safety and self-management support tools that may be adapted for local patient education and clinical protocols.

7. Checklists for Special Populations

The 2025 Canadian Clinical Practice Guidelines Quick Reference Guide includes structured guidance for specific populations, such as individuals planning pregnancy, outlining glycemic

targets, medication adjustments, screening requirements, and referral considerations. This functions as a checklist-style implementation tool for managing defined clinical scenarios.

Collectively, these examples illustrate that actual implementation tools are embedded within guideline materials used in other countries, providing concrete models for operationalizing recommendations. While these tools are not prescriptive for local use, they demonstrate approaches that may be considered for adoption or adaptation in the Philippine context, consistent with AGREE II principles on applicability, feasibility, and implementation support.

The PCEDM will also offer training on monitoring and auditing techniques to healthcare providers and encourage benchmarking against national and international standards to drive continuous improvement. Standardized reporting requirements will be established to promote adherence to evidence-based practices, particularly in facilities implementing diabetes screening programs.

To ensure ongoing refinement and alignment with clinical practice needs, the PCEDM plans to implement robust feedback mechanisms. Feedback will be gathered through online surveys, email correspondence, and dedicated feedback forms, allowing healthcare professionals to provide comments, suggestions, and share their experiences. These insights will be invaluable for the continuous improvement of the guidelines and their practical application in real-world settings.

Criteria to Assess Implementation and Adherence

To assess implementation and adherence to the CPG recommendations, the following key performance indicators (KPIs) and operational definitions may be used. These indicators are derived from strong recommendations and key steps in the screening and diagnostic process:

Indicator	Operational Definition	Level of Measurement
% of adults ≥40 years screened for diabetes	Numerator: Adults ≥40 years screened using FBG or HbA1c within the last 3 years; Denominator: All adults ≥40 years seen in outpatient settings	Facility-level
% of patients with impaired FBG who received confirmatory testing (OGTT or HbA1c)	Numerator: Patients with FBG 100–125 mg/dL who underwent OGTT/HbA1c; Denominator: All patients with impaired FBG	Individual-level
% of new T2DM diagnoses based on recommended tests	Numerator: Patients diagnosed with T2DM using FBG, 2h-OGTT, or HbA1c; Denominator: All newly diagnosed patients with T2DM	Facility-level
% of primary care providers trained on the CPG	Numerator: Providers who completed CPG training; Denominator: All target providers in participating regions	Regional/national

These indicators may be refined through pilot testing and finalized in consultation with the DOH.

Evaluation of Impact

The PCEDM, in coordination with academic partners, will assess the real-world impact of guideline adoption through both qualitative and quantitative approaches. Impact may be evaluated through:

- Trends in early detection rates of diabetes and pre-diabetes
- Changes in HbA1c values or diagnostic delays pre- vs post-CPG dissemination
- Provider adherence to recommended screening pathways
- Reduction in missed or delayed diagnoses
- Patient outcomes and satisfaction (through surveys or interviews)

Tools and Resources to Support Implementation

To improve adherence and uptake, the Task Force will develop and disseminate:

- Quick reference guides and decision aids tailored to primary care settings
- Training modules and webinars for clinicians on how to implement key recommendations
- Patient education materials (e.g., leaflets, posters, digital content) in both English and Filipino
- EMR-integrated prompts and order sets based on CPG pathways

Equity, Acceptability, and Feasibility Assessment

Initial and ongoing evaluations of the equity, acceptability, and feasibility of the CPG across diverse healthcare settings (rural vs urban, public vs private) will be conducted through key informant interviews and focus group discussions with:

- Frontline healthcare workers
- Patients and caregivers
- Local health system administrators

Findings will inform revisions to future versions of the guideline and highlight areas requiring targeted implementation support.

Monitoring, Auditing, and Feedback Mechanisms

The PCEDM will offer training on monitoring, evaluation, and audit techniques to healthcare providers, enabling self-assessment and benchmarking. Recommended strategies include:

- Use of audit templates for facility-level review of screening practices
- Annual or bi-annual benchmarking reports aligned with national targets
- Development of a CPG implementation dashboard for internal quality improvement

A set of standardized reporting requirements may be developed for programs and institutions conducting systematic diabetes screening, in alignment with DOH monitoring protocols.

Feedback and Continuous Improvement

Robust feedback mechanisms will be integrated to support iterative refinement of both guideline content and implementation strategies. Channels include:

- Anonymous online feedback forms embedded in training platforms
- Periodic surveys targeting healthcare professionals and program implementers
- Stakeholder consultations and key informant interviews to identify barriers and enablers

Findings from the feedback process will be reviewed annually and used to inform guideline updates and improvements in implementation support.

Facilitators and barriers to application or compliance to the recommendations

After discussions with the multistakeholder group throughout the CPG development process, information on the types of facilitators and barriers emerged. Additional feedback was sought from other key stakeholders through dialogues or external communications. These methods included identifying facilitators (e.g., government support for medications and services, advocacy group activities) and barriers (e.g., out-of-pocket healthcare payments, availability of specific machines/laboratory tests/reagents). This information influenced the value judgments of the consensus panel during discussions and the formulation of recommendation statements and the strength of each recommendation.

Monitoring and evaluation

The Task Force will distribute an annual questionnaire to identify best practices among relevant stakeholders regarding the screening and diagnosis of diabetes and pre-diabetes. Monitoring the use of this clinical practice guideline may also serve as a topic for research by interested parties. For monitoring and auditing, the CPG group will use the final strength of recommendation to establish key performance indicators. Recommendations classified as “Strong Recommendation” will serve as the primary indicators. These indicators will be identified once the strength of recommendations has been finalized for each guideline review question.

Specific operational definitions for measuring the criteria will be aligned with the recommended standards outlined in the DOH Manual for CPG Development. These definitions will also be determined following consultations with the Department of Health (DOH), as local literature on these measurements is currently limited.

Frequency and Interval of Monitoring and Audit

- Routine monitoring of guideline uptake and implementation may be undertaken as part of existing quality assurance and clinical audit activities at the facility, institutional, or program level.
- Audit of adherence to selected key recommendations, particularly those rated as strong recommendations, may be conducted periodically, such as annually or biennially, depending on local capacity, data availability, and alignment with institutional audit cycles.
- Feedback from end users (e.g., primary care physicians, specialists, and other stakeholders) may be collected on an ongoing basis, particularly during the early phases of dissemination, to identify barriers to implementation and areas requiring clarification or contextualization.
- Findings from monitoring and audit activities are intended to inform quality improvement initiatives and guide refinements in dissemination strategies.

Updating procedure

This Diabetes CPG will be updated every three years (i.e., 2028). The Steering Committee is responsible for initiating such updates or revisions when new evidence from large-scale studies becomes available on certain priority topics or when there are changes in available resources, national policies, approval status of new interventions or health technologies, or values placed on outcomes. The Department of Health (DOH) will also regularly assess the relevance and applicability of these guidelines and issue an update if necessary.

Triggers for Early Updates

While scheduled updates are planned every three years, earlier review and updating may be undertaken if any of the following occur:

- Release of new international guidelines (e.g., from WHO or other authoritative bodies) that substantially alter standards of care
- Availability of new diagnostic tests, technologies, or therapeutic options relevant to diabetes management in the Philippine setting
- Significant changes in national health policy, DOH programs, or PhilHealth coverage that affect delivery of diabetes care
- Emergence of important new evidence that may materially change the balance of benefits and harms of existing recommendations
- Substantial epidemiologic shifts or public health developments that impact diabetes prevention, diagnosis, or management

Access to the guideline

The guideline will be publicly accessible through official websites of healthcare societies and through the Compendium of DOH-Approved Clinical Practice Guidelines (<https://doh.gov.ph/dpcb/doh-approved-cpg/>). The evidence base and final manuscript will be made accessible through both print and electronic media platforms managed by the DOH, the PCEDM, and other contributing organizations.

Recommendations for future research

There is a paucity of data for most aspects of diabetes in the country. Available epidemiology data pertains mainly to the prevalence of the condition. The following have been identified as the research gaps:

Screening

- The cost-effectiveness of screening based on each risk factor for developing diabetes and the long-term follow-up of people screened with the clinical risk score questionnaire in the Philippines.
- The identification of prevalent risk factors in people diagnosed with diabetes during national health surveys to monitor changes in the population's risk profile.
- The optimal frequency for screening diabetes using fasting blood sugar levels.

Diagnosis

- The utility of 1-hour post-75-gram glucose load in diagnosing glucose intolerance in non-pregnant adults.
- The accuracy of HbA1c testing in the Philippines given the lack of standardization of reporting of results, that is, the lack of indicating the specific laboratory assay used and specifying if the methodology used is certified by the National Glycohemoglobin Standardization Program (NGSP).
- The correlation of HbA1c with plasma glucose in the Philippines.
- The validity of point-of-care HbA1c kits.
- An adequately powered cross-sectional study comparing the accuracy of CBG with OGTT.

- An adequately powered cross-sectional study among patients in geographically inaccessible and disadvantaged areas comparing the accuracy of CBG using the government-issued glucometers with OGTT.
- Validated studies to assess the commonly used criterion of random plasma glucose ≥ 200 mg/dL (≥ 11.1 mmol/L) when classic hyperglycemic symptoms are present.

Screening for Complications

- The lack of prevalence data on microvascular complications (neuropathy, nephropathy, and retinopathy) and diabetes-related cardiovascular diseases, including among newly diagnosed people with diabetes.
- The diagnostic accuracy of retinal photograph interpretation by non-ophthalmologist credentialed personnel.
- The most accurate among the existing eGFR calculators for Filipino adults with chronic kidney disease.
- The optimal monitoring frequency for albumin:creatinine ratio and serum creatinine.
- The accuracy of microalbumin test and urine protein-creatinine ratio.
- The optimal cut-off for the Semmes-Weinstein monofilament test that would predict foot ulceration and amputation.

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Guideline Question 1: Prevention (Nonpharmacological management) among at risk of T2DM or prediabetes

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Guideline Question 2. Pharmacologic treatment among those at risk for T2DM or prediabetes

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Guideline Question 6: Non-nutritive Sweeteners

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Guideline Question 15: Aspirin among patients with type 2 diabetes mellitus for primary prevention of cardiovascular events

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Guideline Question 18: Low dose combination drugs for the control of hypertension among persons with Type 2 DM and hypertension

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Guideline Question 20: Sodium-glucose cotransporter-2 (SGLT2) inhibitors among patients with type 2 diabetes mellitus with established atherosclerotic cardiovascular disease (ASCVD)

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Guideline Question 21: GLP-1 Receptor agonists (GLP-1 RA) among patients with Type 2 Diabetes Mellitus (T2DM) and established atherosclerotic cardiovascular disease (ASCVD)

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Annex A. Tools for Assessing Conflicts of Interest

Primary:	Relates to company products in the same therapeutic areas.
Secondary:	Relates to products in other therapeutic areas.
Major:	Huge potential impact on panel decisions.
Minor:	Small impact on panel decisions.
Direct:	Immediate economic returns (e.g. payments)
Indirect:	Delayed material returns (e.g. academic promotions)
Personal:	Arises from a relationship with a person.
Institutional:	Arises from a relationship with an institution.
Financial:	Leads to gains in a material sense.
Intellectual:	Leads to gains in terms of academic status, recognition, or popularity.
Financial:	Leads to material/financial gain.
Non financial:	Leads to intellectual and other gains.

Annex B: Clinical Questions

A. Nonpharmacological Management Among Patients with Diabetes Mellitus

Q1. Should we recommend non-pharmacologic interventions — including weight loss, dietary interventions, physical activity and exercise, proper sleep hygiene, smoking cessation, and avoidance of excessive alcohol intake — among adults at risk for diabetes or with pre-diabetes?

Q1.1 Should we recommend non-pharmacologic intervention (weight loss) among adults at risk for diabetes or with pre-diabetes?

P	Filipino adults at risk of type 2 Diabetes or with pre-diabetes
I	Intensive Lifestyle changes including weight loss
C	No lifestyle change
O	Prevention of diabetes and reduction of glucose to normal or non-overt diabetes levels

Q1.2 Should we recommend non-pharmacologic intervention (dietary intervention) among adults at risk for diabetes or with pre-diabetes?

P	Filipino adults at risk of type 2 Diabetes or with pre-diabetes
I	Intensive Lifestyle changes including dietary interventions
C	No lifestyle change
O	Prevention of diabetes and reduction of glucose to normal or non-overt diabetes levels

Q1.3 Should we recommend non-pharmacologic intervention (physical activity and exercise) among adults at risk for diabetes or with pre-diabetes?

P	Filipino adults at risk of type 2 Diabetes or with pre-diabetes
I	Intensive Lifestyle changes including physical activity and exercise
C	No lifestyle change
O	Prevention of diabetes and reduction of glucose to normal or non-overt diabetes levels

Q1.4 Should we recommend non-pharmacologic intervention (proper sleep hygiene) among adults at risk for diabetes or with pre-diabetes?

P	Filipino adults at risk of type 2 Diabetes or with pre-diabetes
I	Intensive Lifestyle changes including proper sleep hygiene
C	No lifestyle change

O	Prevention of diabetes and reduction of glucose to normal or non-overt diabetes levels
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Q1.5 Should we recommend non-pharmacologic intervention (smoking cessation) among adults at risk for diabetes or with pre-diabetes?

P	Filipino adults at risk of type 2 Diabetes or with pre-diabetes
I	Intensive Lifestyle changes including smoking cessation
C	No lifestyle change
O	Prevention of diabetes and reduction of glucose to normal or non-overt diabetes levels

Q1.6 Should we recommend non-pharmacologic intervention (avoidance of excessive alcohol intake) among adults at risk for diabetes or with pre-diabetes?

P	Filipino adults at risk of type 2 Diabetes or with pre-diabetes
I	Intensive Lifestyle changes including avoidance of excessive alcohol intake
C	No lifestyle change
O	Prevention of diabetes and reduction of glucose to normal or non-overt diabetes levels

Q2. Should we recommend pharmacologic therapy (metformin, pioglitazone, acarbose, orlistat, SGLT-2 inhibitors, GLP-1 RAs) among adults with pre-diabetes?

Q2.1 Should we recommend pharmacologic therapy (metformin, pioglitazone) among adults with pre-diabetes?

P	Filipino adults at risk of type 2 DM or with pre-diabetes
I	pharmacologic therapy (metformin, pioglitazone)
C	Non-use of pharmacologic therapy
O	Prevention of diabetes

Q2.2 Should we recommend pharmacologic therapy (acarbose, orlistat) among adults with pre-diabetes?

P	Filipino adults at risk of type 2 DM or with pre-diabetes
I	pharmacologic therapy (acarbose, orlistat)
C	Non-use of pharmacologic therapy
O	Prevention of diabetes

Q2.3 Should we recommend pharmacologic therapy (SGLT-2 inhibitors, GLP-1 RAs) among adults with pre-diabetes?

P	Filipino adults at risk of type 2 DM or with pre-diabetes
I	pharmacologic therapy (SGLT-2 inhibitors, GLP-1 RAs)
C	Non-use of pharmacologic therapy
O	Prevention of diabetes

Q3. Should we recommend dietary interventions among adults with Type 2 DM?**Q3.1 Should we recommend dietary interventions among adults with Type 2 DM**

P	Filipino adults with type 2 DM
I	Use of dietary interventions
C	Non-use of dietary interventions
O	HbA1c and weight reduction Reduction of Microvascular and macrovascular

Q4. Should we recommend physical activity and exercise for glucose control, weight management and overall cardiovascular health among adults with Type 2 DM?**Q4.1 Should we recommend physical activity and exercise for glucose control, weight management and overall cardiovascular health among adults with Type 2 DM?**

P	Filipino adults with type 2 DM
I	Structured exercise and general physical activity
C	Non-structured exercise and general physical activity/usual care
O	HbA1c and weight reduction Reduction of Microvascular and macrovascular

Q5. Should we recommend limitation of alcohol intake among persons with Type 2 DM?**Q5.1 Should we recommend limitation of alcohol intake among persons with Type 2 DM?**

P	Filipino adults with type 2 DM
I	Limitation of alcohol intake
C	Non-use of alcohol intake limitations
O	HbA1c and weight reduction Reduction of Microvascular and macrovascular

Q6. Should we recommend the use of non-nutritive sweeteners among patients with Type 2 DM?**Q6.1 Should we recommend the use of non-nutritive sweeteners among patients with Type 2 DM?**

P	Filipino adults with type 2 DM
I	Consumption of non nutritive sweeteners; intake of any type of NSS (excluding sugar alcohols and natural caloric sweeteners), whether specified by name or not, and whether used alone or in combination with other NSS (within the Acceptable Daily Intake (ADI) values)
C	Non-use of non-nutritive sweeteners
O	improvement in glycemic control weight reduction; cardiovascular disease; mortality

Q7. Should we use diabetes-specific nutrition formulas among adults with Type 2 DM?**Q7.1 Should we use diabetes-specific nutrition formulas among adults with Type 2 DM?**

P	Filipino adults with type 2 DM
I	use diabetes-specific nutrition formulas
C	Non-use diabetes-specific nutrition formulas
O	improvement in glycemic control weight reduction; cardiovascular disease; mortality

Q8. Should we recommend referral for bariatric surgery to achieve sustained glycemic control and weight loss among obese adults with type 2 DM?

Q8.1 Should we recommend referral for bariatric surgery to achieve sustained glycemic control and weight loss among obese adults with type 2 DM?	
P	Filipino adults with type 2 DM and obesity
I	Bariatric Surgery
C	Usual Care/Medical Management
O	HbA1c and weight reduction Reduction of Microvascular and macrovascular complications

Q9. Should we recommend HbA1c to monitor glycemic control among nonpregnant adults with T2DM?

Q9.1 Should we recommend HbA1c to monitor glycemic control among nonpregnant adults with T2DM?	
P	Filipino adults with type 2 DM
I	HbA1c
C	Usual care
O	HbA1c and weight reduction Reduction of Microvascular and macrovascular complications

Q10. Should we recommend FPG to monitor glycemic control among nonpregnant adults with T2DM?

Q10.1 Should we recommend FPG to monitor glycemic control among nonpregnant adults with T2DM?	
P	Nonpregnant Filipino adults with type 2 DM
I	Fasting plasma glucose
C	Usual care
O	Glycemic control Reduction of Microvascular and macrovascular complications

Q11. Should we recommend the use of 1,5-anhydroglucitol (1,5-AG) to monitor glycemic control among nonpregnant adults with T2DM?

Q11.1 Should we recommend the use of 1,5-anhydroglucitol (1,5-AG) to monitor glycemic control among nonpregnant adults with T2DM?	
P	Nonpregnant Filipino adults with type 2 DM
I	1,5-anhydroglucitol (1,5-AG)
C	Usual care
O	Glycemic control Reduction of Microvascular and macrovascular complications

Q12. Should we recommend the use of fructosamine to monitor glycemic control among nonpregnant adults with T2DM?

Q12.1 Should we recommend the use of fructosamine to monitor glycemic control among nonpregnant adults with T2DM?	
P	Nonpregnant Filipino adults with type 2 DM
I	1,5-anhydroglucitol (1,5-AG)
C	Usual care
O	Glycemic control Reduction of Microvascular and macrovascular complications

Q13. Should we recommend self-monitoring of blood glucose (SMBG) to monitor glycemic control among adults with type 2 DM?

Q13.1. Should we recommend self-monitoring of blood glucose (SMBG) to monitor glycemic control among adults with type 2 DM?	
P	Nonpregnant Filipino adults with type 2 DM
I	self-monitoring of blood glucose (SMBG)
C	Usual care
O	Glycemic control Reduction of Microvascular and macrovascular complications

Q14. Should we use a continuous glucose monitor (CGM) to monitor glycemic control among nonpregnant adults with T2DM?

Q14.1. Should we use a continuous glucose monitor (CGM) to monitor glycemic control among nonpregnant adults with T2DM?	
P	Nonpregnant Filipino adults with type 2 DM
I	continuous glucose monitor (CGM)
C	Usual care
O	Glycemic control Reduction of Microvascular and macrovascular complications

Q15. Should we use aspirin among patients with type 2 diabetes mellitus for primary prevention of cardiovascular events?

Q15.1. Should we use aspirin among patients with type 2 diabetes mellitus for primary prevention of cardiovascular events?	
P	Filipino adults with type 2 DM
I	Use of aspirin
C	Nonuse of aspirin
O	Primary prevention/ Reduction of Microvascular and macrovascular complications

Q16. Should we use aspirin among patients with type 2 diabetes mellitus for secondary prevention of cardiovascular events?

Q16.1. Should we use aspirin among patients with type 2 diabetes mellitus for secondary prevention of cardiovascular events?	
P	Filipino adults with type 2 DM
I	Use of aspirin
C	Nonuse of aspirin
O	Secondary prevention/ Reduction of Microvascular and macrovascular complications

Q17. Should we use clopidogrel among patients with type 2 diabetes mellitus for secondary prevention of cardiovascular events?

Q17.1. Should we use clopidogrel among patients with type 2 diabetes mellitus for secondary prevention of cardiovascular events?	
P	Filipino adults with type 2 DM
I	Use of aspirin
C	Nonuse of aspirin
O	Secondary prevention/ Reduction of Microvascular and macrovascular complications

Q18. Should we recommend low dose combination drugs for the control of hypertension among persons with Type 2 DM and hypertension?

Q18.1. Should we recommend low dose combination drugs for the control of hypertension among persons with Type 2 DM and hypertension?	
P	Filipino adults with type 2 DM and hypertension
I	low dose combination drugs
C	Usual care
O	Hypertension Control Reduction of Microvascular and macrovascular complications

Q19. Should we recommend statins for primary prevention of cardiovascular (CV) events among patients with type 2 diabetes mellitus?

Q19.1. Should we recommend statins for primary prevention of cardiovascular (CV) events among patients with type 2 diabetes mellitus?	
P	Filipino adults with type 2 DM and hypertension
I	Statins
C	Usual care
O	Reduction of Microvascular and macrovascular complications

Q20. Should we use sodium-glucose cotransporter-2 (SGLT2) inhibitors among patients with type 2 diabetes mellitus with established atherosclerotic cardiovascular disease (ASCVD)?

Q20.1. Should we use sodium-glucose cotransporter-2 (SGLT2) inhibitors among patients with type 2 diabetes mellitus with established atherosclerotic cardiovascular disease (ASCVD)?	
P	Filipino adults with type 2 DM and hypertension
I	sodium-glucose cotransporter-2 (SGLT2) inhibitors
C	Usual care/non-use of sodium-glucose cotransporter-2 (SGLT2) inhibitors
O	Reduction of Microvascular and macrovascular complications; improvement in glycemic control

Q21. Should we use GLP-1 Receptor agonists (GLP-1 RA) among patients with Type 2 Diabetes Mellitus (T2DM) and established atherosclerotic cardiovascular disease (ASCVD)?

Q21.1. Should we use GLP-1 Receptor agonists (GLP-1 RA) among patients with Type 2 Diabetes Mellitus (T2DM) and established atherosclerotic cardiovascular disease (ASCVD)?	
P	Filipino adults with type 2 DM and hypertension
I	GLP-1 Receptor agonists (GLP-1 RA)
C	Usual care/non-use of GLP-1 Receptor agonists (GLP-1 RA)
O	Reduction of Microvascular and macrovascular complications; improvement in glycemic control

Annex C: Final List of Guidelines Appraised with AGREE II

1. Ivers NM, Jiang M, Alloo J, et al. Diabetes Canada 2018 clinical practice guidelines: Key messages for family physicians caring for patients living with type 2 diabetes. *Can Fam Physician*. 2019;65(1):14-24. <https://www.cfp.ca/content/65/1/14.long> [**Diabetes Canada Family Medicine**]
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7. International Diabetes Federation. Recommendations For Managing Type 2 Diabetes In Primary Care, 2017. www.idf.org/managing-type2-diabetes. [**IDF Primary Care**]
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9. Chan Siew Pheng et al. Management of Type 2 Diabetes Mellitus (6th edition) [**Malaysian Endocrine and Metabolic Society**]
10. Qaseem A, Barry MJ, Humphrey LL, Forciea MA et al for Clinical Guidelines Committee of the American College of Physicians. Oral Pharmacologic Treatment of Type 2 Diabetes Mellitus: A Clinical Practice Guideline Update From the American College of Physicians. *Ann Intern Med*. 2017 Feb 21;166(4):279-290. doi: 10.7326/M16-1860. Epub 2017 Jan 3. [**US American College of Physicians**]
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13. Schorr SG, Hammes HP, Müller UA, Abholz HH, Landgraf R, Bertram B. The Prevention and Treatment of Retinal Complications in Diabetes. *Dtsch Arztebl Int.* 2016 Dec 2;113(48):816-823. doi: 10.3238/arztebl.2016.0816. [**German Medical Association**]
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Annex E: Detailed Methods and Results of Evidence Syntheses

A. Nonpharmacological Management Among Patients with Diabetes Mellitus

Guideline Question 1. Should we recommend non-pharmacologic interventions — including weight loss, dietary interventions, physical activity and exercise, proper sleep hygiene, smoking cessation, and avoidance of excessive alcohol intake — among adults at risk for diabetes or with pre-diabetes?

Q1.1. Methods

The evidence review was conducted from 2021 to March 2024 and subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024. The initial phase of the review involved examining the recommendations and all references cited in the most recent versions of guidelines selected by the Technical Review Committee using the ADAPTE framework. Full-text articles of these references were retrieved through PubMed. When more updated or methodologically stronger studies related to the topic were identified through citation tracking and PubMed's "similar articles" recommendations, these were also incorporated.

To determine the applicability of international recommendations to the Philippine context, we additionally reviewed the most recent epidemiologic reports from Philippine National Health Surveys published in the Philippine Journal of Internal Medicine and publicly available datasets from the Department of Science and Technology–Food and Nutrition Research Institute (DOST–FNRI). We also searched for relevant Asian consensus statements and their supporting references, particularly those in which the Philippines contributed to regional recommendations.

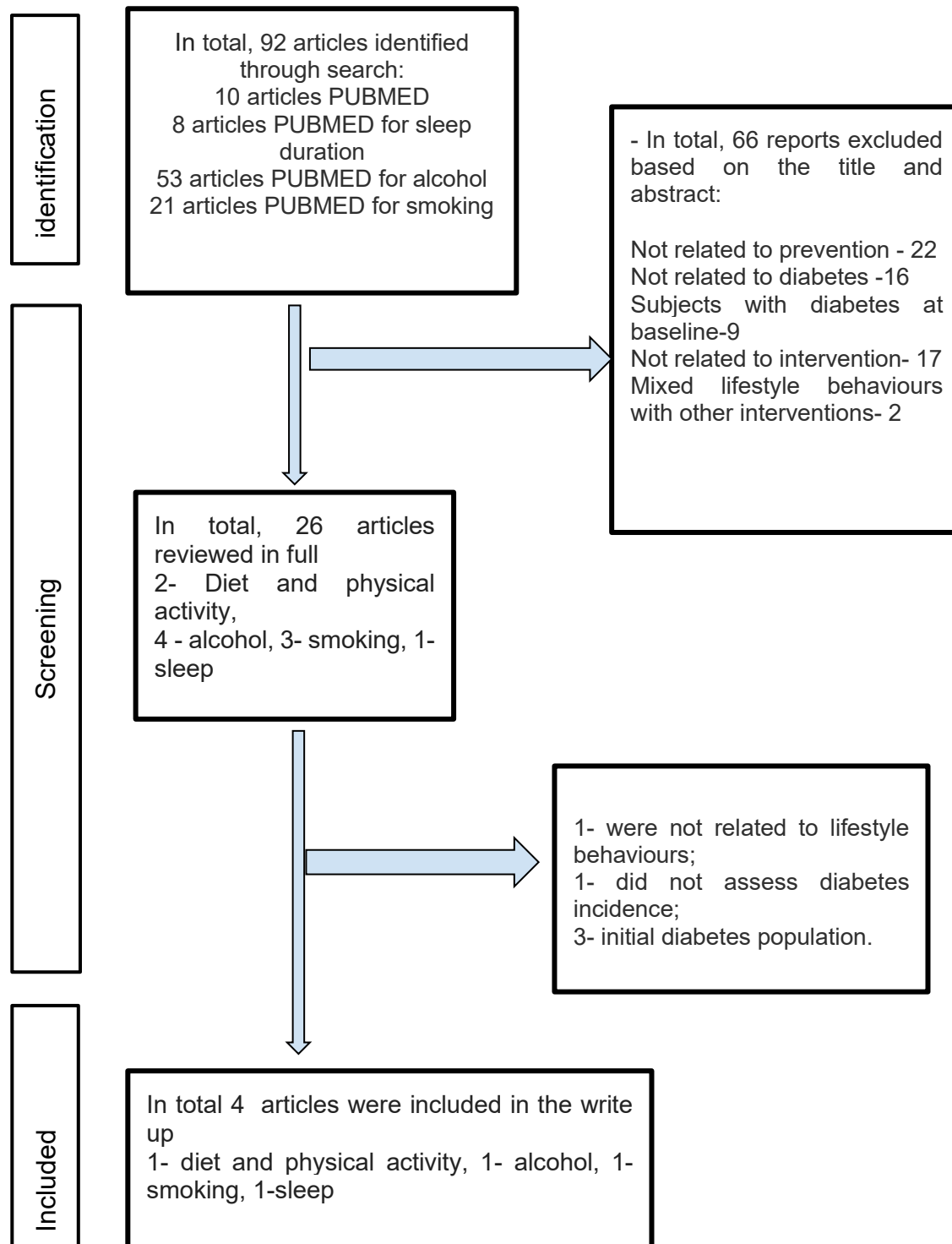
In parallel, reviewers conducted a comprehensive systematic search for the latest evidence from MEDLINE, PubMed Central, and Google Scholar. The search strategy included the following terms: "meta-analysis," "systematic reviews," "type 2 diabetes mellitus," "prevention," "lifestyle intervention," "diet and exercise," "weight reduction," "weight loss," "dietary pattern," "nutrition patterns," "nutrition," "sleep duration," "smoking," and "alcohol intake." Searches were limited to studies published in English between 2010 and 2024, with a preference for Asian cohorts where applicable. Citation chaining was also used to identify additional relevant literature.

Q1.2. Search strategy

DATABASE	SEARCH STRATEGY / SEARCH TERMS	DATE AND TIME OF SEARCH	RESULTS	
			Yield	Eligible
Pubmed	"Prevention" AND "Type 2 diabetes" AND "lifestyle changes" Additional filter: metaanalysis, last 10 years	March 10, 2024	10	1
Cochrane Database of Systematic Reviews	"prevention" AND "type 2 diabetes" AND "lifestyle changes"	March 10, 2024	4	0

Medline via PubMed	((sleep duration[Title] AND (diabetes[Title])) (meta-analysis[Title])) AND (("2014"[Date - Publication] : "2024"[Date - Publication]))	March 30, 2024	8	2
Pubmed	"Prevention" AND "Type 2 diabetes" AND "alcohol" Additional filter: meta analysis, last 10 years	March 10, 2024	53	1
PubMed	"Prevention" AND "type 2 diabetes" AND "smoking" Additional filter: metaanalysis, last 10 years	March 10, 2024	20	1

Q1.3. PRISMA flow diagram



Q1.4. Characteristics of the Included Studies

Table of the Characteristics of the Studies Included

Summary results on the randomized controlled trials on type 2 diabetes prevention in people with impaired glucose tolerance or in people at high increased risk for diabetes. through diet and exercise

Studies	N, characteristics	Study Duration	Risk Reduction of T2DM with Lifestyle versus Control	Dietary Goals/Changes	Physical Activity, Goals/Changes
Da Qing	In total, 577; all had IGT; 33 health care clinics	6 yrs	Diet 33%; exercise 47%; diet + exercise 38%	Weight reduction in overweight; calorie restriction	Increase, e.g., walking
Finnish Diabetes Prevention Study	In total, 522; IGT; five centers	3.2 yrs; median 4 yrs	In total, 58%, weight loss; difference 3.5 and 2.6 kg after 1 and 3 yrs, respectively.	Weight reduction >5%; reduce total and SFA; increase dietary fiber	4 h/wk, sedentary people at yr 3: 17% vs. 29% for intervention and control groups, respectively
Diabetes Prevention Program	In total, 3234; IGT; 27 centers	2.8 yrs	Lifestyle 58%; Metformin 31%; weight loss at yr 1: ~5.6 vs. ~0.1 kg for intervention vs. control, respectively.	NCEP Step 1; weight loss goal 7%	150 min/wk
Japanese trial in IGT males	In total, 458 IGT; 356 in control, 102 in intervention, OGTT (100 g glucose dose)	4 yrs	Incidence of T2D 3.0% vs. 9.3%; risk reduction 67.4%; weight loss ~2.18 kg	BMI goal 22 kg/m ² ; increase vegetables; reduce food intake by 10%; fat < 50 g/d; alcohol restriction	30–40 min walking/d
Indian Diabetes Prevention Trial	In total, 531; IGT; lifestyle 133; metformin 133; lifestyle-plus-metformin 129; control 136	30 mos	Lifestyle 28.5%; Metformin 26.4%; lifestyle-plus-Metformin 28.2%; no change in body weight	Reduce total calories, refined CHO, fat and sugar; increase high fiber-rich foods	Walking 30 min a day
Lifestyle intervention	In total, 375 with dysmetabolism; 169	1 yr,	Risk reduction for T2D 77%, (OR 0.23; 95% CI 0.06–0.85) at year 1	General recommendations	Increase

metabolic syndrome	intervention; 166 control; focus on metabolic syndrome			for lose weight and decrease SFA and increase PUFA and fiber	
European Diabetes Prevention RCT	In total, 102; IGT; 51 in intervention and control, respectively	3 yrs	Diabetes incidence 5% vs. 11, 1% yr. body weight change ~2.5 kg	Like in FDPS, decrease fat and SFA; increase fiber; body weight reduction	Like in FDPS

Summary results on the randomized controlled trials on type 2 diabetes prevention in people with impaired glucose tolerance or in people at high increased risk for diabetes. through quitting smoking

Studies	N, characteristics	Study Duration	Risk for diabetes	Goals/ Changes
Pan et al	6M participants (88 prospective studies)	> 10 years	1.21 (1.10-1.33) for light smokers, 1.34 (1.27-1.41) for moderate smoker and 1.57 (1.47-1.66) for heavy smokers	Smoking cessation

Summary results on the randomized controlled trials on type 2 diabetes prevention in people with impaired glucose tolerance or in people at high increased risk for diabetes. through alcohol intake reduction

Studies	N, characteristics	Study Duration	Risk for diabetes	Goals/ Changes
Knott et al	1,902,605 participants (38 studies)	4 to 20 years	Relative to combined abstainers, reductions in the risk of type 2 diabetes were present at all levels of alcohol intake <63 g/day , but this effect was not found in Asian populations	Alcohol intake cessation

Summary results on the randomized controlled trials on type 2 diabetes prevention in people with impaired glucose tolerance or in people at high increased risk for diabetes. through adequate sleep

Studies	N, characteristics	Study Duration	Risk for diabetes	Goals/ Changes
Lu et al	737002 adults (17 studies)	2.5 to 16 years	short sleep duration <6h vs normal sleep duration (6 h < t < 9 h) (RR = 1.22, 95% CI: 1.15-1.29, P < 0.001)	Sleep between 7-8 hrs
Shan Z et al	482,502 participants	2.5 to 16 years	7 hrs Sleep vs <7 h 1.09 (95% CI 1.04-1.15) and 7 hrs sleep vs >7 h 1.14 (1.03-1.26)	sleep of 7-8 hrs

Q1.5. Assessment of Risk of Bias

Assessment of Risk of Bias of Included Studies

Appraisal of Meta-analysis

Author(s): Uusitupa M, Khan TA, Vigiouliouk E, Kahleova H, Rivellese AA, Hermansen K, Pfeiffer A, Thanopoulou A, Salas-Salvadó J, Schwab U, Sievenpiper JL. Prevention of Type 2 Diabetes by Lifestyle Changes: A Systematic Review and Meta-Analysis. *Nutrients*. 2019 Nov 1;11(11):2611. doi: 10.3390/nu11112611. PMID: 31683759; PMCID: PMC6893436.

Question: Should lifestyle changes be used for prevention of diabetes?

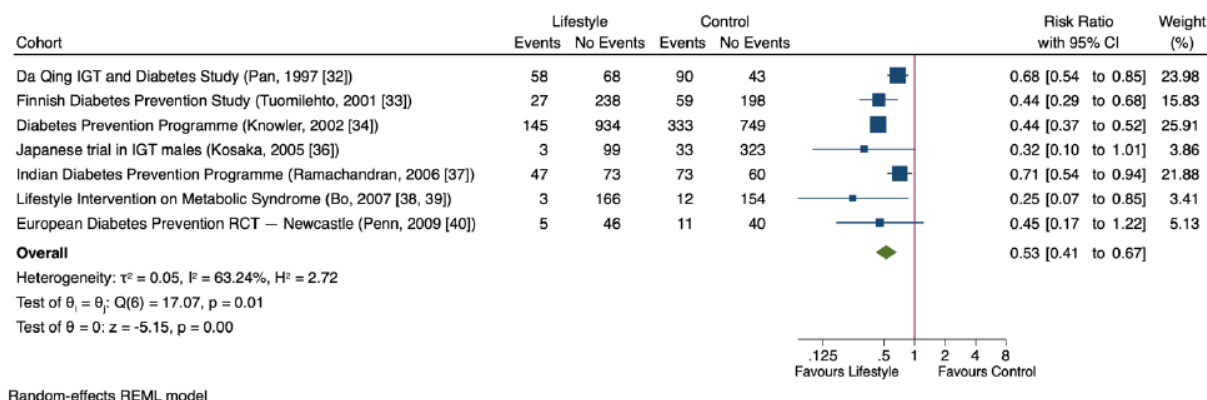
The figure below shows the individual Cochrane Risk of Bias assessments of seven trials included in the current meta-analysis. The majority of trials were judged as having unclear or low risk of bias across domains. No evidence of a serious risk of bias was detected.

	Risk of Bias					
Cohort	A	B	C	D	E	Overall
Da Qing IGT and Diabetes Study (Pan, 1997)	?	+	?	?	?	Unclear
Finnish Diabetes Prevention Study (Tuomilehto)	+	?	+	+	+	Low
Diabetes Prevention Program	?	?	+	+	+	Low
Japanese trial in IGT males	?	?	?	?	+	Unclear
Indian Diabetes Prevention Program (Ramachandran, 2006)	?	?	+	?	+	Unclear
Lifestyle intervention on Metabolic Syndrome	+	+	+	?	+	Low
European Diabetes Prevention	+	+	+	+	+	Low
A- Random sequence generation (selection bias) B- Allocation concealment (selection bias) C- Blinding of participants, personnel and outcome assessment (performance and detection bias) D- Incomplete outcome data (attribution bias) E- Selective reporting (reporting bias)						

Q1.6. Results

Meta-analysis Forest Plot: Lifestyle Intervention and T2 DM risk

Forrest plot of RCTs investigating the effect of lifestyle changes and T2DM risk. The green diamond shows the pooled estimate effect. Data are expressed as weighted risk ratios with 95% confidence intervals (CIs) using the restricted maximum likelihood REML) random-effects model. Inter-study heterogeneity was tested by the Cochrane Q-statistic at a significance level of $p < 0.10$ and quantified by I^2 , where a level of $\geq 50\%$ represented substantial heterogeneity.



Da Qing Study: Effect of Lifestyle Intervention on Important Outcomes

Forest plot of the 30-year follow up of the Da Qing Study showed that the combined intervention group had 26% decrease in cardiovascular events (HR 0.74, 95% CI 0.59–0.92; $p=0.0060$), 35% decrease in microvascular complications (HR 0.65, 0.45–0.95; $p=0.025$), 33% in cardiovascular deaths (HR 0.67, 0.48–0.94; $p=0.022$), and 26% in all-cause mortality (HR 0.74, 0.61–0.89; $p=0.0015$), leading to an increase of 4.82 years in median survival and a mean increase of 1.44 years in life expectancy in the intervention group compared with control.¹³

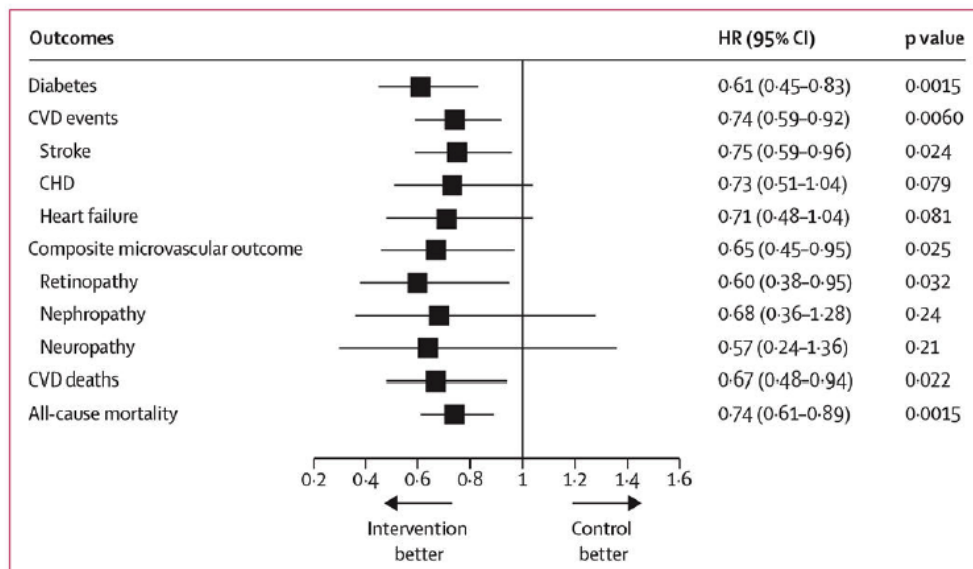
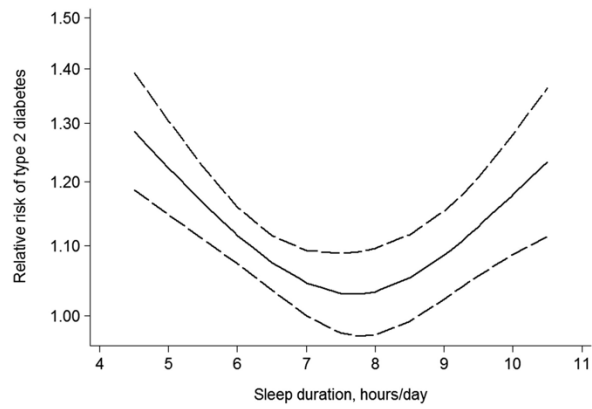


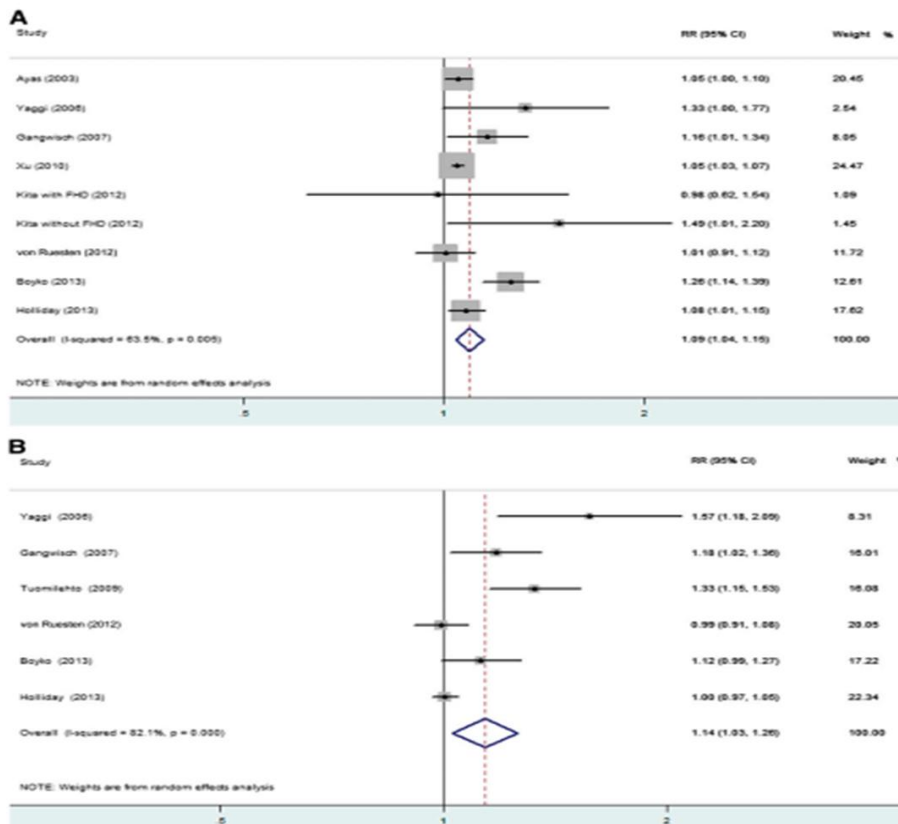
Figure 3: Forest plot of primary and secondary outcome events at 30-year follow-up
The reference category is the control group. Hazard ratios (HRs) are derived from proportional-hazards models, controlled for clinic randomisation. CVD=cardiovascular disease. CHD=coronary heart disease.

APPENDIX 8.3: The relationship between sleep duration and risk of type 2 diabetes.



From: *Sleep Duration and Risk of Type 2 Diabetes: A Meta-analysis of Prospective Studies. Diabetes Care. 2015;38(3):529-537. doi:10.2337/dc14-2073*

APPENDIX 8.4: Forest Plot: Link between sleep duration and Type 2 DM



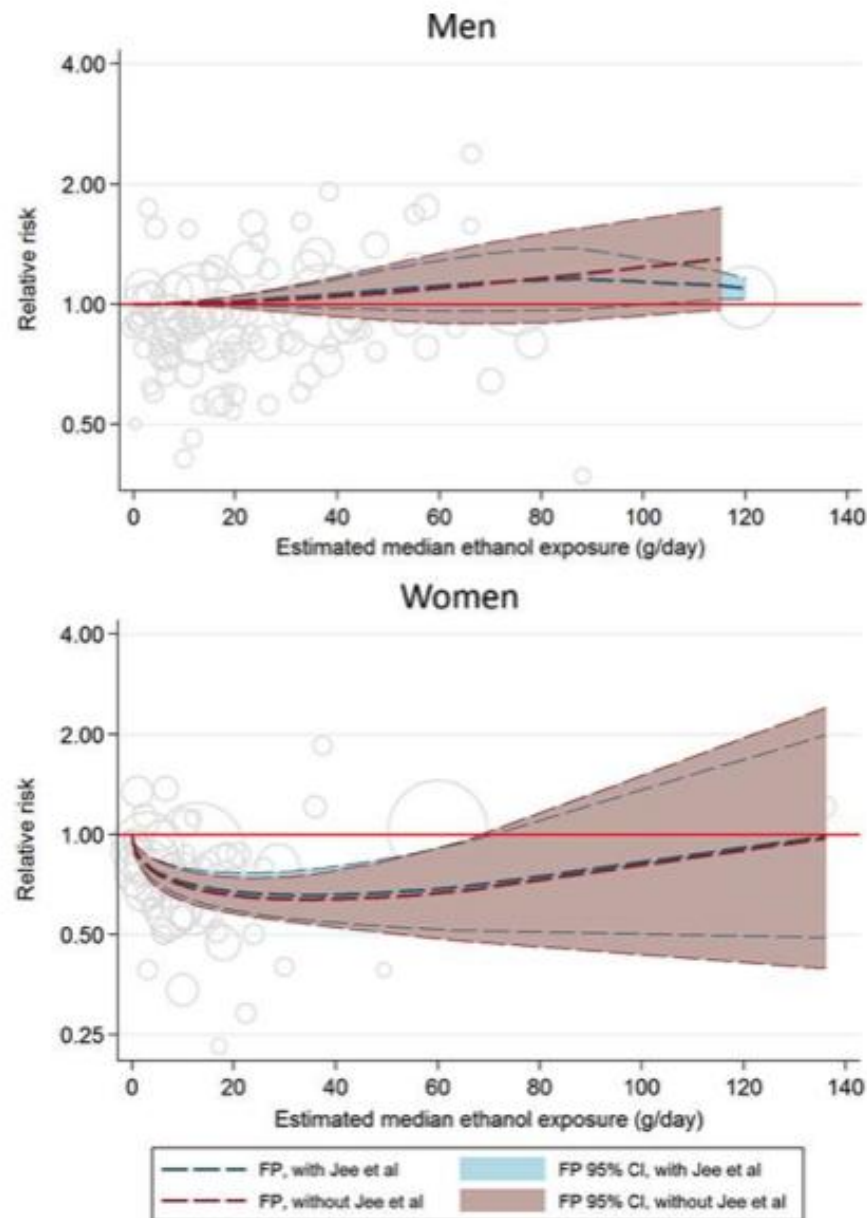
9 data sets
on short sleep duration
and risk of diabetes

7 data sets
on long sleep duration
and risk of diabetes

Figure 3—The dose-response relationship plot between sleep duration (per hour) and risk of type 2 diabetes for short sleep (A) and long sleep (B).

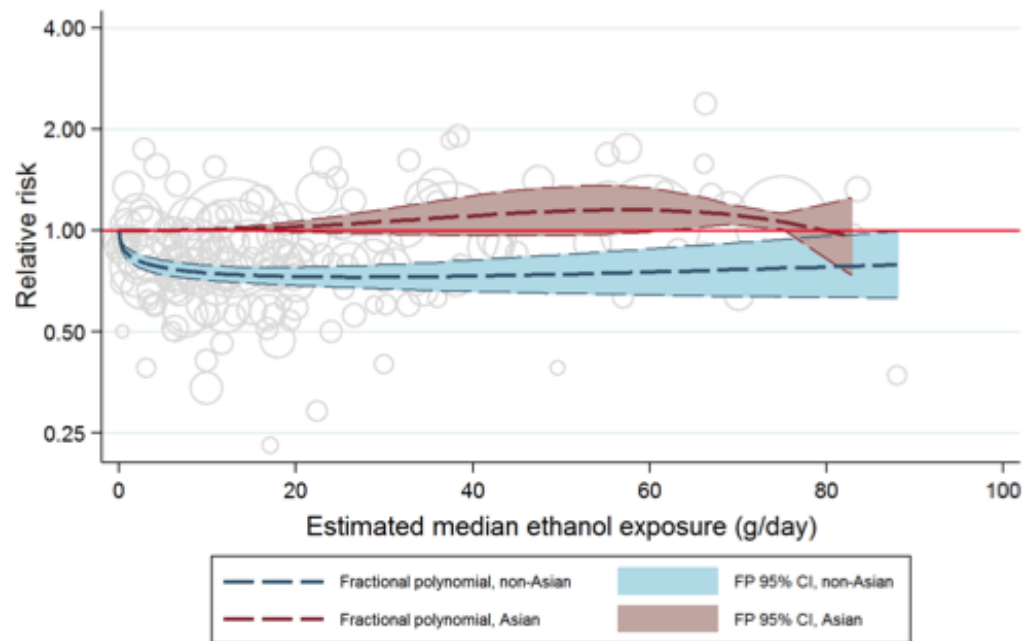
APPENDIX 8.5: Dose-response relationship between average daily alcohol consumption and incident type 2 diabetes mellitus (T2DM) shows no benefit with moderate alcohol consumption among men, and a higher risk of T2DM as exposure to alcohol increased

Source: Craig Knott, Steven Bell, Annie Britton; *Alcohol Consumption and the Risk of Type 2 Diabetes: A Systematic Review and Dose-Response Meta-analysis of More Than 1.9 Million Individuals From 38*



APPENDIX 8.6: Dose-response relationship between average daily alcohol consumption and incident type 2 diabetes mellitus (T2DM) shows that no benefit with moderate alcohol consumption at all levels among Asians, and continued higher risk for T2DM at all levels of alcohol consumption.

Source: Craig Knott, Steven Bell, Annie Britton; *Alcohol Consumption and the Risk of Type 2 Diabetes: A Systematic Review and Dose-Response Meta-analysis of More Than 1.9 Million Individuals From 38 Observational Studies*. *Diabetes Care* 1 September 2015; 38 (9): 1804–1812. <https://doi.org/10.2337/dc15-0710>



Q1.7. GRADE Evidence Profile

GRADE: Meta-analysis of Lifestyle Changes for Prevention of Type 2 DM

Author(s): Uusitupa M, Khan TA, Vigiouk E, Kahleova H, Rivellese AA, Hermansen K, Pfeiffer A, Thanopoulou A, Salas-Salvadó J, Schwab U, Sievenpiper JL. Prevention of Type 2 Diabetes by Lifestyle Changes: A Systematic Review and Meta-Analysis. *Nutrients*. 2019 Nov 1;11(11):2611. doi: 10.3390/nu11112611. PMID: 31683759; PMCID: PMC6893436.

Question: Should lifestyle changes be used for prevention of diabetes?

No of studies	Certainty assessment						Anticipated absolute effects* (95% CI)			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Risk with [comparison]	Risk with lifestyle changes	RR (95% CI)		
Outcome: new onset diabetes, follow up: 6-10 years											
7	RCTs (n=4088)	not serious	not serious	not serious	not serious	none	281 per 1,000	149 per 1,000 (115 to 188)	RR 0.53 (0.41 to 0.67)	⊕⊕⊕⊕ High	CRITICAL

*The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; RR: risk ratio

Question: Course of short or prolonged sleep duration in development of type 2 DM over time

Setting: various countries

Bibliography: ¹

Author(s): Andag-Silva

№ of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	№ of events	№ of individuals	Rate (95% CI)		
short sleep duration and development of type 2 DM (follow-up: range 1 years to 12 years; assessed with: FPG, OGTT, self report)											
17	non-randomised studies	not serious	not serious	not serious	not serious	all plausible residual confounding would reduce the demonstrated effect	10380	219678	event rate 1.2% (1.15 to 1.29)	⊕⊕ ⊕⊕ High	CRITICAL
long sleep duration and development of type 2 DM (follow-up: range 1 years to 12 years; assessed with: FPR, OGTT, self report)											
17	non-randomised studies	not serious	not serious	not serious	not serious	all plausible residual confounding would reduce the demonstrated effect	2270	21652	event rate 1.3% (1.15 to 1.39)	⊕⊕ ⊕⊕ High	CRITICAL

References

1. Huapeng Lu, Qinling Yang, Fang Tian, Yi Lyu, Hairong He, Xia Xin, Xuemei Zheng.. A Meta-Analysis of a Cohort Study on the Association between Sleep Duration and Type 2 Diabetes Mellitus.. *Hindawi Journal of Diabetes Research*; 2021.

APPENDIX 7.3: Alcohol intake and Type 2 DM risk

Question: What is the association between alcohol intake and the risk of Type 2 Diabetes Mellitus

Bibliography: Craig Knott, Steven Bell, Annie Britton; Alcohol Consumption and the Risk of Type 2 Diabetes: A Systematic Review and Dose-Response Meta-analysis of More Than 1.9 Million Individuals From 38 Observational Studies. Diabetes Care 1 September 2015; 38 (9): 1804–1812. <https://doi.org/10.2337/dc15-0710>

№ of studies	Certainty assessment						Impact	Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations			
Type 2 Diabetes									
38 (n=1,902,605)	observational studies	serious ^a	serious ^b	not serious	not serious	publication bias strongly suspected dose response gradient ^c	A reduction in the risk of Type 2 Diabetes appeared in less than 63 g/day levels of alcohol consumption (peak reduction at 10-14 g/day), with risks increasing above the threshold. Subanalyses shows that these reductions are confined to women and non asian populations	⊕⊕⊕○ Moderate	CRITICAL

Explanations

a. possible reporting bias due to self-reported alcohol consumption data; Confounders may be present

b. Considerable heterogeneity I² of 75%

c. analyses hinted at the possibility that existing studies overestimated the degree to which the risk of T2DM is reduced among moderate consumers of alcohol; noted funnel plot asymmetry among female data points

APPENDIX 7.4: GRADE evidence profile for smoking and the association of T2DM

Question: What is the risk of T2DM among smokers ?

Bibliography: Pan A, Wang Y, Talaei M, Hu FB, Wu T. Relation of active, passive, and quitting smoking with incident type 2 diabetes: a systematic review and meta-analysis. Lancet Diabetes Endocrinol. 2015 Dec;3(12):958-67. doi: 10.1016/S2213-8587(15)00316-2. Epub 2015 Sep 18. PMID: 26388413; PMCID: PMC4656094.

№ of studies	Certainty assessment						Effect			Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	№ of events	№ of individuals	RR (95% CI)		
Type 2 Diabetes and active smoking (follow-up: range 5 years to 30 years)											
84	observational studies	not serious	serious ^b	not serious	not serious	publication bias suspected very strong association dose response gradient ^c	294081	5853952	RR= 1.37 (95% CI, 1.33-1.42)	⊕⊕⊕⊕ High	CRITICAL

Explanations

a. substantial heterogeneity detected

b. publication bias- asymmetrical funnel plot

c. risk of reporting bias

APPENDIX 7.5: Smoking Cessation Weight Gain and Type 2 DM

Question: Is post-cessation weight gain associated with the development of type 2 diabetes mellitus?

Bibliography:

Choi JW, Kim TH, Han E. Smoking Cessation, Weight Change, Diabetes, and Hypertension in Korean Adults. Am J Prev Med. 2021 Feb;60(2):205-212. doi: 10.1016/j.amepre.2020.08.024. Epub 2020 Nov 3. PMID: 33153837.

Wu L, Wang X, Dong JY, Zhao YT, Lou H. Smoking Cessation, Weight Gain, and Risk for Type 2 Diabetes: A Prospective Study. Int J Public Health. 2022 Apr 14;67:1604654. doi: 10.3389/ijph.2022.1604654. PMID: 35496941; PMCID: PMC9046538.

№ of studies	Certainty assessment						Effect	Certainty	Importance
	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Rate (95% CI)		
Type 2 Diabetes Mellitus (follow-up: median 6.9 years) n= 96,524									
1	observational studies	very serious ^{a,b}	not serious	not serious ^c	not serious	all plausible residual confounding would suggest spurious effect, while no effect was observed	0.86 (95% CI=0.80, 0.93) in recent quitters with no weight gain 0.94 (95% CI=0.87, 1.03) in those with 0.1–5 kg weight gain 0.93 (95% CI=0.73, 1.19) in those with 5.1–10 kg weight gain 1.49 (95% CI=0.84, 2.62) in those with a weight gain of >10 kg	⊕⊕⊕○ Moderate	CRITICAL
Type 2 Diabetes Mellitus (follow up: 2 years) n=8,951									
1	observational studies	very serious ^d	not serious	not serious ^e	serious ^f	all plausible residual confounding would reduce the demonstrated effect	2.25 (1.02,4.95) T2DM and weight gain more than 2 kg 1.4 (0.86, 2.26) T2Dm and weight gain <2 kg	⊕⊕○○ Low	CRITICAL

Explanations

- Sampling bias- population is exclusively from national korean health screening hence might have been more health conscious than those not having health examinations
- reporting bias, recall bias due to self-reported questionnaire
- Samples only included korean adults.
- Recall bias due to self reported questionnaire; reverse causation (participants who quite smoking were more likely to have clinical symptoms)
- Samples only included Chinese population
- wide confidence interval

Q1.8. Recommendations from other groups

Recommendations from other guidelines:

A review of the guidelines acknowledges the importance of lifestyle management in people at high risk for developing Type 2 Diabetes.

GROUP or AGENCY	RECOMMENDATIONS	STRENGTH OF RECOMMENDATION/ CERTAINTY/QUALITY OF EVIDENCE
American Diabetes Association Professional Practice Committee Standards of Medical Care in Diabetes-2024. ²⁸	The strongest evidence supporting lifestyle intervention for diabetes prevention in the U.S. comes from the Diabetes Prevention Program (DPP) trial. It is recommended that adults with overweight/obesity at high risk of type 2 diabetes should be referred to an intensive lifestyle behavior change program consistent with the DPP to achieve and maintain 7% loss of initial body weight and increase moderate-intensity physical activity (such as brisk walking) to at least 150 minutes/week. A variety of eating patterns can be considered to prevent diabetes in individuals with prediabetes.	Strong Level A evidence
U.K. NICE Public health guideline: Type 2 diabetes: prevention in people at high risk ²⁹	A health economic model showed that lifestyle-change programmes are cost effective for all people at high risk of diabetes, providing support to the existing recommendation to offer the intervention to this group of people. The model also showed that lifestyle-change programmes are particularly cost effective for people with higher HbA1c or fasting plasma glucose levels. Intensive lifestyle-change programmes should offer ongoing tailored advice, support and encouragement to help people: • undertake at least a level of physical activity that is in line with UK government recommendations (moderate to vigorous) • gradually lose 5 to 10% baseline weight in 1 year to reach and maintain a BMI within the healthy range • increase their consumption of wholegrains, vegetables and other foods that are high in dietary fibre • reduce the total amount of fat in their diet • eat less saturated fat; processed meat	Strong
Diabetes Canada	If you have prediabetes, healthy behavior changes that result in a loss of 5% of your initial	Strong

B. Prevention (Pharmacologic Therapy)

Guideline Question 2.

Should we recommend pharmacologic therapy (metformin, pioglitazone, acarbose, orlistat, SGLT-2 inhibitors, GLP-1 RAs) among adults with pre-diabetes?

Q2.1. Methods

The evidence review was conducted from 2021 to January 2024 and was subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through September 2025. The initial phase involved reviewing the recommendations and all references cited in the most recent versions of the guidelines selected by the Technical Review Committee, following the ADAPTE methodology. Full-text articles for these references were retrieved, and when citation links or database suggestions identified similar or more updated studies relevant to the topic, these were also evaluated and incorporated.

Three reviewers independently performed a comprehensive systematic search of related literature using MEDLINE via PubMed, OpenAthens, CENTRAL, and the Cochrane Database of Systematic Reviews. Additional freehand searches were conducted through Google, and the reference lists of all relevant articles were screened to identify further sources of evidence.

Search terms included: type 2 diabetes mellitus prevention, pharmacotherapy (with specific medications detailed in the respective sections), and related concepts. Because the Technical Review Committee prioritized hard clinical endpoints, the reviewers also searched for meta-analyses, systematic reviews, and individual studies assessing the impact of each pharmacologic intervention on diabetes prevention and on major comorbidities such as obesity and metabolic dysfunction—associated steatotic liver disease (formerly non-alcoholic fatty liver disease). Critical clinical outcomes—including cardiovascular events, heart failure, stroke, chronic kidney disease, and mortality—were specifically sought to ensure comprehensiveness and relevance to clinical decision-making.

Q2.2. Search strategy

Search strategy including dates and data base

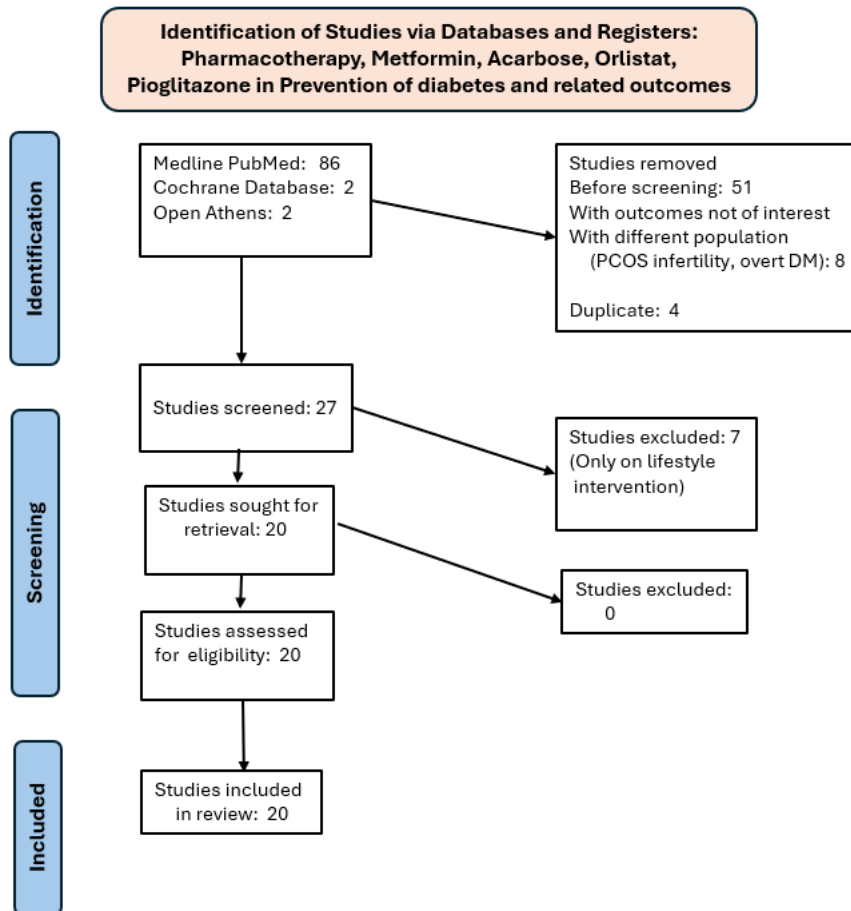
DATABASE	SEARCH STRATEGY / SEARCH TERMS	DATE AND TIME OF SEARCH	RESULTS	
			Yield	Eligible
Medline via PubMed	("diabetes prevention") [Title/ Abstract] AND ("meta-analysis") [Title/ Abstract] AND ("long-term") [Title/ Abstract] , OR ("metformin") [Title] AND "2013/01/01" to "2023/31/05" [Date-publication]	March 2023	9	2
Medline via PubMed	("US Diabetes Prevention Program") [Title] OR ("Diabetes Prevention Program Outcomes Studies") [Title] AND ("2002" [date-publication] to ("2023") Date-publication]	March 2023	62	9
Cochrane Database of Systematic Reviews	("insulin resistance") [Population] OR ("Impaired fasting glucose") [Population] OR ("Impaired Glucose tolerance") [Population] AND ("Metformin") [Intervention]	Nov 2023	1	1
Medline via PubMed Open Athens	"prediabetes", "Acarbose", "impaired glucose tolerance", "diabetes prevention", "meta-analysis" and "randomized controlled trials"	August 2023	2	2
Cochrane Database of Systematic Reviews	("insulin resistance") [Population] OR ("Impaired fasting glucose") [Population] OR ("Impaired Glucose tolerance") [Population] AND ("Pioglitazone") [Intervention]	May 2023	1	1
Medline via PubMed	("pioglitazone") [Title/Abstract] AND ("meta-analysis") [Title/Abstract], AND ("insulin resistance") [title/ abstract] AND ("2000/01/01" to 2023/04/31" [date publication]	May 2023	12	4
Medline via PubMed Open Athens	"prediabetes", "Orlistat", "diabetes prevention", "impaired glucose tolerance", "randomized controlled trial" and "meta-analysis"	May 2023	1	1
Medline via PubMed CENTRAL	("glucagon-like peptide-1 receptor agonist"[Title/Abstract] OR "liraglutide"[Title/Abstract] OR "dulaglutide"[Title/Abstract] OR "exenatide"[Title/Abstract] OR "lixisenatide"[Title/Abstract]) OR "semaglutide"[Title/Abstract] AND ("prediabetes"[Title/Abstract] OR "prediabetic	October 2023	105	8

Google Scholar	state"[MeSH Terms]) AND (("randomized controlled trial"[Publication Type] OR "controlled clinical trial"[Publication Type] OR "randomized"[Title/Abstract] OR "placebo"[Title/Abstract] OR "drug therapy"[MeSH Subheading] OR "randomly"[Title/Abstract] OR "trial"[Title/Abstract] OR "groups"[Title/Abstract]) NOT ("animals"[MeSH Terms] NOT "humans"[MeSH Terms]))			
Medline via PubMed	2 ("sodium glucose cotransporter inhibitors"[Title/Abstract] OR "dapagliflozin"[Title/Abstract] OR "empagliflozin"[Title/Abstract] OR "canagliflozin"[Title/Abstract] OR "ertugliflozin"[Title/Abstract]) AND ("prediabetes"[Title/Abstract] OR "prediabetic state"[MeSH Terms]) AND (("randomized controlled trial"[Publication Type] OR "controlled clinical trial"[Publication Type] OR "randomized"[Title/Abstract] OR "placebo"[Title/Abstract] OR "drug therapy"[MeSH Subheading] OR "randomly"[Title/Abstract] OR "trial"[Title/Abstract] OR "groups"[Title/Abstract]) NOT ("animals"[MeSH Terms] NOT "humans"[MeSH Terms])).	October 2023	41	4
CENTRAL				
Google Scholar				

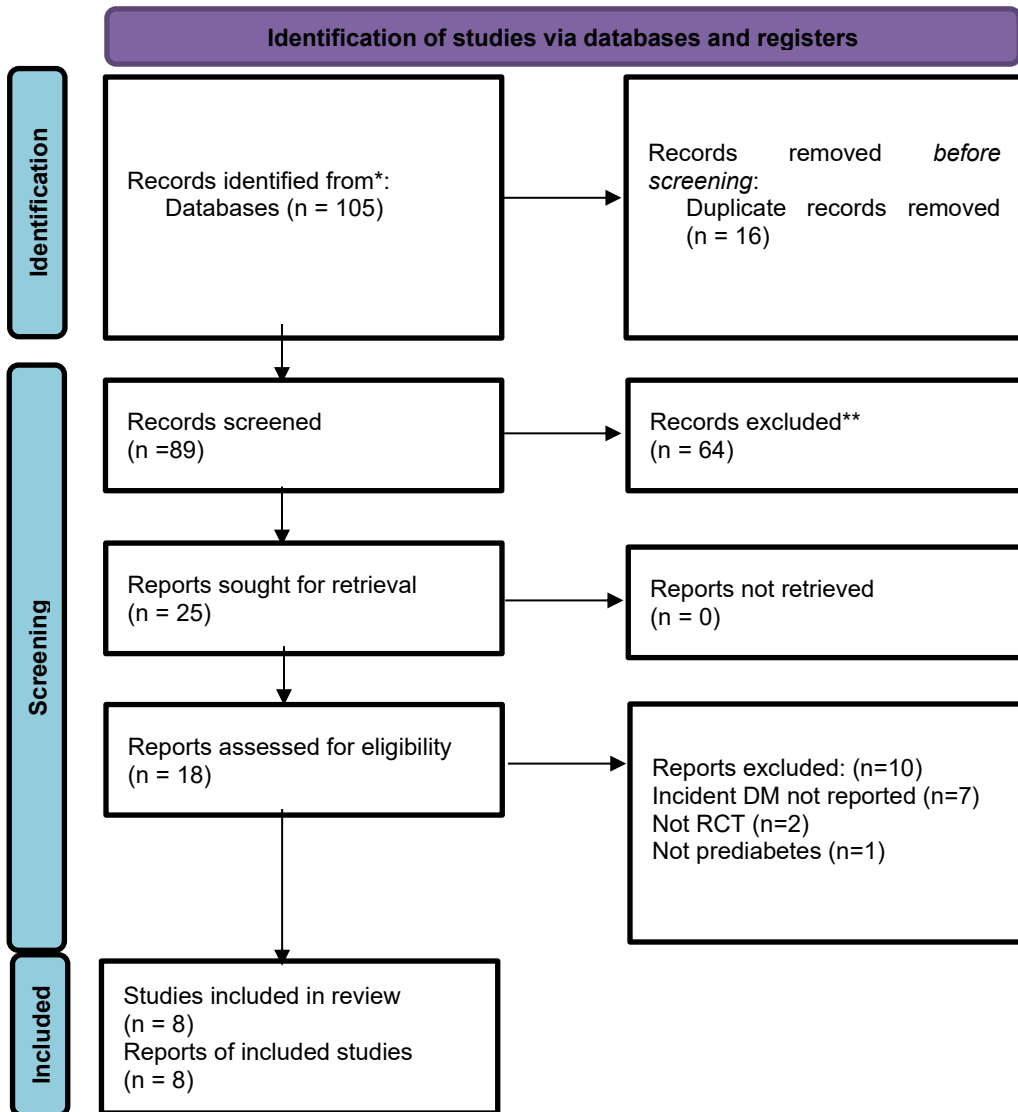
Q1.3. PRISMA flow diagram

PRISMA flow diagram

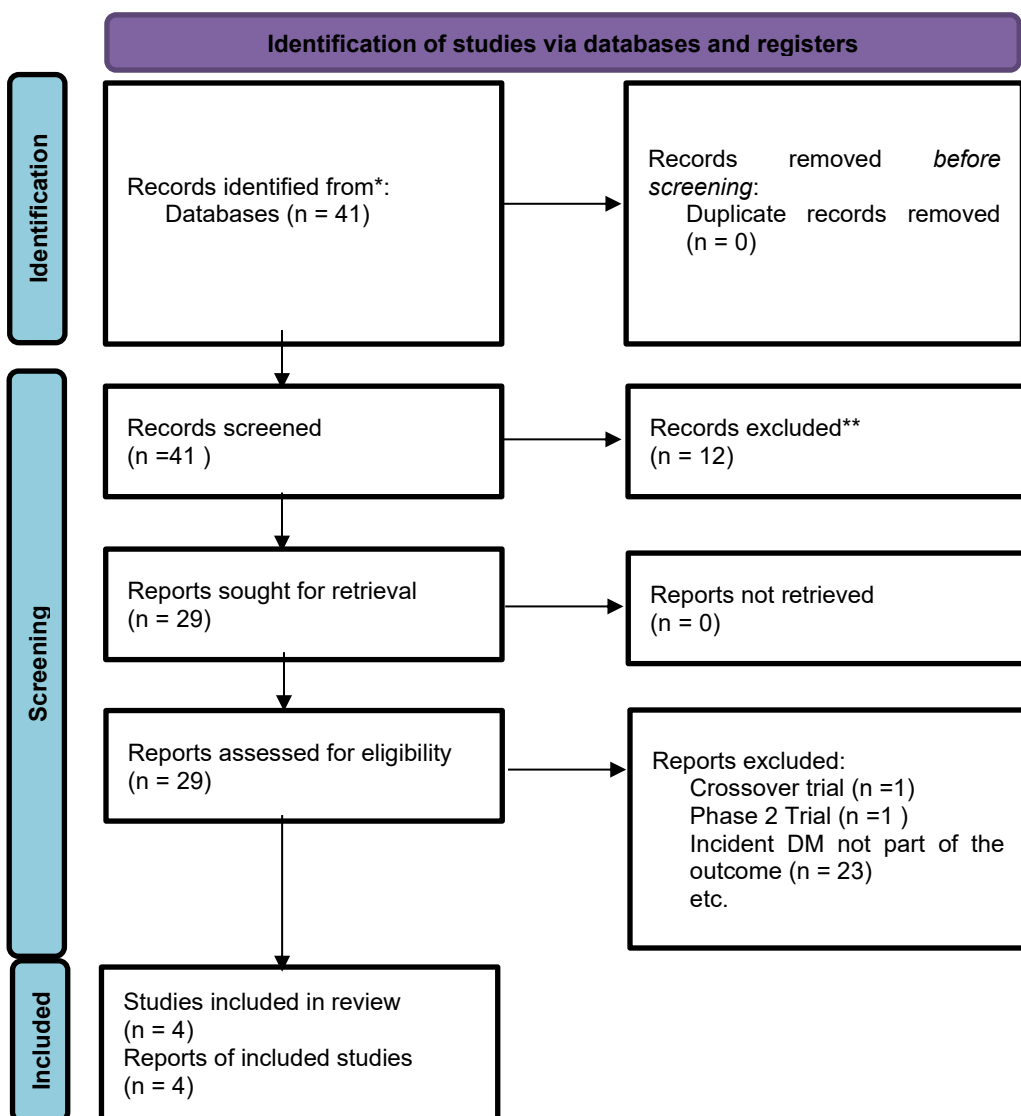
APPENDIX 3.1: Search for Long term effects of Traditional Pharmacotherapy (metformin, acarbose, pioglitazone, orlistat) for Prevention of Type 2 Diabetes and Outcomes of interest



APPENDIX 3.2: GLP-1 RA for Prevention of Type 2 Diabetes



APPENDIX 3.2: SGLT-2 inhibitors for Prevention of Type 2 Diabetes and other outcomes of interest



Q1.4. Characteristics of the Included Studies

Table of the Characteristics of the Studies Included

Summary results on the meta-analysis and randomized controlled trials of metformin, acarbose, pioglitazone and orlistat on type 2 diabetes prevention in people with impaired glucose tolerance or in people at high increased risk for diabetes.

Studies	N, characteristics	Study Duration	Risk Reduction of T2DM with Drug versus Control	Benefit groups/ Advantages	Precautions/ Disadvantages
Haw JS, Galaviz KI, Straus AN, Kowalski AJ, Magee MJ, Weber MB, Wei J, Narayan KMV, Ali MK. Long term Sustainability of Diabetes Prevention Approaches. JAMA Intern Med. 2017 Dec; 177(12): 1808–1817. doi: 10.1001/jamainternmed.2017.6040	Meta-analysis of Metformin, Rosiglitazone, AGI, RAS blocker glipizide, insulin	Mean follow-up 3.1 years	36% Relative risk reduction [RR, 0.64; 95% CI, 0.54-0.76]	Drugs were effective only while being taken; subjects with weight loss responded best	Drug molecule specific established side effects
Barry E, Roberts S, Oke J, Vijayaraghavan S, Normansell R, Greenhalgh T. Efficacy and effectiveness of screen and treat policies in prevention of type 2 diabetes: systematic review and meta-analysis of screening tests and interventions. BMJ 2017;356	Meta-analysis of 4 RCTs Metformin (US, Asian studies)	1 to 3 years	26% risk reduction P < 0.001	Women with prior history of GDM, obese, age less than 60,	GI side effects
Madsen KS, Chi Y, Metzendorf MI, Richter B, Hemmingsen B. Metformin for prevention or delay of type 2 diabetes mellitus and its associated complications in persons at increased risk for the development of type 2 diabetes mellitus. Cochrane Database of Systematic Reviews 2019, Issue 12: 101 – 106 Art. No.: CD008558. DOI:10.1002/14651858.CD008558.pub2.	Meta-analysis of 20 trials (12 RCT's published in Chinese)	1 to 5 years	50% [HR:0.50 95% CI (0.38, 0.65)] in DM progression (moderate-quality evidence)	Effect was independent of the duration of study, ethnicity, & treatment blinding	Sparse reporting of side effects
US Diabetes Prevention Program Study Group: Nathan DM, Barrett-Connor E, et al, Tempresa M. Long-term Effects of Lifestyle Intervention or Metformin on Diabetes Development and Microvascular Complications: the DPP Outcomes Study. Lancet Diabetes Endocrinol. 2015 Nov; 3(11): 866–875.	RCT 2002 with long term follow outcomes study 2013 to 2021	22 years	18% risk reduction P= 0.001	Women with prior history of GDM, obese, age less than 60, FPG 110 mg/dl and higher, A1c 6-6.4%	GI side effects
Coleman R, Scott C, et al. Meta-analysis of the impact of alpha-glucosidase inhibitors on incident diabetes and cardiovascular outcomes. Cardiovasc Diabetol 2019; 18:1-7.	Meta-analysis of 3 RCTS	1 to 3.3 years	23% p < 0.0001	Patients with IGT	GI side effects, 3x a day dosing

Ipsen EØ, Madsen KS, Chi Y, Pedersen-Bjergaard U, Richter B, Metzendorf MI, Hemmingsen B. Pioglitazone for prevention or delay of type 2 diabetes mellitus and its associated complications in people at risk for the development of type 2 diabetes mellitus (Review). Cochrane Database of systematic Reviews 2020, Issue 11. Art. No.: CD013516.	Meta-analysis of 6 RCTs versus placebo 16 RCT's versus no intervention	44 weeks to 2.2 years 24 weeks to 2 years	60% reduction P= 0.04 69% reduction P<0.001	Patients with Pre-DM various ethnicities	Edema, weight gain, fracture Overall sparse data on serious side effects
Inzucchi SE, Viscoli CM, Young LH, Furie KL, Gorman M, Lovejoy AM, Dagogo-Jack S, et al Pioglitazone Prevents Diabetes in Patients with Insulin Resistance and Cerebrovascular Disease Diabetes Care 2016;39:1684–1692	3876 adults high HOMA-IR scores RCT	4.8 years	52% reduction P< 0.0001	Patients with pre-diabetes and prior history of ischemic stroke or TIA	Edema, weight gain, fractures
Kernan WN, Viscoli CM, Furie KL, Young LH, Inzucchi SE, et al. Pioglitazone after Ischemic Stroke or Transient Ischemic Attack. N Engl J Med 2016; 374:1321-31.	3876 adults high HOMA-IR scores RCT RCT	4.8 years	CV outcome: 24% reduction in ischemic stroke and MI	Patients with insulin resistance and prior history of ischemic stroke or TIA	Edema, weight gain, fractures
Mantovani A, et al. Efficacy and safety of anti-hyperglycaemic drugs in patients with non-alcoholic fatty liver disease with or without diabetes: An updated systematic review of randomized controlled trials. Diabetes Metab (2020)	Systematic review of 6 RCT's	4 months to 3 years	Metabolic outcome: Improvement in liver enzymes, risk scores, fibrosis activity and histology	Patients with pre-diabetes and metabolic associated steatohepatitis	Edema, weight gain
Zhao Y, Zhao W, Wang H, Zhao Y, Bu H, Takahashi H. Pioglitazone on nonalcoholic steatohepatitis: A systematic review and metaanalysis of 15 RCTs. Medicine 2022;101:46(e31508). http://dx.doi.org/10.1097/MD.00000000000031508	Meta-analysis 15 RCTS (USA and Asian studies)	3 to 24 months	Metabolic outcomes, liver enzymes, fibrosis and histology Overall efficacy 78% p<0.00003	Patients with MASH including pre-diabetes	Weight gain
Heymsfield S, Segal K, et al. Effects of Weight Loss with Orlistat on Glucose Tolerance and Progression to Type 2 Diabetes in Obese Adults. Arch Intern Med 2000; 160:1321-1326	Pooled analysis of patients in 3 RCTS	19 months	71.6% had normalization of blood sugar vs 41% in placebo p = 0.04	Patients with obesity and IGT	GI side effects

Torgerson J, Boldren M, et al. XENical in the Prevention of Diabetes in Obese Subjects (XENDOS) Study. Diabetes Care, 2004;27(1): 157.	RCT	4 years	37.3% reduction P=0.0032	Patients with obesity and IGT	GI side effects
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Studies on GLP-Ra based incretin therapies with prevention of type 2 diabetes in people with impaired glucose tolerance or in people at high risk of diabetes

Author	N	Study Design	Population	Study Duration	Experimental Group	Control Group
Le Roux 2017	2254	RCT	Overweight/Obese Prediabetes	160 weeks	Liraglutide	Placebo
Kim 2013A	68	RCT	Overweight/Obese Prediabetes	14 weeks	Liraglutide	Placebo
Santilli 2017	62	RCT	Overweight/Obese Prediabetes	16 weeks	Liraglutide	Lifestyle counselling
Kim 2013B	49	RCT	Overweight/Obese Prediabetes	14 weeks	Liraglutide	Placebo
Larsen 2017	97	RCT	Overweight/Obese Prediabetes	16 weeks	Liraglutide	Placebo
Rosenstock 2010	152	RCT	Overweight/Obese Prediabetes	24 weeks	Exenatide	Placebo
Perreault 2022	1536	RCT	Overweight/Obese Prediabetes	68 weeks	Semaglutide	Placebo
Wilding 2015	3731	RCT	Overweight/Obese Prediabetes	56 weeks	Liraglutide	Placebo

Studies on SGLT-2 inhibitor-based therapies with prevention of type 2 diabetes in people with impaired glucose tolerance or in people at high risk of diabetes

Author	N preDM/total	Study Design	Population	Median Follow-up months	Experimental Group	Control Group
DAPA-HF	1748/4744	RCT	Age $\geq$ 18 y, LVEF $\leq$ 40%, NYHA class II, III, or IV Receiving standard heart failure therapy NT-proBNP $\geq$ 600 pg ($\geq$ 400 pg if heart failure hospitalization within 12 mo, or $\geq$ 900 pg if patients had atrial fibrillation or atrial flutter)	18.2	Dapagliflozin 10mg	Placebo
DAPA-CKD	660/4304	RCT	Age $\geq$ 18 y, eGFR of 25-75 mL/min per 1.73m ² U-alb/U-Cre of 200-5000 mg/g Stable dose of ACE inhibitor or ARB for at least 4 wk before screening, if tolerated	28.8	Dapagliflozin 10mg	Placebo
EMPEROR-Reduced	1268/3730	RCT	Age $\geq$ 18 y, LVEF $\leq$ 40%, NYHA class II, III, or IV Receiving standard heart failure therapy NT-proBNP $>$ 2500 pg for LVEF 36-40%, $>$ 1000 pg for LVEF 31-35%, or $>$ 600 pg for LVEF $\leq$ 30% (those thresholds were doubled if patients had atrial fibrillation)	16	Empagliflozin 10mg	Placebo
EMPEROR-Preserved	1979/5988	RCT	Age $\geq$ 18 y, LVEF $\leq$ 40%, NYHA class II, III, or IV Left atrial enlargement, left ventricular hypertrophy, or heart failure hospitalization within 12 mo NT-proBNP level $>$ 300 pg ($>$ 900 pg for patients with atrial fibrillation)	26.2	Empagliflozin 10mg	Placebo

Q1.5. Assessment of Risk of Bias

Assessment of Risk of Bias of Included Studies

Risk of bias in 2017 Meta-analysis of Metformin studies: (and Forest plot)

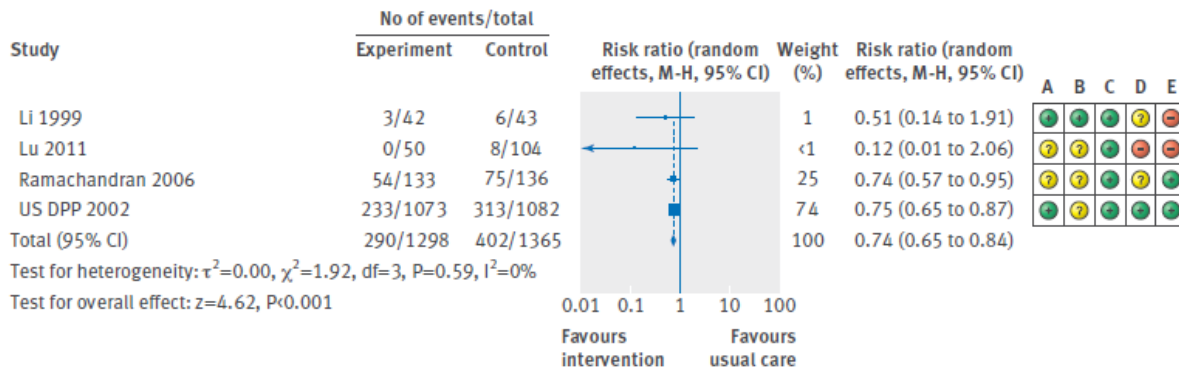


Fig 8 | Relative reduction in risk of diabetes at end of metformin trial. A=random sequence generation (selection bias); **B=**allocation concealment (selection bias); **C=**blinding of outcome assessment (detection bias); **D=**incomplete outcome data (attrition bias); **E=**selective reporting (reporting bias)

Barry E, Roberts S, Oke J, Vijayaraghavan S, Normansell R, Greenhalgh T. Efficacy and effectiveness of screen and treat policies in prevention of type 2 diabetes: systematic review and meta-analysis of screening tests and interventions. *BMJ* 2017;356: i6538 <http://dx.doi.org/10.1136/bmj.i6538>.

Risk of bias in 2019 Cochrane Review of Metformin studies:

Figure 2. 'Risk of bias' graph: review authors' judgements about each risk of bias item presented as percentages across all included studies (blank cells indicate that the particular outcome was not measured in some trials).



Madsen KS, Chi Y, Metzendorf MI, Richter B, Hemmingsen B. Metformin for prevention or delay of type 2 diabetes mellitus and its associated complications in persons at increased risk for the development of type 2 diabetes mellitus. Cochrane Database of Systematic Reviews 2019, Issue 12: 101 – 106 Art. No.: CD008558. DOI:10.1002/14651858.CD008558.pub2

Figure 3. 'Risk of bias' summary: review authors' judgements about each risk of bias item for each included study (blank cells indicate that the trial did not measure that particular outcome).

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): all-cause mortality/cardiovascular mortality	Blinding of participants and personnel (performance bias): amputation, blindness/severe vision loss, end-stage renal disease	Blinding of participants and personnel (performance bias): hypoglycaemia	Blinding of participants and personnel (performance bias): incidence of T2DM	Blinding of participants and personnel (performance bias): measures of blood glucose control	Blinding of participants and personnel (performance bias): serious adverse events	Blinding of participants and personnel (performance bias): time to progression to T2DM	Blinding of participants and personnel (performance bias): socioeconomic effects	Blinding of participants and personnel (performance bias): health-related quality of life	Blinding of participants and personnel (performance bias): non-fatal myocardial infarction/stroke	Blinding of participants and personnel (performance bias): non-serious adverse events	Blinding of outcome assessment (detection bias): all-cause mortality/cardiovascular mortality	Blinding of outcome assessment (detection bias): amputation, blindness/severe vision loss, end-stage renal disease	Blinding of outcome assessment (detection bias): hypoglycaemia	Blinding of outcome assessment (detection bias): incidence of T2DM	Blinding of outcome assessment (detection bias): measures of blood glucose control	Blinding of outcome assessment (detection bias): non-fatal myocardial infarction/stroke	Blinding of outcome assessment (detection bias): non-serious adverse events	Blinding of outcome assessment (detection bias): serious adverse events	Blinding of outcome assessment (detection bias): time to progression to T2DM	Blinding of outcome assessment (detection bias): socioeconomic effects	Blinding of outcome assessment (detection bias): health-related quality of life	Incomplete outcome data (attrition bias): all-cause mortality/cardiovascular mortality	Incomplete outcome data (attrition bias): amputation, blindness/severe vision loss, end-stage renal disease	Incomplete outcome data (attrition bias): hypoglycaemia	Incomplete outcome data (attrition bias): incidence of T2DM	Incomplete outcome data (attrition bias): measures of blood glucose control	Incomplete outcome data (attrition bias): non-fatal myocardial infarction/stroke	Incomplete outcome data (attrition bias): non-serious adverse events	Incomplete outcome data (attrition bias): serious adverse events	Incomplete outcome data (attrition bias): time to progression to T2DM	Incomplete outcome data (attrition bias): socioeconomic effects	Incomplete outcome data (attrition bias): health-related quality of life	Selective reporting (reporting bias)	Other bias				
Attawaz 2018	●	●																																							
BIGPRO-1 2009	●	?												●																											
Chen 2009	?	?																																							
DPP/DPPOS 2002	●	●	●											●	●																										
Fang 2004	●	?	●																																						
IDPP-1 2006	?	?	●																																						
Iqbal Hydrie 2012	?	?	●																																						
Ji 2011	?	?																																							
Jin 2009	?	?																																							
Li 1999	?	?																																							
Li 2009	?	?																																							
Liao 2012	?	?																																							
Lu 2002	?	?	●																																						
Lu 2010	?	?																																							
Maji 2005	?	?																																							
Papoz 1978	?	?																																							
PREVENT-DM 2017	●	●	●																																						
Wang 2009	?	?																																							
Zeng 2013	●	?																																							
Zhao 2013	?	?																																							

Madsen KS, Chi Y, Metzendorf MI, Richter B, Hemmingsen B. Metformin for prevention or delay of type 2 diabetes mellitus and its associated complications in persons at increased risk for the development of type 2 diabetes mellitus. Cochrane Database of Systematic Reviews 2019, Issue 12: 101 – 106 Art. No.: CD008558. DOI:10.1002/14651858.CD008558.pub2

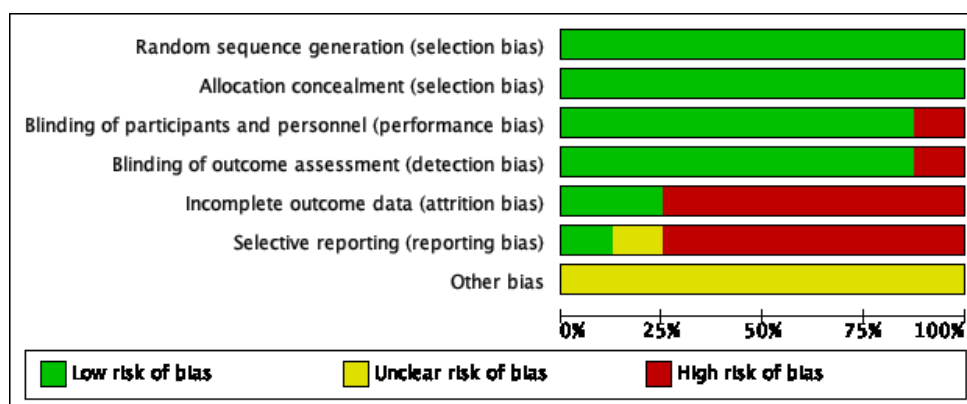
Risk of bias for the 2020 Cochrane Meta-analysis of Pioglitazone in Pre-diabetes

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies (blank cells indicate that the particular outcome was not measured in some studies)



Ipsen EØ, Madsen KS, Chi Y, Pedersen-Bjergaard U, Richter B, Metzendorf MI, Hemmingsen B. Pioglitazone for prevention or delay of type 2 diabetes mellitus and its associated complications in people at risk for the development of type 2 diabetes mellitus (Review). Cochrane Database of systematic Reviews 2020, Issue 11. Art. No.: CD013516. DOI: 10.1002/14651858.CD013516.pub2.

Risk of Bias Assessment for GLP-1 RA based incretin therapies

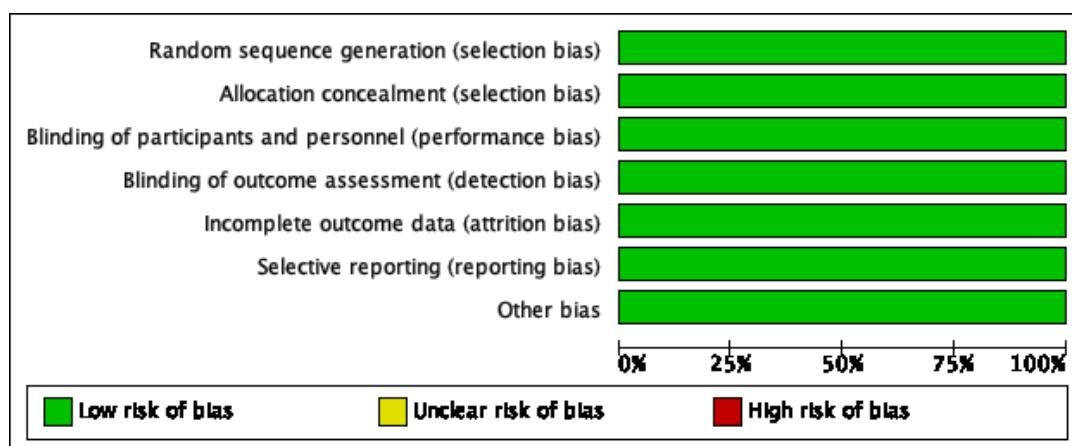


Risk of bias and applicability concerns graph: review authors' judgements about each domain presented as percentages across included studies

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Kim 2013 A	+	+	+	+	-	-	?
Kim 2013 B	+	+	+	+	-	-	?
Larsen 2017	+	+	+	+	+	-	?
Le Roux 2017	+	+	+	+	+	?	?
Perreault 2022	+	+	+	+	-	-	?
Rosenstock 2010	+	+	+	-	-	-	?
Santilli 2017	+	+	-	+	-	-	?
Wilding 2015	+	+	+	+	-	+	?

Risk of bias and applicability concerns summary: review authors' judgements about each domain for each included study

Risk of Bias Assessment for SGLT-2 inhibitor therapies



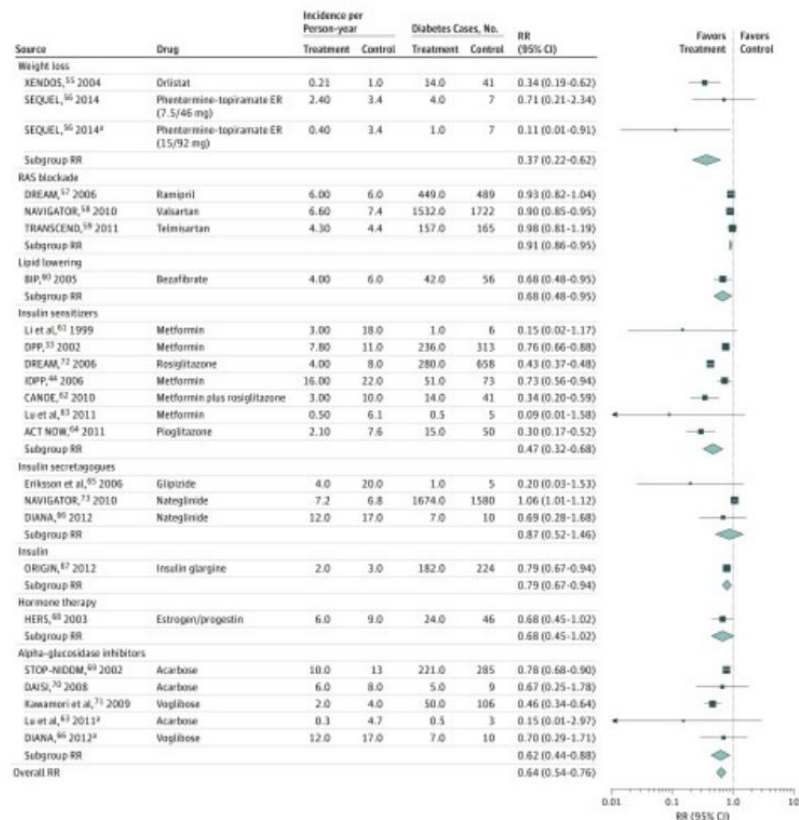
Risk of bias and applicability concerns graph: review authors' judgements about each domain presented as percentages across included studies

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
DAPA-CKD 2020	+	+	+	+	+	+	+
DAPA-HF 2019	+	+	+	+	+	+	+
EMPEROR-Reduced 2020	+	+	+	+	+	+	+
EMPEROR Preserved 2021	+	+	+	+	+	+	+

Risk of bias and applicability concerns summary: review authors' judgements about each domain for each included study

Q2.6. Results

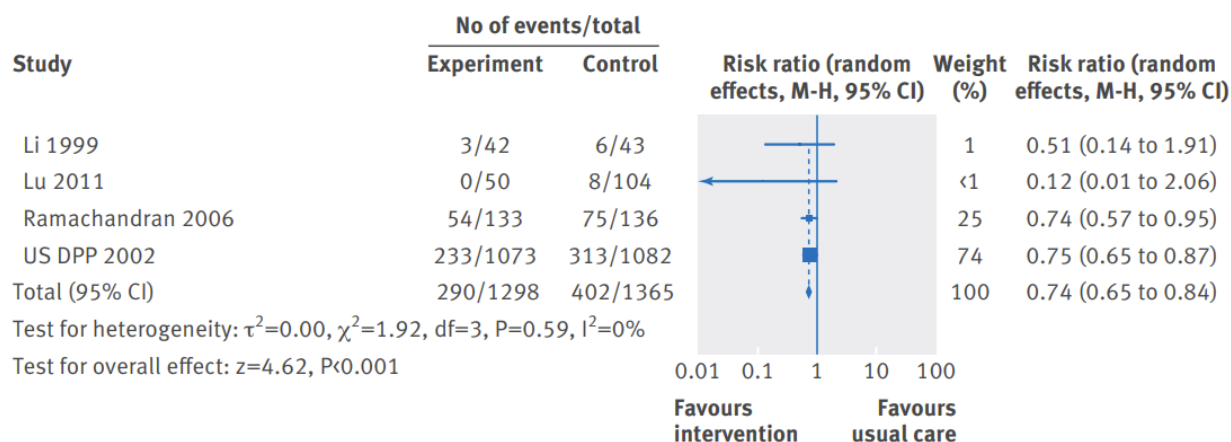
Meta-analysis of sustainability of Pharmacotherapy in Type 2 DM Prevention as of 2017:



Relative Risks (RRs) and Diabetes Incidence Rates Among Medication Studies Stratified by Drug Class at the End of the Active Intervention Period

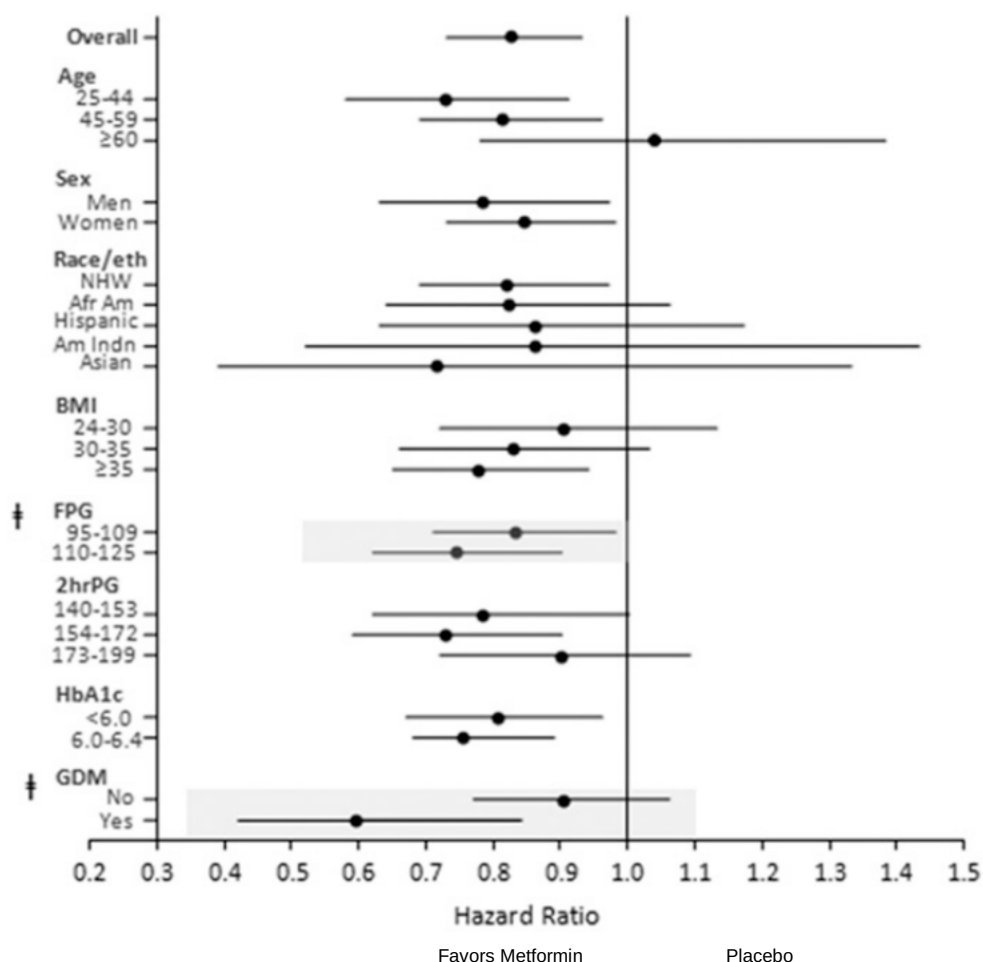
Haw JS, Galaviz KI, Straus AN, Kowalski AJ, Magee MJ, Weber MB, Wei J, Narayan KMV, Ali MK. Long term Sustainability of Diabetes Prevention Approaches. JAMA Intern Med. 2017 Dec; 177(12): 1808–1817. doi: 10.1001/jamainternmed.2017.6040

Metformin versus usual care in diabetes prevention



Barry E, Roberts S, Oke J, Vijayaraghavan S, Normansell R, Greenhalgh T. Efficacy and effectiveness of screen and treat policies in prevention of type 2 diabetes: systematic review and meta-analysis of screening tests and interventions. *BMJ* 2017;356: i6538 <http://dx.doi.org/10.1136/bmj.i6538>.

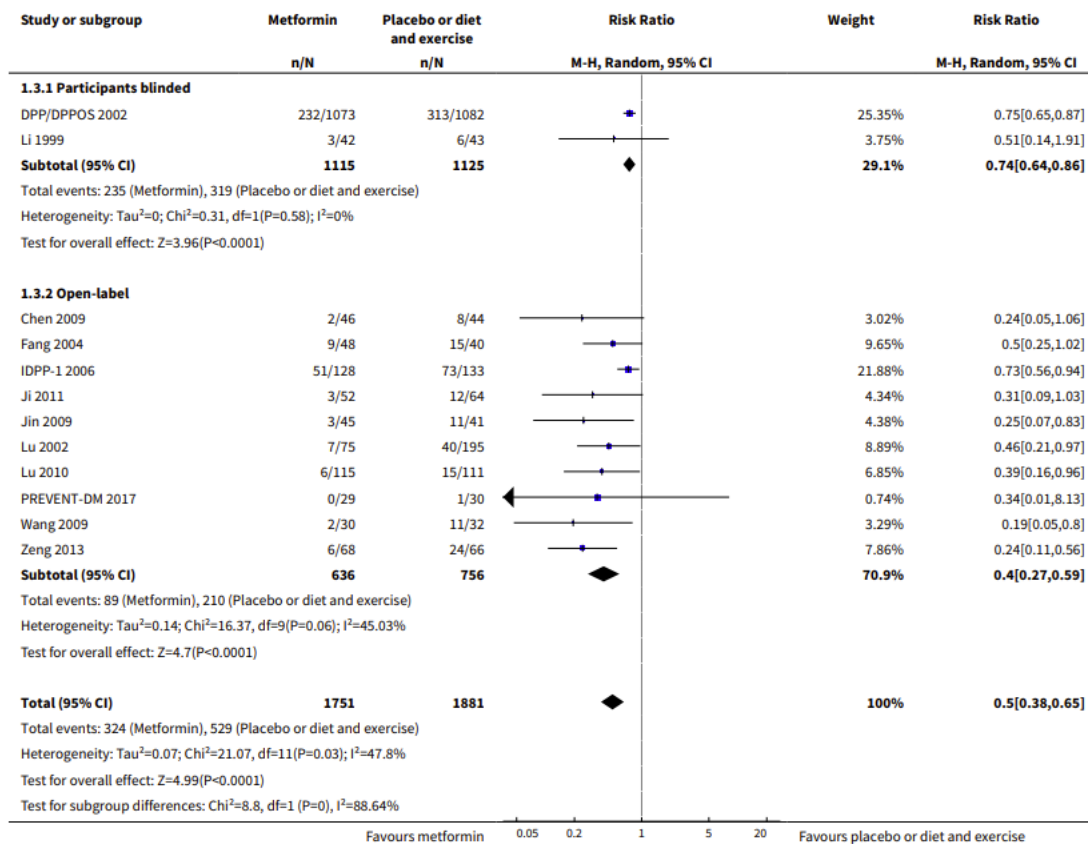
Long term (15 years) follow up Diabetes Prevention Program outcomes Study (DPPOS) showing groups that benefit the most: high BMI, IFG and post GDM



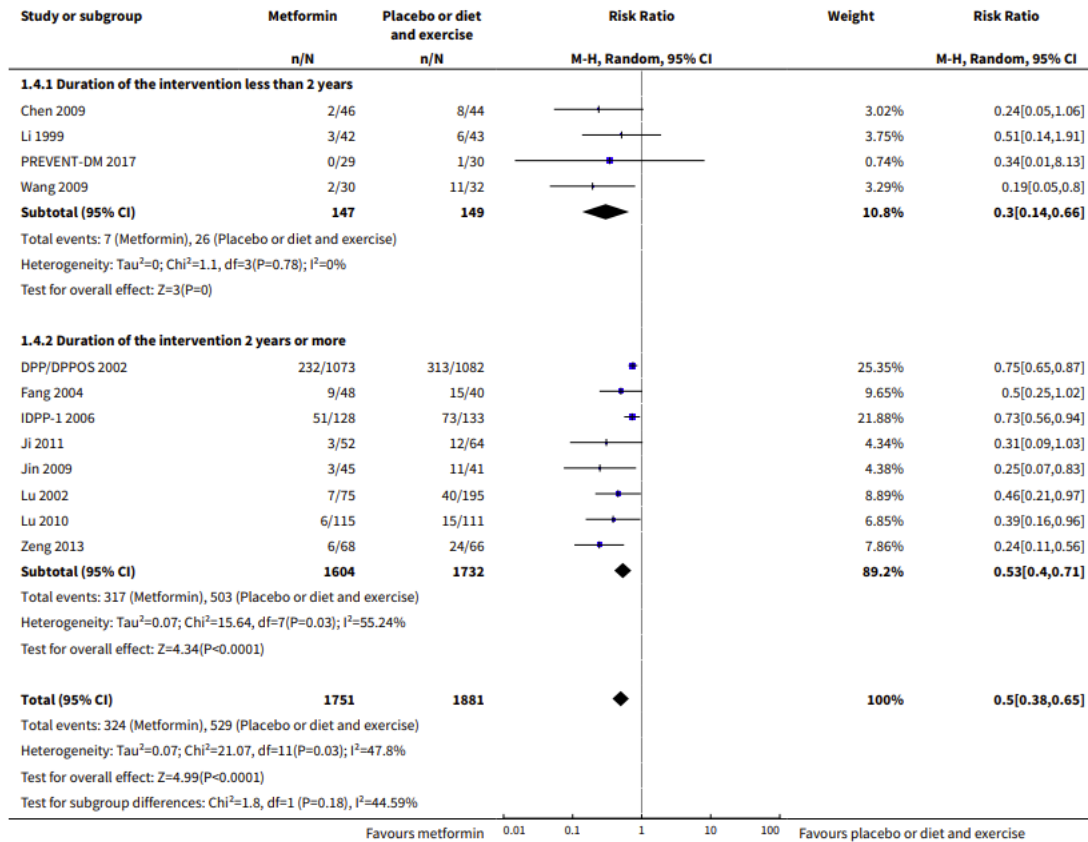
US Diabetes Prevention Program: Nathan DM, Knowler WC, Edelstein SL, Crandall JP, Dabelea D, Goldberg RB, Kahn SE, Mather KJ, Pi-Sunyer X, Trandafirescu G, Walker EA, and Tempresa M. Long-term Effects of Metformin on Diabetes Prevention: Identification of Subgroups That Benefited Most in the DPP and DPPOS. Diabetes Care Publish Ahead of Print published online March 15, 2019. <https://doi.org/10.2337/dc18-1970>

Metformin on Prevention of Type 2 DM (Cochrane review 20 RCTs)

Analysis 1.3. Comparison 1 Metformin versus placebo or diet and exercise, Outcome 3 Incidence of type 2 diabetes (blinded vs open-label).

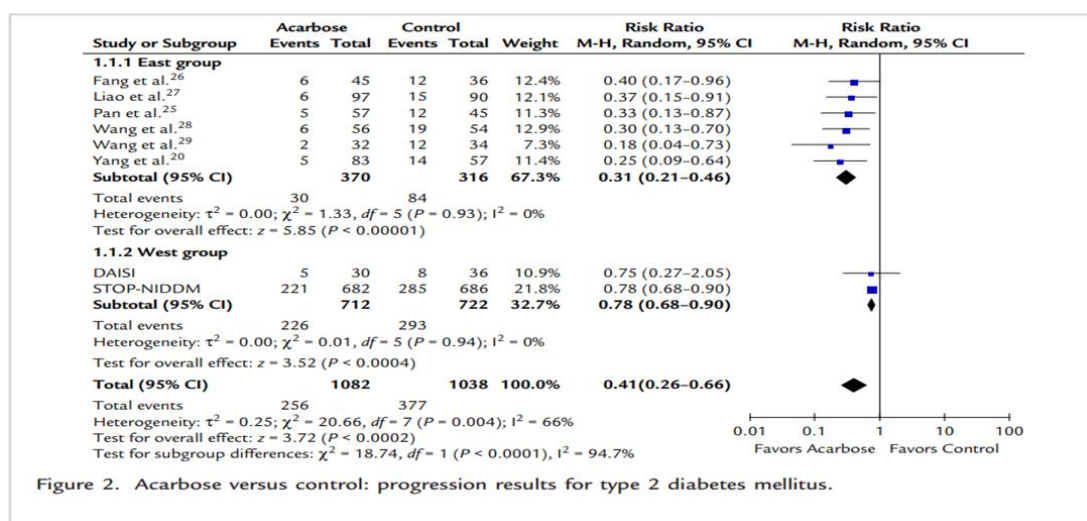


**Analysis 1.4. Comparison 1 Metformin versus placebo or diet and exercise,
Outcome 4 Incidence of type 2 diabetes (duration of the intervention).**

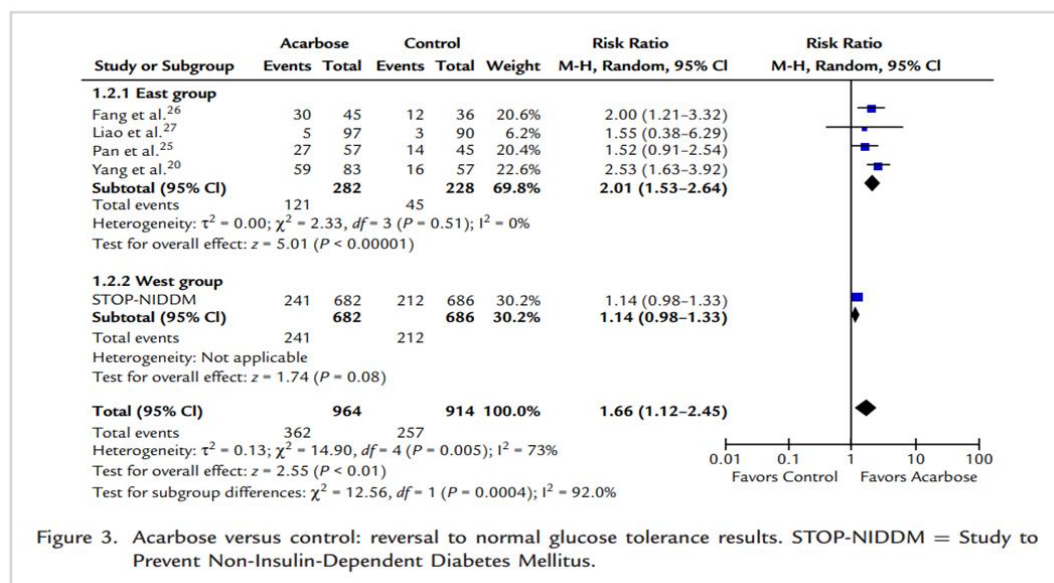


Madsen KS, Chi Y, Metzendorf MI, Richter B, Hemmingsen B. Metformin for prevention or delay of type 2 diabetes mellitus and its associated complications in persons at increased risk for the development of type 2 diabetes mellitus. Cochrane Database of Systematic Reviews 2019, Issue 12: 101 – 106 Art. No.: CD008558. DOI:10.1002/14651858.CD008558.pub2.

Acarbose in reduction of progression to diabetes:



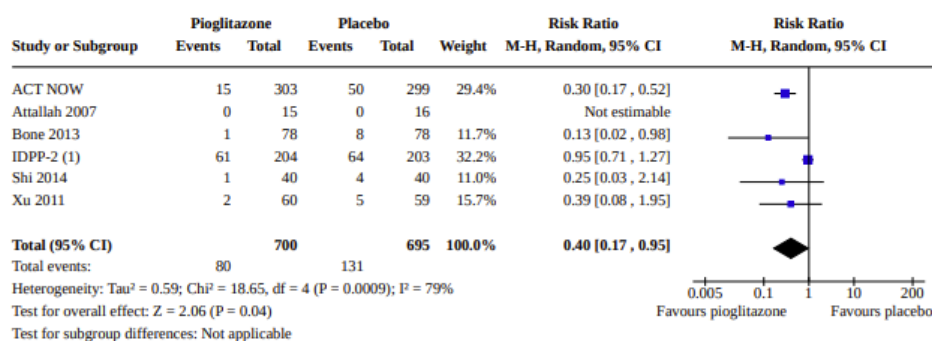
Acarbose in reversion to normoglycemia



Hu R, Li Yi, et al. Acarbose Monotherapy and type 2 Diabetes Prevention in Eastern and Western Prediabetes: An Ethnicity-Specific Meta-Analysis. 2015;8:10.

Pioglitazone versus placebo in prevention of diabetes

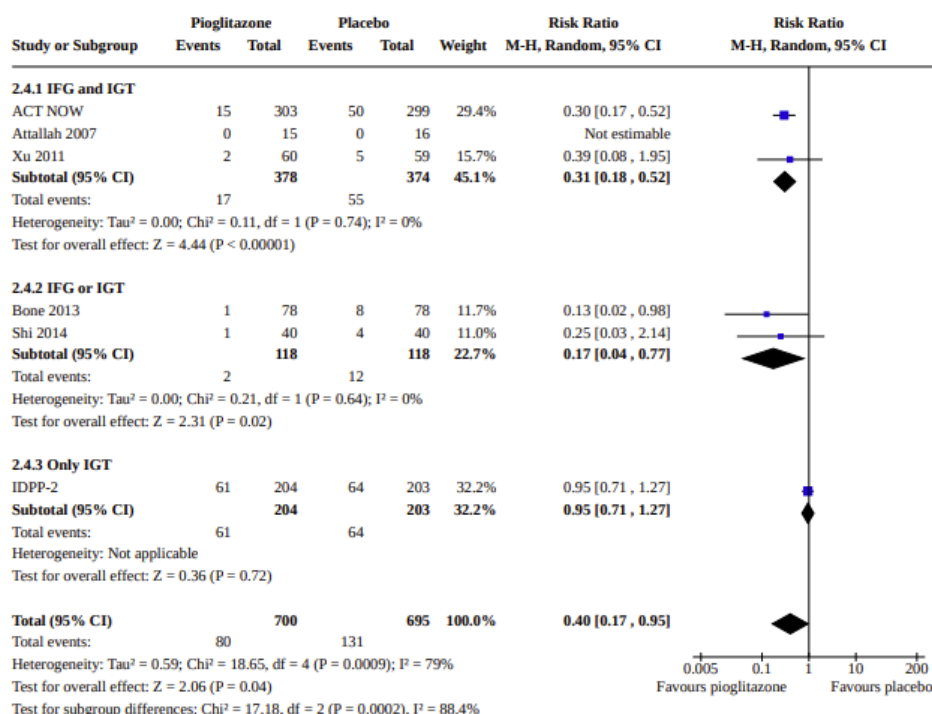
Analysis 2.3. Comparison 2: Pioglitazone versus placebo, Outcome 3: Incidence of T2DM



Footnotes

(1) Study reported cumulative incidence as percentages without specifying the population used for calculation. We calculated the incidence with populati

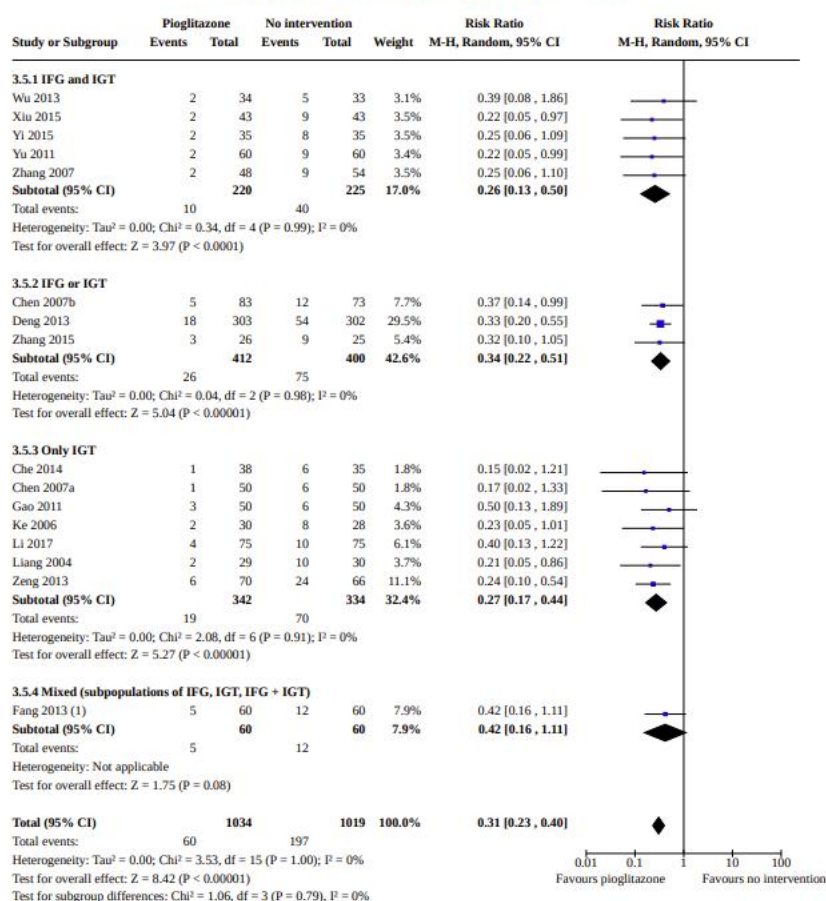
Analysis 2.4. Comparison 2: Pioglitazone versus placebo, Outcome 4: Incidence of T2DM by 'prediabetes' criteria



Ipsen EØ, Madsen KS, Chi Y, Pedersen-Bjergaard U, Richter B, Metzendorf MI, Hemmingsen B. Pioglitazone for prevention or delay of type 2 diabetes mellitus and its associated complications in people at risk for the development of type 2 diabetes mellitus (Review). Cochrane Database of systematic Reviews 2020, Issue 11. Art. No.: CD013516.
DOI: 10.1002/14651858.CD013516.pub2

Pioglitazone in prevention of diabetes versus no intervention

Analysis 3.5. Comparison 3: Pioglitazone versus no intervention, Outcome 5: Incidence of T2DM by 'prediabetes' criteria



Footnotes

(1) Study divides population up to only IFG group, IGT group and IFG + IGT group. Does not state individual incidences of T2DM

Ipsen EØ, Madsen KS, Chi Y, Pedersen-Bjergaard U, Richter B, Metzendorf MI, Hemmingsen B. Pioglitazone for prevention or delay of type 2 diabetes mellitus and its associated complications in people at risk for the development of type 2 diabetes mellitus (Review). Cochrane Database of systematic Reviews 2020, Issue 11. Art. No.: CD013516.

DOI: 10.1002/14651858.CD013516.pub2

Pioglitazone efficacy rate in treatment of MASH

Zhao et al. • Medicine (2022) 101:46

Medicine

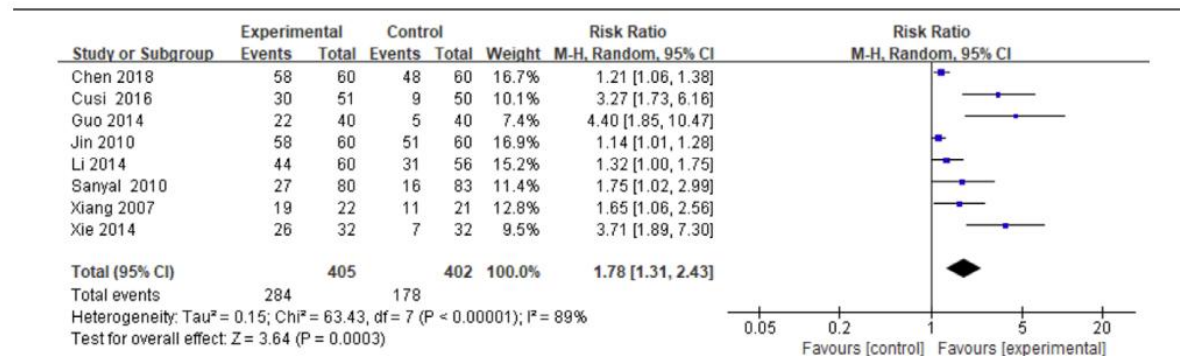


Figure 2. Changes in total effective rate of the experimental group compared with the control group.

Zhao Y, Zhao W, Wang H, Zhao Y, Bu H, Takahashi H. Pioglitazone on nonalcoholic steatohepatitis: A systematic review and meta analysis of 15 RCTs. Medicine 2022; 101:46(e31508).

<http://dx.doi.org/10.1097/MD.00000000000031508>

Pioglitazone effect on MACE in patients with insulin resistance/ pre-diabetes

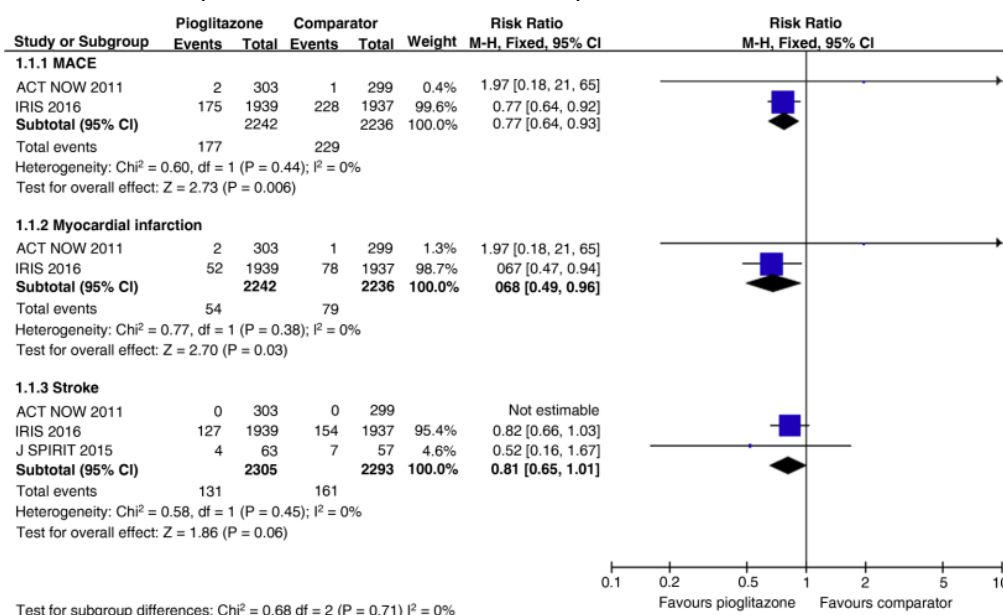
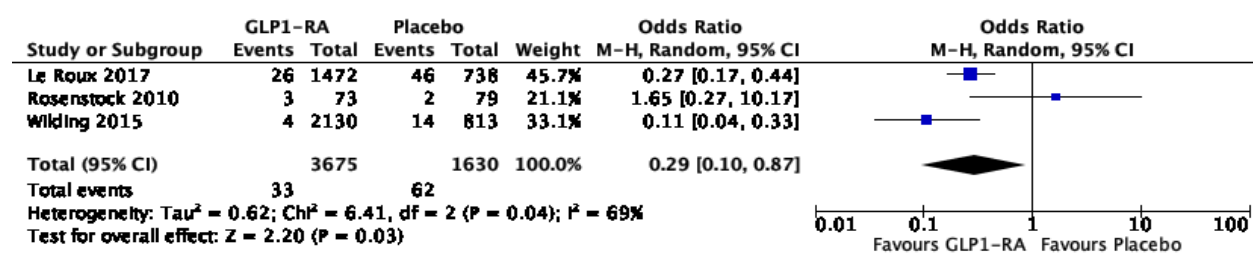


Figure 1 Separate and pooled relative risk and 95% CIs estimates for cardiovascular outcomes among patients with pre-diabetes or insulin resistance (pioglitazone vs control).

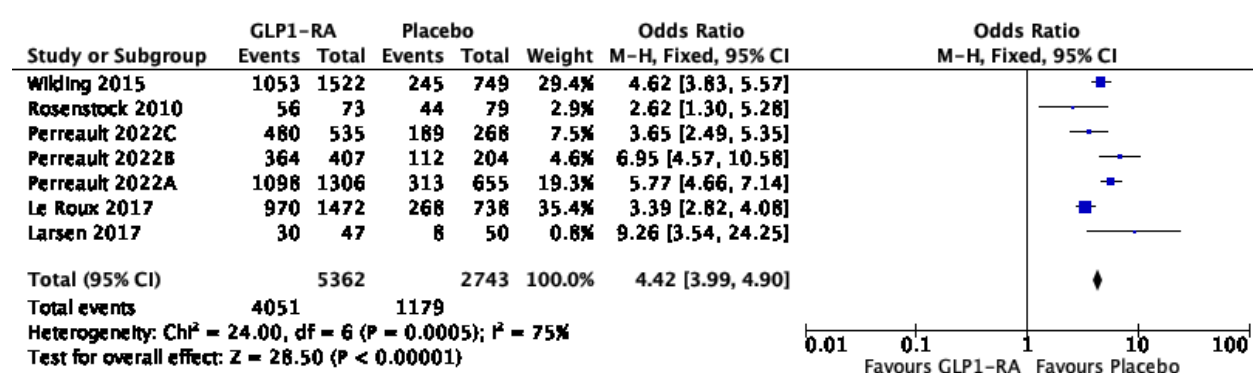
: Liao H-W, Saver JL, Wu Y-L, et al. Pioglitazone and cardiovascular outcomes in patients with insulin resistance, pre-diabetes and type 2 diabetes: a systematic review and meta-analysis. *BMJ Open* 2017;7:e013927. doi:10.1136/bmjopen-2016- 013927

GLP1- Receptor Agonists in Prevention of diabetes

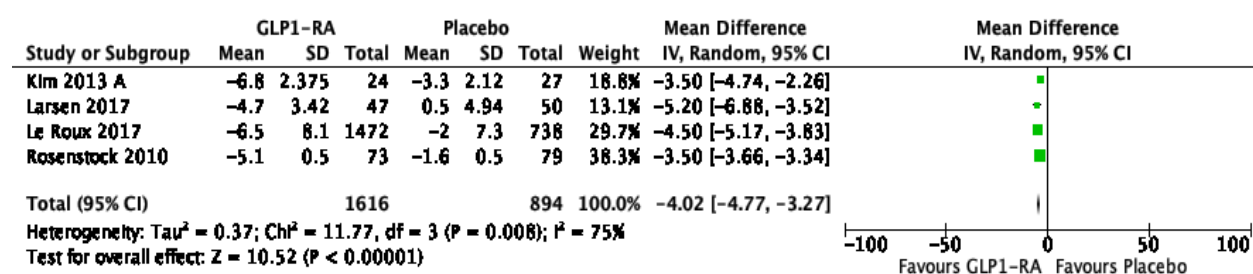
8.11.1 Progression of Prediabetes to Diabetes



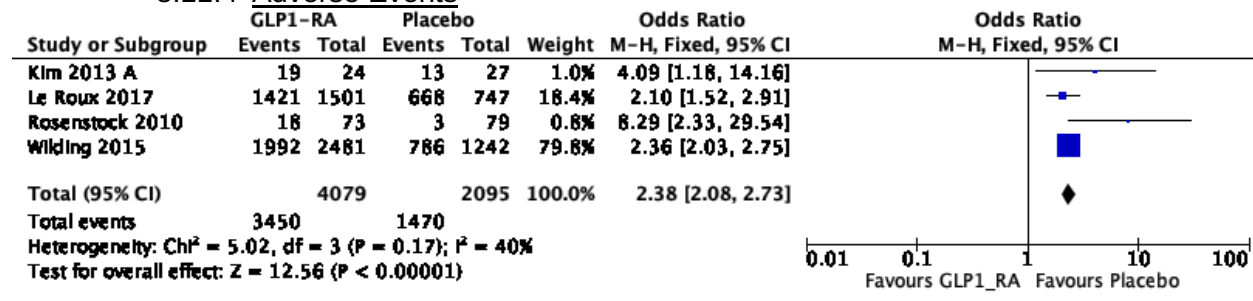
8.11.2 Regression of prediabetes to normoglycemia



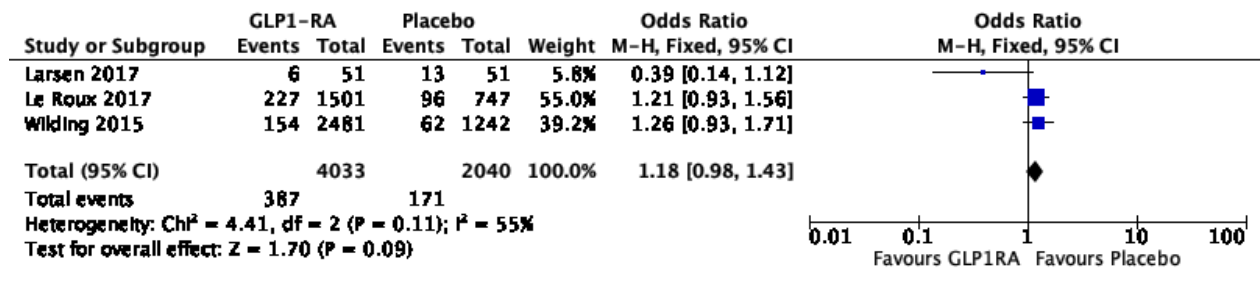
8.11.3 Weight loss



8.11.4 Adverse Events



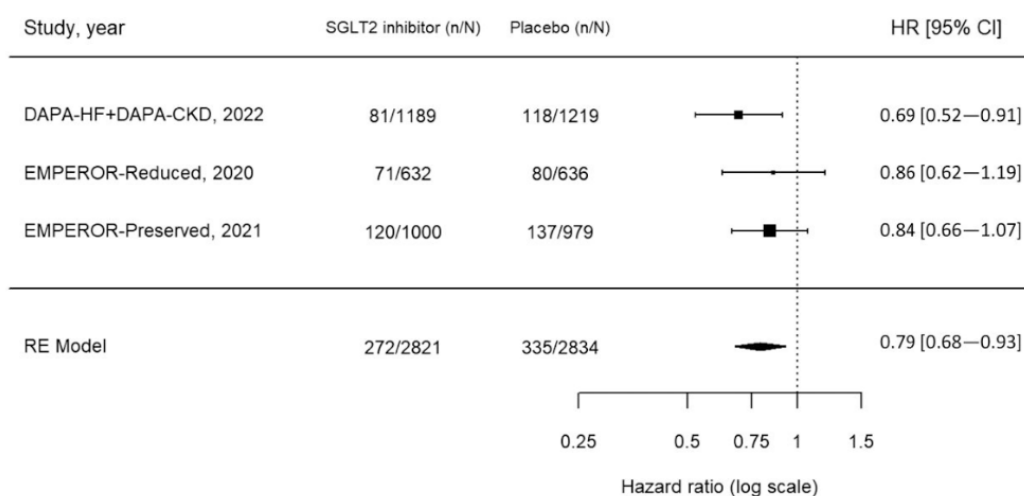
8.11.5 Serious Adverse Events



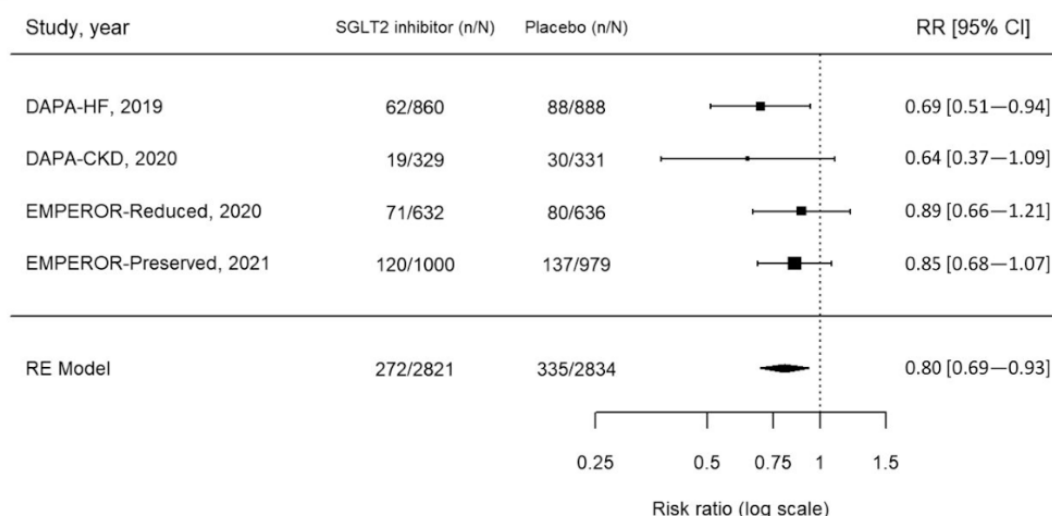
Wang W, Wei R, et al., Effects of treatment with Glucagon-like peptide-1 receptor agonist on prediabetes with overweight/obesity: A systematic review and meta-analysis. Diab Metab Res Ver. 2023;e3680

SGLT-2 inhibitor in the Prevention of Diabetes

A



B



Panel A shows the summary estimates of hazard ratio (HR) reported in the pooled sample of DAPA-HF and DAPA-CKD trials, EMPEROR-Reduced trial, and EMPEROR-Preserved trial.

Panel B shows the summary estimates of the risk ratio (RR) calculated from the raw number of events in each of the 4 trials. All 4 trials conducted randomizations with stratification by participants' diabetes status in randomization.

Q2.7. GRADE Evidence Profile

GRADE Evidence Profile:

Meta-analysis of Pharmacotherapy for the Prevention of Diabetes in Patients with Pre-DM:

Author(s): A. Andag-Silva

Question: Pharmacotherapy compared to placebo for the prevention of diabetes among various population

Setting: United States, Asia

Bibliography: ¹

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	pharmacotherapy	placebo	Relative (95% CI)	Absolute (95% CI)		

Development of overt diabetes (follow-up: range 0.5 years to 6.2 years; assessed with: development of overt DM)

24	randomised trials	not serious ^a	serious	serious ^b	not serious	publication bias strongly suspected all plausible residual confounding would reduce the demonstrated effect dose response gradient	10306/0	5910/0	RR 0.64 (0.54 to 0.76)	1,000 fewer per 1,000 (from 1,000 fewer to 1,000 fewer)	⊕⊕⊕○ Moderate	CRITICAL
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CI: confidence interval; RR: risk ratio

Explanations

a. Publication bias was noted in this meta-analysis

b. Studies were heterogenous. this analysis is on all types of meds for weight loss, hyperglycemia, blood pressure, triglycerides

References

1. Haw JS, Galaviz KI, Straus AN, Kowalski AJ, Magee MJ, Weber MB, Wei J, Narayan KMV, Ali MK.. Long term Sustainability of Diabetes Prevention Approaches . JAMA Intern Med; 2017 Dec.

Meta-analysis of Metformin in Pre-DM

Author(s): A. Andag-Silva

Question: Metformin compared to placebo for pre-diabetes

Setting: USA, China, India

Bibliography: ¹

Certainty assessment							N _e of patients		Effect		Certainty	Importance
N _e of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	metformin	placebo	Relative (95% CI)	Absolute (95% CI)		

development of overt diabetes (follow-up: range 0.5 years to 6.2 years; assessed with: development of overt DM)

4	randomised trials	not serious ^a	not serious	not serious	not serious	dose response gradient	580/2596 (22.3%)	804/2730 (29.5%)	RR 0.74 (0.65 to 0.84)	77 fewer per 1,000 (from 103 fewer to 47 fewer)	⊕⊕⊕⊕ ⊕ High	CRITICAL
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CI: confidence interval; RR: risk ratio

References

Barry E, Roberts S, Oke J, Vijayaraghavan S, Normansell R, Greenhalgh T. Efficacy and effectiveness of screen and treat policies in prevention of type 2 diabetes: systematic review and meta-analysis of screening tests and interventions. . BMJ; 2017.

Metformin in Prevention of DM in Pre-DM 2.8 years of follow-up

Author(s): A. Andag-Silva

Question: Metformin compared to standard lifestyle for prevention of diabetes

Setting: USA

Bibliography: ¹

Certainty assessment							N _e of patients		Effect		Certainty	Importance
N _e of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	metformin	standard lifestyle	Relative (95% CI)	Absolute (95% CI)		

development of diabetes (follow-up: mean 2.8 years; assessed with: OGTT, FPG)

1	randomized trials	not serious	not serious	serious ^a	not serious	none	232/1073 (21.6%)	313/1082 (28.9%)	RR 0.69 (0.57 to 0.83)	9 fewer per 100 (from 12 fewer to 5 fewer)	⊕⊕⊕⊕ ○ Moderate	CRITICAL
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CI: confidence interval; RR: risk ratio

Explanations

a. Very few Asian subjects on the trial

References

1. US Diabetes Prevention Program Study Group: Knowler WC, Barrett-Conner E, Fowler SE, Hamman RF, Lachin JM, Walker E, Nathan D. Reduction in the Incidence of type 2 diabetes mellitus with lifestyle intervention and metformin. . New Engl J Med; 2002 February 7.

Metformin in Prevention of DM in Pre-DM with long term follow-up

Author(s): A. Andag-Silva

Question: Metformin compared to standard lifestyle for long term prevention of diabetes

Setting: USA

Bibliography: ^{1,2}

Certainty assessment							N ₂ of patients		Effect		Certainty	Importance
N ₂ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	metformin	standard lifestyle	Relative (95% CI)	Absolute (95% CI)		

Development of diabetes (follow-up: mean 15 years; assessed with: OGTT, FPG, HbA1c)

1	randomised trials	not serious	not serious	serious ^a	not serious	none	506/772 (65.5%)	564/780 (72.3%)	RR 0.82 (0.72 to 0.93)	13 fewer per 100 (from 20 fewer to 5 fewer)	⊕⊕⊕ ○ Moderate	CRITICAL
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CI: confidence interval; RR: risk ratio

Explanations

a. Very few Asian subjects

References

1. US Diabetes Prevention Program Study Group: Knowler WC, Barrett-Conner E, Fowler SE, Hamman RF, Lachin JM, Walker E, Nathan D. Reduction in the Incidence of type 2 diabetes mellitus with lifestyle intervention and metformin. . New Engl J Med; 2002 February 7.

2. US Diabetes Prevention Program Study Group: Nathan DM, Barrett-Connor E, Crandall JP, Edelstein SL, Goldberg RB, Horton ES, Knowler WC, Mather KJ, Orchard TJ, Pi-Sunyer X, Schade D, and Tempresa M.. Long-term Effects of Lifestyle Intervention or Metformin on Diabetes Development and Microvascular Complications: the DPP Outcomes Study. . Lancet Diabetes Endocrinol; 2015 Nov.

Metformin in Prevention of DM in Pre-DM with prior history of GDM

Author(s): A. Andag-Silva

Question: Metformin compared to standard lifestyle for prevention of diabetes among women with current pre-diabetes and prior history of gestational diabetes

Setting: USA

Bibliography: ¹

Certainty assessment							N ₂ of patients		Effect		Certainty	Importance
N ₂ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	metformin	standard lifestyle	Relative (95% CI)	Absolute (95% CI)		

Development of diabetes (follow-up: mean 3 years; assessed with: OGTT, FPG)

1	randomised trials	not serious	not serious	serious ^a	not serious	none	21/111 (18.9%)	47/122 (38.5%)	RR 0.5 (– to –)	19 fewer per 100 (from – to –)	⊕⊕⊕ ○ Moderate	CRITICAL
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CI: confidence interval; RR: risk ratio

Explanations

a. Very few Asian subjects

References

1. Ratner RE, Christophi CA, Metzger BE, Dabelea D, Bennett PH, Pi-Sunyer X, Fowler S, Kahn SE, and The Diabetes Prevention Program Research Group. Prevention of Diabetes in Women with a History of Gestational Diabetes: Effects of Metformin and Lifestyle Interventions. J Clin Endocrinol Metab.; 2008 December. 93(12):4774–4779. doi: 10.1210/jc.2008-0772

Acarbose in Pre-DM of various ethnicities

Summary of findings:

Acarbose Compared to Placebo and/or Detailed Advice on Diet and Exercise for Type 2 Diabetes Prevention Among Patients with Prediabetes

Patient or population: Type 2 diabetes prevention among patients with prediabetes

Setting: China, Canada, Germany, Austria, Norway, Denmark, Sweden, Finland, Israel, Spain, Netherlands

Intervention: Acarbose

Comparison: placebo and/or detailed advice on diet and exercise

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N _e of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo and/or detailed advice on diet and exercise	Risk with Acarbose				
Type 2 diabetes mellitus (T2DM) assessed with: 75g OGTT follow-up: mean 3 years	Study population		RR 0.41 (0.26 to 0.66)	2120 (8 RCTs)	⊕⊕⊕⊕ Moderate ^a	Patients with IGT taking Acarbose for 3 years have 59% less chance of developing diabetes mellitus than patients with IGT given placebo and/or advice on diet and exercise. Critical level of importance
	36 per 100	15 per 100 (9 to 24)				
	Moderate					
	6 per 100	2 per 100 (2 to 4)				

*The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; RR: risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Explanations

a. Most of the RCTs have unclear sequence generation, allocation concealment and blinding process.

Hu R, Li Yi, et al. Acarbose Monotherapy and type 2 Diabetes Prevention in Eastern and Western Prediabetes: An Ethnicity-Specific Meta-analysis. Clinical Therapeutics 2015;8:1-15.

Acarbose in Pre-DM of various ethnicities

Summary of findings:

Acarbose Monotherapy and Reversal to Normal Glucose Tolerance

Patient or population: reversal to normoglycemia among IGT patients

Setting: China, Canada, Germany, Austria, Norway, Denmark, Sweden, Finland, Israel, Spain, Netherlands

Intervention: Acarbose

Comparison: placebo and/or lifestyle modification

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N _o of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo and/or lifestyle modification	Risk with Acarbose				
Reversal to NGT (NGT) assessed with: 75g OGTT follow-up: mean 3 years	Study population		RR 1.66 (1.12 to 2.45)	1878 (5 RCTs)	⊕⊕⊕○ Moderate ^a	Patients with IGT who are taking Acarbose for 3 years have a 66% chance of reverting to normoglycemia compared to patients with IGT on placebo and/or had detailed advice on diet and exercise. Critical level of importance
	28 per 100	47 per 100 (31 to 69)				
	High					
	40 per 100	66 per 100 (45 to 98)				

*The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; RR: risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Explanations

a. Most of the RCTs involved have unclear sequence generation, allocation concealment and blinding process.

Meta-analysis of Acarbose in Pre-DM- larger RCT

Summary of findings:

Alpha glucosidase inhibitors and incident diabetes among IGT patients

Patient or population: prevention of Type 2 diabetes among IGT patients

Setting: Europe, Canada, Israel, Japan, China

Intervention: an alpha glucosidase inhibitor

Comparison: placebo

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N _o of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo	Risk with an alpha glucosidase inhibitor				
Type 2 diabetes mellitus (T2DM) assessed with: 75g OGTT follow-up: mean 12 months	Study population		HR 0.77 (0.67 to 0.88)	9729 (3 RCTs)	⊕⊕⊕⊕ High ^a	Allocation of patients with IGT to an alpha glucosidase inhibitor significantly reduced their incidence of diabetes by 23%
	146 per 1,000	114 per 1,000 (100 to 130)				
	Moderate					
	60 per 1,000	47 per 1,000 (41 to 53)				

*The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; HR: hazard Ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Explanations

a. The 3 RCTs included have different study population characteristics. STOP-NIDDM involved IGT patients given either Acarbose 100mg TID or placebo. Voglibose ph 3 involved IGT patients assigned to either Voglibose 0.2 mg TID or placebo. ACE trial involved patients with IGT and coronary heart disease.

Coleman R, Scott C, et al. Meta-analysis of the impact of alpha-glucosidase inhibitors on incident diabetes and cardiovascular outcomes. Cardiovasc Diabetol 2019;18:5.

Pioglitazone in pre-diabetes with MASH

Author(s): A. Andag-Silva

Question: Pioglitazone compared to placebo for prevention of diabetes among Filipinos with pre-diabetes and current metabolic associated steatohepatitis

Setting: USA, Asia, EU

Bibliography: ¹

Certainty assessment							Impact	Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations			
Resolution of metabolic associated steatohepatitis (follow-up: median 12 months; assessed with: liver transaminases, fibrosis score, histology)									
6	randomised trials		serious ^a	serious ^{a,b}	serious ^c	all plausible residual confounding would reduce the demonstrated effect	^d (see narrative text below)	-	CRITICAL

CI: confidence interval

Explanations

a. various ethnicities were involved: EU, America and Asia

b. a minority of the subjects have diabetes type 2 already and not just pre-diabetes

c. Various parameters to assess steatohepatitis was used- UTS, transaminases, risk scores and histology

d. When compared to placebo or reference therapy, both pioglitazone and rosiglitazone significantly improved liver fat content and even NASH. With regard to a possible improvement of liver fibrosis, glitazones were not superior to placebo or other active molecules, except for one RCT using pioglitazone 45 mg/day for 18 months in patients with biopsy-proven NASH and T2DM/ prediabetes. A significant reduction of serum aminotransferase levels was observed in most patients treated with glitazones, when compared to placebo or reference therapy. Glitazones had a similar adverse event profile to placebo or reference therapy, with the exception of a weight gain (2–3 kg at a dosage of 45 mg/day of pioglitazone for 18 months). Withdrawals due to serious adverse effects were not increased in the glitazone group compared to either placebo or other active agents.

References

1. Mantovani A, Byrne CD, Scorletti E, Mantzoros CS, Targher G. Efficacy and safety of anti-hyperglycaemic drugs in patients with non-alcoholic fatty liver disease with or without diabetes: An updated systematic review of randomized controlled trials. . Diabetes Metab; 2020. <https://doi.org/10.1016/j.diabet.2019.12.007>

Pioglitazone in Pre-DM with prior history of Ischemic Stroke and TIA

Author(s): A. Andag-Silva

Question: Pioglitazone compared to placebo for the prevention of diabetes among patients with pre-diabetes and prior history of ischemic stroke or transient ischemic attack

Setting: USA

Bibliography: ^{1,2}

Certainty assessment							N ₂ of patients		Effect		Certainty	Importance
N ₂ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	pioglitazone	placebo	Relative (95% CI)	Absolute (95% CI)		

Development of diabetes (follow-up: median 4.8 years; assessed with: FPG, HbA1c)

1	randomised trials	not serious	not serious	serious ^a	not serious	none	73/1939 (3.8%)	149/1937 (7.7%)	HR 0.48 (0.33 to 0.69)	39 fewer per 1,000 (from 51 fewer to 23 fewer)	⊕⊕⊕ ○ Moderate	CRITICAL
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Ischemic stroke or myocardial infarction (follow-up: median 4.8 years; assessed with: documented events and admissions)

1	randomised trials	not serious	not serious	serious ^b	not serious	none	213/1939 (11.0%)	228/1937 (11.8%)	HR 0.76 (0.62 to 0.93)	27 fewer per 1,000 (from 43 fewer to 8 fewer)	⊕⊕⊕ ○ Moderate	CRITICAL
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CI: confidence interval; HR: hazard Ratio

Explanations

a. Very few Asian subjects in the study

b. Different health care setting- the study was not done in Asia.

References

1. Inzucchi SE, Viscoli CM, Young LH, Furie KL, Gorman M, Lovejoy AM, Dagogo-Jack S, et al. Pioglitazone Prevents Diabetes in Patients with Insulin Resistance and Cerebrovascular Disease. *Diabetes Care*; 2016; 39:1684–1692 | DOI: 10.2337/dc16-0798
2. Kernan WN, Viscoli CM, Furie KL, Young LH, Inzucchi SE, et al. Pioglitazone after Ischemic Stroke or Transient Ischemic Attack. *N Engl J Med* 2016; 374:1321-31. DOI: 10.1056/NEJMoa1506930

GRADE table for Orlistat

Summary of findings:

Orlistat with Lifestyle Changes in Diabetes Prevention

Patient or population: diabetes prevention in obese patients as adjunct to lifestyle changes
Setting: Sweden
Intervention: Orlistat
Comparison: placebo

Outcomes	Anticipated absolute effects ^a (95% CI)		Relative effect (95% CI)	N of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo	Risk with Orlistat				
Type 2 diabetes mellitus (T2DM) assessed with: 75g OGTT follow-up: mean 4 years	Study population		HR 0.627 (0.455 to 0.863)	3277 (1 RCT)	 Moderate ^a	Over a period of 4 years, in obese patients, Orlistat with lifestyle changes resulted to a greater reduction in the incidence of type 2 diabetes as compared to lifestyle changes alone.
	90 per 1,000	57 per 1,000 (42 to 78)				
	Moderate					
	440 per 1,000	305 per 1,000 (232 to 394)				

*The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI).

CI: confidence interval; HR: hazard Ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Explanations

a. The subjects were from 22 Swedish centers.

Author(s): Olivia P. Roderos MD, Aimee A. Silva MD
Question: Orlistat with Lifestyle Changes in Diabetes Prevention
Setting: Sweden
Bibliography: Torgerson J, Boldrin M, et al. XENical in the Prevention of Diabetes in Obese Subjects (XENDOS) Study. Diabetes Care, 2004;27(1):157.

Certainty assessment							N of patients		Effect		Certainty	Importance
N of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Orlistat	placebo	Relative (95% CI)	Absolute (95% CI)		
1	randomised trials	not serious	not serious	serious ^a	not serious	none	102/1640 (6.2%)	147/1637 (9.0%)	HR 0.627 (0.455 to 0.863)	33 fewer per 1,000 (from 48 fewer to 12 fewer)	 Moderate	IMPORTANT
								44.0%		135 fewer per 1,000 (from 208 fewer to 46 fewer)		

CI: confidence interval; HR: hazard Ratio

Explanations

a. The subjects were from 22 Swedish centers.

Torgerson J, Boldrin M, et al. XENical in the Prevention of Diabetes in Obese Subjects (XENDOS) Study. Diabetes Care, 2004;27(1): 157.

GRADE table for GLP-1 Receptor Agonists

Author(s): Antonio L. Faltado Jr.
Question: GLP1-RA compared to Placebo in Prediabetes
Setting:
Bibliography:

Certainty assessment							N: of patients		Effect		Certainty	Importance
N: of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	GLP1-RA	Placebo	Relative (95% CI)	Absolute (95% CI)		
Progression to diabetes mellitus (follow-up: range 14 weeks to 160 weeks)												
3	randomised trials	serious ^{a,b,c}	serious ^d	not serious	not serious	none	33/3675 (0.9%)	62/1630 (3.8%)	OR 0.29 (0.10 to 0.87)	27 fewer per 1,000 (from 34 fewer to 5 fewer)	⊕⊕○○ Low	CRITICAL
Regression from prediabetes to normoglycemia (follow-up: range 14 weeks to 160 weeks)												
7	randomised trials	serious ^{a,b,c}	not serious	not serious	not serious	none	4051/5362 (75.6%)	1179/2743 (43.0%)	OR 4.42 (3.99 to 4.90)	339 more per 1,000 (from 321 more to 357 more)	⊕⊕⊕○ Moderate	CRITICAL
Weight loss (follow-up: range 14 weeks to 160 weeks)												
4	randomised trials	serious ^{a,b,c}	not serious	not serious	not serious	none	1616	894	-	MD 4.02 kg lower (4.77 lower to 3.27 lower)	⊕⊕⊕○ Moderate	IMPORTANT
Adverse Events (follow-up: range 14 weeks to 160 weeks)												
4	randomised trials	serious ^{a,b,c}	not serious	not serious	not serious	none	3450/4079 (84.6%)	1470/2095 (70.2%)	OR 2.38 (2.08 to 2.73)	147 more per 1,000 (from 129 more to 164 more)	⊕⊕⊕○ Moderate	CRITICAL
Serious Adverse Events (follow-up: range 14 weeks to 160 weeks)												
3	randomised trials	serious ^{a,b,c}	not serious	not serious	not serious	none	187/4033 (4.6%)	171/2040 (8.4%)	OR 1.18 (0.98 to 1.43)	14 more per 1,000 (from 2 fewer to 32 more)	⊕⊕⊕○ Moderate	CRITICAL

CI: confidence interval; MD: mean difference; OR: odds ratio

Explanations

- a. Reporting Bias
- b. Attrition Bias
- c. Detection bias
- d. The study of Rosenstock had an OR of 1.65

GRADE table for SGLT-2 inhibitors

Author(s): Antonio L. Faltado Jr.
Question: SGLT2i compared to Placebo for Prediabetes
Setting:
Bibliography:

Certainty assessment							N: of patients		Effect		Certainty	Importance
N: of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	SGLT2i	Placebo	Relative (95% CI)	Absolute (95% CI)		
Onset of new diabetes mellitus (follow-up: range 16 months to 28.8 months)												
4	randomised trials	not serious	not serious	not serious	not serious	none	272/2821 (9.6%)	335/2834 (11.8%)	RR 0.82 (0.70 to 0.95)	21 fewer per 1,000 (from 35 fewer to 6 fewer)	⊕⊕⊕⊕ High	CRITICAL
Hypoglycemia Events												
4	randomised trials	not serious	not serious	not serious	not serious	none	There were no hypoglycemic events in adults without diabetes in the DAPA-HF and DAPA CKD Trials. The hypoglycemic events in the EMPEROR-Reduced and EMPEROR-Preserved were low (7 (0.7%) in the SGLT2i group and 6 (0.6%) in the placebo group) and (10 (0.7%) in the SGLT2i group and 12 (0.8%) in the placebo group), respectively				⊕⊕⊕⊕ High	CRITICAL

CI: confidence interval; RR: risk ratio

Q2.8. Recommendations from other groups

Recommendations from other groups:

GROUP or AGENCY	RECOMMENDATIONS	STRENGTH OF RECOMMENDATION/ CERTAINTY/QUALITY OF EVIDENCE
American Diabetes Association Professional Practice Committee Standards of Medical Care in Diabetes-2024. ⁴⁸	Metformin therapy for the prevention of diabetes should be considered in the following population typified in the DPP: aged 25 to 59 years, BMI ≥ 35 kg/m², fasting glucose ≥ 110 m/dL or A1c $\geq 6\%$ and women with prior history of gestational diabetes. In people with history of stroke and evidence of insulin resistance, pioglitazone may be used to lower the risk of stroke and myocardial infarction. Benefits must be balanced with increased weight gain, edema and fracture. Lower doses may mitigate adverse events but may be less effective. Pharmacotherapy for weight management, minimizing progression of hyperglycemia and CV risk reduction, may be considered to support patient-centered care goals.	Level A Level A Level C for low dose Level B
U.K. NICE Public health guideline: Type 2 diabetes: prevention in people at high risk ⁴⁹	Consider to offer metformin to support lifestyle change for people whose HbA1c or fasting plasma glucose blood test results have deteriorated if: • this has happened despite their participation in intensive lifestyle-change programmes or • they are unable to participate in an intensive lifestyle-change programme particularly if they have a BMI greater than 35. Start with a low dose (for example, 500 mg once daily) and then increase gradually as tolerated, to 1500 to 2000 mg daily. If the person is intolerant of standard metformin consider using modified-release metformin. Prescribe metformin for 6 to 12 months initially. Monitor the person's fasting plasma glucose or HbA1c levels at 3 month intervals and stop the drug if no effect is seen. The results of the economic modelling also showed that, in the high-risk population overall and in most subgroups, lifestyle-	Strong

	change programmes are more clinically and cost effective than metformin. They also showed that, compared with control alone, metformin is cost effective in the high-risk population overall, and for most subgroups. metformin could be used for people whose BMI is over 35 when their blood test results have deteriorated because the model showed that metformin is particularly clinically and cost effective for this group. Offer orlistat to people with a BMI of 28.0 kg/m² or more, as part of an overall plan for managing obesity. Take into account the person's risk and the level of weight loss and lifestyle change required to reduce this risk. Review the use of orlistat after 12 weeks. If the person has not lost at least 5% of their original body weight, use clinical judgement to decide whether to stop the orlistat. Use orlistat for more than 12 months (usually for weight maintenance) only after discussing the potential benefits, limitations and side effects	
Diabetes Canada 2018; (with section updates in 2023) ⁵⁰	Metformin may be considered as a strategy to prevent type 2 diabetes in people with IGT (especially in combination with IFG or with elevated A1C between 6.0% to 6.4% range). Metformin may be more effective among younger individuals (<60 years) with significant obesity (>35 kg/m²) and among women with a history of GDM. Other drugs with studies in IFG/ IGT mentioned: pioglitazone, acarbose, voglibose, orlistat there is no straight recommendation for the wide use of these drugs. Liraglutide could also prevent diabetes among obese patients however there were issues on cost effectiveness and long term safety data.	Level 1 Grade A
Malaysian Clinical Practice Guidelines Management Of Type 2 Diabetes Mellitus (6 th Edition) ⁵¹	Consider metformin in those at very high risk of diabetes, pre-diabetes, and women with prior history of gestational diabetes, or for those who failed lifestyle therapy after 6 months.	Grade A

	Newer therapeutic agents, e.g. high dose liraglutide, in obese prediabetic individuals have been shown to result in weight reduction and delay progression to T2DM.	No grading
Japanese Clinical Practice Guidelines 2019 ⁵²	Biguanides, α -glucosidase inhibitors, thiazolidinediones (TZDs), basal insulin formulations, and GLP-1 receptor agonists (GLP-1RA) are shown to have inhibitory effects on the onset of type 2 diabetes.	No grading

C. Nonpharmacologic management Among Patients with Diabetes Mellitus

Guideline Question 3.

Should we recommend dietary interventions among individuals with Type 2 diabetes?

Q3.1. Methods

A systematic search was conducted from database inception to March 31, 2024 using MEDLINE and HERDIN Plus and was subsequently updated to include all relevant randomized trials and related evidence published through September 2025. A combined MeSH and free-text strategy was applied using the terms: “type 2 diabetes,” “primary care,” “nutrition therapy,” “lifestyle change,” and “randomized controlled trial.” All identified studies were assessed for risk of bias using the Cochrane Risk of Bias assessment criteria.

Only randomized controlled trials comparing nutritional therapy with conventional treatment were included. To ensure consistency in methodological quality and applicability, only full-text published articles written in English were considered.

Q3.2. Search strategy

DATABASE	SEARCH STRATEGY / SEARCH TERMS	DATE AND TIME OF SEARCH	RESULTS	
			Yield	Eligible
Medline	(((((("diabetes mellitus, type 2"[MeSH Terms] OR ("diabetes mellitus"[MeSH Terms] OR ("diabetes"[All Fields] AND "mellitus"[All Fields]) OR "diabetes mellitus"[All Fields]) AND "Noninsulin-Dependent"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("diabetes"[All Fields] AND "mellitus"[All Fields] AND "ketosis"[All Fields] AND "resistant"[All Fields])) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("ketosis"[All Fields] AND "resistant"[All Fields] AND "diabetes"[All Fields] AND "mellitus"[All Fields]) OR "ketosis resistant diabetes mellitus"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("diabetes"[All Fields] AND "mellitus"[All Fields] AND "non"[All Fields] AND "insulin"[All Fields] AND "dependent"[All Fields]) OR "diabetes mellitus non insulin dependent"[All Fields]) OR ("diabetes mellitus"[MeSH Terms] OR ("diabetes"[All Fields] AND "mellitus"[All Fields]) OR "diabetes mellitus"[All Fields]) AND "Non-InsulinDependent"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("non"[All Fields] AND "insulin"[All Fields] AND "dependent"[All Fields] AND "diabetes"[All Fields] AND "mellitus"[All Fields]) OR "non insulin dependent diabetes mellitus"[All Fields]) OR "diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("diabetes"[All Fields] AND "mellitus"[All Fields] AND "stable"[All Fields]) OR "diabetes mellitus stable"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("stable"[All Fields] AND "diabetes"[All Fields] AND "mellitus"[All Fields]) OR "stable diabetes mellitus"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("diabetes"[All Fields] AND "mellitus"[All Fields] AND "type"[All Fields] AND "ii"[All Fields]) OR "diabetes mellitus type ii"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "niddm"[All Fields] OR "niddms"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("diabetes"[All Fields] AND "mellitus"[All Fields] AND "noninsulin"[All Fields] AND "dependent"[All Fields]) OR "diabetes mellitus noninsulin dependent"[All Fields]) OR ("diabetes mellitus"[MeSH Terms] OR ("diabetes"[All Fields]	March 31, 2024	8,942	30

	AND "mellitus"[All Fields] OR "diabetes mellitus"[All Fields] AND "Maturity-Onset"[All Fields] OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("diabetes"[All Fields] AND "mellitus"[All Fields] AND "maturity"[All Fields] AND "onset"[All Fields]) OR "diabetes mellitus maturity onset"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("maturity"[All Fields] AND "onset"[All Fields] AND "diabetes"[All Fields] AND "mellitus"[All Fields]) OR "maturity onset diabetes mellitus"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("maturity"[All Fields] AND "onset"[All Fields] AND "diabetes"[All Fields] AND "mellitus"[All Fields]) OR "maturity onset diabetes mellitus"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "mody"[All Fields]) OR (("diabetes mellitus"[MeSH Terms] OR ("diabetes"[All Fields] AND "mellitus"[All Fields]) OR "diabetes mellitus"[All Fields]) AND "Slow-Onset"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("diabetes"[All Fields] AND "mellitus"[All Fields] AND "slow"[All Fields] AND "onset"[All Fields]) OR "diabetes mellitus slow onset"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("slow"[All Fields] AND "onset"[All Fields] AND "diabetes"[All Fields] AND "mellitus"[All Fields])) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("noninsulin"[All Fields] AND "dependent"[All Fields] AND "diabetes"[All Fields] AND "mellitus"[All Fields]) OR "noninsulin dependent diabetes mellitus"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("noninsulin"[All Fields] AND "dependent"[All Fields] AND "diabetes"[All Fields] AND "mellitus"[All Fields]) OR "noninsulin dependent diabetes mellitus"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("maturity"[All Fields] AND "onset"[All Fields] AND "diabetes"[All Fields]) OR "maturity onset diabetes"[All Fields]) OR ("diabetes"[All Fields] OR "diabetes mellitus"[MeSH Terms] OR ("diabetes"[All Fields] AND "mellitus"[All Fields]) OR "diabetes mellitus"[All Fields] OR "diabetes insipidus"[MeSH Terms] OR ("diabetes"[All Fields] AND "insipidus"[All Fields]) OR "diabetes insipidus"[All Fields] OR "diabetic"[All Fields] OR "diabetics"[All Fields] OR "diabets"[All Fields]) AND "Maturity-Onset"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("maturity"[All Fields] AND "onset"[All Fields] AND "diabetes"[All Fields]) OR "maturity onset diabetes"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "type 2 diabetes"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("diabetes"[All Fields] AND "mellitus"[All Fields] AND "adult"[All Fields] AND "onset"[All Fields]) OR "diabetes mellitus adult onset"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "adult"[All Fields] AND "onset"[All Fields] AND "diabetes"[All Fields] AND "mellitus"[All Fields]) OR "adult onset diabetes mellitus"[All Fields]) OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR ("diabetes"[All Fields] AND "mellitus"[All Fields] AND "adult"[All Fields] AND "onset"[All Fields]) OR "diabetes mellitus adult onset"[All Fields]) OR ("Hypertension"[MeSH Terms] OR ("Hypertension"[MeSH Terms] OR "Hypertension"[All Fields] OR ("blood"[All Fields] AND "pressure"[All Fields] AND "high"[All Fields]) OR "blood pressure high"[All Fields]) OR ("Hypertension"[MeSH Terms] OR "Hypertension"[All Fields] OR ("blood"[All Fields] AND "pressures"[All Fields] AND "high"[All Fields]) OR "blood pressures high"[All Fields]) OR ("Hypertension"[MeSH Terms] OR "Hypertension"[All Fields] OR ("high"[All Fields] AND "blood"[All Fields] AND "pressure"[All Fields]) OR "high blood pressure"[All Fields]) OR ("Hypertension"[MeSH Terms] OR "Hypertension"[All Fields] OR ("high"[All Fields] AND "blood"[All Fields] AND "pressures"[All Fields]) OR "high blood pressures"[All Fields])) AND ("Primary Health Care"[MeSH Terms] OR ("Primary Health Care"[MeSH Terms] OR ("primary"[All Fields] AND "health"[All Fields] AND "care"[All Fields]) OR "Primary Health Care"[All Fields] OR ("care"[All Fields] AND "primary"[All Fields] AND "health"[All Fields]) OR "care primary health"[All Fields]) OR ("Primary Health Care"[MeSH Terms] OR "primary"[All Fields] AND "health"[All Fields] AND "care"[All Fields]) OR "Primary Health Care"[All Fields] OR "health"[All Fields] AND "care"[All Fields] AND "primary"[All Fields]) OR "health care primary"[All Fields]) OR ("Primary Health			
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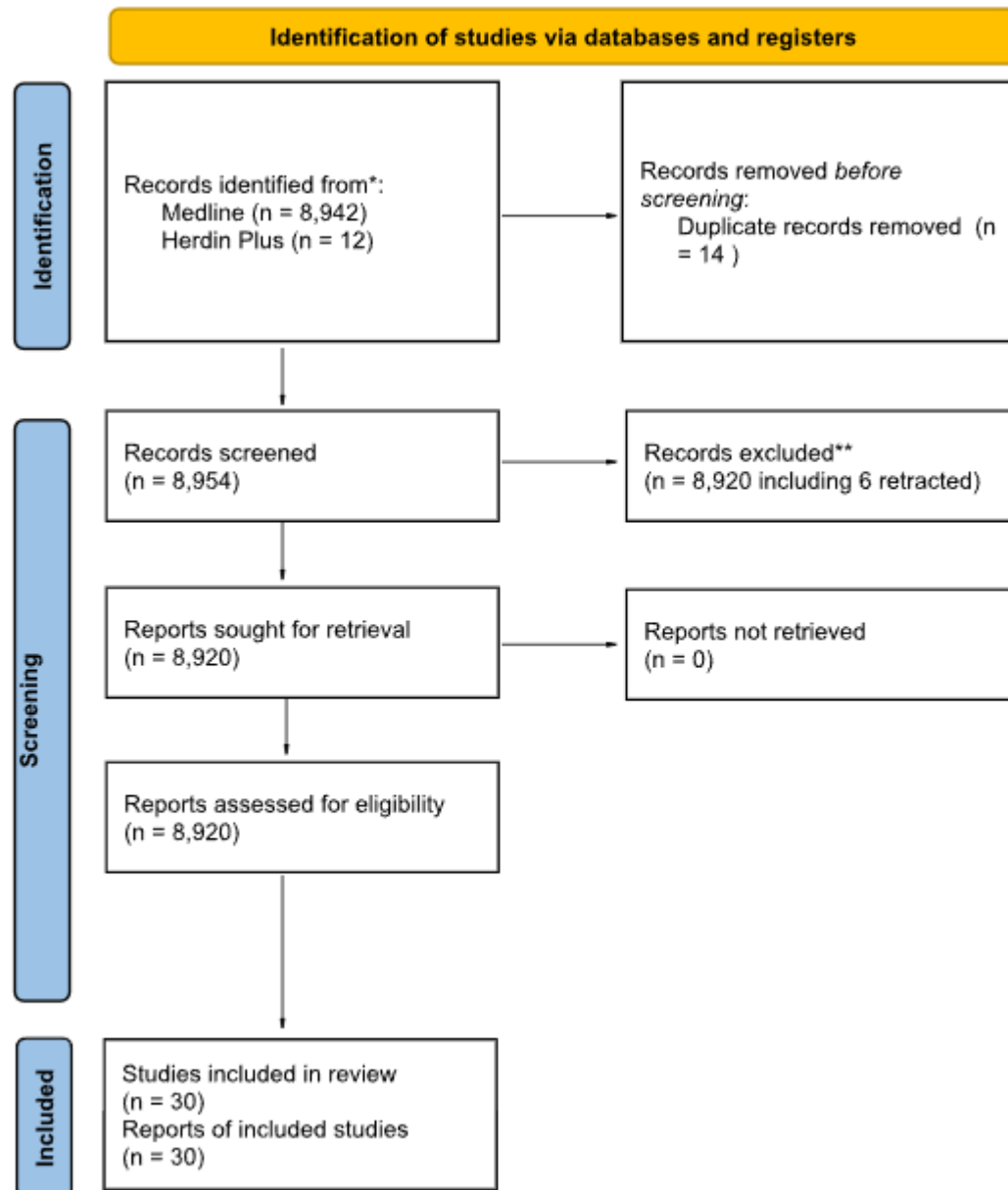
	Care"[MeSH Terms] OR ("primary"[All Fields] AND "health"[All Fields] AND "care"[All Fields]) OR "Primary Health Care"[All Fields] OR ("primary"[All Fields] AND "healthcare"[All Fields]) OR "primary healthcare"[All Fields] OR ("Primary Health Care"[MeSH Terms] OR ("primary"[All Fields] AND "health"[All Fields] AND "care"[All Fields]) OR "Primary Health Care"[All Fields] OR ("healthcare"[All Fields] AND "primary"[All Fields]) OR "healthcare primary"[All Fields]) OR ("Primary Health Care"[MeSH Terms] OR ("primary"[All Fields] AND "health"[All Fields] AND "care"[All Fields]) OR "Primary Health Care"[All Fields] OR ("primary"[All Fields] AND "care"[All Fields]) OR "primary care"[All Fields]) OR ("Primary Health Care"[MeSH Terms] OR ("primary"[All Fields] AND "health"[All Fields] AND "care"[All Fields]) OR "Primary Health Care"[All Fields] OR ("care"[All Fields] AND "primary"[All Fields]) OR "care primary"[All Fields]) OR "physicians, primary care"[MeSH Terms] OR ("physicians, primary care"[MeSH Terms] OR ("physicians"[All Fields] AND "primary"[All Fields] AND "care"[All Fields]) OR "primary care physicians"[All Fields] OR ("physician"[All Fields] AND "primary"[All Fields] AND "care"[All Fields]) OR "physician primary care"[All Fields]) OR ("physicians, primary care"[MeSH Terms] OR ("physicians"[All Fields] AND "primary"[All Fields] AND "care"[All Fields]) OR "primary care physicians"[All Fields] OR ("primary"[All Fields] AND "care"[All Fields] AND "physician"[All Fields]) OR "primary care physician"[All Fields]) OR ("physicians, primary care"[MeSH Terms] OR ("physicians"[All Fields] AND "primary"[All Fields] AND "care"[All Fields]) OR "primary care physicians"[All Fields] OR ("primary"[All Fields] AND "care"[All Fields] AND "physicians"[All Fields])) OR ("healthies"[All Fields] OR "healthy"[All Fields]) AND ("Primary Health Care"[MeSH Terms] OR ("primary"[All Fields] AND "health"[All Fields] AND "care"[All Fields]) OR "Primary Health Care"[All Fields] OR ("primary"[All Fields] AND "care"[All Fields]) OR "primary care"[All Fields])) OR "Primary Care Nursing"[MeSH Terms] OR ("Primary Care Nursing"[MeSH Terms] OR ("primary"[All Fields] AND "care"[All Fields] AND "nursing"[All Fields]) OR "Primary Care Nursing"[All Fields] OR ("care"[All Fields] AND "nursing"[All Fields] AND "primary"[All Fields]) OR "care nursing primary"[All Fields]) OR ("Primary Care Nursing"[MeSH Terms] OR ("primary"[All Fields] AND "care"[All Fields] AND "nursing"[All Fields]) OR "Primary Care Nursing"[All Fields] OR ("nursing"[All Fields] AND "primary"[All Fields] AND "care"[All Fields]) OR "nursing primary care"[All Fields])) OR ("Community Health Planning"[MeSH Terms] OR ("Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR ("community"[All Fields] AND "health"[All Fields] AND "plannings"[All Fields])) OR ("Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR ("health"[All Fields] AND "planning"[All Fields] AND "community"[All Fields]) OR "health planning community"[All Fields]) OR ("Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR ("planning"[All Fields] AND "community"[All Fields] AND "health"[All Fields]) OR ("Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR ("population"[All Fields] AND "based"[All Fields] AND "planning"[All Fields]) OR "population based planning"[All Fields]) OR ("plan s"[All Fields] OR "planned"[All Fields] OR "planning"[All Fields] OR "plannings"[All Fields] OR "plans"[All Fields]) AND "Population-Based"[All Fields]) OR ("plan s"[All Fields] OR "planned"[All Fields] OR "planning"[All Fields] OR "plannings"[All Fields] OR "plans"[All Fields]) AND "Population-Based"[All Fields]) OR ("Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR ("population"[All Fields] AND "based"[All Fields] AND "planning"[All Fields]) OR "population based planning"[All Fields]) OR "Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR "Community Health Planning"[All			
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Fields] OR ("population"[All Fields] AND "based"[All Fields] AND "plannings"[All Fields])) OR ("Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR ("community"[All Fields] AND "health"[All Fields] AND "systems"[All Fields]) OR "community health systems"[All Fields]) OR ("Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR ("community"[All Fields] AND "health"[All Fields] AND "system"[All Fields]) OR "community health system"[All Fields]) OR ("Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR ("health"[All Fields] AND "system"[All Fields] AND "community"[All Fields]) OR "health system community"[All Fields]) OR ("Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR "health systems community"[All Fields]) OR ("Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR ("system"[All Fields] AND "community"[All Fields] AND "health"[All Fields]) OR "system community health"[All Fields]) OR ("Community Health Planning"[MeSH Terms] OR ("community"[All Fields] AND "health"[All Fields] AND "planning"[All Fields]) OR "Community Health Planning"[All Fields] OR ("systems"[All Fields] AND "community"[All Fields] AND "health"[All Fields]) OR "systems community health"[All Fields])))) AND ("Nutrition Therapy"[MeSH Terms] OR ("Nutrition Therapy"[MeSH Terms] OR ("nutrition"[All Fields] AND "therapy"[All Fields]) OR "Nutrition Therapy"[All Fields] OR ("medical"[All Fields] AND "nutrition"[All Fields] AND "therapy"[All Fields]) OR "medical nutrition therapy"[All Fields]) OR ("Nutrition Therapy"[MeSH Terms] OR ("nutrition"[All Fields] AND "therapy"[All Fields]) OR "Nutrition Therapy"[All Fields] OR ("therapy"[All Fields] AND "nutrition"[All Fields]) OR "therapy nutrition"[All Fields]) OR ("Nutrition Therapy"[MeSH Terms] OR ("nutrition"[All Fields] AND "therapy"[All Fields]) OR "Nutrition Therapy"[All Fields] OR ("nutrition"[All Fields] AND "therapy"[All Fields] AND "medical"[All Fields] AND "nutrition"[All Fields]) OR "therapy medical nutrition"[All Fields]) OR ("nutrition s"[All Fields] OR "nutritional status"[MeSH Terms] OR ("nutritional"[All Fields] AND "status"[All Fields]) OR "nutritional status"[All Fields] OR "nutrition"[All Fields] OR "nutritional sciences"[MeSH Terms] OR ("nutritional"[All Fields] AND "sciences"[All Fields]) OR "nutritional sciences"[All Fields] OR "nutritional"[All Fields] OR "nutritionals"[All Fields] OR "nutritions"[All Fields] OR "nutritive"[All Fields]) AND ("manage"[All Fields] OR "managed"[All Fields] OR "management s"[All Fields] OR "managements"[All Fields] OR "manager"[All Fields] OR "manager s"[All Fields] OR "managers"[All Fields] OR "manages"[All Fields] OR "managing"[All Fields] OR "managment"[All Fields] OR "organization and administration"[MeSH Terms] OR ("organization"[All Fields] AND "administration"[All Fields]) OR "organization and administration"[All Fields] OR "management"[All Fields] OR "disease management"[MeSH Terms] OR ("disease"[All Fields] AND "management"[All Fields]) OR "disease management"[All Fields]) OR ("nutrition s"[All Fields] OR "nutritional status"[MeSH Terms] OR ("nutritional"[All Fields] AND "status"[All Fields]) OR "nutritional status"[All Fields] OR "nutrition"[All Fields] OR "nutritional sciences"[MeSH Terms] OR ("nutritional"[All Fields] AND "sciences"[All Fields]) OR "nutritional sciences"[All Fields] OR "nutritional"[All Fields] OR "nutritionals"[All Fields] OR "nutritions"[All Fields] OR "nutritive"[All Fields]) AND ("strategie"[All Fields] OR "strategies"[All Fields] OR "strategy"[All Fields] OR "strategy s"[All Fields])) OR ("nutrition s"[All Fields] OR "nutritional status"[MeSH Terms] OR ("nutritional"[All Fields] AND "status"[All Fields]) OR "nutritional status"[All Fields] OR "nutrition"[All Fields] OR "nutritional sciences"[MeSH Terms] OR ("nutritional"[All Fields] AND "sciences"[All Fields]) OR "nutritional sciences"[All Fields] OR "nutritional"[All Fields] OR "nutritionals"[All Fields] OR "nutritions"[All Fields] OR "nutritive"[All Fields]) AND ("strategie"[All Fields] OR "strategies"[All Fields] OR "strategy"[All Fields] OR "strategy s"[All Fields])) OR "Nutritionists"[MeSH Terms] OR ("nutritionist s"[All Fields] OR "Nutritionists"[MeSH Terms] OR "Nutritionists"[All Fields] OR "nutritionist"[All Fields]) OR ("Nutritionists"[MeSH Terms] OR "Nutritionists"[All Fields] OR			
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	Type] OR "randomized"[Title/Abstract] OR "placebo"[Title/Abstract] OR "clinical trials as topic"[MeSH Terms:noexp] OR "randomly"[Title/Abstract] OR "trial"[Title]) NOT ("animals"[MeSH Terms] NOT "humans"[MeSH Terms])) AND ((randomizedcontrolledtrial[Filter])			
Herdin Plus	lifestyle OR nutrition AND diabetes	March 31, 2024	11	0

#1	"Diabetes Mellitus, Type 2"[Mesh] OR (Diabetes Mellitus, Noninsulin-Dependent) OR (Diabetes Mellitus, Ketosis Resistant) OR (Ketosis-Resistant Diabetes Mellitus) OR (Diabetes Mellitus, Non Insulin Dependent) OR (Diabetes Mellitus, Non-Insulin-Dependent) OR (Non-Insulin-Dependent Diabetes Mellitus) OR (Diabetes Mellitus, Stable) OR (Stable Diabetes Mellitus) OR (Diabetes Mellitus, Type II) OR (NIDDM) OR (Diabetes Mellitus, Noninsulin Dependent) OR (Diabetes Mellitus, Maturity-Onset) OR (Diabetes Mellitus, Maturity Onset) OR (Maturity-Onset Diabetes Mellitus) OR (Maturity Onset Diabetes Mellitus) OR (MODY) OR (Diabetes Mellitus, Slow-Onset) OR (Diabetes Mellitus, Slow Onset) OR (Slow-Onset Diabetes Mellitus) OR (Type 2 Diabetes Mellitus) OR (Noninsulin-Dependent Diabetes Mellitus) OR (Noninsulin Dependent Diabetes Mellitus) OR (Maturity-Onset Diabetes) OR (Diabetes, Maturity-Onset) OR (Maturity Onset Diabetes) OR (Type 2 Diabetes) OR (Diabetes, Type 2) OR (Diabetes Mellitus, Adult-Onset) OR (Adult-Onset Diabetes Mellitus) OR (Diabetes Mellitus, Adult Onset)
#2	"Hypertension"[Mesh] OR (Blood Pressure, High) OR (Blood Pressures, High) OR (High Blood Pressure) OR (High Blood Pressures)
#3	"Primary Health Care"[Mesh] OR (Care, Primary Health) OR (Health Care, Primary) OR (Primary Healthcare) OR (Healthcare, Primary) OR (Primary Care) OR (Care, Primary) or "Physicians, Primary Care"[Mesh] OR (Physician, Primary Care) OR (Primary Care Physician) OR (Primary Care Physicians) OR (Healthy Primary Care) OR "Primary Care Nursing"[Mesh] OR (Care Nursing, Primary) OR (Nursing, Primary Care)
#4	"Community Health Planning"[Mesh] OR (Community Health Plannings) OR (Health Planning, Community) OR (Health Plannings, Community) OR (Planning, Community Health) OR (Plannings, Community Health) OR (Population-Based Planning) OR (Planning, Population-Based) OR (Plannings, Population-Based) OR (Population Based Planning) OR (Population-Based Plannings) OR (Community Health Systems) OR (Community Health System) OR (Health System, Community) OR (Health Systems, Community) OR (System, Community Health) OR (Systems, Community Health)
#5	"Nutrition Therapy"[Mesh] OR (Medical Nutrition Therapy) OR () OR (Therapy, Nutrition) OR (Nutrition Therapy, Medical) OR (Therapy, Medical Nutrition) OR (Nutrition Management) OR (Nutrition Strategy) OR (Nutrition Strategies) OR "Nutritionists"[Mesh] OR (Nutritionist) OR (Dieticians) OR (Dietician) OR (Dietitians) OR (Dietitian) OR (Registered Dietitian)
#6	"Life Style" [Mesh] OR (Life Styles) OR (Lifestyle) OR (Lifestyles) OR (Life Style Induced Illness) OR (Life Style Intervention) OR (Life Style Interventions) OR (Change of Life Style) OR "Health Behavior"[Mesh] OR (Behavior, Health) OR (Behaviors, Health) OR (Health Behaviors) OR "Healthy Lifestyle"[Mesh] OR (Lifestyle, Healthy) OR (Lifestyles, Healthy) OR (Healthy Life Styles) OR (Healthy Lifestyles) OR (Healthy Life Style) OR (Life Style, Healthy) OR (Life Styles, Healthy)
#7	randomized controlled trial[pt] OR controlled clinical trial[pt] OR randomized[tiab] OR placebo[tiab] OR clinical trials as topic[mesh:noexp] OR randomly[tiab] OR trial[ti] NOT (animals[mh] NOT humans [mh])
#1 OR #2 AND #3 OR #4 AND #5 OR #6 AND #7	

Q3.3. PRISMA flow diagram



Q3.4. Characteristics of the Included Studies

Study	Study Design	Country	Number of patients	Population (Analyzed)	Intervention	Control	Outcomes	Follow-up
Adachi 2013	RCT	Japan	193	DM2 HbA1c>6.5	SILE program (4 sessions, registered dietitian, Structured individual-based lifestyle	usual care, Registered dietitian once general advice given	Usual care, Registered dietitian once general advice given	6mo
Adolfsson 2007	RCT	Sweden	88	Aged < 75, HbA1c between 6.5% and 10%, diagnosed with T2DM at least 1 year before inclusion	empowerment group education n:	usual care	Confidence in diabetes knowledge, self-efficacy and satisfaction with daily life questionnaire 2. HbA1c, BMI	12 months
Baer 2020	RCT	USA	6624	Aged 20 and <70, BMI between 27 and 40, diagnosed with either HTN or T2DM	Educational program delivered by a non-clinical population health manager	usual care	weight change, change in SBP, DBP, HbA1c	12 months

Benson 2019	RCT	USA	224	Aged 20 and <70, BMI between 27 and 40, diagnosed with either HTN or T2DM	Educational program delivered by dietitian	usual care	changes in BMI, LDL, HbA1c	12 months
Brown 2002	RCT	USA Mexico	224	Aged >35 and <70, DM2 after 35 yo	Educational program delivered by nurses, dietitians, and community workers	usual care	changes in HbA1c, fasting blood glucose, lipid profile, BMI	12 mos
Browning 2016	RCT	China	577	≥50 yo DM2	Educational program delivered by community physicians, nurses and psychologists	usual care	changes in HbA1c; SBP, DBP, BMI, waist and hip circumference	12 months
Clancy 2007	RCT	USA	186	>18 yo HbA1c >8%	Educational program delivered by internal medicine physicians and practice nurses	usual care	Changes in HbA1c, SBP, DBP, Lipid profile	12 months
Dela Fuente 2020	RCT	Spain	236	18-80 y/o DM2	Educational program delivered by a trained practice nurse	usual care	changes in HbA1c, SBP, DBP, lipid profile	12 months and 24 months

Do Rosario Pinto 2017	RCT	Portugal	122	35-57 yo DM2 >7.5% HbA1c	Educational program delivered by doctors, nurses, psychologists, dieticians, pharmacists	usual care	Changes in HbA1c, BMI, BP, self-care activities	6 months
Edelman 2015	RCT	USA	377	>18 up DM2 w HbA1c >7.5 w/ HTN	"Tailored Case Management for Diabetes and Hypertension (T EACH-DM) study (tailored telephone-delivered behavioral nurse intervention)"	usual care	changes in HbA1c, SBP, DBP, weight, lipid profile	24 months
Eriksson 2009	RCT	Sweden	25		intensive lifestyle intervention. three-month intervention period was administered in the primary care setting and consisted of supervised exercise sessions and diet counselling, followed by regular group meetings during three years	usual care and printed materials	changes in BMI, weight, waist, self-reported physical activity, blood pressure, glucose	3 years
Gabbay 2013	RCT	USA	421	DM2 HbA1c >8.5%,	counseling by trained nurses	usual care	changes in HbA1c,	24 months

				18- <75yo			LDL, BP, QOL	
Hornst en 2005	RCT	Sweden	99	40-80 yo last 2 yrsDM2	education by diabetes experts and nurses	usual care	changes in HbA1c, lipid profile, overall well- being	12 months
Huang 2010	RCT	Taiwan	154	30-70 yo	educational program by dietitian	usual care	changes in HbA1c, FPG, SBP, DBP, lipid profile	12 months
Javaid 2019	RCT	Pakista n	244	>18 yo DM2 HbA1c >8%	educational program by pharmacist	usua l care	changes in fasting glucose, HbA1c, BP, Lipid profile	9 mon ths
Lean 2019	RCT	UK		aged 20- 65 yo DM2 within 6 years, BMI 27- 45 kg/m	Food substitution program ff by reintroduction of meals	usual care	reduction in body weight, HbA1c <6.5%	24 months
Liss 2018	RCT	USA	331	aged >18 w/ BMI $\geq$ 24 kg/m ²	educational program by wellness instructors	usual	change in weight, HbA1c SBP, DBP, lipid profile	12 months
Mash 2014	RCT	South Africa	1,570	adults w DM2	Educational program by health educators	usual	improved diabetes self-care activities, 5% weight loss and 1% reduction in HbA1c;	12 months

							(ii) changes in blood pressure, weight, waist circumfer ence, HbA1c, lipid profile and quality of life	
McDer mott 2015	RCT	Australi a	213	> 18 yo HbA1c >8.5%w/ 1 major comorbid ity	education de livered to case workers from community	usual	changes in HbA1c, BP, weight, height,lip[id profile, QOL	18 months
Mehuy s 2011	RCT	Belgium	288	>45- to <74 yo , BMI >25 kg/m2	counselling by pharmacists	usual	changes in fasting glucose, hbA1c	6 months
Moncri eft 2016	RCT	USA	111	Age >18- <70, BMI> 27 w/ deprese d symptom s	educational program by therapists	usual care	changes in weight, HbA1c, GFR	12 months
Muchiri 2015	RCT	South Africa	82	Age >40 and <70 DM2 at least 1 yr without insulin	Educational program by dietitians	usual care	change in HbA1c, BP, lipid profile, dietary behaviors	12 months
Rosal 2011	RCT		252	Age >18	Educational program by nutritioninsts and health educators	usual care	changes in HbA1c and BMI	12 months

Siaw 2017	RCT	Singapore	313	Age >21 HbA1c >7%, multiple comorbidities	educational programs by pharmacists, nurses educators and dietitians	usual care	change in HbA1c, SBP, lipid, Diabetes satisfaction questionnaire	6 months
Smith 2011	RCT	Ireland	335	Age >18	educational program by peer supporters	usual care	change in HbA1c, BP, cholesterol, BMI, well-being	24 months
Taheri 2020	RCT	Qatar	75	Age >18 and <50, DM2 at least 3 yrs BMI >27	food substitution program	usual care	changes in weight, HbA1c, proportion w/ diabetes remission	12 months
Tejada, Tayabas 2006	RCT	Mexico	75	Age >18 DM2 at least 4 yrs, w/o comorbidities	educational program by investigators	usual care	changes in HbA1c	9 months
Toober 2011	RCT	USA		Age >30 and <75. Latinos only, DM2 at least 6 mo	Mediterranean diet	usual care	self-efficacy and behavior change, BMI HbA1c	24 months
Vos 2019	RCT	Netherlands	106	Age >18 and <75 DM2 at least 3 mos, at most 5 yrs	educational program by nurse	usual care	change in BMI, HbA1c, SBP, lipid profile, medication adherence	30 months

							e, cost-effectiveness	
Wolf 2004	RCT	USA	144	Age >20 BMI >27	counselling by dietitian	usual care w/ printed materials	changes in weight, waist circumf, HbA1c, lipid profile	12 months

Q3.5. Assessment of Risk of Bias

Assessment of Risk of Bias of Included Studies

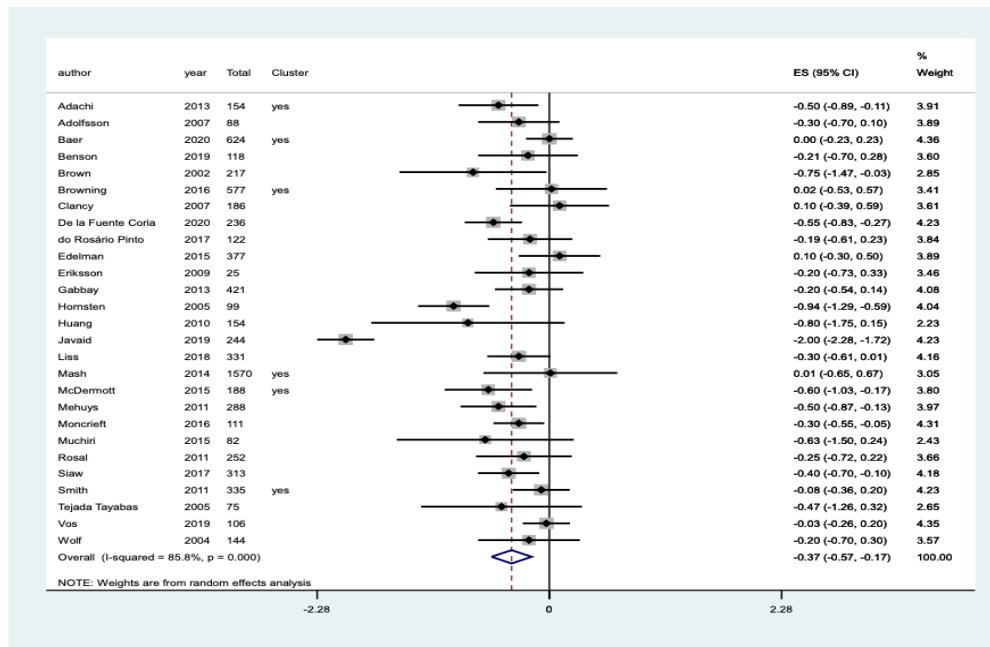
Study ID	Outcome	Randomization process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported results	Overall
Beune 2014	Blood Pressure	+	+	+	+	+	+
Hachisanoglu 2011	Blood pressure	+	?	+	+	?	+
Hochmann 2021	Blood pressure	+	?	+	+	+	+
Kastarhien 2012	Blood pressure	+	?	+	+	?	+
Ogedegbe 2014	Blood pressure	+	+	?	+	+	?
Rodriguez-Martin 2009	Blood pressure	+	?	+	+	?	?
Schoenfelder 2016	Blood pressure	+	?	+	+	+	?
Tonstad 2007	Blood pressure	+	?	+	+	+	+
Woolard 2003	Blood pressure	+	?	+	+	?	+
Adachi 2013	HbA1c	+	+	?	+	+	+
Adolfsson 2007	HbA1c	+	?	+	?	+	?
Baer 2020	HbA1c	+	+	+	+	+	+
Benson 2019	HbA1c	+	+	+	+	+	+
Brown 2002	HbA1c	+	?	+	+	+	?
Browning 2016	HbA1c	+	?	?	+	+	?
Clancy 2007	HbA1c	+	?	+	+	+	?
De la Fuente Coria 2020	HbA1c	+	?	+	+	+	+
De Rosario Pinto 2017	HbA1c	?	+	+	+	+	+
Edelman 2015	HbA1c	+	+	?	+	+	+
Erksson 2009	HbA1c	?	?	+	+	+	+
Gabbay 2013	HbA1c	?	?	+	+	+	+
Hochmann 2021	HbA1c	?	?	+	+	+	?
Hörmsten 2005	HbA1c	?	?	+	+	?	?
Huang 2010	HbA1c	+	?	+	+	?	+
Javadi 2016	HbA1c	+	?	+	+	+	+
Lean 2019	HbA1c	+	?	+	+	+	?
Liss 2018	HbA1c	+	?	+	+	+	?
Mash 2014	HbA1c	+	+	+	+	+	+
McEmott 2015	HbA1c	+	?	+	+	+	?
Mhuys 2011	HbA1c	+	?	+	+	+	+
Moncrieff 2016	HbA1c	+	?	+	+	+	?
Muchiri 2015	HbA1c	+	?	+	+	+	?
Rosal 2011	HbA1c	+	?	+	+	+	+
Saw 2017	HbA1c	+	?	+	+	+	+
Smith 2011	HbA1c	+	?	+	+	+	?
Taheri 2020	HbA1c	+	?	+	+	+	?
Tekda Teyabas 2006	HbA1c	+	?	+	+	?	?
Toobert 2011	HbA1c	?	?	+	+	+	?
Vos 2019	HbA1c	+	?	+	+	+	+
Wolf 2004	HbA1c	+	?	+	+	+	?

+ Low risk
 ? Some concerns
 - High risk

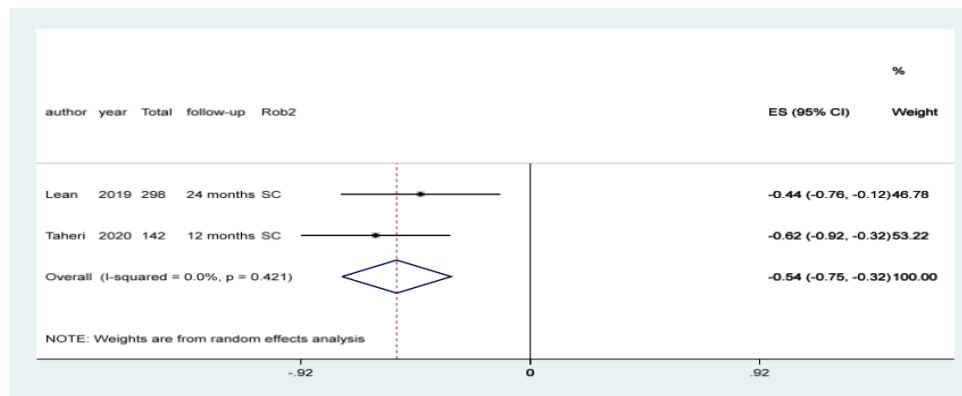
	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)
Andrews 2011	●	●	●	●	●
Barnard 2009	●	●	●	●	●
Brand 1994	●	●	●	●	●
Brehm 2009	●	●	●	●	●
Brinkworth 2004	●	●	●	●	●
Bruneirova 2007	●	●	●	●	●
Ceriello 2014	●	●	●	●	●
Coppell 2010	●	●	●	●	●
Daly 2006	●	●	●	●	●
Davis 2009	●	●	●	●	●
de Bont 1981	●	●	●	●	●
Dyson 2007	●	●	●	●	●
Ethayaw 2010	●	●	●	●	●
Esposito 2009	●	●	●	●	●
Fabricatore 2011	●	●	●	●	●
Guldbrand 2012	●	●	●	●	●
Halbrosen 1999	●	●	●	●	●
Hockaday 1978	●	●	●	●	●
Huang 2010	●	●	●	●	●
Isaai 2005	●	●	●	●	●
Itatopoulos 2011	●	●	●	●	●
Jenkins 2009	●	●	●	●	●
Jenkins 2012	●	●	●	●	●
Jenkins 2014	●	●	●	●	●
Jönsson 2009	●	●	●	●	●
Kanileva 2011	●	●	●	●	●
Kaplan 1987	●	●	●	●	●
Krebs 2012	●	●	●	●	●
Larsen 2011	●	●	●	●	●
Lea 2014	●	●	●	●	●
Lee 2016	●	●	●	●	●
Li 2016	●	●	●	●	●
Liu 2016	●	●	●	●	●
Luger 2013	●	●	●	●	●
Ma 2008	●	●	●	●	●
McLaughlin 2007	●	●	●	●	●
Milne 1994	●	●	●	●	●
Mishra 2013	●	●	●	●	●
Nicholson 1999	●	●	●	●	●
Parker 2002	●	●	●	●	●
Pedersen 2014	●	●	●	●	●
Pritchard 1996	●	●	●	●	●
Rock 2014	●	●	●	●	●
Sisilew 2014	●	●	●	●	●
Sato 2016	●	●	●	●	●
Shai 2008	●	●	●	●	●
Shige 2000	●	●	●	●	●
Stem 2004	●	●	●	●	●
Tay 2015	●	●	●	●	●
Uusitupa 1993	●	●	●	●	●
Walker 1995	●	●	●	●	●
Watson 2016	●	●	●	●	●
Westman 2008	●	●	●	●	●
Wolfever 2006	●	●	●	●	●
Wycheley 2010	●	●	●	●	●
Yamada 2014	●	●	●	●	●

Supplementary Figure S2: Risk of bias of individual trials

Q3.6. Results

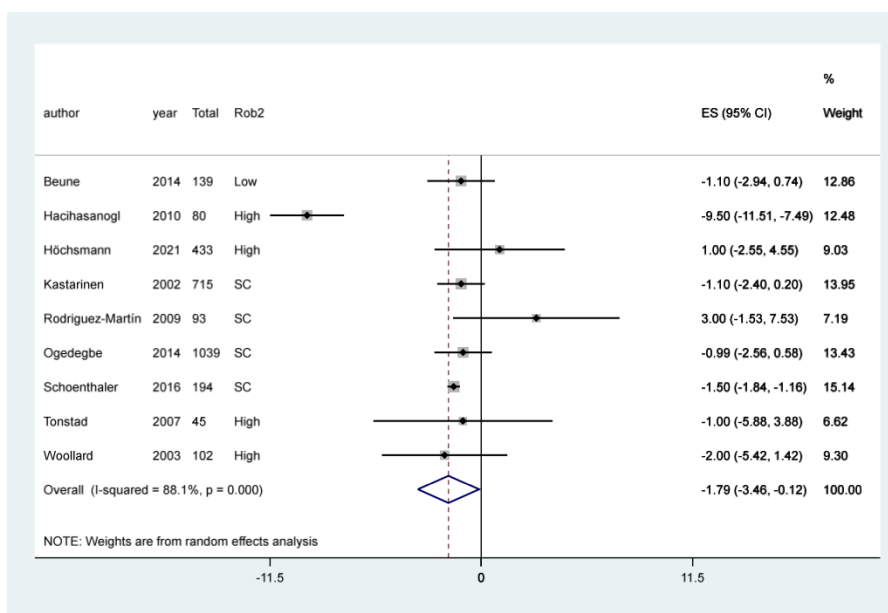


(A)

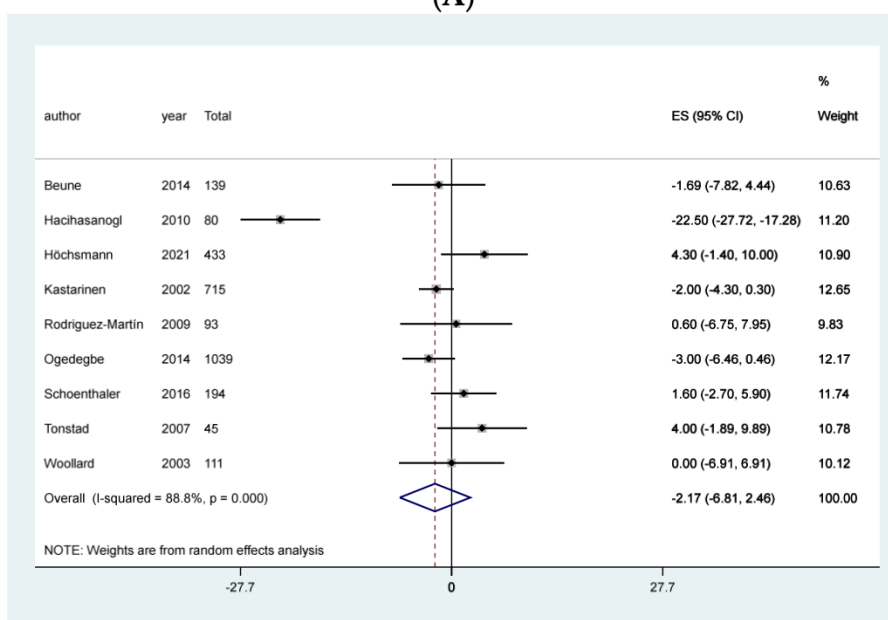


(B)

Fig 2: Metaanalysis of the effect of NT on HbA1c among patients with DM2 according to type of intervention (A.)counselling and educational programs, (B.) food replacement Galendi J, et al (1)



(A)



(B)

Fig 3 Meta-analysis of the effect of NT on blood pressure in patients diagnosed with hypertension. (A) diastolic BP, (B) Systolic BP. (1)

Network Diagram for HbA1c

Fig. 2 Network diagram for HbA1c: The size of the nodes is proportional to the total number of participants allocated to each dietary approach and the thickness of the lines proportional to the number of studies evaluating each direct comparison

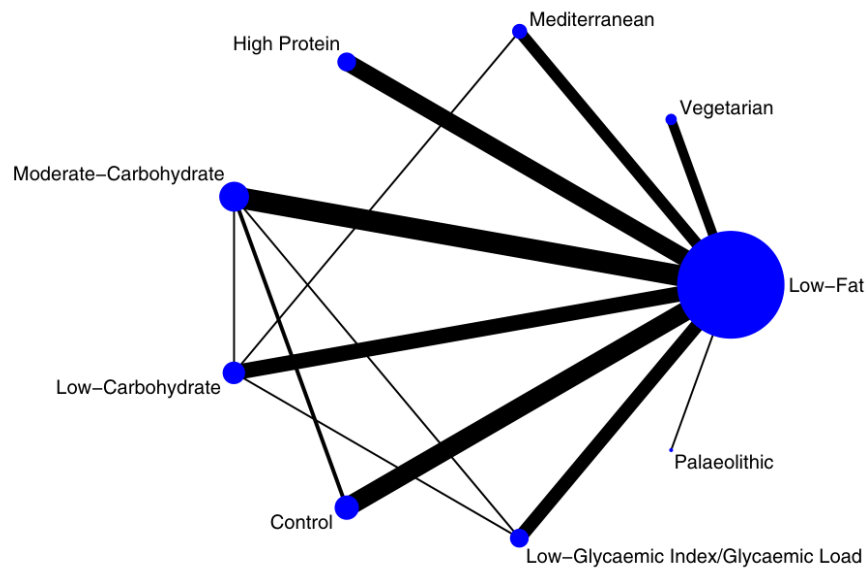


Table 6. Percentage Contribution of each direct estimate derived from direct (blue) and indirect (red) comparisons

Table 1 Percentage contribution of each direct estimate derived from direct (blue) and indirect (red) comparisons (the colour corresponds to the percentage of contribution)

HbA1c																
Low-Carb	8%	92%	0%	100%	0%	100%	2%	98%	23%	77%	0%	100%	83%	17%	0%	100%
14%	86%	Mediterranean	0%	100%	0%	100%	0%	100%	0%	100%	0%	100%	98%	2%	0%	100%
0%	100%	0%	100%	Palaeolithic	0%	100%	0%	100%	0%	100%	0%	100%	100%	0%	0%	100%
0%	100%	0%	100%	0%	100%	Vegetarian	0%	100%	0%	100%	0%	100%	100%	0%	0%	100%
27%	73%	0%	100%	0%	100%	0%	100%	Low-GI/GL	30%	70%	0%	100%	90%	10%	0%	100%
20%	80%	0%	100%	0%	100%	0%	100%	25%	75%	Moderate-Carb	0%	100%	57%	43%	19%	81%
0%	100%	0%	100%	0%	100%	0%	100%	0%	100%	0%	100%	High-Protein	100%	0%	0%	100%
57%	43%	94%	6%	100%	0%	100%	0%	87%	13%	61%	39%	100%	0%	Low-Fat	92%	8%
0%	100%	0%	100%	0%	100%	0%	100%	0%	100%	25%	75%	0%	100%	90%	10%	Control
Fasting glucose																

The values above the dietary approaches correspond to the percentage contribution of direct and indirect comparisons between the row and columns for HbA1c (e.g., the percentage contribution of direct comparisons for HbA1c between Low-Carb and Low-Fat diet is 83%, and 17% for the indirect comparisons). The values below the dietary approaches correspond to the percentage contribution of direct and indirect comparisons between the column and the rows for fasting glucose (e.g., the percentage contribution of direct comparisons for fasting glucose between Low-Carb and Low-Fat diet is 57%, and 43% for the indirect comparisons). GI/GL, glycaemic index/load. (Color table online)

Table 7. League table showing the results of the network meta-analysis comparing the effects (mean difference: MD) of all dietary approaches and 95% confidence intervals (95% CI)

HbA1c (%)								
Low-Carb [84%, 58%]	-0.03 (-0.33, 0.27)	0.05 (-0.65, 0.75)	-0.15 (-0.51, 0.21)	-0.19 (-0.45, 0.07)	-0.23 (-0.50, 0.04)	-0.33 (-0.61, -0.05)	-0.35 (-0.56, -0.14)	-0.82 (-1.11, -0.53)
0.38 (-0.32, 1.08)	Mediterranean [80%, 88%]	0.08 (-0.62, 0.78)	-0.12 (-0.48, 0.24)	-0.16 (-0.43, 0.11)	-0.20 (-0.49, 0.09)	-0.30 (-0.59, -0.01)	-0.32 (-0.53, -0.11)	-0.79 (-1.07, -0.52)
0.26 (-1.12, 1.65)	-0.11 (-1.43, 1.21)	Palaeolithic [76%, 71%]	-0.20 (-0.93, 0.52)	-0.24 (-0.93, 0.44)	-0.28 (-0.98, 0.41)	-0.38 (-1.07, 0.31)	-0.40 (-1.07, 0.27)	-0.87 (-1.56, -0.18)
0.06 (-0.81, 0.92)	-0.32 (-1.09, 0.45)	-0.21 (-1.62, 1.20)	Vegetarian [60%, 63%]	-0.04 (-0.37, 0.28)	-0.08 (-0.43, 0.27)	-0.18 (-0.52, 0.16)	-0.20 (-0.49, 0.09)	-0.67 (-1.01, -0.33)
-0.21 (-0.84, 0.43)	-0.59 (-1.13, -0.04)	-0.47 (-1.78, 0.83)	-0.27 (-1.00, 0.47)	Low-GI/GL [54%, 37%]	-0.04 (-0.28, 0.20)	-0.14 (-0.38, 0.11)	-0.16 (-0.31, 0.00)	-0.63 (-0.88, -0.38)
-0.03 (-0.68, 0.62)	-0.41 (-0.96, 0.14)	-0.30 (-1.60, 1.01)	-0.09 (-0.82, 0.64)	0.18 (-0.30, 0.66)	Moderate- Carb [46%, 57%]	-0.10 (-0.37, 0.17)	-0.12 (-0.31, 0.08)	-0.59 (-0.85, -0.32)
-0.16 (-0.88, 0.57)	-0.53 (-1.14, 0.07)	-0.42 (-1.75, 0.91)	-0.21 (-0.99, 0.56)	0.05 (-0.51, 0.61)	-0.13 (-0.69, 0.44)	High-Protein [29%, 44%]	-0.02 (-0.21, 0.17)	-0.49 (-0.76, -0.22)
-0.24 (-0.82, 0.35)	-0.61 (-1.03, -0.20)	-0.50 (-1.75, 0.75)	-0.29 (-0.93, 0.35)	-0.03 (-0.38, 0.31)	-0.20 (-0.56, 0.15)	-0.08 (-0.52, 0.36)	Low-Fat [22%, 32%]	-0.47 (-0.66, -0.28)
-1.23 (-1.91, -0.55)	-1.61 (-2.16, -1.06)	-1.50 (-2.80, -0.19)	-1.29 (-2.02, -0.55)	-1.02 (-1.52, -0.52)	-1.20 (-1.69, -0.71)	-1.07 (-1.64, -0.51)	-1.00 (-1.36, -0.63)	Control [0%, 0%]
Fasting glucose (mmol/l)								

The values above the dietary approaches correspond to the MD and 95% CI in HbA1c (%) between the row and columns (e.g., the MD in HbA1c between Low-Carb and Low-Fat diet is -0.35%). The value below the dietary approaches correspond to the MD in fasting glucose (mmol/l) between the column and the row (e.g., the MD fasting glucose between Low-Carb and Low-Fat diet is -0.24 mmol/l). The values in square brackets represent the SUCRA for HbA1c and fasting glucose (e.g., the LC diet was ranked as the best dietary approach for reducing HbA1c, SUCRA: 84%; the Mediterranean diet was ranked as the best dietary approach for reducing fasting glucose, SUCRA: 88%). GI/GL, glycaemic index/load

Q3.7. GRADE Evidence Profile

Nutrition Therapy Compared to Usual Care for Type 2 Diabetes and/or Hypertension						
Patient or population: Patients with type 2 diabetes and/or hypertension.						
Setting: Primary care						
Intervention: Nutrition therapy (counseling/educational programs in type 2 diabetes; food substitution).						
Comparison: Usual care						
Outcomes	Anticipated Absolute Effects * (95% CI)		Relative Effect (95% CI)	N of Participants (Studies)	Certainty of the Evidence (GRADE)	Comments
	Risk with Usual Care	Risk with Nutrition Therapy				
HbA1c (%) in counseling/educational programs follow-up: mean 14 months		MD 0.37 lower (0.57 lower to 0.17 lower)	-	7437 (27 RCTs)	⊕⊕⊕○ Moderate ^{a,b}	In primary care, counseling/educational programs likely result in a reduction in HbA1c (%) in type 2 diabetes patients.
HbA1c (%) in food replacement follow-up: mean 18 months		MD 0.54 lower (0.75 lower to 0.32 lower)	-	440 (2 RCTs)	⊕⊕⊕○ Moderate ^c	In primary care, food replacement likely results in reduction in HbA1c (%) in type 2 diabetes patients.
Systolic blood pressure in participants with hypertension follow-up: mean 14 months		MD 2.17 lower (6.81 lower to 2.46 higher)	-	4518 (9 RCTs)	⊕⊕○○ Low ^{d,e}	
Diastolic blood pressure in participants with hypertension follow-up: mean 14 months		MD 1.79 lower (3.46 lower to 0.12 lower)	-	2840 (9 RCTs)	⊕⊕⊕○ Moderate ^{d,f}	In primary care, counseling/educational programs likely reduce slightly diastolic blood pressure in participants with hypertension.

Table 1. Summary of findings and grading of evidence. Meta-analysis of Galendi, et al. (1)

Explanations

- All included studies were classified as having either some concerns or high risk of bias according to RoB2.
- Although the meta-analysis showed a high statistic heterogeneity (I-squared = 85.8%), it was attributed to two studies that overestimated the intervention effect. The meta-analysis without these two studies maintained the effect in favor of the intervention (supplementary file). As a result of this the quality of evidence was not rate down due to inconsistency.
- The two included studies were classified as having some concerns according to RoB2
- Most included studies were classified as having high risk of bias according to RoB2. The 95% CI overlaps no effect, but it fails to exclude important benefit.
- The summary effect crossed the line of the null effect. Although the meta-analysis showed a high statistic heterogeneity (I-squared = 88.8%), it was attributed to a study that overestimated the intervention effect. The meta-analysis without this study maintained the null effect (supplementary file). As a result of this the quality of evidence was not rate down due to inconsistency.
- Although the meta-analysis showed a high statistic heterogeneity (I-squared = 88.1%), it was attributed to a study that overestimated the intervention effect. The meta-analysis without this study maintained the effect in favor of the intervention (supplementary file). As a result of this the quality of evidence was not rate down due to inconsistency.

Table 2. GRADE Evidence of Juri K. et al meta-analysis

Author(s): Juri K. 2021

Question: Dietary education compared to no dietary counselling for HbA1c reduction

Setting: among type 2 DM patients

Bibliography:

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	dietary education	no dietary counselling	Relative (95% CI)	Absolute (95% CI)		

HbA1c reduction in 4 to 6 months (follow-up: range 4 months to 6 months)

15	randomised trials	serious ^a	serious ^b	not serious	not serious	none	1376	1366	-	mean 0.51 lower (0.71 lower to 0.32 lower)	⊕⊕ ○ ○ Low	
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HbA1c reduction (follow-up: range 10 months to 12 months)

11	randomised trials	serious ^a	serious ^b	not serious	not serious	none	946	1052	-	mean 0.42 lower (0.53 lower to 0.32 lower)	⊕⊕ ○ ○ Low	
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CI: confidence interval

Explanations

- a. lack of blinding in most studies, but lack of blinding may not affect outcome.
- b. attrition bias with incomplete outcome data

Other Data:

Table 4. Assessment of Certainty of Evidence for Network Estimate

Comparison (vs Control)	GRADE Certainty of Evidence Assessment of Studies	INDIRECT ESTIMATE ASSESSMENT					NETWORK ESTIMATE ASSESSMENT			
		Dominant Direct Estimate Loop 1	Dominant Direct Estimate Loop 2 (vs Control)	Lowest Rating of the Direct Estimates from the loop	Examine for intransitivity	Final Rating For Indirect Estimate	Rating (Contribute the most OR direct estimate OR highest between direct and indirect)	Incoherence	Imprecision	Final Rating
Low Carbohydrate (100% from indirect)	Low	Very Low (vs Low Fat)	Low	Very Low	Not serious	Very Low	Low	Only indirect evidence	Not serious	Low
Mediterranean (100% from indirect)	Low	Very Low (vs Low Fat)	Low	Very Low	Not serious	Very Low	Low	Only indirect evidence	Not serious	Low
Paleolithic (100% from indirect)	Low	Moderate (vs Low Fat)	Low	Low	Not serious	Low	Low	Only indirect evidence	Not serious	Low
Vegetarian (100% from indirect)	Low	Low (vs Low Fat)	Low	Low	Not serious	Low	Low	Only indirect evidence	Not serious	Low
Low Glycemic Index / Load (100% from indirect)	Low	Very Low (vs Low Fat)	Low	Very Low	Not serious	Very Low	Low	Only indirect evidence	Not serious	Low
Moderate Carbohydrate (81% from indirect)	Low	Low (vs Low Fat)	Low	Low	Not serious	Low	Low	Not serious	Not serious	Low
High Protein (100% from indirect)	Low	Low (vs Low Fat)	Low	Low	Not serious	Low	Low	Only indirect evidence	Not serious	Low
Low Fat (92% from direct)	Low	Low (vs Mod Carbohydrate)	Low	Low	Not serious	Low	Low	Not serious	Not serious	Low

NMA Summary of Findings Table

Estimates of effects, credible intervals and certainty of evidence for comparison of dietary interventions in patients with Type 2 Diabetes Mellitus

Patients: Patients with Type 2 Diabetes Mellitus

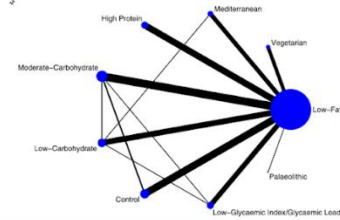
Interventions: Low Carbohydrate, Mediterranean, Paleolithic, Low Glycemic Index / Load, Moderate Carbohydrate, High Protein, Low Fat

Comparator (Reference): Control

Outcome: Glycosylated Hemoglobin (HbA1c) (%)

Setting: Outpatient

Bibliography: Schwingshackl L, Chaimani A, Hoffmann G, Schwedhelm C, Kamp B, Boeing H (2018) A network meta-analysis on the comparative efficacy of different dietary approaches on glycosylated hemoglobin in patients with type 2 diabetes mellitus. *European Journal of Epidemiology*, 33(2), 157–170. <https://doi.org/10.1007/s10654-017-0332-x>



Total Studies: 56 RCTs Total Participants: 4,937	Relative Effect (Mean Difference) (95% CrI)	Difference	Certainty of Evidence	Ranking (SUCRA) (%)	Interpretation of Findings
Low Carbohydrate No direct evidence; Indirect evidence only	-0.82 (-1.11, -0.53) Network Estimate	MD 0.82 % lower (1.11 lower to 0.53 lower)	⊕⊕○○ Low Due to study limitations ¹	1 (84%)	Probable result in moderate reduction in HbA1c
Mediterranean No direct evidence; Indirect evidence only	-0.79 (-1.07, -0.52) Network Estimate	MD 0.79 % lower (1.07 lower to 0.52 lower)	⊕⊕○○ Low Due to study limitations ¹	2 (80%)	Probable result in moderate reduction in HbA1c
Paleolithic No direct evidence; Indirect evidence only	-0.87 (-1.56, -0.18) Network Estimate	MD 0.87 % lower (1.56 lower to 0.18 lower)	⊕⊕○○ Low Due to study limitations ¹	3 (76%)	Probable result in moderate reduction in HbA1c
Vegetarian No direct evidence; Indirect evidence only	-0.67 (-1.01, -0.33) Network Estimate	MD 0.67 % lower (1.01 lower to 0.33 lower)	⊕⊕○○ Low Due to study limitations ¹	4 (60%)	Probable result in moderate reduction in HbA1c
Low Glycemic Index / Load No direct evidence; Indirect evidence only	-0.63 (-0.88, -0.38) Network Estimate	MD 0.63 % lower (0.88 lower to 0.38 lower)	⊕⊕○○ Low Due to study limitations ¹	5 (54%)	Probable result in moderate reduction in HbA1c
Moderate Carbohydrate (2 RCT; 297 participants)	-0.59 (-0.85, -0.32) Network Estimate	MD 0.59 % lower (0.85 lower to 0.32 lower)	⊕⊕○○ Low Due to study limitations ¹	6 (46%)	Probable result in moderate reduction in HbA1c
High Protein No direct evidence; Indirect evidence only	-0.49 (-0.76, -0.22) Network Estimate	MD 0.49 % lower (0.76 lower to 0.22 lower)	⊕⊕○○ Low Due to study limitations ¹	7 (29%)	Probable small reduction in HbA1c
Low Fat (8 RCT; 886 participants)	-0.47 (-0.66, -0.28) Network Estimate	MD 0.47 % lower (0.66 lower to 0.28 lower)	⊕⊕○○ Low Due to study limitations ¹	8 (22%)	Probable small reduction in HbA1c
Control	Reference Comparator	No estimate	Reference Comparator	Control (0%)	Reference Comparator
GRADE Working Group grades of evidence (or certainty in the evidence) High quality: We are very confident that the true effect lies close to that of the estimate of the effect Moderate quality: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different Low quality: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect Very low quality: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect					
Explanatory Footnotes ¹ The study has serious concern for risk of bias and publication bias.					

Annex 3.1: GRADE Evidence Profile of Pairwise comparison of the studies



Comparison	Risk of bias	Indirectness	Inconsistency	Imprecision	Publications bias	Rating
1:2	Serious concerns	Serious concerns	No serious concerns	Serious concerns	Serious concerns	Low
1:3	Serious concerns	Serious concerns	Very serious concerns	No serious concerns	Serious concerns	Very low
1:4	Serious concerns	Serious concerns	No serious concerns	Serious concerns	Serious concerns	Low
1:5	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
1:6	Serious concerns	Serious concerns	Very serious concerns	No serious concerns	Serious concerns	Very low
1:7	Serious concerns	Serious concerns	Serious concerns	No serious concerns	Serious concerns	Low
1:8	Serious concerns	Serious concerns	Very serious concerns	Serious concerns	Serious concerns	Very low
1:9	No serious concerns	Serious concerns	No serious concerns	Serious concerns	Serious concerns	Moderate
3:6	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
5:6	Serious concerns	Serious concerns	Very serious concerns	Serious concerns	Serious concerns	Very low
5:7	Serious concerns	Serious concerns	Serious concerns	No serious concerns	Serious concerns	Low
5:8	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
6:8	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
2:3	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
2:4	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
2:5	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
2:6	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low

4

2:7	Serious concerns	Serious concerns	Serious concerns	No serious concerns	Serious concerns	Low
2:8	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
2:9	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
3:4	Serious concerns	Serious concerns	Very serious concerns	No serious concerns	Serious concerns	Very low
3:5	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
3:7	Serious concerns	Serious concerns	Serious concerns	No serious concerns	Serious concerns	Low
3:8	Serious concerns	Serious concerns	Serious concerns	No serious concerns	Serious concerns	Low
3:9	No serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Moderate
4:5	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
4:6	Serious concerns	Serious concerns	Very serious concerns	Serious concerns	Serious concerns	Very low
4:7	Serious concerns	Serious concerns	Serious concerns	No serious concerns	Serious concerns	Low
4:8	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
4:9	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
5:9	No serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Moderate
6:7	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Low
6:9	No serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Moderate
7:8	Serious concerns	Serious concerns	Serious concerns	No serious concerns	Serious concerns	Low
8:9	Serious concerns	Serious concerns	Serious concerns	No serious concerns	Serious concerns	Low
8:9	No serious concerns	Serious concerns	Serious concerns	Serious concerns	Serious concerns	Moderate

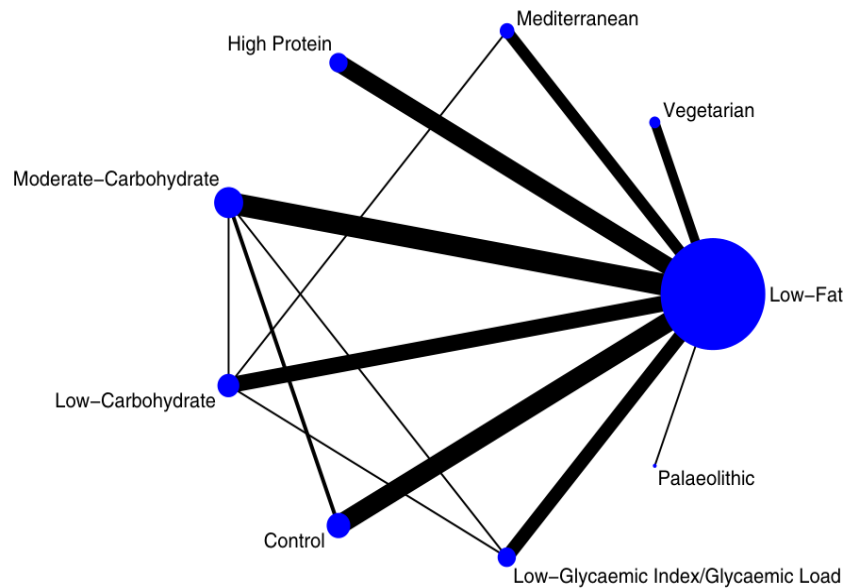
[1=Low-fat, 2=Vegetarian, 3=Mediterranean, 4=High-Protein, 5=Moderate-carbohydrate, 6=Low-carbohydrate, 7=control, 8=Low GI/GL, 9=Palaeolithic].

Annex 3.2: Assessment of Certainty of Evidence for Network Estimate

Comparison (vs Control)	GRADE Certainty of Evidence Assessment of Studies	INDIRECT ESTIMATE ASSESSMENT					NETWORK ESTIMATE ASSESSMENT			
		Dominant Direct Estimate Loop 1	Dominant Direct Estimate Loop 2 (vs Control)	Lowest Rating of the Direct Estimates from the loop	Examine for Intransitivity	Final Rating For Indirect Estimate	Rating (Contribute the most OR direct estimate OR highest between direct and indirect)	Incoherence	Imprecision	Final Rating
Low Carbohydrate (100% from indirect)	Low	Very Low (vs Low Fat)	Low	Very Low	Not serious	Very Low	Low	Only indirect evidence	Not serious	Low
Mediterranean (100% from indirect)	Low	Very Low (vs Low Fat)	Low	Very Low	Not serious	Very Low	Low	Only indirect evidence	Not serious	Low
Paleolithic (100% from indirect)	Low	Moderate (vs Low Fat)	Low	Low	Not serious	Low	Low	Only indirect evidence	Not serious	Low
Vegetarian (100% from indirect)	Low	Low (vs Low Fat)	Low	Low	Not serious	Low	Low	Only indirect evidence	Not serious	Low
Low Glycemic Index / Load (100% from indirect)	Low	Very Low (vs Low Fat)	Low	Very Low	Not serious	Very Low	Low	Only indirect evidence	Not serious	Low
Moderate Carbohydrate (91% from indirect)	Low	Low (vs Low Fat)	Low	Low	Not serious	Low	Low	Not serious	Not serious	Low
High Protein (100% from indirect)	Low	Low (vs Low Fat)	Low	Low	Not serious	Low	Low	Only indirect evidence	Not serious	Low
Low Fat (92% from direct)	Low	Low (vs Mod Carbohydrate)	Low	Low	Not serious	Low	Low	Not serious	Not serious	Low

Annex 4.1: Network Diagram

Fig. 2 Network diagram for HbA1c: The size of the nodes is proportional to the total number of participants allocated to each dietary approach and the thickness of the lines proportional to the number of studies evaluating each direct comparison



Annex 4.2: Percentage Contribution of each direct estimate derived from direct (blue) and indirect (red) comparisons

Table 1 Percentage contribution of each direct estimate derived from direct (blue) and indirect (red) comparisons (the colour corresponds to the percentage of contribution)

HbA1c																
Low-Carb	8%	92%	0%	100%	0%	100%	2%	98%	23%	77%	0%	100%	83%	17%	0%	100%
14%	86%	Mediterranean	0%	100%	0%	100%	0%	100%	0%	100%	0%	100%	98%	2%	0%	100%
0%	100%	0%	100%	Palaeolithic	0%	100%	0%	100%	0%	100%	0%	100%	100%	0%	0%	100%
0%	100%	0%	100%	0%	100%	Vegetarian	0%	100%	0%	100%	0%	100%	100%	0%	0%	100%
27%	73%	0%	100%	0%	100%	0%	100%	Low-GI/GL	30%	70%	0%	100%	90%	10%	0%	100%
20%	80%	0%	100%	0%	100%	0%	100%	25%	75%	Moderate-Carb	0%	100%	57%	43%	19%	81%
0%	100%	0%	100%	0%	100%	0%	100%	0%	100%	0%	100%	High-Protein	100%	0%	0%	100%
57%	43%	94%	6%	100%	0%	100%	0%	87%	13%	61%	39%	100%	0%	Low-Fat	92%	8%
0%	100%	0%	100%	0%	100%	0%	100%	0%	100%	25%	75%	0%	100%	90%	10%	Control
Fasting glucose																

The values above the dietary approaches correspond to the percentage contribution of direct and indirect comparisons between the row and columns for HbA1c (e.g., the percentage contribution of direct comparisons for HbA1c between Low-Carb and Low-Fat diet is 83%, and 17% for the indirect comparisons). The values below the dietary approaches correspond to the percentage contribution of direct and indirect comparisons between the column and the rows for fasting glucose (e.g., the percentage contribution of direct comparisons for fasting glucose between Low-Carb and Low-Fat diet is 57%, and 43% for the indirect comparisons). GI/GL, glycaemic index/load. (Color table online)

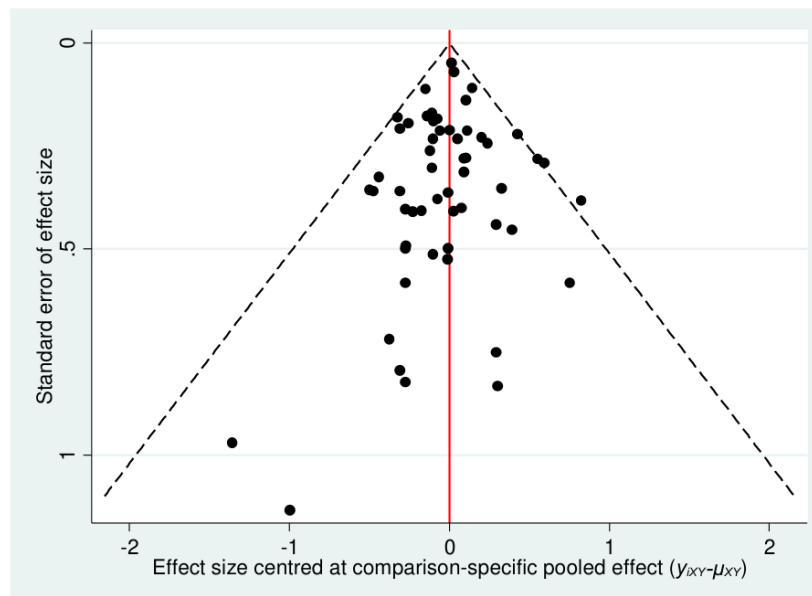
Annex 4.3. League table showing the results of the network meta-analysis comparing the effects (mean difference: MD) of all dietary approaches and 95% confidence intervals (95% CI)

Table 2 League table showing the results of the network meta-analysis comparing the effects (mean difference: MD) of all dietary approaches and 95% confidence intervals (95% CI)

HbA1c (%)								
Low-Carb [84%, 58%]	-0.03 (-0.33, 0.27)	0.05 (-0.65, 0.75)	-0.15 (-0.51, 0.21)	-0.19 (-0.45, 0.07)	-0.23 (-0.50, 0.04)	-0.33 (-0.61, -0.05)	-0.35 (-0.56, -0.14)	-0.82 (-1.11, -0.53)
0.38 (-0.32, 1.08)	Mediterranean [80%, 88%]	0.08 (-0.62, 0.78)	-0.12 (-0.48, 0.24)	-0.16 (-0.43, 0.11)	-0.20 (-0.49, 0.09)	-0.30 (-0.59, -0.01)	-0.32 (-0.53, -0.11)	-0.79 (-1.07, -0.52)
0.26 (-1.12, 1.65)	-0.11 (-1.43, 1.21)	Palaeolithic [76%, 71%]	-0.20 (-0.93, 0.52)	-0.24 (-0.93, 0.44)	-0.28 (-0.98, 0.41)	-0.38 (-1.07, 0.31)	-0.40 (-1.07, 0.27)	-0.87 (-1.56, -0.18)
0.06 (-0.81, 0.92)	-0.32 (-1.09, 0.45)	-0.21 (-1.62, 1.20)	Vegetarian [60%, 63%]	-0.04 (-0.37, 0.28)	-0.08 (-0.43, 0.27)	-0.18 (-0.52, 0.16)	-0.20 (-0.49, 0.09)	-0.67 (-1.01, -0.33)
-0.21 (-0.84, 0.43)	-0.59 (-1.13, -0.04)	-0.47 (-1.78, 0.83)	-0.27 (-1.00, 0.47)	Low-GI/GL [54%, 37%]	-0.04 (-0.28, 0.20)	-0.14 (-0.38, 0.11)	-0.16 (-0.31, 0.00)	-0.63 (-0.88, -0.38)
-0.03 (-0.68, 0.62)	-0.41 (-0.96, 0.14)	-0.30 (-1.60, 1.01)	-0.09 (-0.82, 0.64)	0.18 (-0.30, 0.66)	Moderate- Carb [46%, 57%]	-0.10 (-0.37, 0.17)	-0.12 (-0.31, 0.08)	-0.59 (-0.85, -0.32)
-0.16 (-0.88, 0.57)	-0.53 (-1.14, 0.07)	-0.42 (-1.75, 0.91)	-0.21 (-0.99, 0.56)	0.05 (-0.51, 0.61)	-0.13 (-0.69, 0.44)	High-Protein [29%, 44%]	-0.02 (-0.21, 0.17)	-0.49 (-0.76, -0.22)
-0.24 (-0.82, 0.35)	-0.61 (-1.03, -0.20)	-0.50 (-1.75, 0.75)	-0.29 (-0.93, 0.35)	-0.03 (-0.38, 0.31)	-0.20 (-0.56, 0.15)	-0.08 (-0.52, 0.36)	Low-Fat [22%, 32%]	-0.47 (-0.66, -0.28)
-1.23 (-1.91, -0.55)	-1.61 (-2.16, -1.06)	-1.50 (-2.80, -0.19)	-1.29 (-2.02, -0.55)	-1.02 (-1.52, -0.52)	-1.20 (-1.69, -0.71)	-1.07 (-1.64, -0.51)	-1.00 (-1.36, -0.63)	Control [0%, 0%]
Fasting glucose (mmol/l)								

The values above the dietary approaches correspond to the MD and 95% CI in HbA1c (%) between the row and columns (e.g., the MD in HbA1c between Low-Carb and Low-Fat diet is -0.35%). The value below the dietary approaches correspond to the MD in fasting glucose (mmol/l) between the column and the row (e.g., the MD fasting glucose between Low-Carb and Low-Fat diet is -0.24 mmol/l). The values in square brackets represent the SUCRA for HbA1c and fasting glucose (e.g., the LC diet was ranked as the best dietary approach for reducing HbA1c, SUCRA: 84%; the Mediterranean diet was ranked as the best dietary approach for reducing fasting glucose, SUCRA: 88%). GI/GL, glycaemic index/load

Annex 5: Funnel Plot for Publication Bias



Supplementary Figure S14: Comparison-adjusted funnel plot for HbA1c involving all studies comparing all low-fat diets vs. all other dietary approaches.

Q3.8. Recommendations from other groups

Group/ Guidelines	Recommendation	Strength of Recommendation and Certainty of Evidence
ADA Nutrition Therapy Consensus Report 2019	• Advice from a dietitian with comprehensive knowledge in diabetes care • An individualized diabetes-specific medical nutrition therapy should be done upon diagnosis and throughout the lifespan depending on the changing health status • Diabetes-focused MNT provided by a specially trained dietitian is a covered Medicare benefit • The Mediterranean-style eating pattern is recommended as “healthy eating” for glycemic control and CVD risk reduction. Reducing overall carbohydrate intake can improve glycemia. • People with diabetes should consume at least 14 g of fiber per 1,000 kcal with at least half of grain consumption being whole, intact grains. • Health care providers should also focus on minimizing added sugars and refined grains and choose whole foods over highly processed foods.	
ADA 2023 Guideline	• An individualized MNT program is needed to achieve treatment goals, provided by a registered dietitian with comprehensive knowledge in diabetes care • There is no ideal macronutrient pattern for people with diabetes but meal plans should be individualized • A variety of eating patterns can be considered for the management of diabetes	A E B B

	- Reducing overall carbohydrate intake demonstrated the most evidence for reducing glycemia among individuals with diabetes. Emphasis on nutrient-dense carbohydrate sources high in fiber (14g fiber/1000 kcal) and minimally processed - An eating plan emphasizing on elements of the Mediterranean eating pattern rich in MUFA and PUFA may improve glucose metabolism and cardiovascular disease risk.	B
European recommendations for dietary management of diabetes, 2023	- Minimally processed whole grains are recommended to improve blood glucose control, CV risk factors. - Legumes are recommended to improve blood glucose control - Nuts and seeds are recommended to improve blood glucose and other cardiometabolic risk factors	Moderate certainty Low Moderate

Guideline Question 4.
Should we recommend physical activity and exercise for glucose control, weight management and over-all cardiovascular health among adult individuals with Type 2 DM?

Q4.1. Methods

A systematic search was conducted using MEDLINE, the Cochrane Library, and HERDIN Plus utilizing the keywords “physical activity,” “exercise,” and “type 2 diabetes mellitus” from database inception until March 31, 2024, and was subsequently updated to include relevant studies published through September 2025. The search was restricted to randomized controlled trials using appropriate database filters. All included studies were appraised for risk of bias using the Cochrane Risk of Bias assessment criteria. Only randomized controlled trials that compared physical activity or exercise interventions with usual or standard care among patients with type 2 diabetes mellitus were included. To ensure methodological consistency, only full-text published articles written in English were considered.

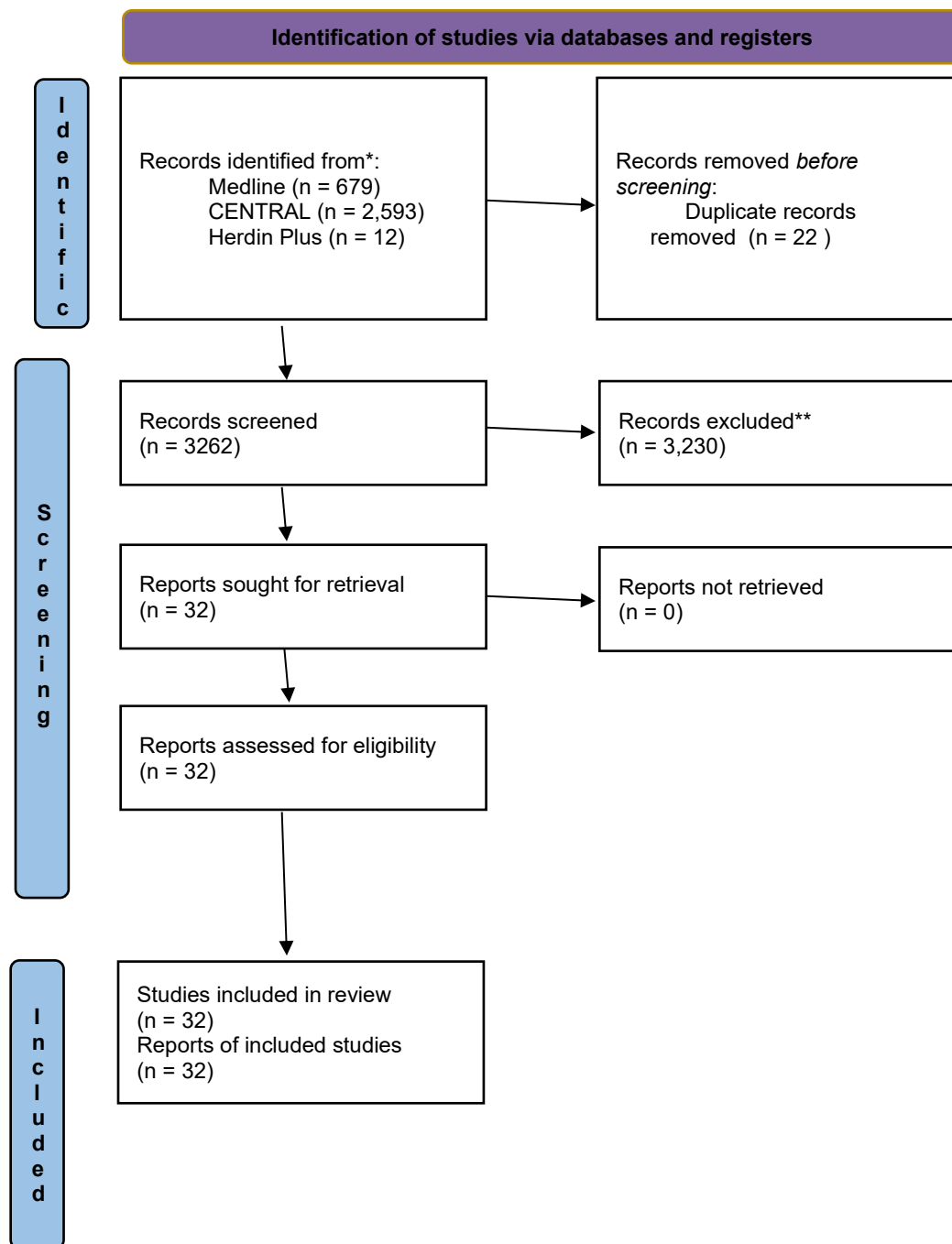
Q4.2. Search strategy

Search strategy including dates and database

DATABASE	SEARCH STRATEGY / SEARCH TERMS	DATE AND TIME OF SEARCH	RESULTS	
			Yield	Eligible
Medline	(("physical activity"[Title/Abstract] OR "exercise"[Title/Abstract]) AND "type 2 diabetes mellitus"[Title/Abstract]) AND (randomizedcontrolledtrial[Filter])	March 31, 2024	679	19
CENTRAL	("physical activity" OR "exercise") AND "type 2 diabetes mellitus"	March 31, 2024	2593	2
Herdin Plus	exercise OR physical activity AND diabetes	March 31, 2024	12	0

Q4.3. PRISMA flow diagram

PRISMA Flow Diagram



Q4.4. Characteristics of the Included Studies

Characteristics of Included Studies

Study no.	Reference	Participant information	Groups	Exercise frequency/duration/volume/intensity/mode	Progression	Adverse events	P value ^a	Conclusion
1	Abdelbasset et al. (2020) [30]	Sample size = 48 G1 n = 16, G2 n = 16, G3 n = 16 Sex: M = 27, F = 21 Age = 40–60 years	G1. MIC G2. HIIT G3. Control	40 min exercise 3 times/week for 8 weeks	Not reported	Not reported	< 0.05	Both MIC and HIIT groups showed a significant decrease in BMI, IHTG, visceral lipids, ALT, HbA1c, and lipid profile
2	Brinkmann et al. (2019) [38]	Sample size = 30 G1 n = 15, G2 n = 15 Sex: M = 19, F = 11 Age = 60 ± 8 years	G1. Training in the overnight-fasted state G2. Training in the fed state	30 min combined endurance/strength training program 3 times/week for 8 weeks	Not reported	Not reported	0.001	Exercising in either the fasted or fed state resulted in improvements in the subjects' physical fitness, body composition, glucose, and fasting triglyceride values
3	Dasgupta et al. (2017) [29]	Sample size = 347 G1 n = 174, G2 n = 173 Sex: M = 157, F = 190 Age = 60 ± 11 years	G1. 3000 steps/day over a year G2. 30–60 min of activity daily	3000 steps/day over 1 year, 30–60 min of activity daily	Number of steps increased with each visit	Not reported	< 0.05	HbA1c and insulin sensitivity significantly improved
4	Magalhães et al. (2019) [47]	Sample size = 80 G1 n = 28, G2 n = 25, G3 n = 27 Sex: M = 38, F = 42 Age = 58.5 ± 7.7 years	G1. MCT with RT G2. HIIT with RT G3. Control group	5.0 ± 7.1 min MCT group, 33.1 ± 6.4 min HIIT group 3 times/wk	Exercise sessions were upgraded every 3 months for VO2 max	Not reported	> 0.05	No significant effects on glycemic control were reported, but the combination of MCT with RT improved body composition and CRF
Study no.	Reference	Participant information	Groups	Exercise frequency/duration/volume/intensity/mode	Progression	Adverse events	P value ^a	Conclusion
5	Vanroy et al. (2017) [39]	Sample size = 48, G1 n = 27, G2 n = 21 Sex: M = 27, F = 21 Age = 59.4 ± 8.2 years	G1. IC G2. CC	60 min walking, aerobic and strength training for 8 weeks	Not reported	Not reported	< 0.05	HbA1c levels decreased significantly
6	Winding et al. (2018) [37]	Sample size = 32 G1 n = 12, G2 n = 13, G3 n = 7 Sex: M = 19, F = 13 Age = 58 ± 8 years	G1. END G2. HIIT G3. Control group	40 min END or 20 min HIIT, 3 days/wk	Not reported	Not reported	< 0.05	Fasting glucose, HbA1c levels, and glycemic variability were reduced in the HIIT group
7	Halvari et al. (2017) [31]	Sample size = 108 G1 n = 57, G2 n = 51 Sex: M = 88, F = 20 Age = 63.1 ± 7.9 years	G1. Physical activity G2. Non-physical activity	60 min walking, swimming, bicycling, resistance training for 2 days/wk	Not reported	Not reported	0.001	Physical activity intervention was effective in decreasing glucose levels relative to a non-PA control group
8	Hangping et al. (2019) [40]	Sample size = 300 G1 n = 200, G2 n = 100 Sex: M = 122, F = 178 Age = 50–70 Years	G1. BioDensity resistance exercise G2. Control group	5–10 min for 1 day/wk	Not reported	Not reported	< 0.05	BioDensity resistance training device lowered fasting plasma glucose and HbA1c
9	Heiskanen et al. (2017) [32]	Sample size = 26 G1 n = 13, G2 n = 13 Sex: M = 16, F = 10 Age = 40–55 Years	G1. SIT G2. MICT	6 sessions in 2 weeks, SIT: 4–6 × 30 s of all-out cycling, MICT: 40–60 min at the intensity of 60% of peak workload	Bouts and duration were increased with each session	Not reported	0.002	Fasting glucose, insulin, and 2 h glucose remained unaltered, but HbA1c decreased after training

Study no.	Reference	Participant information	Groups	Exercise frequency/duration/volume/intensity/mode	Progression	Adverse events	P value ^a	Conclusion
10	Akinci et al. (2018) [27]	Sample size = 65 G1 n = 22, G2 n = 21, G3 n = 22 Sex: M = 18, F = 47 Age = 40–65 years	G1. Control group G2. Supervised exercise group G3. Internet-based exercise	50–60 min of physical activity, exercise, aerobic and resistance 3 days/wk for 8 weeks	Not reported	Not reported	0.003	Supervised group-based and Internet-based exercise can similarly improve glycemic control, waist circumference, and quality of life
11	Alonso-Domínguez et al. (2019) [46]	Sample size = 204 G1 n = 102, G2 n = 102 Sex: M = 111, F = 93 Age = 25–70 years	G1. Control G2. Intervention (food workshop, 5 walks, smartphone application)	Once a week for 5 weeks, 4 km walk	Not reported	Not reported	0.241	Improvements in postprandial glucose, waist circumference, and systolic BP were noted, but they did not reach statistical significance ($P < 0.05$)
12	Di Loreto et al. (2005) [44]	Sample size = 182 G1 n = 28, G2 n = 27, G3 n = 31, G4 n = 27, G5 n = 32, G6 n = 34 Sex: M = 88, F = 94 Age = 62 ± 0.7 years	Counseling session with at least 30 min of physical activity, followed by 15-min sessions every 3 months for 2 years	Physical activity	Not reported	Not reported	< 0.05	Bodyweight, waist circumference, heart rate, fasting plasma glucose, serum LDL, HDL cholesterol, and HbA1c were improved by the intervention
13	Church et al. (2010) [41]	Sample size = 262 G1 n = 41, G2 n = 73, G3 n = 72, G4 n = 76 Sex: M = 97, F = 165 Age = 55.8 years	G1. Control G2. Resistance exercise G3. Aerobic exercise G4. Both resistance and aerobic exercise	Resistance training for 3 days/wk	During weeks 12 and 24, exercise dose was reduced by one-third	Not reported	0.03	A combination of aerobic and resistance training improved HbA1c levels
Study no.	Reference	Participant information	Groups	Exercise frequency/duration/volume/intensity/mode	Progression	Adverse events	P value ^a	Conclusion
14	Balducci et al. (2012) [52]	Sample size = 606 G1 n = 303, G2 n = 303 Sex: M = 351, F = 254 Age = 58.8 years	G1. Exercise + counselling group G2. Counselling alone	Exercise 150 min/wk in 2 sessions of mixed aerobic and resistance	Not reported	Not reported	0.002	Physical activity improved HbA1c, BMI, and waist circumference in subjects with type 2 diabetes
15	Castaneda et al. (2002) [53]	Sample size = 62 G1 n = 31, G2 n = 31 Sex: M = 22, F = 40 Age = 66 ± 8 years	G1. PRT (progressive resistance training) G2. Control group	45 min session for 3 days/wk	Not reported	3 incidents of chest pain in subjects with CAD	< 0.05	PRT was effective at improving glycemic control and some of the abnormalities associated with metabolic syndrome in type 2 diabetes adults
16	Earnest et al. (2014) [45]	Sample size = 262 G1 n = 41, G2 n = 72, G3 n = 73, G4 n = 76 Sex: M = 97, F = 165 Age = 30–75 years	G1. Control G2. Aerobic G3. Resistance training G4. Aerobic + resistance training	Treadmill for 3–5 days/wk, 10–12 repetitions; AER + RES = 2 sessions of RES/wk, 10–12 repetitions	Not reported	Not reported	< 0.02	Aerobic and aerobic plus resistance training significantly improved metabolic syndrome scores in patients with T2DM
17	Andrews et al. (2011) [26]	Sample size = 593 G1 n = 99, G2 n = 248, G3 n = 246 Sex: M = 385, F = 208 Age = 30–80 years	G1. Control G2. Intensive dietary support G3. Intensive dietary support and activity	30 min brisk walk 5 days/wk	Increased over 5 weeks then maintained	Not reported	0.013	An intensive diet intervention soon after diagnosis improved glycaemic control; the addition of an activity intervention conferred no additional benefit
18	Cox et al. (2016) [33]	Sample size = 47 G1 n = 21, G2 n = 18 Sex: M = 18, F = 21 Age = 24–80 years	G1. Routine care G2. GEM intervention	Routine activities of daily living and food choices	Not reported	Not reported	0.01	GEM intervention led to significant improvements in HbA1c

Study no.	Reference	Participant information	Groups	Exercise frequency/duration/volume/intensity/mode	Progression	Adverse events	P value ^a	Conclusion
19	Hemmati (2017) [60]	Sample size = 90 G1 n = 30, G2 n = 30, G3 n = 30 Sex: M = 51, F = 39 Age = 50.6 years	G1. FTF G2. TEG G3. CTG	30 min of moderate-to-vigorous intensity aerobic exercise	Initially twice/month then once/month for 2 months	Not reported	0.21	Improvements in BMI, HbA1c, FBS, cholesterol, and triglyceride were noted, but were not statistically significant
20	Hsieh et al. (2018) [48]	Sample size = 30 G1 n = 15, G2 n = 15 Sex: M = 11, F = 19 Age = 35–65 years	G1. RTA G2. CG	RT 2 times/wk, walking or cycling (60–80% aerobic capacity) 5 times/wk	Not reported	Not reported	0.407	Supervised RT exercises improved waist circumference, fasting glucose levels, and peak diastolic blood pressure; no significant changes in overall QoL
21	Miyamoto et al. (2017) [42]	Sample size = 32 G1 n = 10, G2 n = 12, G3 n = 10 Age = 60.0 ± 3.1 years	G1. LPA G2. NLPA G3. Control group	Wear triaxial accelerometer and increase LPA by verbal instruction from PT for 12 weeks	Not reported	Not reported	0.684	There were marked improvements in both HbA1c and insulin sensitivity
22	Mensberg et al. (2016) [36]	Sample size = 34 G1 n = 18, G2 n = 16 Sex: F = 10, M = 24 Age = 56.5 years	G1. GLP-1RA G2. Placebo group (PG)	Supervise spinning and resistance exercises 1 h 3 times/wk for 16 weeks	Not reported	Not reported	< 0.001	A combination of supervised exercise training and liraglutide induced major improvements in glycemic control
Study no.	Reference	Participant information	Groups	Exercise frequency/duration/volume/intensity/mode	Progression	Adverse events	P value ^a	Conclusion
23	Jibril (2017) [35]	Sample size = 60 G1 n = 30, G2 n = 30 Sex: M = 28, F = 32 Age = 65.3 ± 8 years	G1. RE G2. CG	RE for 12 weeks of 20 min RE thrice or three times/wk	Not reported	Not reported	> 0.05	The PA intervention as a whole did not produce the expected effects in patients with T2DM, but time effects suggest that certain aspects are effective
24	Shakil-ur-Rehman et al. (2018) [51]	Sample size = 102 G1 n = 51, G2 n = 51 Sex: M = 55, F = 47 Age = 58 ± 8 years	G1. SSAET G2. CG	25 weeks of SSAET exercises, routine medication, and dietary plan	Not reported	Not reported	P < 0.05	SSAET was equally effective, irrespective of gender, as compared to the nonexercise control group in T2DM patients
25	Sreedevi et al. (2017) [34]	Sample size = 124 G1 n = 41, G2 n = 42, G3 n = 41 Age = 30–65 years	G1. YG G2. PIG G3. CG	Yoga for 60 min 2 days/wk, peer mentorship of 13–14 T2DM patients with 45- to 60-min educational class on diabetes/wk	Not reported	Not reported	0.5	A longer follow-up is needed to observe the effects of yoga and peer mentorship on glycemic control
26	Cassidy et al. (2019) [57]	Sample size = 22 G1 n = 11, G2 n = 11 Sex: M = 17, F = 5 Age = 60 ± 2 years	G1. HIIT G2. Control group	36 cycle sessions over 12 weeks, 3 sessions/wk	High-intensity intervals increased by 10 s each week	Not reported	0.03	HIIT improved glycemic control
27	Dixit et al. (2017) [28]	Sample size = 87 G1 n = 47, G2 n = 40 Sex: M = 53, F = 34 Age = 50 years	G1. Moderate-intensity aerobic exercise G2. Control group	Moderate-intensity aerobic exercise for 8 weeks	Not reported	Not reported	< 0.001	Moderate-intensity aerobic exercise enhanced glycemic control

Study no.	Reference	Participant information	Groups	Exercise frequency/duration/volume/intensity/mode	Progression	Adverse events	P value ^a	Conclusion
28	Fayh et al. (2018) [49]	Sample size = 30 G1 n = 15, G2 n = 15 Sex: M = 12, F = 18 Age = 50.57 ± 6.38 years	G1. Placebo plus exercise G2. n-3 PUFA capsules plus exercise	High-intensity exercise, 30 min, two sessions during the study	Exercise until exhaustion	Not reported	0.752	PUFA n-3 supplementation reduced triglyceride as well as TRAP levels after exercise, but its effects on HbA1c, inflammatory, and oxidative stress markers were not significant
29	Francois et al. (2018) [55]	Sample size = 53 G1 n = 18, G2 n = 16, G3 n = 19 Sex: M = 19, F = 34 Age = 56 ± 9 years	G1. Milk + HIIT G2. Macronutrient control + HIIT G3. Placebo + HIIT	12 weeks of cardio and resistance-based HIIT thrice weekly	Not reported	Not reported	< 0.01	HIIT strongly improved HbA1c and cardiovascular risk factors, so there is potential for reduced cardiovascular mortality and morbidity
30	Bock et al. (2019) [43]	Sample size = 48 G1 n = 24, G2 n = 24 Sex: M = 18, F = 30 Age = 32–74 years	G1. Lyengar yoga G2. Program of standard exercise	Two 60-min sessions weekly for 12 weeks, moderate-intensity exercise	Not reported	Musculoskeletal pain observed	< 0.05	Yoga intervention produced improvements in blood glucose and psychosocial measures of diabetes management
31	Wycherley et al. (2014) [54]	Sample size = 84 G1 n = 33, G2 n = 51 Sex: M = 49, F = 35 Age = 55.6	G1. Calorie-restricted diet CRD G2. CRD + exercise	Moderate energy-restricted diet for 16 weeks, 8 weeks resistance exercise 3 days/wk	Not reported	Not reported	0.04	Lifestyle modification and participation in structured weight-loss programs were beneficial for the management of type 2 diabetes
32	Zhang et al. (2017) [56]	Sample size = 32 G1 n = 17, G2 n = 15 Sex: M = 15, F = 17	G1. Exercise group G2. CG	60-min aerobic bicycle training 5 times/wk for 12 weeks	Not reported	Not reported	0.099	Aerobic exercise significantly decreased serum fetuin-A levels in type 2 diabetes mellitus. There was no significant difference in fasting glucose, OGTT 2-h glucose, HbA1c, fasting insulin, and HOMA-IR levels between the exercise and control groups

Table shows the characteristics of the studies included in this systematic and meta-analysis review

T2DM type 2 diabetes mellitus, *HIIT* high-intensity interval training, *RT* resistance training, *MCT* continuous moderate-intensity training, *END* endurance, *MIC* moderate-intensity continuous, *SIT* sprint interval training, *MICT* moderate-intensity continuous training, *wk* week, *M* male, *F* female, *n* sample size, *FTF* face-to-face group, *TEG* telephonic education group, *CTG* control group, *RTG* resistance training group, *GLP-1RA* glucagon-like peptide-1 receptor agonist, *RE* rebound exercise, *SSAE* supervised structured aerobic exercise training, *YG* yoga group, *PIG* peer intervention group, *CRD* calorie-restricted diet, *PA* physical activity, *LPA* locomotive physical activity, *NPLA* nonlocomotive physical activity, *TRAP* total reactive antioxidant potential

^a P values are only for glycemic control variables (HbA1c)

Q4.5. Assessment of Risk of Bias

Risk of bias graph

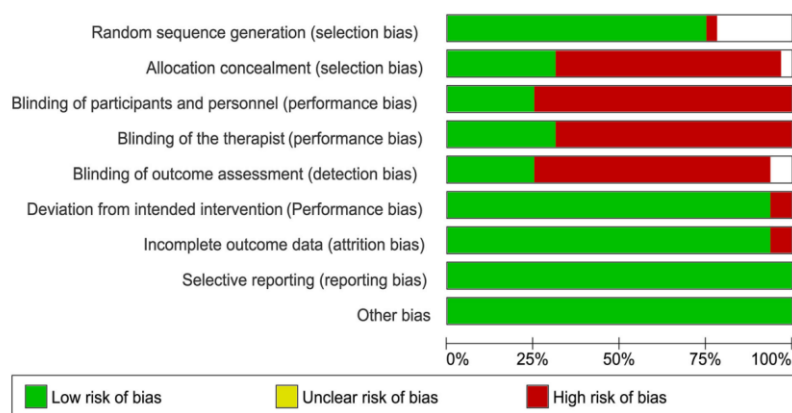


Fig. 2 Risk of bias graph. The review authors' judgments about each risk of bias item are presented as percentages across all included studies

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of the therapist (performance bias)	Blinding of outcome assessment (detection bias)	Deviation from intended intervention (Performance bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Abdelbasset, Walid Kamal, et al. 2020	●	●	●	●	●	●	●	●	●
Akinci, Buket, et al. 2018	●	●	●	●	●	●	●	●	●
Alonso-Dominguez, Rosario, et al. 2019	●	●	●	●	●	●	●	●	●
Andrews, R. C., et al. 2011	●	●	●	●	●	●	●	●	●
Balducci, Stefano, et al. 2012	●	●	●	●	●	●	●	●	●
Bock, B. C., et al. 2019	●	●	●	●	●	●	●	●	●
Brinkmann, Christian, et al. 2019	●	●	●	●	●	●	●	●	●
Cassidy, Sophie, et al. 2019	●	●	●	●	●	●	●	●	●
Castaneda, Carmen, et al. 2002	●	●	●	●	●	●	●	●	●
Church, Timothy S., et al. 2010	●	●	●	●	●	●	●	●	●
Cox, Daniel J., et al. 2016	●	●	●	●	●	●	●	●	●
Dasgupta, Kaberi, et al. 2017	●	●	●	●	●	●	●	●	●
Di Loreto, Chiara, et al. 2005	●	●	●	●	●	●	●	●	●
Dixit, Snehil. 2017	●	●	●	●	●	●	●	●	●
Earnest, Conrad P., et al. 2014	●	●	●	●	●	●	●	●	●
Fayh, Ana Paula Trussardi, et al. 2018	●	●	●	●	●	●	●	●	●
Francois, Monique E., et al. 2018	●	●	●	●	●	●	●	●	●
Halvari, Hallgeir, et al. 2017	●	●	●	●	●	●	●	●	●
Hangping, Zheng, et al. 2019	●	●	●	●	●	●	●	●	●
Heiskanen, Marja A., et al. 2017	●	●	●	●	●	●	●	●	●
Hemmati Maslarpak. 2017	●	●	●	●	●	●	●	●	●
Hsieh, Ping-Lun, et al. 2018	●	●	●	●	●	●	●	●	●
Jibril M. 2017	●	●	●	●	●	●	●	●	●
Magalhães, João P., et al. 2019	●	●	●	●	●	●	●	●	●
Mensberg, Pernille, et al. 2016	●	●	●	●	●	●	●	●	●
Miyamoto, Toshiaki, et al. 2017	●	●	●	●	●	●	●	●	●
Shakil-ur-Rehman, Syed, et al. 2018	●	●	●	●	●	●	●	●	●
Sreedevi, Aswathy, et al. 2017	●	●	●	●	●	●	●	●	●
Vanroy, Jari, et al. 2017	●	●	●	●	●	●	●	●	●
Winding, Kamilla M., et al. 2018	●	●	●	●	●	●	●	●	●
Wycherley, Thomas Philip, et al. 2014	●	●	●	●	●	●	●	●	●
Zhang, Ling-Yun, et al. 2018	●	●	●	●	●	●	●	●	●

Fig. 3 Risk of bias summary table that presents the authors' judgments about each risk of bias item for each included study

Q4.6. Results

Forest plots

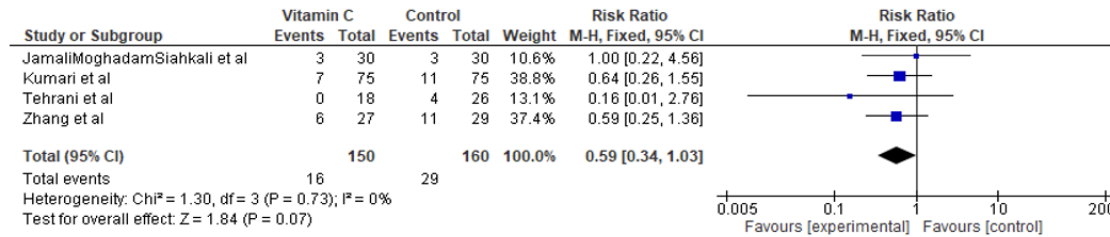
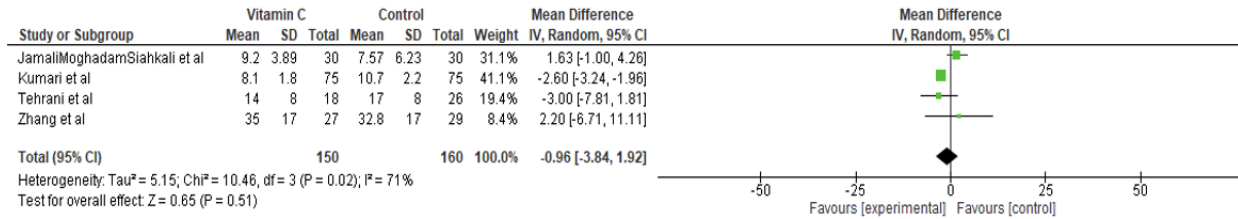


Figure 1. Mortality Outcome



Q4.7. GRADE Evidence Profile

GRADE evidence profile

Question: Exercise compared to no exercise for reduction of HbA1c and weight

Setting: among type 2 DM patients

Bibliography: 3. Shah SZA, Karam JA, Zeb A, Ullah R, Shah A, Haq IU, Ali I, Darain H, Chen H. Movement is Improvement: The Therapeutic Effects of Exercise and General Physical Activity on Glycemic Control in Patients with Type 2 Diabetes Mellitus: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. Diabetes Ther. 2021 Mar;12(3):707-732. doi: 10.1007/s13300-021-01005-1. Epub 2021 Feb 5. PMID: 33547579; PMCID: PMC7947168.

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	exercise	no exercise	Relative (95% CI)	Absolute (95% CI)		

HbA1c reduction

26	randomised trials	not serious	serious ^a	not serious	not serious	none	1324	1132	-	mean 0.34 lower (0.5 lower to 0.19 lower)	⊕⊕⊕○ Moderate	
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weight reduction (assessed with: kg)

11	randomised trials	not serious	serious ^b	not serious	not serious	none	872	725	-	mean 0.73 kg lower (1.62 lower to 0.16 higher)	⊕⊕⊕○ Moderate	
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reduction in waist circumference (Scale from: 0.07 to 103.5)

10	randomised trials	not serious	not serious	not serious	serious ^c	none	661	477	-	mean 2.63 cm lower (4.54 lower to 0.73 lower)	⊕⊕⊕○ Moderate	
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CI: confidence interval

Explanations

a. i283%

- b. i2 78%
- c. i2 91%, heterogenous studies

Author(s):

Question: Physical activity compared to no physical activity for reducing HbA1c

Setting: among type 2 DM individuals

Bibliography: Umpierre D, Ribeiro PA, Kramer CK, Leitão CB, Zucatti AT, Azevedo MJ, PGross JL, Ribeiro JP, Schaan BD. Physical activity advice only or structured exercise training and association with HbA1c levels in type 2 diabetes: a systematic review and meta-analysis. JAMA. 2011 May 4;305(17):1790-9. doi: 10.1001/jama.2011.576. PMID: 21540423

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	physical activity	no physical activity	Relative (95% CI)	Absolute (95% CI)		

HbA1c reduction (follow-up: range 12 weeks to 104 weeks)

24	randomised trials	serious ^a	not serious	not serious	serious ^b	none	3529	3496	-	MD 0.43 lower (0.59 lower to 0.28 lower)	⊕⊕○○ Low	
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CI: confidence interval; **MD:** mean difference

Explanations

- a. no allocation concealment, not blinded
- b. high heterogeneity i2 62.9%

Author(s): Umpierre, et al

Question: Structured exercise compared to no exercise for HbA1c reduction

Setting: among type 2 DM individuals

Bibliography: 4. Umpierre D, Ribeiro PA, Kramer CK, Leitão CB, Zucatti AT, Azevedo MJ, PGross JL, Ribeiro JP, Schaan BD. Physical activity advice only or structured exercise training and association with HbA1c levels in type 2 diabetes: a systematic review and meta-analysis. JAMA. 2011 May 4;305(17):1790-9. doi: 10.1001/jama.2011.576. PMID: 21540423

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	structured exercise	no exercise	Relative (95% CI)	Absolute (95% CI)		
23	randomised trials	serious ^a	not serious	not serious	serious ^b	none	933	825	-	MD 0.67% lower (0.84 lower to 0.49 lower)	⊕⊕○○ Low	

CI: confidence interval; MD: mean difference

Explanations

- a. allocation concealment unclear
- b. high heterogeneity $i^2 = 91.3\%$

Author(s):**Question:** Structured exercise compared to no exercise for HbA1c reduction**Setting:** among type 2 DM individuals**Bibliography:** Thomas D, Elliott EJ, Naughton GA. Exercise for type 2 diabetes mellitus. Cochrane Database of Systematic Reviews 2006, Issue 3. Art. No.: CD002968. DOI: 10.1002/14651858.CD002968.pub2.

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	structured exercise	no exercise	Relative (95% CI)	Absolute (95% CI)		

HbA1c reduction (follow-up: range 8 weeks to 12 months)

13	randomised trials	serious	not serious	not serious	not serious	none	185	176	-	0.62 % lower (0.91 lower to 0.33 lower)	⊕⊕⊕○ Moderate	
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weight reduction (follow-up: range 8 weeks to 12 months)

10	randomised trials	serious	not serious	not serious	not serious	none	126	122	-	MD 0.04 kg lower (3.83 lower to 3.76 higher)	⊕⊕⊕○ Moderate	
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triglycerides (follow-up: range 8 weeks to 16 weeks)

5	randomised trials	serious	not serious	not serious	not serious	none	69	70	-	MD 0.25 mmol/l lower (0.48 lower to 0.02 lower)	⊕⊕⊕○ Moderate	
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Systolic blood pressure (follow-up: range 8 weeks to 6 months)

4	randomised trials	serious ^a	not serious	not serious	not serious	none	64	63	-	MD 4.16 mmHg lower (9.46 lower to 1.14 higher)	⊕⊕⊕○ Moderate	
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Total chole (follow-up: range 8 weeks to 16 weeks)

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	structured exercise	no exercise	Relative (95% CI)	Absolute (95% CI)		
5	randomised trials	serious ^a	not serious	not serious	not serious	none	69	70	-	MD 0.11 mmol/l lower (0.41 lower to 0.18 higher)	⊕⊕⊕○ Moderate	

LDL chole (follow-up: range 8 weeks to 16 weeks)

3	randomised trials	serious ^a	not serious	not serious	not serious	none	37	36	-	MD 0.12 mmol/l higher (0.29 lower to 0.53 higher)	⊕⊕⊕○ Moderate	
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HDL chole (follow-up: range 8 weeks to 12 weeks)

5	randomised trials	serious ^a	not serious	not serious	not serious	none	69	70	-	MD 0.02 mmol/l lower (0.1 lower to 0.06 higher)	⊕⊕⊕○ Moderate	
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CI: confidence interval; MD: mean difference

Q4.8. Recommendations from other groups

Based on the ADA guideline¹, most adults w/ type 2 diabetes should engage in 150 min or more of moderate to vigorous intensity aerobic activity per week with no more than 2 consecutive days without exercise. Shorter durations (minimum of 75 min/week) of vigorous-intensity or interval training may be adequate for younger and physically fit individuals. Two to three sessions of resistance exercise on non-consecutive days is advised as well. For older adults, flexibility and balance training like Yoga and tai chi are recommended 2-3x/week to increase muscle strength.

Recommendations of the 2018 Canadian Guidelines² emphasize on doing aerobic and resistance exercise at least 150 mins per week w/ at least 2 sessions of resistance exercise. Strategies to enhance self-motivation to increase physical activity can be maintained by setting specific physical activity goals using self-monitoring tools like pedometers. Supervised exercise programs should be strongly encouraged. The 2018 Canadian Clinical Practice Guidelines defined several types of exercise as ff:

- Aerobic exercise involves continuous, rhythmic movements of large muscle groups at least 10 mins at a time, exemplified by walking, bicycling, swimming, jogging.
- Resistance exercise involves brief repetitive exercises with weights, resistance bands or one's own body weight (e.g., push-ups) to increase muscle strength.
- Flexibility exercise enhances the ability to move through fuller ranges of motion like for the lower back or hamstring stretching. Some exercises like yoga can incorporate both resistance and flexibility exercise.

CPG	Recommendation Duration /week	Type of activity	Structure	Frequency	Physical activity and glycemic control
ADA ¹	Type 2 diabetic individuals should engage in 150 min or more of moderate to vigorous-intensity aerobic activity per week, spread over at least 3 days/week, w/ no more than 2 consecutive days without activity. (B) Shorter duration (minimum 75 min/week) of vigorous-intensity or interval training may be sufficient for younger, physically fit individuals.	Flexibility training and balance training are recommended 2-3x/week for older people. Yoga and tai chi may be included to increase flexibility, muscular strength, and balance. (C)	Type 2 diabetics should engage in 2-3 sessions/week of resistance exercise on nonconsecutive days. (B) Type 2 diabetics should decrease time spent in sedentary behavior. (B) Prolonged sitting should be interrupted every 30 min for blood glucose benefits (C)	Aerobic activity bouts at least 10 min., goal of 30 mins/day or more	At least 2 weekly sessions of resistance exercise with free weights on nonconsecutive days. (B)

CPG	Recommendation Duration /week	Type of activity	Structure	Frequency	Physical activity and glycemic control
Scottish Intercollegiate Guidelines Network	75 min of vigorous intensity aerobic physical activity	Aerobic activity of at least 10 mins per episode	30 mins 5x a week		
2019 Japanese CPG for Diabetes ⁶	Moderate-intensity aerobic exercise 150 min or longer	Aerobic & resistance exercise are effective individually and more in combination in improving glycemic control (Grade A)		3 or more days in a week w/o more than 2 consecutive days off exercise	Resistance exercise 2 to 3 times a week or every other day of the week
2018 Diabetes Canada ²	At least 150 mins per week of moderate- to vigorous-intensity aerobic exercise (walking, bicycling, jogging) w/ no more than 2 consecutive days without exercise to improve glycemic control (Grade B, Level 2) and reduce risk of CVD and overall mortality Grade C, Level 3)	Diabetics including elderly people should perform at least 2 sessions per week of resistance exercise (Grade B, Level 2)	Supervised exercise programs are effective in improving glycemic control. People w/ diabetes should minimize the amount of time spent in sedentary activities and break up long periods of sitting. (Grade C, level 3) Interrupt sitting time by getting up every 20-30 mins.		2 sessions/week of strength training w/ weights
IDF Recommendations 2017	Walking at least 150 min per week at intervals no longer than 48 hours				

Guideline Question 5.

Should we recommend limitation of alcohol intake among patients with Type 2 DM?

Q5.1. Methods and search strategy

A systematic review of the literature was conducted by independent reviewers. A targeted search strategy was employed to identify relevant clinical practice guidelines (CPGs) on type 2 diabetes mellitus from international databases, including MEDLINE/PubMed, the National Guidelines Clearinghouse, the Guideline International Network, and the National Institute for Clinical Evidence, and was subsequently updated to include new and relevant publications through September 2025. Additional searches of other databases and medical specialty society websites were performed by the Technical Working Group (TWG) to identify guidelines and studies not captured in the primary search. The ADA 2022 and AACE 2022 CPGs were adapted, and cross-referencing of their cited studies was undertaken.

PubMed was also utilized to search for evidence specifically related to alcohol intake and diabetes using the following terms: “diabetes mellitus alcohol consumption,” “diabetes mellitus ethanol consumption,” “alcohol consumption,” “alcohol intake,” “alcoholic beverage consumption,” “alcoholic beverage intake,” “diabetes alcohol intake,” “alcohol consumption glycemic control,” and “alcohol intake glycemic control.” The search covered publications from January 2006 to January 2023. Full texts of all eligible studies were retrieved.

The outcomes of interest included the impact of alcohol consumption on:

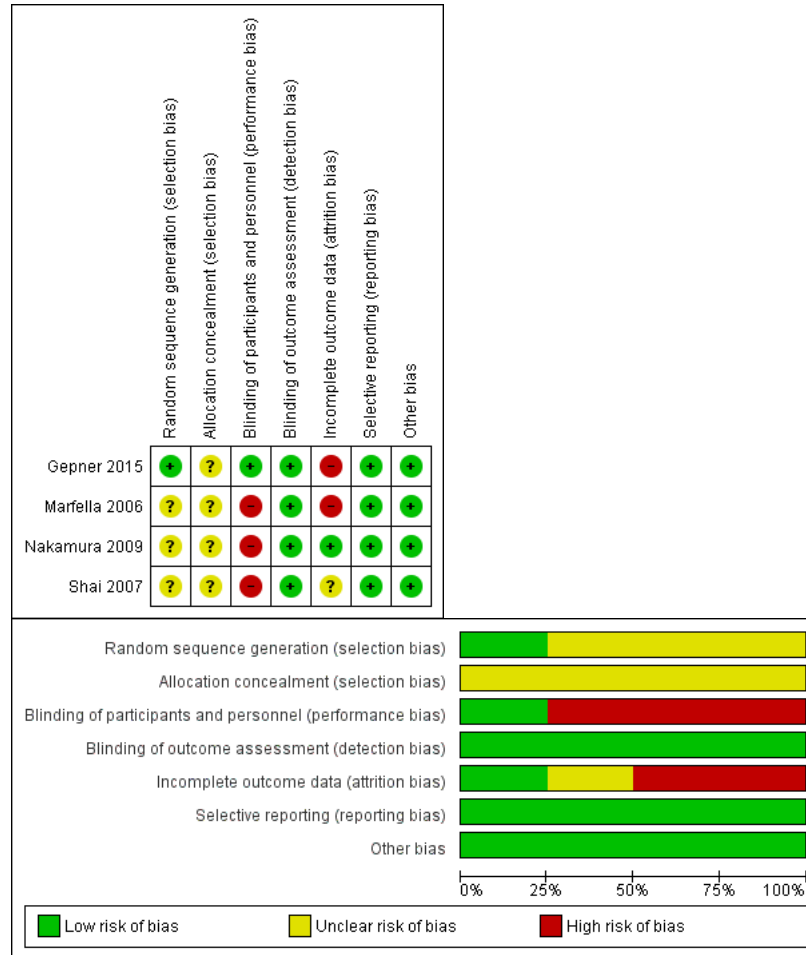
1. glycemic control (defined as achieving glycosylated hemoglobin [HbA1c] < 7.0%);
2. fasting plasma glucose or postprandial glucose levels;
3. blood pressure among individuals with diabetes;
4. lipid profile;
5. cardiovascular events; and
6. the degree and frequency of hypoglycemia.

Adverse effects of alcohol consumption were also reviewed.

Randomized controlled trials were prioritized; however, cohort studies, population-based studies, and meta-analyses were also included, particularly those referenced by the World Health Organization in support of its position that “No level of alcohol consumption is safe for our health.”

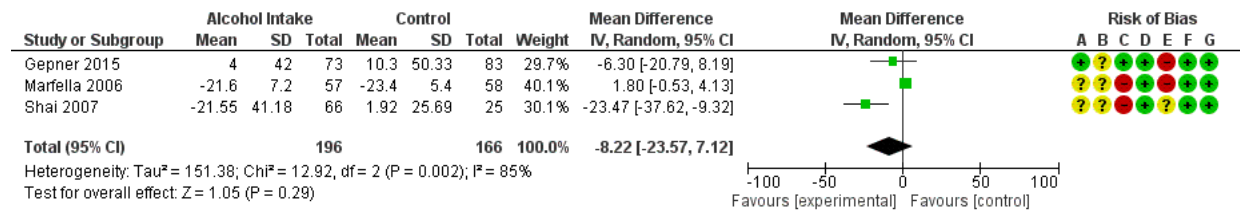
Q5.5. Assessment of Risk of Bias

RISK OF BIAS SUMMARY AND GRAPH



Q5.6. Results

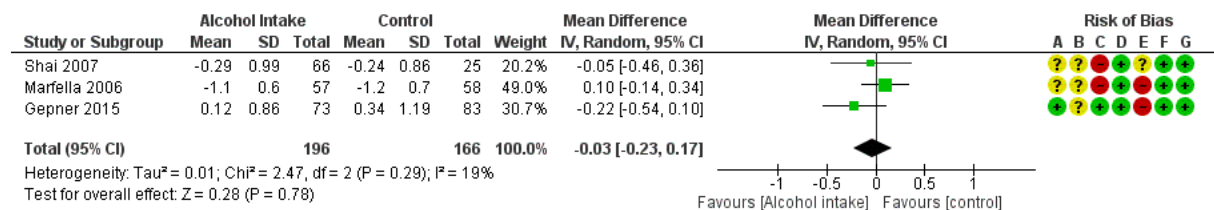
Alcohol and FPG:



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Alcohol and HbA1c:



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Alcohol Intake and Lipid Profile

The above mentioned randomized controlled trials showed that with regards to lipid levels, overall, there was no significant differences in levels of LDL cholesterol, but significant improvement in triglyceride levels and HDL levels, the latter especially among those who took red wine.

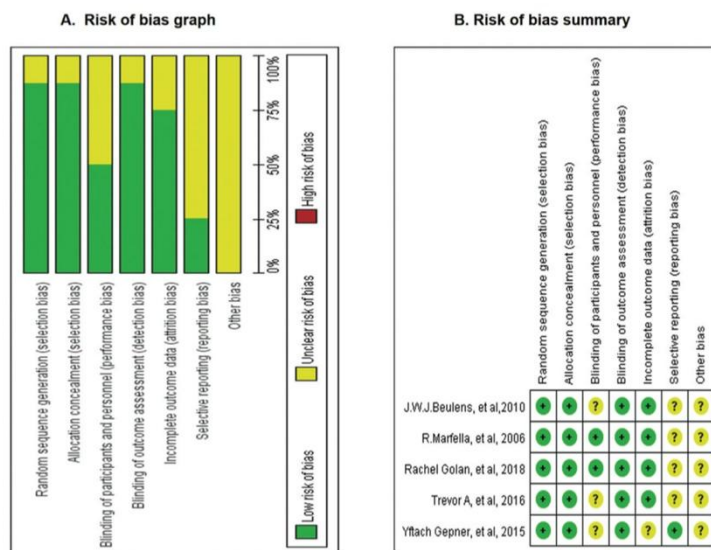
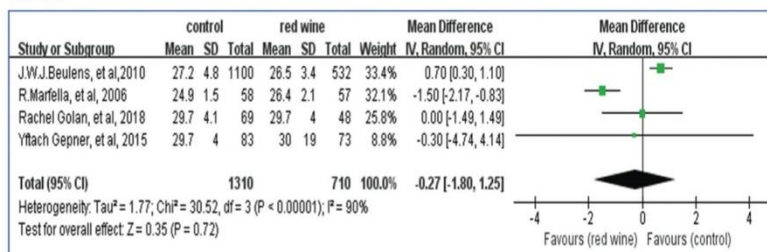


Fig. 2 Analysis the quality of enrolled studies for meta-analysis (A) risk of bias graph, with each risk of bias item presented as a percentage across all the included studies. (B) Risk of bias summary, with each risk of bias item for each included study.

A. BMI



B. Total cholesterol (mmol/l)

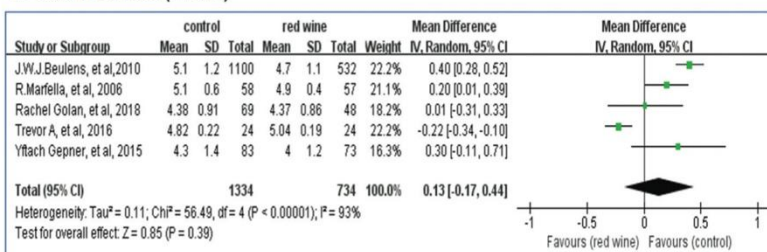
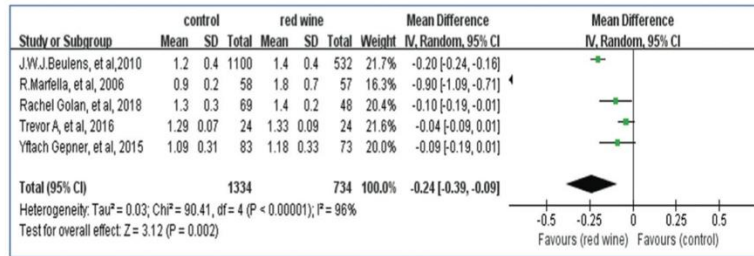


Fig. 3 MD scoring with moderate red wine intervention compared to control intervention. (A) MD of BMI with moderate red wine intervention compared to non-red wine intervention group. (B) MD of total cholesterol with moderate red wine intervention compared to non-red wine intervention group.

A. HDL (mmol/l)



B. LDL (mmol/l)

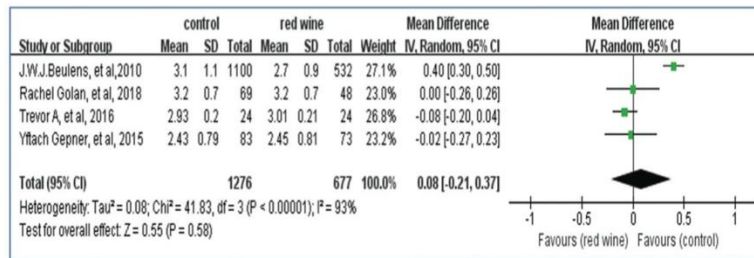
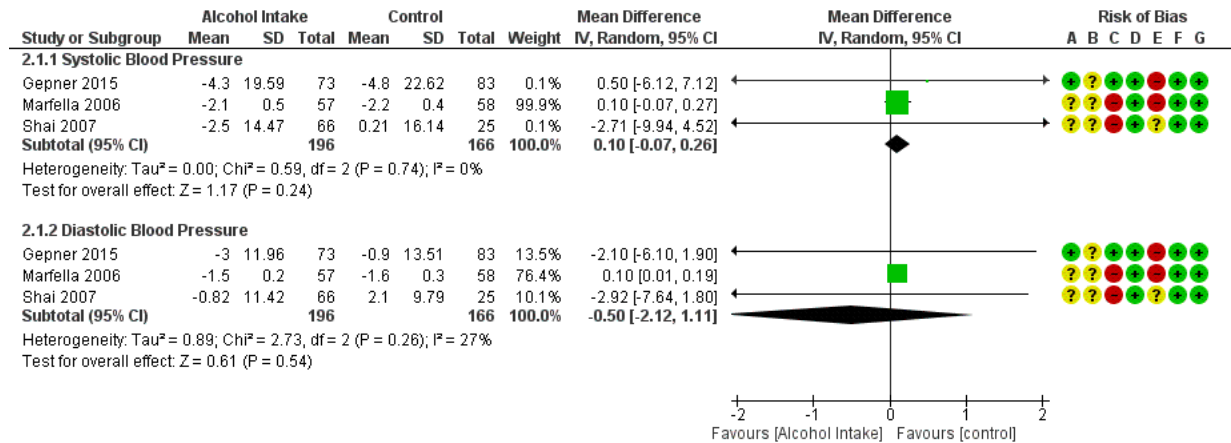


Fig. 5 MD scoring with moderate red wine intervention compared to control intervention. (A) MD of HDL with moderate red wine intervention compared to non-red wine intervention group. (B) MD of diastolic LDL with moderate red wine intervention compared to non-red wine intervention group.

ALCOHOL AND BP

The randomized controlled trials by Gepner, Marfella and Shai showed that moderate intake of alcohol did not significantly impact blood pressure.^{6,7,8}



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

On the other hand, according to a meta-analysis by S.A. Naame, et al of randomized controlled trials that included those of R.Marfella, et al and Y. Gepner, et al cited above, had favorable effects on systolic and diastolic blood pressures.⁹

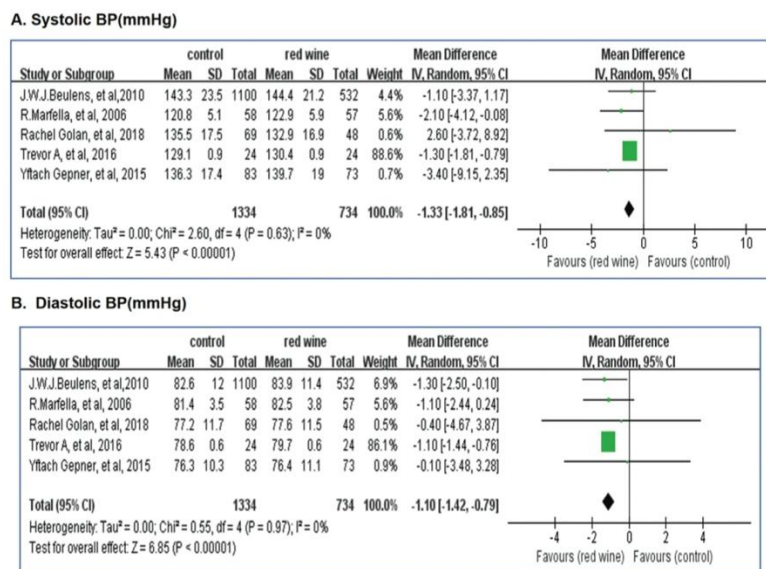
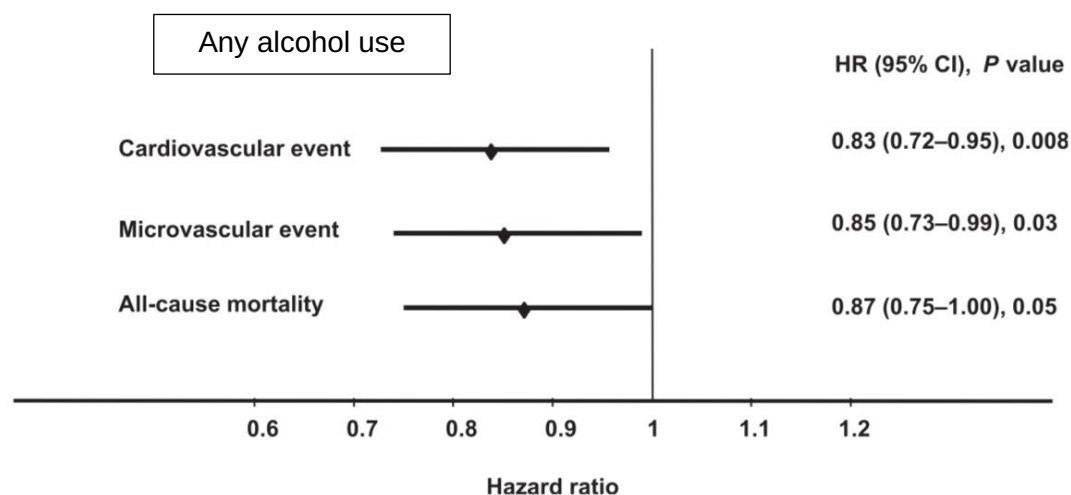


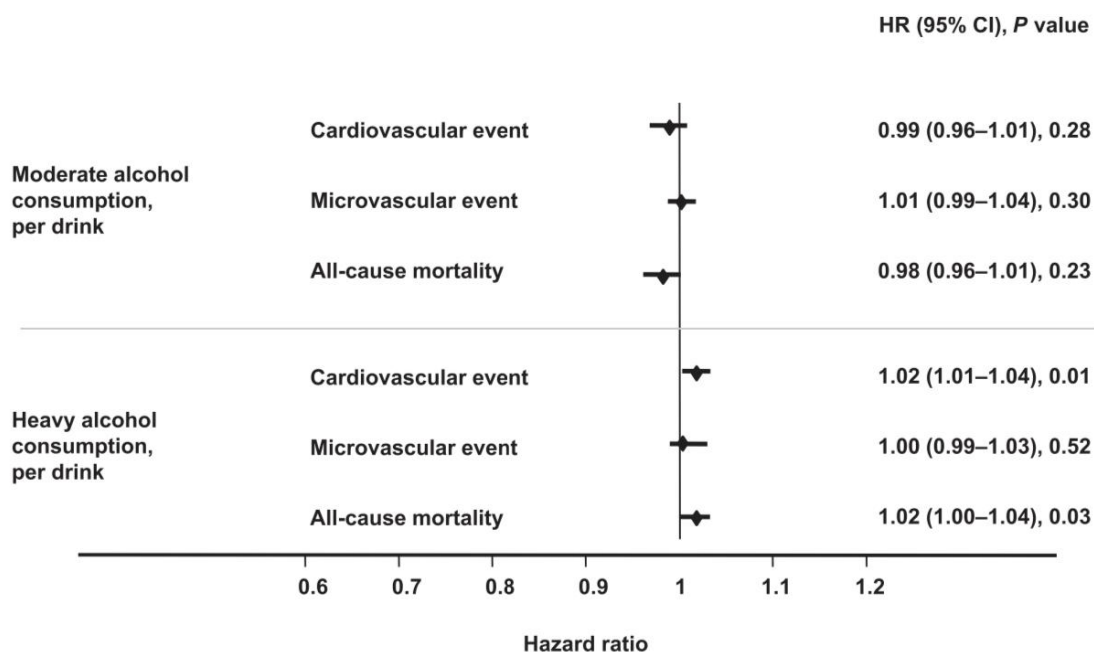
Fig. 4 MD scoring with moderate red wine intervention compared to control intervention. (A) MD of systolic BP with moderate red wine intervention compared to non-red wine intervention group. (B) MD of diastolic BP with moderate red wine intervention compared to non-red wine intervention group.

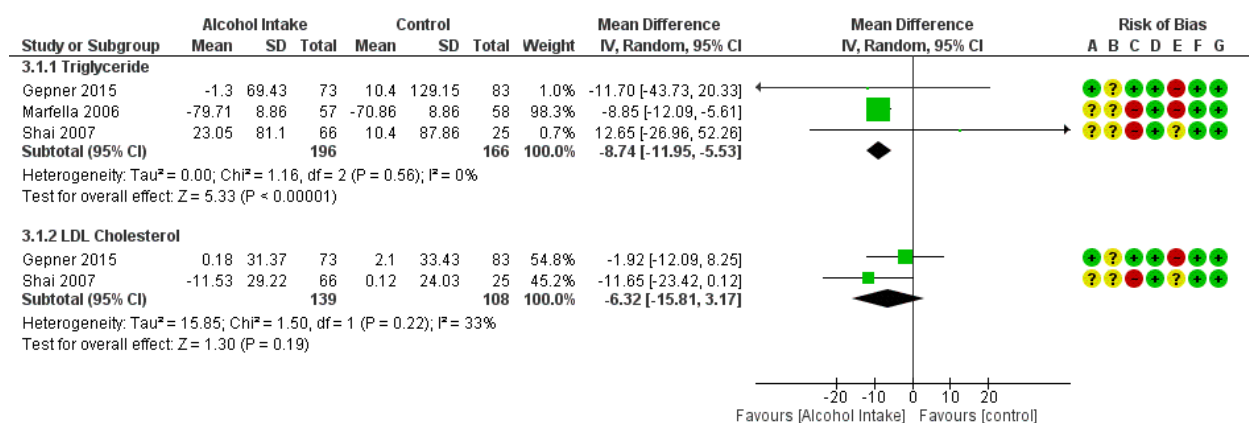
Alcohol and CV outcomes: [total patients 11,140]

A post hoc analysis of the Action in Diabetes and Vascular Disease: Preterax and Diamicon Modified-Release Controlled Evaluation (ADVANCE) trial¹⁰ examined the relationship of alcohol use in the patients enrolled in this trial. This was an observational analysis based on self-reporting of alcohol use at the baseline visit, at the 24-month follow-up visit and at the 5-year final study visit. There was a total of 11,140 patients and they reported that patients who took alcoholic beverages had fewer cardiovascular events, less microvascular complications and less all-cause mortality, as noted in the figure below. These benefits were primarily seen in those who drank moderate amounts of alcohol.



However, moderate alcohol use did not show any significant relationship with cardiovascular and microvascular events, as well as all-cause mortality, when examined separately from heavy alcohol intake. On the other hand, high alcohol intake, defined as more than 21 standard dinks weekly for men and more than 14 standard drinks weekly for women, was inversely related to cardiovascular events and all-cause mortality, with a similar trend for microvascular complications, as shown below.





Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Q5.7. GRADE Evidence Profile

Author(s):
 Question: Moderate alcohol consumption compared to no alcohol consumption for patients with type 2 diabetes mellitus
 Setting:
 Bibliography:

No. of studies		Certainty assessment					No. of patients		Effect		Certainty	Importance
Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	moderate alcohol consumption	no alcohol consumption	Relative (95% CI)	Absolute (95% CI)			
Cardiovascular Events (Death from cardiovascular disease, nonfatal stroke, or nonfatal myocardial infarction) (follow-up: 5 years)												
1	observational studies	serious ^a	not serious	not serious	not serious	none	292/2976 (9.8%)	817/7751 (10.5%)	HR 0.82 (0.71 to 0.95)	18 fewer per 1,000 (from 29 fewer to 5 fewer)	⊕○○○ Very low	
Microvascular complications (follow-up: 5 years)												
1	observational studies	serious ^a	not serious	not serious	not serious	none	247/2976 (8.3%)	846/7751 (10.9%)	HR 0.83 (0.72 to 0.97)	18 fewer per 1,000 (from 29 fewer to 3 fewer)	⊕○○○ Very low	
All Cause mortality (follow-up: 5 years)												
1	observational studies	serious ^a	not serious	not serious	not serious	none	276/2976 (9.3%)	718/7751 (9.3%)	HR 0.86 (0.74 to 0.99)	12 fewer per 1,000 (from 23 fewer to 1 fewer)	⊕○○○ Very low	
Glycemic Control (follow-up: range 3 months to 24 months; assessed with: HbA1c (%))												
3	randomised trials	serious ^b	not serious	not serious	not serious	none	196	166	-	MD 0.03 % lower (0.23 lower to 0.17 higher)	⊕⊕⊕○ Moderate	
Glycemic Control (follow-up: range 3 months to 24 months; assessed with: Fasting Blood Sugar (mg/dl))												
3	randomised trials	serious ^b	serious ^c	not serious	not serious	none	196	166	-	MD 8.22 mg/dl lower (23.57 lower to 7.12 higher)	⊕○○○ Low	
Systolic Blood Pressure (follow-up: range 3 months to 24 months)												
3	randomised trials	serious ^b	not serious	not serious	not serious	none	196	166	-	MD 0.1 mm Hg higher (0.07 lower to 0.26 higher)	⊕⊕⊕○ Moderate	
Diastolic Blood Pressure (follow-up: range 3 months to 24 months; assessed with: mm Hg)												
3	randomised trials	serious ^b	not serious	not serious	serious ^d	none	196	166	-	MD 0.5 mm Hg lower (2.12 lower to 1.11 higher)	⊕⊕○○ Low	
Lipid Profile (follow-up: range 3 months to 24 months; assessed with: Triglyceride (mg/dl))												
3	randomised trials	serious ^b	not serious	not serious	not serious	none	196	166	-	MD 8.74 mg/dl lower (11.05 lower to 5.53 lower)	⊕⊕⊕○ Moderate	
Lipid Profile (follow-up: range 3 months to 24 months; assessed with: LDL)												
2	randomised trials	serious ^d	not serious	serious ^d	not serious	none	139	108	-	MD 6.32 mm Hg lower (15.81 lower to 3.17 higher)	⊕⊕○○ Low	
Hypoglycemia (follow-up: range 3 months to 6 months)												
2	randomised trials	serious ^f	not serious	not serious	not serious	none	Only two studies reported hypoglycemia events. Both intervention and control have no events except for the intervention group (3/66) in the study of Shai et al.			⊕⊕⊕○ Moderate		

CI: confidence interval; HR: hazard Ratio; MD: mean difference

Guideline Question 6.

Should we recommend the use of non-nutritive sweeteners among patients with Type 2 DM?

Q6.1. Methods

The evidence review was conducted from July 2021 to November 2023 and subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024...

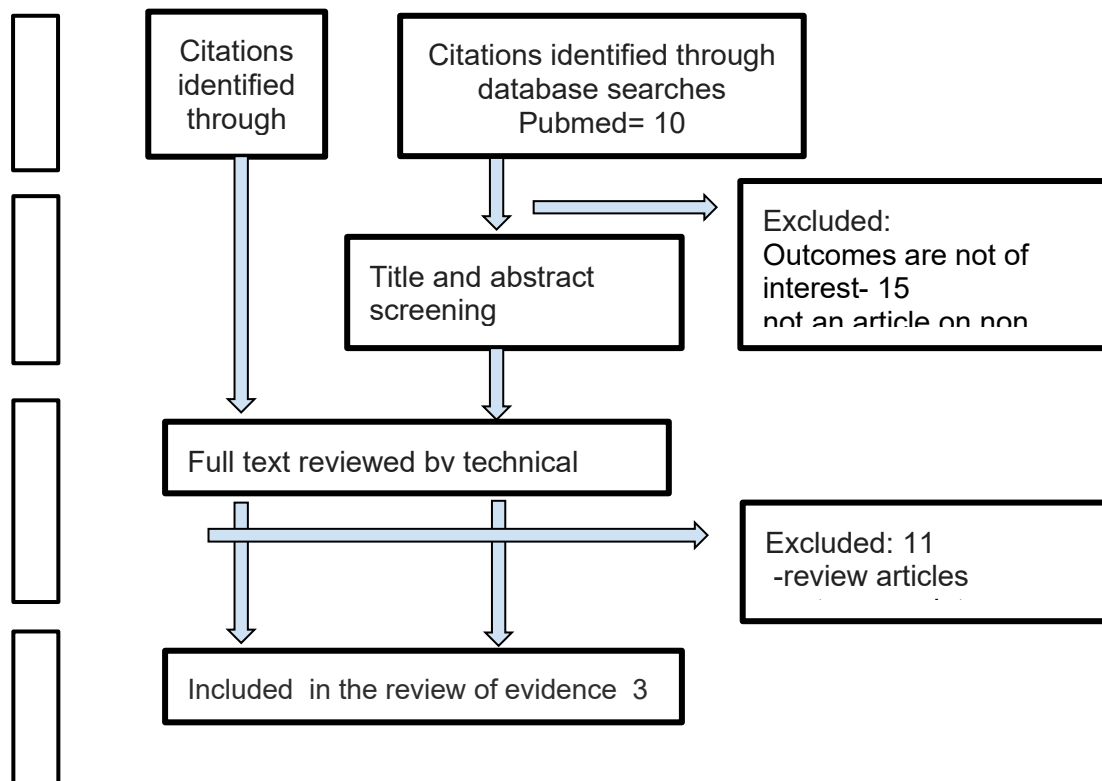
Q6.2. Search strategy

Search strategy including dates and data base

Database	Search strategy / Search terms	Date and time of search	Results	
			Yield	Eligible
PubMed	("artificial sweeteners" OR "Non nutritive sweeteners" OR "sugar substitutes") AND ("glycemic control" Or diabetes or cancer or "cardiovascular risk") Additional filter: metaanalysis, past 10 years	August 27, 2024 16: 58	10	1
Cochrane	("artificial sweeteners" OR "Non nutritive sweeteners" OR "sugar substitutes") AND ("glycemic control" Or diabetes or cancer or "cardiovascular risk")	August 27, 2024 16: 58	2	0
Google scholar	"artificial sweeteners" "Non nutritive sweeteners" "sugar substitutes" "glycemic control" Or diabetes or cancer or "cardiovascular risk" Additional filter: metaanalysis, past 10 years Citations removed	August 27, 2024 16: 58	28	0

Q6.3. PRISMA flow diagram

Literature review flow diagram



Q6.4. Characteristics of the Included Studies

Table of the characteristics of the studies included

Title/Author	Study design	Country	Number of patients	Population	Intervention Group(s)	Control	Outcomes
Nichol et al	Systemic review and metaanalyses	USA	741	526 healthy adults 69 with T2DM 146 unknown health status	LCS	comparator arm: This included sugar-sweetened beverages (SSBs) in 2 studies (23, 30) and a usual diet that included 280-kcal caloric beverages/d other than milk in the third study	Change in blood glucose levels relative to the baseline
Miller et al	RCT	USA	1951	healthy adults	Food and beverages with non-sugar sweeteners	sugar-sweetened beverages; usual diet	Body weight and body mass index
Health effects of the use of non-sugar sweeteners: a / Rios-Leyvraz M, Montez J.	systematic review and meta-analysis	Switzerland	1857	healthy adults	individuals who consumed non-sugar sweeteners (NSS) as part of their diet	individuals who do not consume non-sugar sweeteners (NSS) as part of their diet	Change in body mass index
Health effects of the use of non-sugar sweeteners: a / Rios-Leyvraz M, Montez J.	RCT	USA	1494	healthy adults	individuals who consumed non-sugar sweeteners (NSS) as part of their diet	individuals who do not consume non-sugar sweeteners (NSS) as part of their diet	Diabetes (FBS)
Health effects of the use of non-sugar sweeteners: a / Rios-Leyvraz M, Montez J.	RCT	USA	411	healthy adults	individuals who consumed non-sugar sweeteners (NSS) as part of their diet	individuals who do not consume non-sugar sweeteners (NSS) as part of their diet	Diabetes (HbA1c)
Health effects of the use of non-sugar sweeteners: a / Rios-Leyvraz M, Montez J.	Observational	USA	860 873	healthy adults	individuals who consumed non-sugar sweeteners (NSS) as part of their diet	individuals who do not consume non-sugar sweeteners (NSS) as part of their diet	All cause mortality
Health effects of the use of non-sugar sweeteners: a / Rios-Leyvraz M, Montez J.	Observational	USA	598 951	healthy adults	individuals who consumed non-sugar sweeteners (NSS) as part of their diet	individuals who do not consume non-sugar sweeteners (NSS) as part of their diet	Cardiovascular disease mortality

Title/Author	Study design	Country	Number of patients	Population	Intervention Group(s)	Control	Outcomes
Health effects of the use of non-sugar sweeteners: a / Rios-Leyvraz M, Montez J.	Observational	USA	166 938	healthy adults	individuals who consumed non-sugar sweeteners (NSS) as part of their diet	individuals who do not consume non-sugar sweeteners (NSS) as part of their diet	Cardiovascular events*
Health effects of the use of non-sugar sweeteners: a / Rios-Leyvraz M, Montez J.	Observational	USA	205 455	healthy adults	individuals who consumed non-sugar sweeteners (NSS) as part of their diet	individuals who do not consume non-sugar sweeteners (NSS) as part of their diet	Coronary heart disease
Health effects of the use of non-sugar sweeteners: a / Rios-Leyvraz M, Montez J.	Observational	USA	655 953	healthy adults	individuals who consumed non-sugar sweeteners (NSS) as part of their diet	individuals who do not consume non-sugar sweeteners (NSS) as part of their diet	Stroke
Health effects of the use of non-sugar sweeteners: a / Rios-Leyvraz M, Montez J.	Observational	USA	18 372	healthy adults	individuals who consumed non-sugar sweeteners (NSS) as part of their diet	individuals who do not consume non-sugar sweeteners (NSS) as part of their diet	Chronic kidney disease
Health effects of the use of non-sugar sweeteners: a / Rios-Leyvraz M, Montez J.	Observational	USA	942 600	healthy adults	individuals who consumed non-sugar sweeteners (NSS) as part of their diet	individuals who do not consume non-sugar sweeteners (NSS) as part of their diet	Cancer incidence (any type)

Q6.5. Assessment of Risk of Bias

Risk of Bias of included studies

Summary of the Risk of Bias of included studies in the meta-analysis “Health effects of the use of non-sugar sweeteners: a systematic review and meta-analysis”. By Rios-Leyvraz M, Montez J. Geneva

Cohort	Risk of Bias							Overall
	A	B	C	D	E	F	G	
Al-Dujaili 2017	?	?	?	?	?	?	-	Unclear
Angelopoulos 2015 & 2016	?	?	-	?	?	-	?	Unclear
Baird 2000	?	?	?	?	+	?	+	Unclear
Ballantyne 2011	?	?	?	?	?	?	?	Unclear
Blackburn 1997	?	?	-	?	-	+	+	Unclear
Bonnet 2018	?	?	-	?	-	+	?	Unclear
Bueno-Hernandez 2020	+	+	-	+	-	+	+	Low
Campos 2015	?	?	-	?	?	+	+	Unclear
Cocco 2019	?	+	?	+	+	+	+	Low
Crutchley 2013	?	?	-	?	?	?	?	Unclear
Dalenberg 2020	?	?	+	?	-	+	+	Unclear
de Ruyter 2012 & 2013	+	+	+	?	-	+	+	Low
Ebbeling 2019	?	?	-	+	+	+	+	Low
Engel 2018	+	?	-	?	-	?	+	Unclear
Fantino 2018	?	?	-	+	+	+	+	Low
Han 2018	+	+	?	?	+	+	?	Low
Higgins 2018	+	?	?	?	+	+	+	Low
Higgins 2019	+	?	?	?	-	+	+	Unclear
Judah 2020	?	?	?	-	?	?	+	Unclear
Kanders 1988	?	?	-	?	+	?	+	Unclear
Kassi 2016	?	?	-	?	?	?	-	Unclear
Kim 2011	?	?	?	?	?	?	?	Unclear
Kim 2020	-	?	-	?	?	?	?	Unclear
Knopp 1976	?	?	+	?	?	?	+	Unclear
Kreuch 2020	?	?	+	?	?	?	?	Unclear

Cohort	Risk of Bias							Overall
	A	B	C	D	E	F	G	
Kuzma 2015	+	+	?	?	+	+	+	Unclear
Lee 2012	?	?	?	?	?	+	?	Unclear
Iertrit 2018	?	?	?	+	?	+	?	Unclear
Madjd 2018	+	+	-	?	?	+	?	Unclear
Markey 2016	+	?	?	?	+	?	+	Unclear
McLay-Cooke 2016 (Ice Tea Study)	+	+	?	?	+	+	+	Low
Nike 2011	+	?	?	+	+	+	?	Low
Peters 2016	+	?	-	?	?	-	+	Unclear
Raben 2002	?	?	?	?	+	+	+	Unclear
Reid 2007	-	-	+	?	+	+	+	Low
Romo-Romo	+	+	-	?	+	+	+	Low
Sagrario Lopez-Meza 2021	?	?	?	?	-	+	?	Unclear
Sanchez- Delgado 2021	?	?	?	?	-	-	+	Unclear
Serrano 2021	?	+	+	+	?	+	+	Low
Spiers 1998	?	+	?	?	+	?	?	Unclear
Stamataki 2020	+	?	-	?	+	+	+	Low
Taljaard 2013	?	+	?	+	+	+	+	Low
Tate 2012	+	?	-	+	+	+	+	Low
Vandana 2017	?	+	+	+	?	+	+	Low
Vazquez-Duran 2013 & 2016	+	+	-	+	?	+	?	Low
Viveros- Watty 2021	-	+	-	?	+	+	?	Unclear
Warrington 2011	?	?	+	?	+	-	+	Unclear
Young 2017	+	+	+	+	?	?	?	Low
A - Random sequence generation (selection bias) B - Allocation concealment (selection bias) C - Blinding of participants, personnel (performance bias) D - Blinding of outcome assessment (detection bias) E - Incomplete outcome data (attribution bias) F - Selective reporting (reporting bias) G - Other bias								

Q1.6. Results

NA (i.e., no forest plots generated)

Q1.7. GRADE Evidence Profile

A. GRADE Summary of Findings: Association between Non nutritive sweeteners (NNS) and glycemic control

Quality Assessment						№ of patients		Effect	Certainty	Importance
№ of studies	Design	Risk of bias	Inconsistency	Indirectness	Imprecision	Baseline	Post NSS consumption	Change in blood glucose relative to the baseline (95% CI)		
Nichol et al (measured with: Change in blood glucose level after 180-210 mins following non nutritive sweetener intake)										
29	RCTs	serious ¹	not serious	serious ²	not serious	741	741	-0.359 mmol/L -0.434, -0.283	⊕⊕○○ Low	IMPORTANT
¹ Publication bias										
² Study Population Only 69 out of 741 has T2DM										

Quality Assessment						№ of patients		Effect	Certainty	Importance
№ of studies	Design	Risk of bias	Inconsistency	Indirectness	Imprecision	Control	Steviol glycoside	Change in A1c (95% CI)		
Nichol et al (Duration: 16 weeks, measured with: HbA1c)										
1	RCT	serious ¹	not serious	not serious	not serious	62	60	p=0.355	⊕⊕○○ Low	IMPORTANT
¹ Study Population <100										

B. GRADE Summary of Findings: Association between non nutritive sweeteners (NNS) and weight loss

Quality Assessment						№ of patients		Effect	Certainty	Importance
№ of studies	Design	Risk of bias	Inconsistency	Indirectness	Imprecision	Baseline	Post NNS intake	Change in BMI (95% CI)		
Miller et al (Duration: 3 to 78 weeks, measured with: change in BMI)										
15	RCTs	not serious	serious ²	serious ³	not serious	--	-- 1951	-0.24; CI: 20.41, 20.07	⊕⊕○ ○ Low	IMPORTANT
¹ Heterogeneity I ² >77% ² LCS intake variable/ extent not documented properly ³ Population: Non-DM adult and children, T1DM										

Quality Assessment						Effect	Quality	Importance
No of studies	Design	Risk of bias	Inconsistency	Indirectness	Imprecision	Change in weight (95% CI)		
Miller et al (Duration: 3 to 78 weeks, measured with: change kg)								
15	RCTs	not serious	serious ¹	serious ²	not serious	-0.80 kg 95% CI: -1.17, -0.43)	⊕⊕○○ Low	IMPORTANT
¹ LCS intake variable/ extent not documented properly								
² Population: Non-DM adult and children, T1DM								

C. GRADE summary of findings: Association of intake of non-nutritive sweeteners and cardiometabolic risks, cancer

ANNEX 7. GRADE evidence profiles

Question: What is the effect of higher vs lower intake of non-sugar sweeteners in adults?

Population: General adult population

Minimum follow up duration: 1 year for disease incidence outcomes (CVD, T2DM)

Assessment							№ of events/participants (Study event rate)		Effect	Absolute – per 1000 (95% CI)	Certainty	Importance
№ of studies	Design	Risk of bias	Inconsistency	Indirectness	Imprecision	Others	Lower /No NNS intake	Higher NNS intake	Relative/Mean Difference (95% CI)			
Diabetes (Incident diabetes)												
13	observational (beverages)	not serious ⁹	not serious	not serious	not serious	none	28222/408609 (6.9%)		HR 1.23 (1.14 to 1.32)	16 more (from 10 more to 22 more)	⊕⊕○○ Low	CRITICAL
2	observational (table top)	not serious ⁹	not serious	not serious	not serious	none	2250/62582 (3.6%)		HR 1.34 (1.21 to 1.48)	12 more (from 8 more to 17 more)	⊕⊕○○ Low	CRITICAL
Diabetes: fasting glucose (mmol/L)												
16	RCT	serious ¹⁰	not serious	not serious	not serious ¹⁰	none	844	650	MD –0.01 (–0.05 to 0.04)	-	⊕⊕⊕○ Moderate	CRITICAL
Diabetes: HbA1c (%)												
6	RCT	serious ²²	not serious	not serious	serious ¹ ₆	none	212	199	MD 0.02 (–0.03 to 0.07)	–	⊕⊕⊕○ Moderate	CRITICAL
All-cause mortality												
8	observational	not serious ⁹	serious ⁸	not serious	not serious	none	102677/860873 (11.9%)		HR 1.12 (1.05 to 1.19)	14 more (from 6 more to 23 more)	⊕○○○ Very low	CRITICAL
Cardiovascular diseases: cardiovascular disease mortality												

Assessment							№ of events/participants (Study event rate)		Effect	Absolute – per 1000 (95% CI)	Certainty	Importance
№ of studies	Design	Risk of bias	Inconsistency	Indirectness	Imprecision	Others	Lower /No NNS intake	Higher NNS intake	Relative/Mean Difference (95% CI)			
5	observational	not serious ⁹	not serious	not serious	not serious	none	13089/598951 (2.2%)		HR 1.19 (1.07 to 1.32)	4 more (from 2 more to 7 more)	⊕⊕○○ Low	CRITICAL
Cardiovascular diseases: cardiovascular events												
3	observational	not serious ⁹	not serious	not serious	not serious	none	6384/166938 (3.8%)		HR 1.32 (1.17 to 1.50)	12 more (from 6 more to 19 more)	⊕⊕○○ Low	CRITICAL
Cardiovascular diseases: coronary heart disease												
4	observational	not serious ⁹	serious ⁸	not serious	serious ¹⁴	none	10104/205455 (4.9%)		HR 1.16 (0.97 to 1.39)	(from 1 fewer to 19 more)	⊕○○○ Very low	CRITICAL
Cardiovascular diseases: stroke												
6	observational	not serious ⁹	not serious	not serious	not serious	none	8346/655953 (1.3%)		HR 1.19 (1.09 to 1.29)	2 more (from 1 more to 4 more)	⊕⊕○○ Low	CRITICAL
Chronic kidney disease: creatinine (μmol/L)												
2	RCT	serious ⁽³⁰⁾	serious ⁽⁸⁾	not serious	very serious ⁽³¹⁾	none	93	52	MD 8.80 (–14.65 to 32.25)	-	⊕○○○ Very low	CRITICAL
Cancer												
	observational	not serious	not serious	not serious	not serious	none	1345/37976		HR 1.13 [1.03 to 1.25]	4 more (from 1 more to 8 more)	⊕⊕○○ Low	-

Assessment							№ of events/participants (Study event rate)		Effect	Absolute – per 1000 (95% CI)	Certainty	Importance
№ of studies	Design	Risk of bias	Inconsistency	Indirectness	Imprecision	Others	Lower /No NNS intake	Higher NNS intake	Relative/Mean Difference (95% CI)			
⁸ $I^2 \geq 50\%$, indicating a significant level of heterogeneity. Where the number of studies was sufficient to explore heterogeneity via subgroup and sensitivity analyses, results of the analysis did not significantly explain the observed heterogeneity. Downgraded once. ⁹ Mean Newcastle–Ottawa Score of >5 with very conservative application of ratings. Not downgraded. ¹⁰ A small mean effect, likely of little to no clinical significance, and neither bound of the 95% CI includes a potentially important benefit or harm. Therefore, consider a sufficiently precise estimate of no effect. ¹⁴ One bound of the 95% CI includes potentially important benefit or harm and the other bound crosses the null in the opposite direction, and/or the sample size is small. Downgraded once. ¹⁶ A small mean effect, likely of little to no clinical significance, and neither bound of the 95% CI includes a potentially important benefit or harm. However, the sample size is small. Downgraded once. ²² In these trials, the majority had adequate randomization, and half had adequate allocation concealment (the remainder were unclear). Half the trials could not blind participants to treatment. One trial had incomplete data, and one had concerns about selective reporting. One trial was an abstract only with overall high risk of bias. The remaining domains were approximately half with low risk of bias and half unclear. Not downgraded ³⁰ One trial was fairly well reported, and the other was mostly unclear, with concerns about selective reporting of outcomes. Downgraded once. ³¹ The 95% CI crosses the null and includes both significant benefit and harm. Downgraded twice.												

Q6.8. Recommendations from other groups

Group or Agency	Recommendation	Strength of Recommendation/ Certainty/Quality of Evidence
The Standards of Care in Diabetes, American Diabetes Association (ADA)	The use of NNS may have the potential to reduce overall calorie and carbohydrate intake if substituted for caloric sweeteners and without compensation by intake of additional calories from other food sources. For those who consume sugar sweetened beverages regularly, NNS may serve as a short term replacement strategy. Overall, people are encouraged to decrease both sweetened and NNS-sweetened beverages and other alternatives, with emphasis on water intake	Moderate recommendation (B)
Diabetes Canada	RCTs show a weight loss benefit when NNS is used to displace excess calories from added sugars. Systematic reviews and metaanalysis of prospective cohorts (high risk for bias, reverse causality) associate NNS with weight gain, CVD, stroke. Health Canada has set acceptable daily intake (ADI) values, which are expressed on a body weight basis and are considered safe daily intake levels over a lifetime.	Grade C, Level 3
Japanese Clinical Practice Guideline for Diabetes 2019	The influence of artificial sweeteners on the risk of diabetes and glycemic control has not been sufficiently elucidated.	Low evidence
Clinical Practice Guidelines (CPG) on Management of Type 2 Diabetes Mellitus (T2DM).Malaysia (2020)	Sugar substitutes do not impact glycemic level. Replacing sugars with sugar substitutes will decrease daily intake of CHO and calories. Intake should not exceed acceptable daily intake (ADI) levels.	Level II evidence- Evidence from at least one well-designed RCT (e.g. large multi-site RCT)
Use of non-sugar sweeteners: WHO guideline, World Health Organization 2022	WHO suggests that non-sugar sweeteners not be used as a means of achieving weight control	(conditional recommendation).

Guideline Question 7.

Should we use diabetes-specific nutrition formulas among patients with type 2 diabetes mellitus?

Q7.1. Methods and search strategies

A systematic review of the literature was conducted by independent reviewers. A targeted search strategy was employed to identify relevant clinical practice guidelines (CPGs) on type 2 diabetes mellitus from international databases, including MEDLINE/PubMed, the National Guidelines Clearinghouse, the Guideline International Network, and the National Institute for Clinical Evidence, and was subsequently updated to include additional publications through September 2025. Additional searches of other databases and medical specialty society websites were also performed by the Technical Working Group (TWG) to identify articles not captured in the primary search. The ADA 2022 and AACE 2022 CPGs were adapted, and cross-referencing of their cited studies was undertaken.

PubMed was further utilized to search for evidence on diabetes-specific nutrition formulas (DSNF) using the terms: “diabetes-specific nutrition formula,” “diabetes-specific enteral formula,” “diabetes specific nutrition,” “glycemia targeted nutrition,” “enteral nutrition and glycemic control,” “enteral nutrition in type 2 diabetes,” “diabetes nutrition formula and cardiovascular risk,” and “enteral nutrition and type 2 diabetes mellitus and cardiometabolic risk.” The initial timeframe from January 1, 2013 to December 2023 was subsequently expanded to January 2005 to December 2023. Full texts of all eligible studies were retrieved.

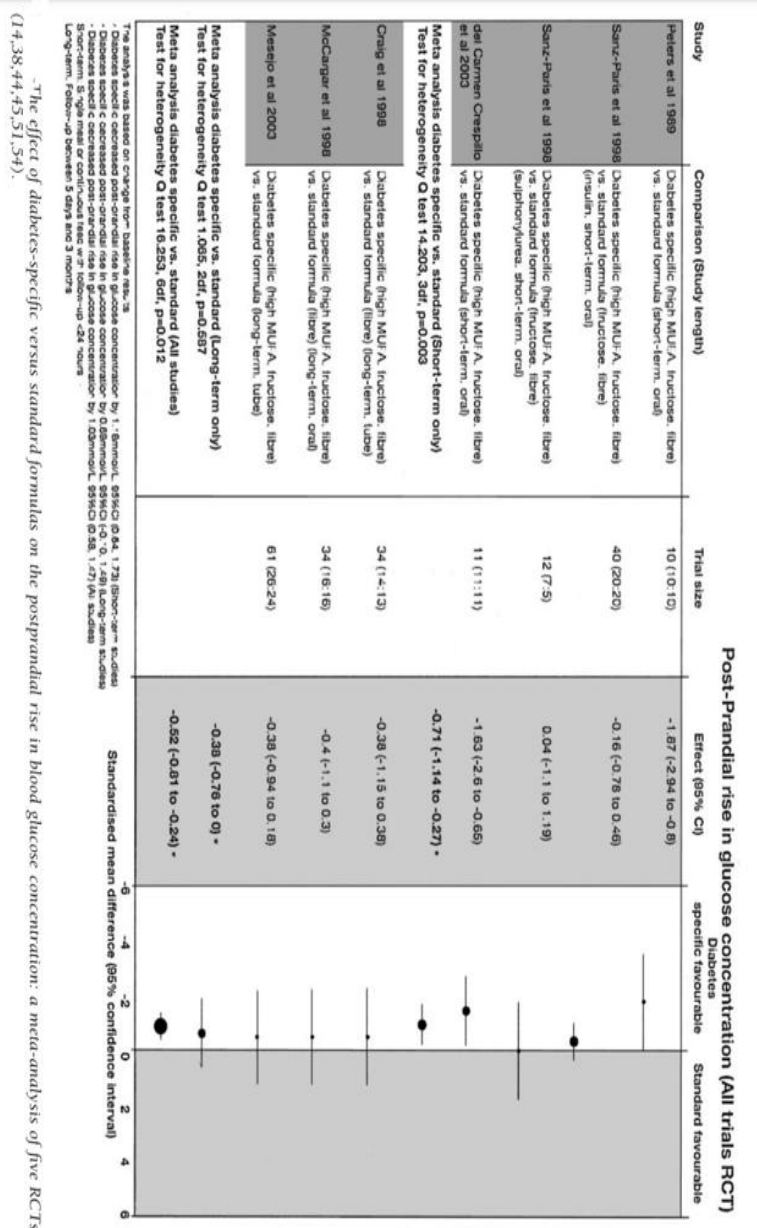
The outcomes of interest included the effects of DSNF on:

1. glycemic control, defined as achieving glycosylated hemoglobin (HbA1c) < 7.0%;
2. fasting plasma glucose or postprandial blood glucose;
3. glycemic variability; and
4. cardiometabolic risk profile.

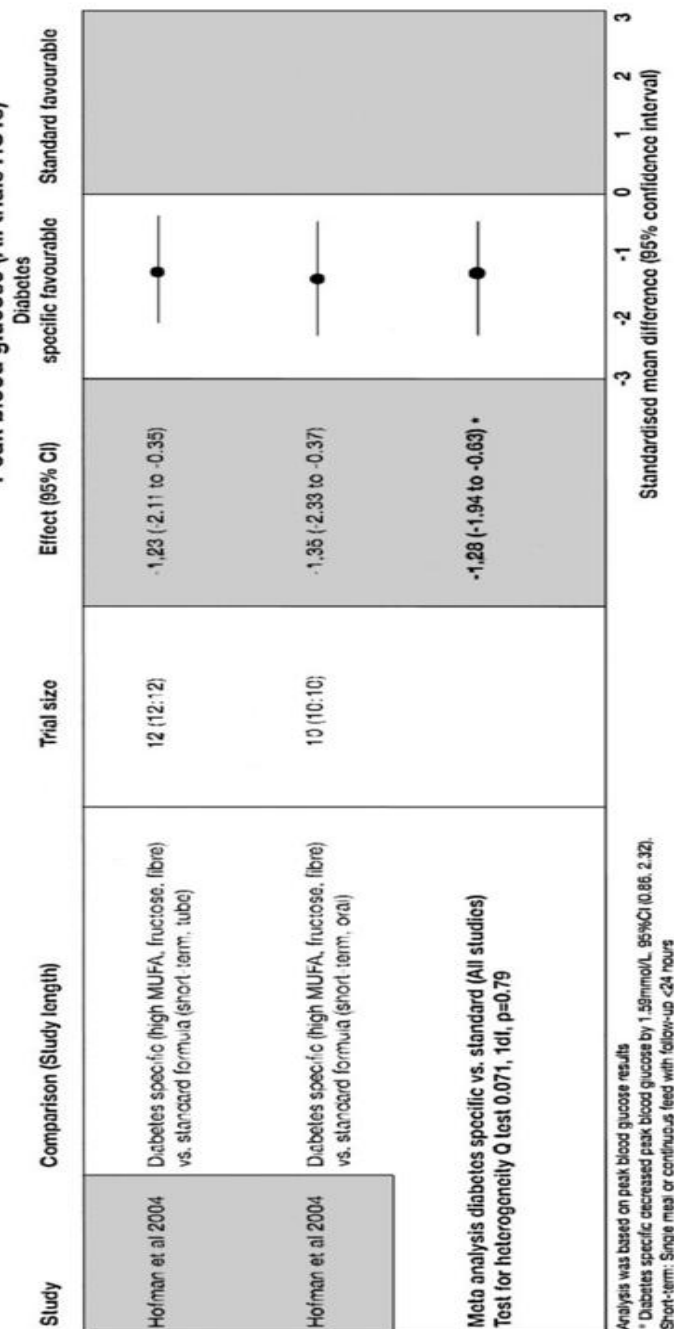
Studies evaluating diabetes-specific nutrition formulas that reported glycemic and/or cardiometabolic outcomes were included. Randomized controlled trials were prioritized, although meta-analyses and narrative reviews were also considered. Studies involving formulas that were not diabetes-specific were excluded unless used as comparators for DSNF.

GRADEpro was used to appraise the included studies.

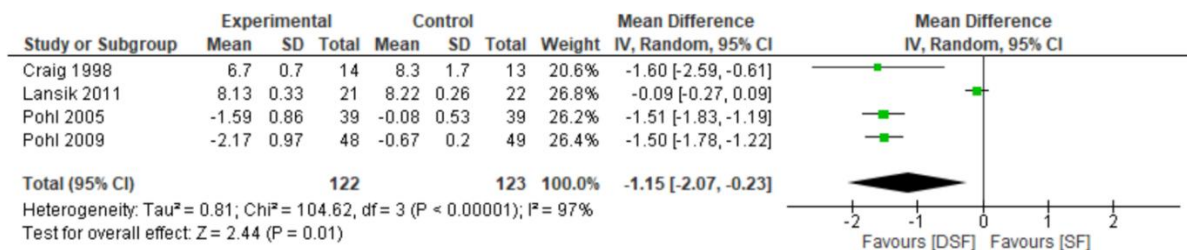
Q1.6. Results



Peak blood glucose (All trials RCTs)

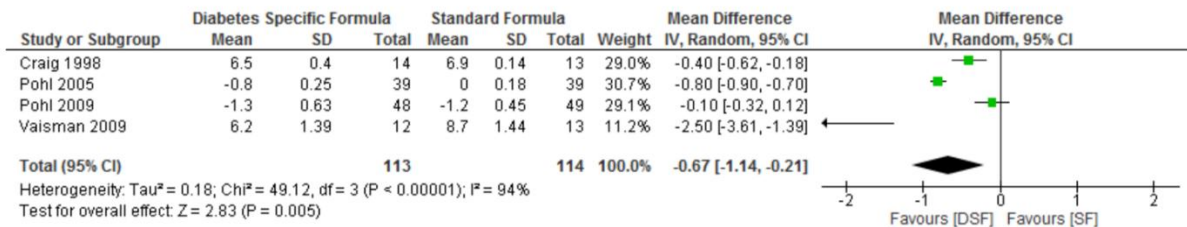


The effect of diabetes-specific versus standard formulas on peak blood glucose concentration in short-term studies: a meta-analysis of two RCTs (47,48).



The effect of DSF on fasting blood glucose (mmol/L).

The glycated haemoglobin was significantly lower ($p = 0.005$) in the DSF group compared to the SF group following meta-analysis (Figure 4) and sensitivity analysis.



The effect of DSF on Glycated Haemoglobin %.

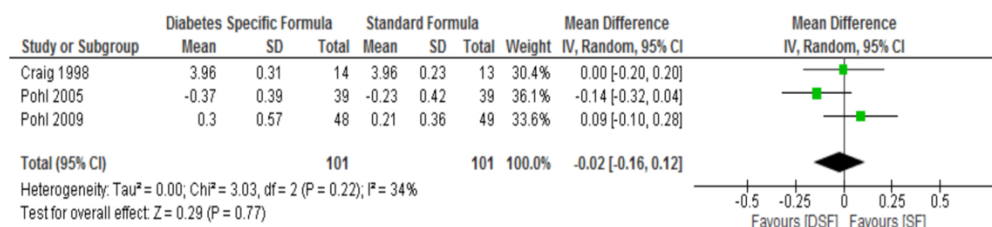


Figure 3. The effect of DSF on total cholesterol (mmol/L).

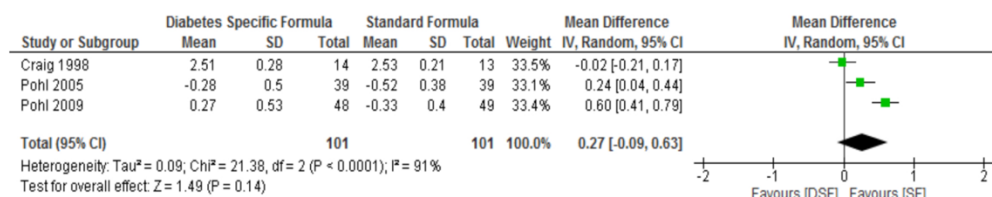


Figure 4. The effect of DSF on LDL cholesterol (mmol/L).

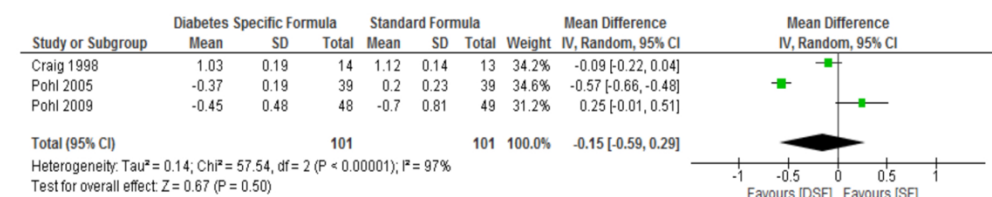


Figure 5. The effect of DSF on Triglycerides (mmol/L).

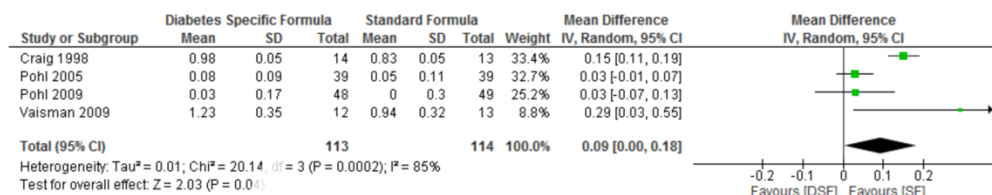


Figure 6. The effect of DSF on HDL cholesterol (mmol/L).

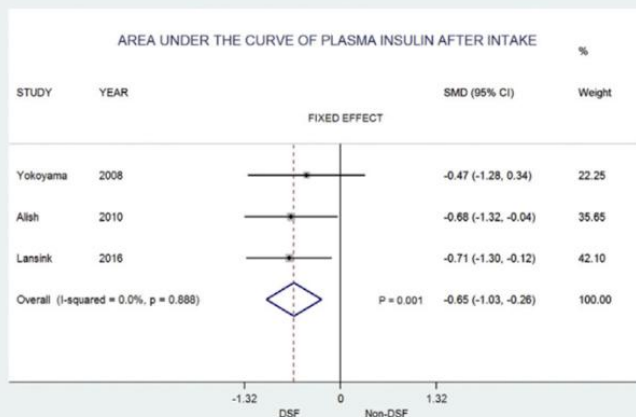
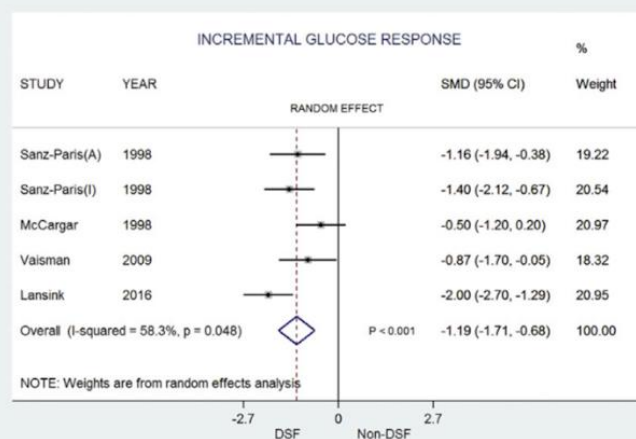
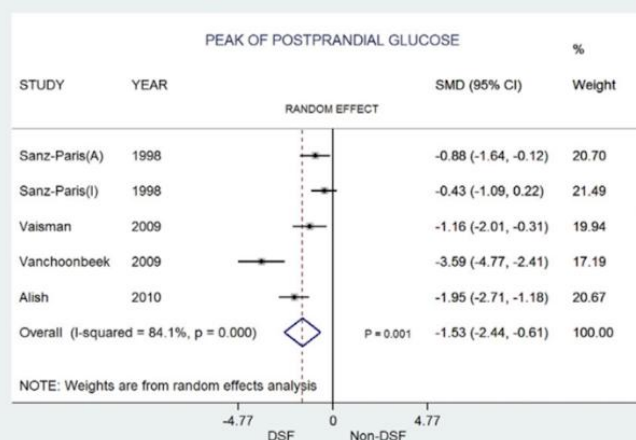


Fig. 2. Forest plots analyses for metabolic outcomes in postprandial studies. DSF versus non-DSF in DM patients. Effect sizes were calculated as standardized mean differences (SMDs) (<0.4 were considered small, 0.4–0.7 moderate, and >0.7 large).

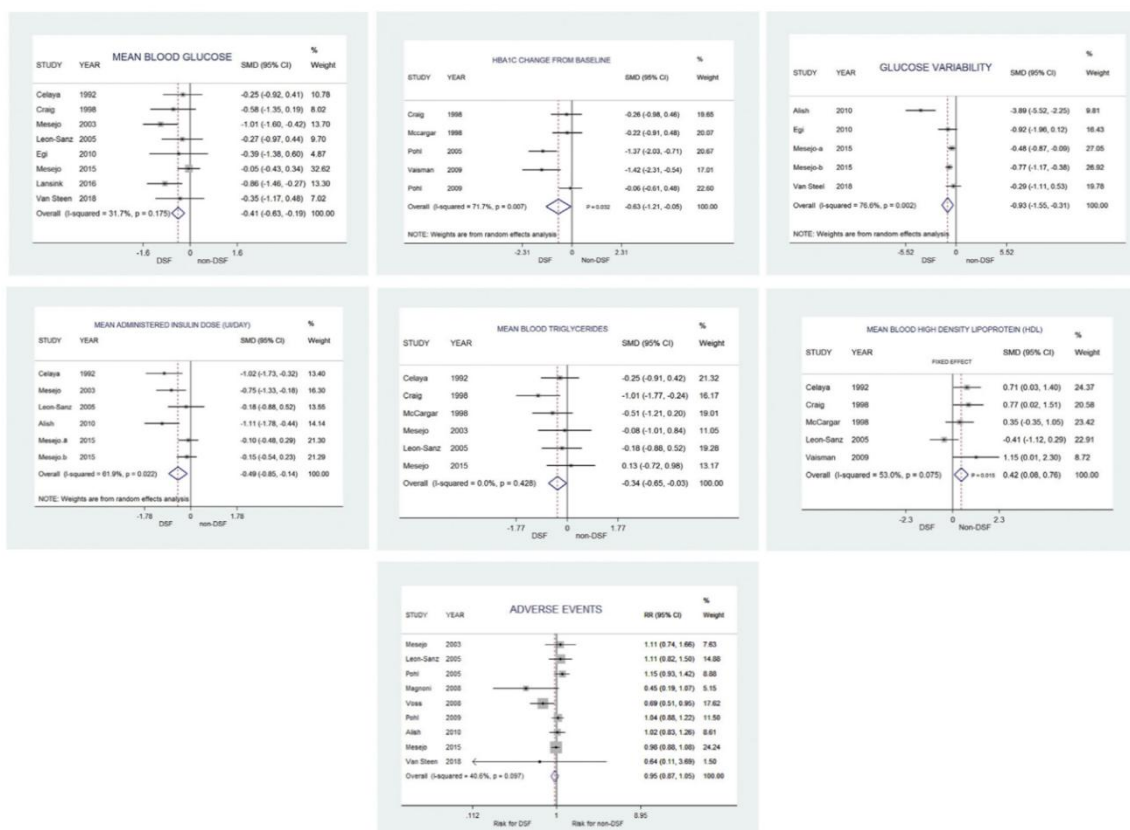


Fig. 3. Forest plots analyses for metabolic outcomes in medium-long term studies. DSF versus non-DSF in DM patients. Effect sizes were calculated as standardized mean differences (SMDs) (<0.4 were considered small, 0.4–0.7 moderate, and >0.7 large).

Q7.7. GRADE Evidence Profile

Author(s): Diabetes specific formula compared to standard formulas for patients with T2DM												
Question: Diabetes specific formula compared to standard formulas for patients with T2DM												
Setting: Outpatient												
Bibliography: Qin, Omorogieva, Sharon Marie Weldon, Trevor Thompson, Rachel Cockrill, and Xiao-Hua Wang. "The Effect of Diabetes-Specific Enteral Nutrition Formula on Cardiometabolic Parameters in Patients with Type 2 Diabetes: A Systematic Review and Meta-Analysis of Randomised Controlled Trials." <i>Nutrients</i> 11, no. 8 (August 15, 2019): 1469. https://doi.org/10.3390/nu11081469 .												
Certainty assessment												
No. of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	diabetes specific formula	standard formulas	Effect		Certainty	Importance
									Relative (95% CI)	Absolute (95% CI)		
Fasting Blood Glucose (mmol/L)												
4	randomised trials	not serious	serious ^a	not serious	serious ^b	none	122	123	-	MD 1.15 mmol/L lower (2.07 lower to 0.23 lower)	⊕⊕○○ Low	CRITICAL
Glycosylated Hemoglobin (%)												
4	randomised trials	serious ^c	serious ^d	not serious	not serious	none	113	114	-	MD 0.67 % lower (1.14 lower to 0.21 lower)	⊕⊕○○ Low	CRITICAL
Total Cholesterol (mmol/L)												
3	randomised trials	serious ^a	not serious	not serious	not serious	none	101	101	-	MD 0.02 mmol/L lower (0.18 lower to 0.12 higher)	⊕⊕⊕○ Moderate	NOT IMPORTANT
Low Density Lipoprotein (mmol/L)												
3	randomised trials	serious ^a	serious ^d	not serious	not serious	none	101	101	-	MD 0.27 mmol/L higher (0.29 lower to 0.63 higher)	⊕⊕○○ Low	NOT IMPORTANT
Triglycerides (mmol/L)												
3	randomised trials	serious ^a	serious ^d	not serious	not serious	none	101	101	-	MD 0.15 mmol/L lower (0.39 lower to 0.29 higher)	⊕⊕○○ Low	NOT IMPORTANT
High Density Lipoprotein (mmol/L)												
4	randomised trials	serious ^c	serious ^d	not serious	not serious	none	113	114	-	MD 0.09 mmol/L higher (0 to 0.18 higher)	⊕⊕○○ Low	NOT IMPORTANT

CI: confidence interval; MD: mean difference

Explanations

a. I² = 97%
b. wide confidence interval
c. two studies with high attrition bias
d. I² = 94%
e. 95% study with high attrition bias
f. I² = 93%
g. I² = 85%

Q7.8. Recommendations from other groups

Recommendations from other groups:

The American Diabetes Association (ADA) Standards of Care in Diabetes – 2024 stated that partial or total meal replacements may be effective tools to aid individuals in attaining glycemic control, to improve nutrient quality and to promote weight loss by reducing calorie intake.

Guideline Question 8.

Should we recommend referral for bariatric surgery to achieve sustained glycemic control and weight loss among obese individuals with type 2 diabetes?

Q8.1. Methods

MEDLINE (via PubMed), the Cochrane Library (CENTRAL), and HERDIN Plus were systematically searched from database inception to March 31, 2024, and were subsequently updated to include relevant publications through September 2025. A combination of Medical Subject Headings (MeSH) and free-text terms—including “bariatric surgery,” “metabolic surgery,” and “type 2 diabetes mellitus”—was used. Only randomized controlled trials comparing any bariatric or metabolic surgical procedure with medical and/or lifestyle management in adults with type 2 diabetes were eligible. Studies were required to report at least one critical or important outcome as defined in the PICO framework. Non-randomized studies, trials without a comparator arm, and articles not written in English were excluded. Two reviewers independently screened studies, extracted data using a standardized form, and assessed risk of bias using the Cochrane Risk of Bias 2 (RoB 2) tool.

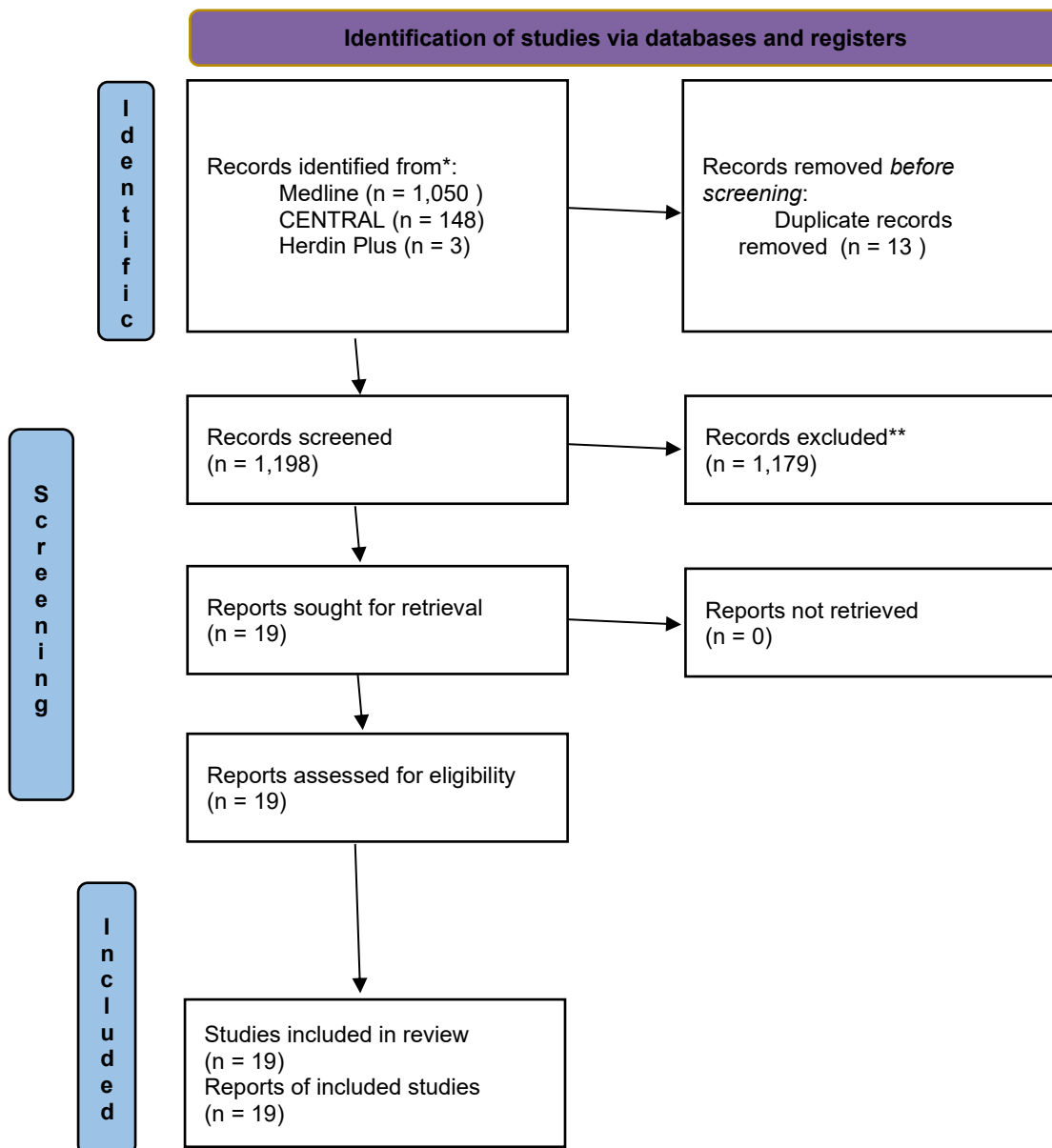
Q8.2. Search strategy

Q8.2: Search strategy including dates and database

Database	Search strategy / Search terms	Date and time of search	Results	
			Yield	Eligible
Medline	("medical nutrition therapy"[Title/Abstract] OR "dietary intervention"[Title/Abstract] OR "nutrition therapy"[MeSH Terms]) AND "type 2 diabetes mellitus"[Title/Abstract] AND (("randomized controlled trial"[Publication Type] OR "controlled clinical trial"[Publication Type] OR "randomized"[Title/Abstract] OR "placebo"[Title/Abstract] OR "drug therapy"[MeSH Subheading] OR "randomly"[Title/Abstract] OR "trial"[Title/Abstract] OR "groups"[Title/Abstract]) NOT ("animals"[MeSH Terms] NOT "humans"[MeSH Terms]))	March 31, 2024	511	
CENTRAL	("medical nutrition therapy" OR "dietary intervention") AND "type 2 diabetes mellitus"	March 31, 2024	164	
Herdin Plus	medical nutrition therapy	March 31, 2024	22	0

Q8.3. PRISMA flow diagram

PRISMA Flow Diagram



Q8.4. Characteristics of the Included Studies

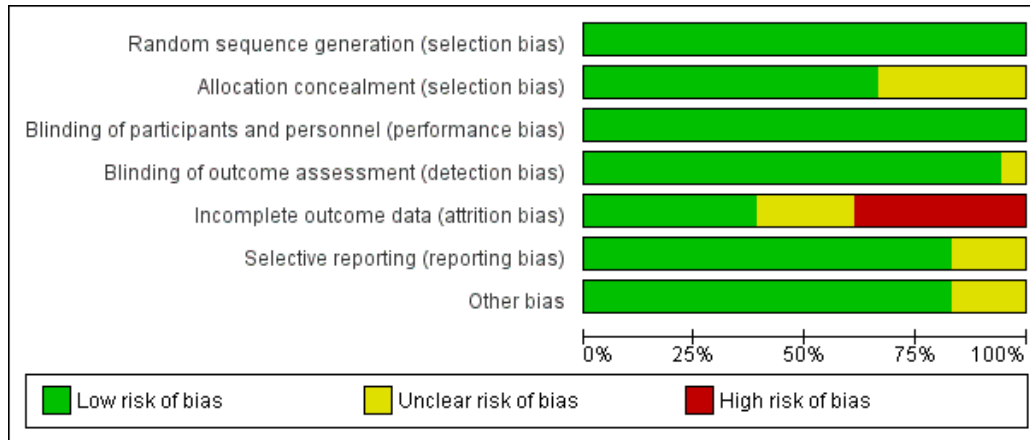
Characteristics of Included Studies

Study	Study Design	Country	Number of patients	Mean baseline BMI (kg/m ²)	Population	Intervention	Control
Courcoula s 2014 / 2015 / 2016	RCT	USA	69	35.5	Patients with T2DM	RYGB and LAGB	Intensive lifestyle weight loss intervention (LWLI) modeled after the Look AHEAD
Cummings 2016	RCT	USA	32	38.3	Patients with T2DM	Laparoscopic RYGB	intensive lifestyle and medical intervention (ILMI)
Ding 2015	RCT	USA	40	36.4	Patients with T2DM	LAGB	Why Wait
Dixon 2008	RCT	Australia	55	36	Patients with T2DM	LAGB	Multidisciplinary diabetes care based on ADA
Halperin 2014	RCT	USA	38	36	Patients with T2DM	RYGB	Why Wait
Ikramuddin 2013 / 2015 / 2017	RCT	USA	114	34.9	Patients with T2DM	Laparoscopic RYGB	Medical / Lifestyle management modeled on DPP and Look AHEAD protocol
Liang 2013	RCT	China	101	30.4	Patients with T2DM	RYGB	Multidisciplinary team
Mingrone 2012 /	RCT	Italy	53	45	Patients with T2DM	BPD / Laparosc	Multidisciplinary team

2015 / 2021						opie RYGB	
Parikh 2014	RCT	USA	44	32.8	Patients with T2DM	bypass, band, or sleeve gastrectomy based on patient preference	Intensive Medical Weight Management (MWM) Protocol
Schauer 2012 / 2014 / 2017	RCT	USA	80	35.5	Patients with T2DM	Laparoscopic RYGB and Sleeve Gastrectomy	intensive lifestyle and weight loss intervention (ILWI) modeled after Look Ahead trial
Wentworth 2014	RCT	Australia	48	29	Patients with T2DM	LAGB	Multidisciplinary team

Q1.5. Assessment of Risk of Bias

Risk of Bias Graph

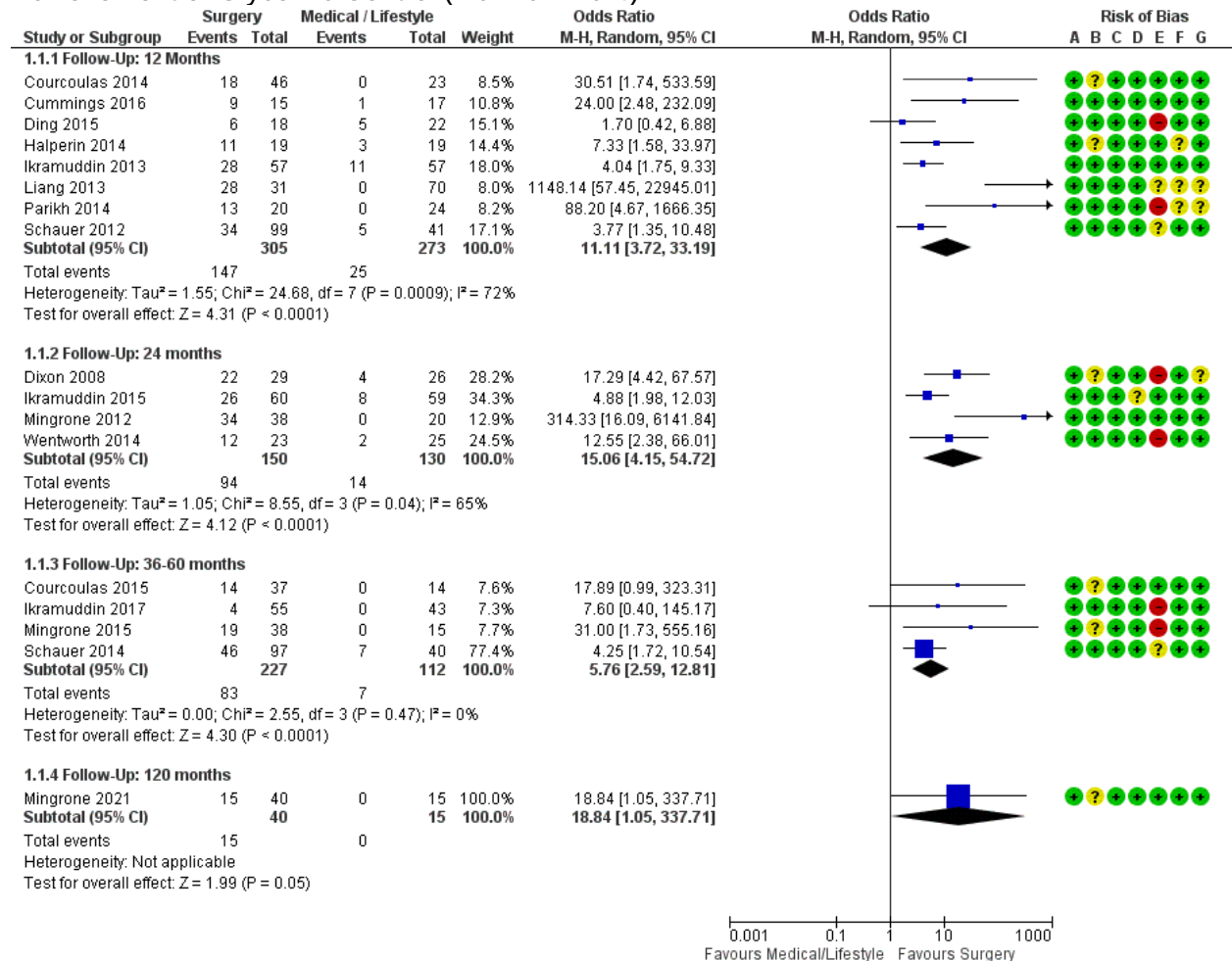


Risk of Bias Summary

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Courcoulas 2014	●	?	●	●	●	●	●
Courcoulas 2015	●	?	●	●	●	●	●
Cummings 2016	●	●	●	●	●	●	●
Ding 2015	●	●	●	●	●	●	●
Dixon 2008	●	?	●	●	●	●	?
Halperin 2014	●	?	●	●	●	?	●
Ikramuddin 2013	●	●	●	●	●	●	●
Ikramuddin 2015	●	●	●	?	●	●	●
Ikramuddin 2017	●	●	●	●	●	●	●
Liang 2013	●	●	●	●	?	?	?
Mingrone 2012	●	●	●	●	●	●	●
Mingrone 2015	●	?	●	●	●	●	●
Mingrone 2021	●	?	●	●	●	●	●
Parikh 2014	●	●	●	●	●	?	?
Schauer 2012	●	●	●	?	●	●	●
Schauer 2014	●	●	●	?	●	●	●
Schauer 2017	●	●	●	?	●	●	●
Wentworth 2014	●	●	●	●	●	●	●

Q1.6. Results

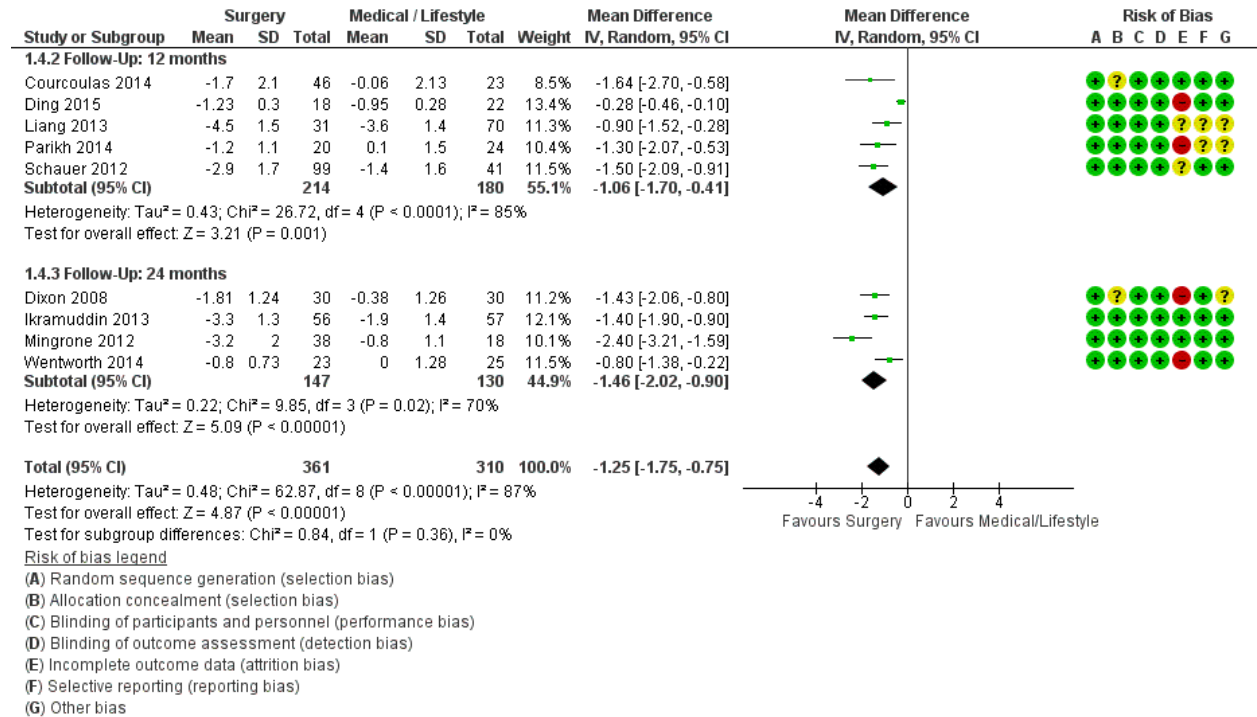
Achievement of Glycemic Control (HbA1c <7.0%)



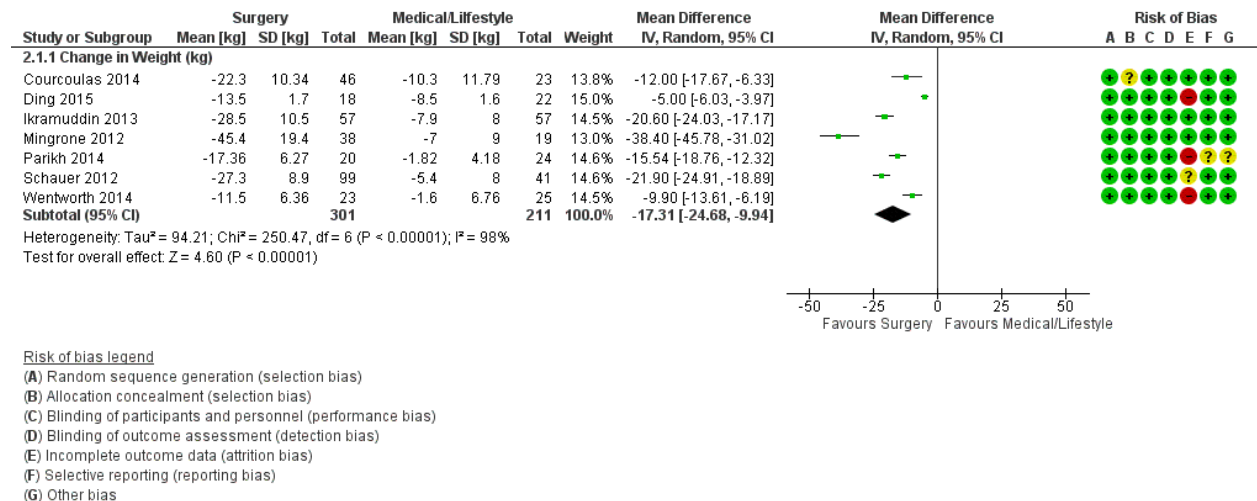
Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

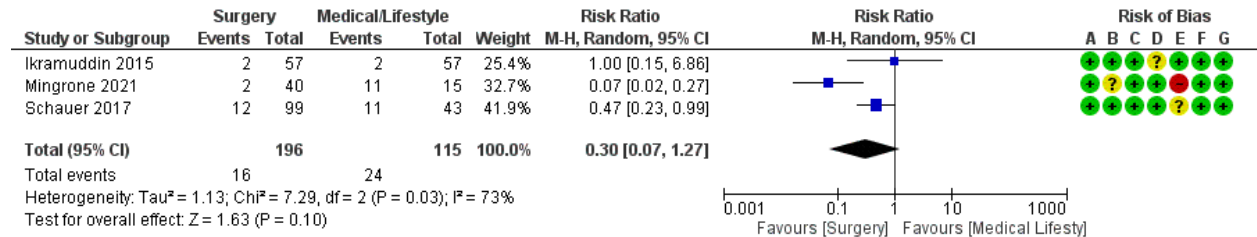
Change in HbA1c (%)



Change in Weight (kg)



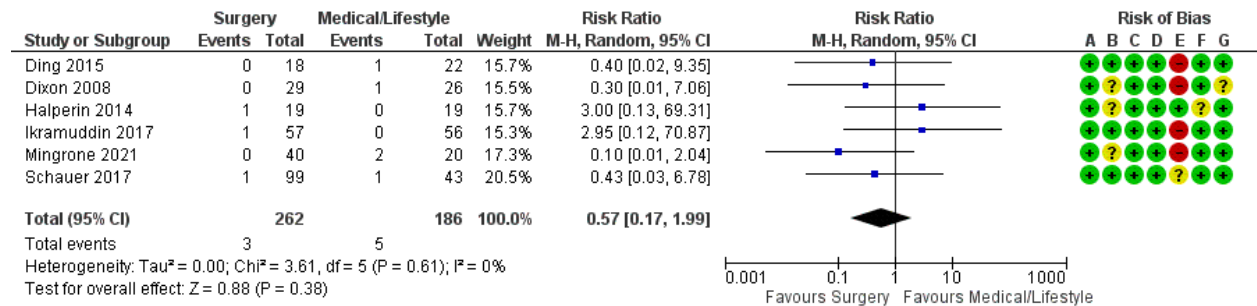
Microvascular Complications



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Annex 4.E. Macrovascular Complications



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Q8.7. GRADE Evidence Profile

GRADE Evidence profile table

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Bariatric Surgery	Lifestyle / Medical Management	Relative (95% CI)	Absolute (95% CI)		
Sustained Glycemic Control (follow-up: range 6 months to 12 months; assessed with: HbA1c < 7.0%)												
8	randomised trials	not serious	serious ^a	not serious	not serious	none	152/305 (49.8%)	25/273 (9.2%)	OR 8.71 (5.55 to 13.65)	38 more per 100 (from 27 more to 49 more)	⊕⊕⊕ ○ Moderate	
Sustained Glycemic Control (follow-up: range 12 months to 24 months; assessed with: HbA1c < 7.0%)												
4	randomised trials	not serious	serious ^b	not serious	serious ^c	none	94/150 (62.7%)	14/130 (10.8%)	OR 11.39 (6.11 to 21.22)	47 more per 100 (from 32 more to 61 more)	⊕⊕○ ○ Low	
Sustained Glycemic Control (follow-up: range 36 months to 60 months; assessed with: HbA1c < 7.0%)												
4	randomised trials	not serious	not serious	not serious	serious ^c	none	83/227 (36.6%)	7/112 (6.3%)	OR 6.90 (3.18 to 14.96)	25 more per 100 (from 11 more to 44 more)	⊕⊕⊕ ○ Moderate	
Weight Reduction (follow-up: range 12 months to 24 months; assessed with: kilogram)												
7	randomised trials	serious ^d	serious ^e	not serious	not serious	none	301	211	-	MD 8.99 kg lower (13.61 lower to 8.61 lower)	⊕⊕○ ○ Low	
Macrovascular Complications (follow-up: range 12 months to 120 months)												
6	randomised trials	serious ^d	not serious	not serious	serious ^c	none	3/262 (1.1%)	5/186 (2.7%)	OR 0.56 (0.19 to 1.69)	1 fewer per 100 (from 2 fewer to 2 more)	⊕⊕○ ○ Low	
Microvascular Complications												

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Bariatric Surgery	Lifestyle / Medical Management	Relative (95% CI)	Absolute (95% CI)		
3	randomised trials	not serious	serious ^f	not serious	serious ^c	none	16/196 (8.2%)	24/115 (20.9%)	OR 0.25 (0.13 to 0.49)	15 fewer per 100 (from 18 fewer to 9 fewer)	⊕⊕○ ○ Low	

CI: confidence interval; **MD:** mean difference; **OR:** odds ratio

Explanations

- a. $i^2 = 71\%$
 - b. $i^2 = 65\%$
 - c. low event rate / sample size and wide confidence interval
 - d. three studies with high risk of attrition bias
 - e. $i^2 = 78\%$
- $i^2 = 81\%$

Q8.8. Recommendations from other groups

Recommendations from Other Groups

Group/ Guidelines	Recommendation	Strength of Recommendation and Certainty of Evidence
ADA 2022 (S114)	Metabolic surgery should be a recommended option to treat adults w/ type 2 DM w/ BMI >40 kg/m² (BMI >37.5 kg/m² in Asian Americans) and adults w/ BMI 35-39.9 kg/m² (32.5-37.4 kg/m² in Asian Americans) who do not achieve durable weight loss and improvement in comorbidities w/ nonsurgical methods. Metabolic surgery may be considered as an option to treat Type 2 diabetes in adults w/ BMI 30-34.9 kg/m² (27.5-32.4 kg/m² in Asian Americans) who do not achieve durable weight loss and improvement in comorbidities (including hyperglycemia) w/ nonsurgical methods.	Grade A
AACE 2022	Persons w/ BMI >35 kg/m² and one or more severe obesity-related complications like OSA, T2DM, NAFLD should be considered for a bariatric surgery. Persons w/ BMI 30-34.9 kg/m² w/ T2DM and inadequate glycemic control despite optimal lifestyle and medical therapy should be considered for bariatric surgery.	Grade C BEL 3 Grade B BEL 2
Canadian CPG 2018	Bariatric surgery is a treatment option for adults w/ BMI >40 or BMI 35-39.9 kg/m ² w/ co-morbidities like T2DM who were not able to achieve weight loss maintenance w/ nonsurgical methods.	no grading
IDF 2017	Bariatric surgery should be considered in patients w/ T2DM and BMI >35 kg/m² (32.5 kg/m² in Asian population) PCPs should inform about bariatric surgery for T2DM adult w/ BMI between 30 and 35 kg/m² (27.5 and 32.5 kg/m² in Asians when metabolic response to regular treatment has been poor.	no grading

D. Monitoring and treatment targets

Guideline Question 9.

Should we recommend HbA1c to monitor glycemic control among nonpregnant adults with T2DM?

Q9.1. Methods and search strategies

A systematic review of the evidence was conducted to address the question, “Should HbA1c be used to monitor glycemic control among nonpregnant adults with type 2 diabetes mellitus (T2DM)?” The search covered MEDLINE (via PubMed), the Cochrane Library (CENTRAL), and HERDIN Plus from database inception to March 31, 2024, and was subsequently updated to include new and relevant publications through September 2025. Both Medical Subject Headings (MeSH) and free-text terms were used, including “hemoglobin A1c,” “HbA1c,” “glycated hemoglobin,” “glycemic monitoring,” “type 2 diabetes,” and “diabetes management.”

Randomized controlled trials, prospective or retrospective cohort studies, cross-sectional diagnostic accuracy studies, and systematic reviews evaluating HbA1c as a tool for monitoring glycemic control in adults with T2DM were included. Studies had to assess at least one critical or important outcome aligned with the PICO framework, such as treatment intensification, risk of hypoglycemia, microvascular or macrovascular complications, mortality, or correlation of HbA1c with short- and long-term glycemic outcomes (e.g., fasting plasma glucose, self-monitoring of blood glucose, or continuous glucose monitoring metrics). Studies involving pregnant populations, type 1 diabetes, gestational diabetes, or pediatric patients were excluded. Only full-text articles written in English were considered.

Two reviewers independently screened titles, abstracts, and full texts, extracted relevant data using a standardized data collection form, and reviewed reference lists of included studies to identify additional sources. Risk of bias was assessed using appropriate tools: the Cochrane Risk of Bias 2 tool for randomized trials, the ROBINS-I tool for nonrandomized studies, and QUADAS-2 for diagnostic accuracy studies. Certainty of evidence was appraised using the GRADE approach.

Guideline Question 10.

Should we use fasting plasma glucose (FPG) to monitor glycemic control among nonpregnant adults with T2DM?

Q10.1. Methods and search strategies

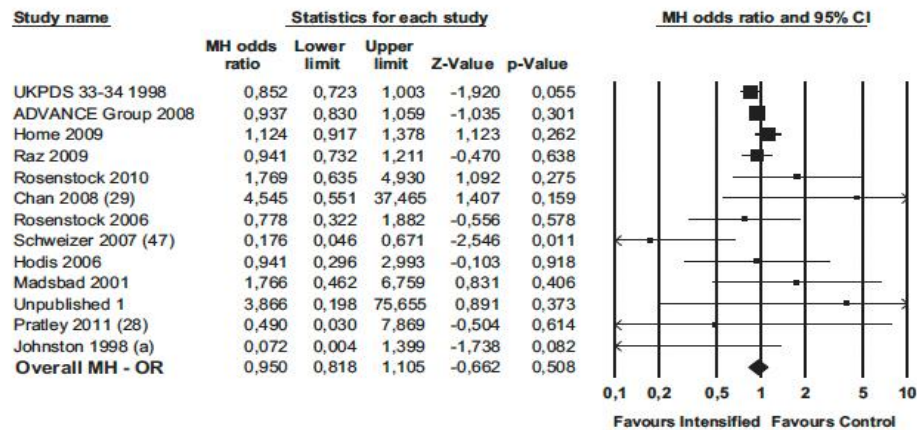
A systematic review of the evidence was conducted to address the question, “*Should fasting plasma glucose (FPG) be used to monitor glycemic control among nonpregnant adults with type 2 diabetes mellitus (T2DM)?*” Searches were performed in MEDLINE (via PubMed), the Cochrane Library (CENTRAL), and HERDIN Plus from database inception to March 31, 2024, and were subsequently updated to include newly published studies through September 2025. A combination of Medical Subject Headings (MeSH) and free-text terms was used, including “fasting plasma glucose,” “FPG,” “glycemic monitoring,” “glycemic control,” “type 2 diabetes,” and “diabetes management.”

Eligible study designs included randomized controlled trials, prospective and retrospective cohort studies, cross-sectional diagnostic accuracy studies, and systematic reviews that evaluated FPG as a monitoring tool in adults with T2DM. Studies were required to report at least one critical or important outcome consistent with the PICO framework, such as correlation of FPG with long-term glycemic control, microvascular or macrovascular complications, treatment decisions or intensification, hypoglycemia, or concordance between FPG and other monitoring modalities (e.g., HbA1c, self-monitoring of blood glucose, continuous glucose monitoring). Studies involving pregnant populations, type 1 diabetes, gestational diabetes, or pediatric patients were excluded. Only full-text, English-language publications were included.

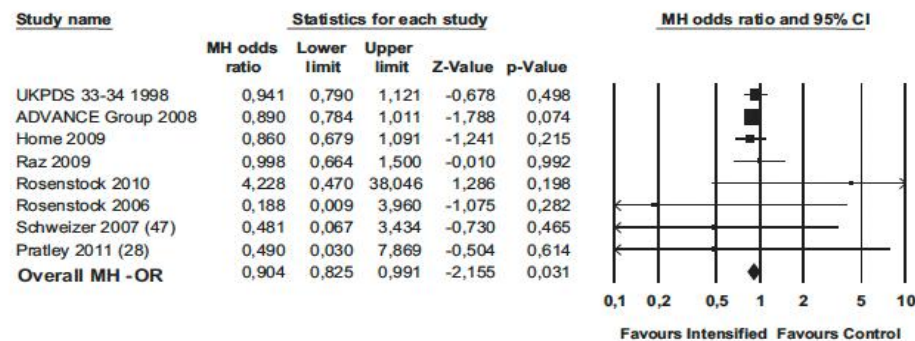
Two reviewers independently screened titles, abstracts, and full-text articles, extracted data using a standardized template, and examined reference lists of included studies to capture additional relevant evidence. Risk of bias was evaluated using appropriate validated tools: the Cochrane Risk of Bias 2 (RoB 2) tool for randomized trials, the ROBINS-I tool for nonrandomized comparative studies, and QUADAS-2 for diagnostic accuracy studies. Certainty of evidence was assessed using the GRADE approach.

Q1.6. Results

A) MAJOR CARDIOVASCULAR EVENTS



B) ALL-CAUSE MORTALITY



C) CARDIOVASCULAR MORTALITY

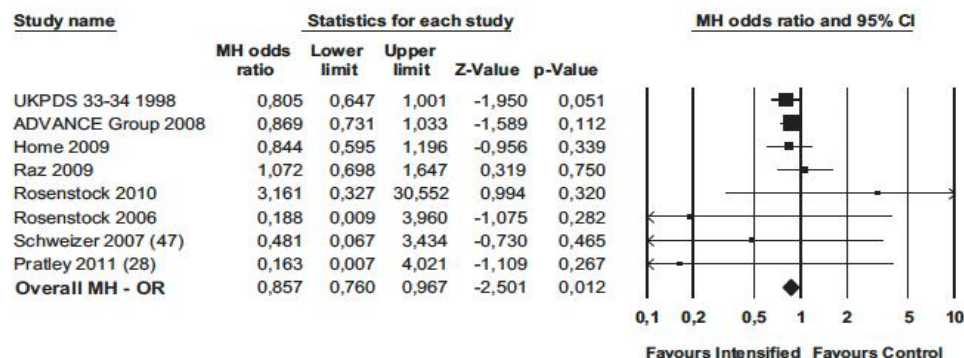


Figure 3 MH-Odds Ratio for major cardiovascular events (Panel A), all-cause mortality (Panel B), and cardiovascular mortality (Panel C) in patients on more intensive treatments, in trials with a between-group difference of FPG greater than 1 mmol/L.

Monami M, Adalsteinsson JE, Desideri CM, Ragghianti B, Dicembrini I, Mannucci E. Fasting and post-prandial glucose and diabetic complication. A meta-analysis. *Nutr Metab Cardiovasc Dis.* 2013 Jul;23(7):591-8.

Q1.7. GRADE Evidence Profile

Certainty of Evidence

Table 10.1: Summary of Evidence: Effect of fasting plasma glucose reduction on major cardiovascular events, all-cause mortality, and cardiovascular mortality.

Certainty assessment							N ₂ of patients		Effect		Certainty	Importance
N ₂ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	More intensive FPG reduction	Less intensive FPG reduction	Relative (95% CI)	Absolute (95% CI)		
Major cardiovascular events												
13	randomised trials	not serious	not serious	not serious	not serious	none			OR 0.950 (0.818 to 1.105)	1 fewer per 1,000	⊕⊕⊕○ Moderate	CRITICAL
All-cause mortality												
8	randomised trials	not serious	not serious	not serious	not serious	none			OR 0.904 (0.825 to 0.991)	1 fewer per 1,000	⊕⊕⊕○ Moderate	CRITICAL
Cardiovascular mortality												
8	randomised trials	not serious	not serious	not serious	not serious	none			OR 0.857 (0.760 to 0.967)	1 fewer per 1,000	⊕⊕⊕○ Moderate	CRITICAL

Monami M, Adalsteinsson JE, Desideri CM, Ragghianti B, Dicembrini I, Mannucci E. Fasting and post-prandial glucose and diabetic complication. A meta-analysis. *Nutr Metab Cardiovasc Dis.* 2013 Jul;23(7):591-8.

Guideline Question 11.

Should we recommend the use of 1,5-anhydroglucitol (1,5-AG) to monitor glycemic control among nonpregnant adults with T2DM?

Q11.1. Methods

The selected guidelines (refer to the main methodology) were carefully reviewed to determine their appropriateness in addressing the clinical question. The scientific evidence underpinning their recommendations, as well as the adaptability of these recommendations to the local context, was thoroughly evaluated. In addition, MEDLINE was systematically searched from database inception to March 10, 2023, and was subsequently updated to include newly published studies through September 2025, using the following search terms: “1,5-anhydroglucitol,” “1,5-AG,” “glycemic control,” “retinopathy,” “neuropathy,” “nephropathy,” “sensitivity,” and “specificity” (Appendix A). Full-text articles of all eligible studies were retrieved to evaluate the diagnostic and clinical utility of 1,5-anhydroglucitol in relation to glycemic control and microvascular complications.

Q11.2. Search strategy

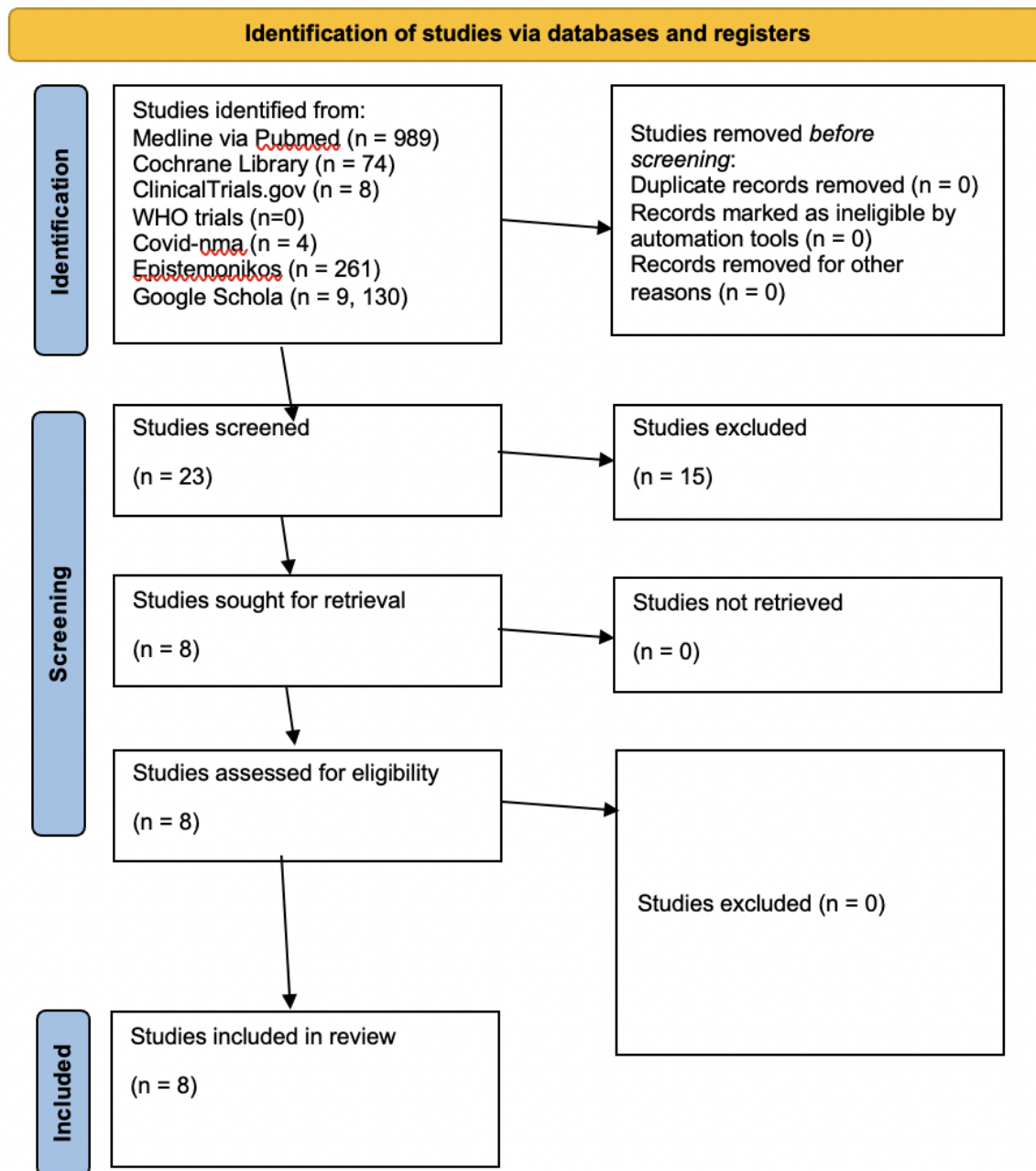
Utilizing the PICO for key search terms of each clinical question, search was conducted through PubMed with the following study type filters: meta-analysis, RCTs, systematic review

DATABASE	SEARCH STRATEGY / SEARCH TERMS	DATE AND TIME OF SEARCH	RESULTS	
			Yield	Eligible
Medline	("1,5-anhydroglucitol" OR "1,5-anhydroglucitol levels") AND ("glycemic control" OR "glycemic index" OR "glycemic management") AND ("sensitivity" OR "specificity") AND ("diabetic complications" OR "diabetes complications" OR "diabetic neuropathy" OR "diabetic retinopathy" OR "diabetic nephropathy")	March 3, 2023 8PM	13	0

Q11.3. PRISMA flow diagram

PRISMA FLOW DIAGRAM FOR UPDATING SYSTEMATIC REVIEWS

PRISMA Flow Diagram



Q11.4. Characteristics of the Included Studies and Results

We didn't find any study that directly assessed the effect of 1,5-AG on diabetic complications or glycemic control. There were also no studies assessing its accuracy.

Q11.5. Recommendations from other groups

None of the clinical guidelines reviewed mentioned 1,5-AG.

Guideline Question 12.

Should we use fructosamine to monitor glycemic control among adults with type 2 DM?

Q12.1. Methods

A systematic search of MEDLINE was conducted from database inception to August 2024 using the terms diabetes mellitus, fructosamine, sensitivity and specificity, and diagnostic accuracy. Only published articles in English were included. Eligible studies assessed the relationship between fructosamine and glycemic control, its correlation with HbA1c, its association with diabetes-related complications, and potential analytical limitations.

Q12.2 Search strategy

Utilizing the PICO for key search terms of each clinical question, search was conducted through PubMed with the following study type filters: meta-analysis, systematic review.

Database	Search Strategy / Search Terms	Date and Time of search	Results	
			Yield	Eligible
Medline	Search: (((diabetes mellitus, type 2[MeSH Terms]) OR (NIDDM)) OR ((Non insulin* AND depe*) OR (Noninsulin* AND depe*) OR (Non insulin depe*))) OR ((Type II diabet*) OR (Type 2 diabet*) OR (diabet* type 2) OR (diabet* type II)) AND fructosamine AND ((sensitivity and specificity[MeSH Terms]) OR (predict* OR diagnos* OR accura*)) Filters: Meta-Analysis, Systematic Review (("diabetes mellitus, type 2"[MeSH Terms] OR ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes mellitus"[All Fields] OR "niddm"[All Fields] OR "niddms"[All Fields]) OR ((("Non"[All Fields] AND "insulin"[All Fields] AND "depe"[All Fields]) OR ("noninsulin"[All Fields] AND "depe"[All Fields]) OR ("Non"[All Fields] AND ("insulin"[MeSH Terms] OR "insulin"[All Fields] OR "insulin s"[All Fields] OR "insuline"[All Fields] OR "insulinic"[All Fields] OR "insulinization"[All Fields] OR "insulinized"[All Fields] OR "insulins"[MeSH Terms] OR "insulins"[All Fields]) AND "depe"[All Fields])) OR ((("Type"[All Fields] AND "II"[All Fields] AND "diabet"[All Fields]) OR ("Type"[All Fields] AND "2"[All Fields] AND "diabet"[All Fields]) OR ("diabet"[All Fields] AND "Type"[All Fields] AND "2"[All Fields]) OR ("diabet"[All Fields] AND "Type"[All Fields] AND "II"[All Fields])))) AND ("fructosamin"[All Fields] OR "fructosamine"[MeSH Terms] OR	August 21, 2024, 11:17pm	6	0

	"fructosamine"[All Fields] OR "fructosamines"[All Fields]) AND ("sensitivity and specificity"[MeSH Terms] OR ("predict*"[All Fields] OR "diagnos*"[All Fields] OR "accura*"[All Fields]))) AND (meta-analysis[Filter] OR systematicreview[Filter])			
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Q12.3. Recommendations from other groups

Various practice guidelines mention fructosamine as a test for monitoring glycemic control but only as an alternative in situations wherein HbA1c cannot be relied upon.

Group or Agency	Recommendation	Strength of Recommendation/ Certainty/Quality of Evidence
Diabetes Canada 2018	Utility of fructosamine and its correlation with A1c has not yet been established	
ADA Standards of Care 2024	Fructosamine may be a useful alternative to monitor glycemic control when interpretation of A1c is problematic or when A1c cannot be measured	
NICE	Fructosamine may be used to estimate trends in blood glucose control if HbA1c monitoring is invalid	
Malaysia	Fructosamine is an alternative to HbA1c in monitoring glycemic control in patients with inaccurate HbA1c measurements	

Guideline Question 13.

Should we do SMBG to monitor glycemic control among nonpregnant adults with type 2 DM?

Q13.1. Methods

The selected clinical practice guidelines (refer to the main methodology) were carefully reviewed to ascertain their appropriateness to address the clinical question. Additionally, the scientific evidence forming the basis for the recommendations, along with their adaptability to the local context, was scrutinized.

To supplement guideline adaptation, a systematic search of the literature was conducted to evaluate the effectiveness of self-monitoring of blood glucose (SMBG) for monitoring glycemic control among nonpregnant adults with type 2 diabetes mellitus (T2DM). Searches were performed in MEDLINE (via PubMed), the Cochrane Library (CENTRAL), and HERDIN Plus from database inception to March 31, 2024, and were subsequently updated to include newly published studies through September 2025. Both Medical Subject Headings (MeSH) and free-text terms were used, including: “self-monitoring of blood glucose,” “SMBG,” “capillary glucose monitoring,” “glycemic monitoring,” “type 2 diabetes,” and “diabetes management.”

Eligible study designs included randomized controlled trials, prospective and retrospective cohort studies, and systematic reviews that compared SMBG with usual care, minimal monitoring, or alternative monitoring methods (e.g., fasting plasma glucose or HbA1c-based monitoring). To be included, studies had to report at least one critical or important outcome consistent with the PICO framework, such as HbA1c reduction, hypoglycemia, treatment intensification, patient adherence, quality of life, or frequency of SMBG use. Studies involving type 1 diabetes, gestational diabetes, pregnant women with T2DM, or pediatric populations were excluded. Only full-text articles written in English were considered.

Two reviewers independently screened all titles, abstracts, and full texts, extracted relevant data using a standardized form, and performed citation tracking to identify additional eligible studies. Risk of bias was assessed using appropriate tools: the Cochrane Risk of Bias 2 (RoB 2) tool for randomized trials, the ROBINS-I tool for nonrandomized studies, and AMSTAR 2 for systematic reviews. Certainty of evidence was appraised using the GRADE approach.

Q13.2. Search strategy

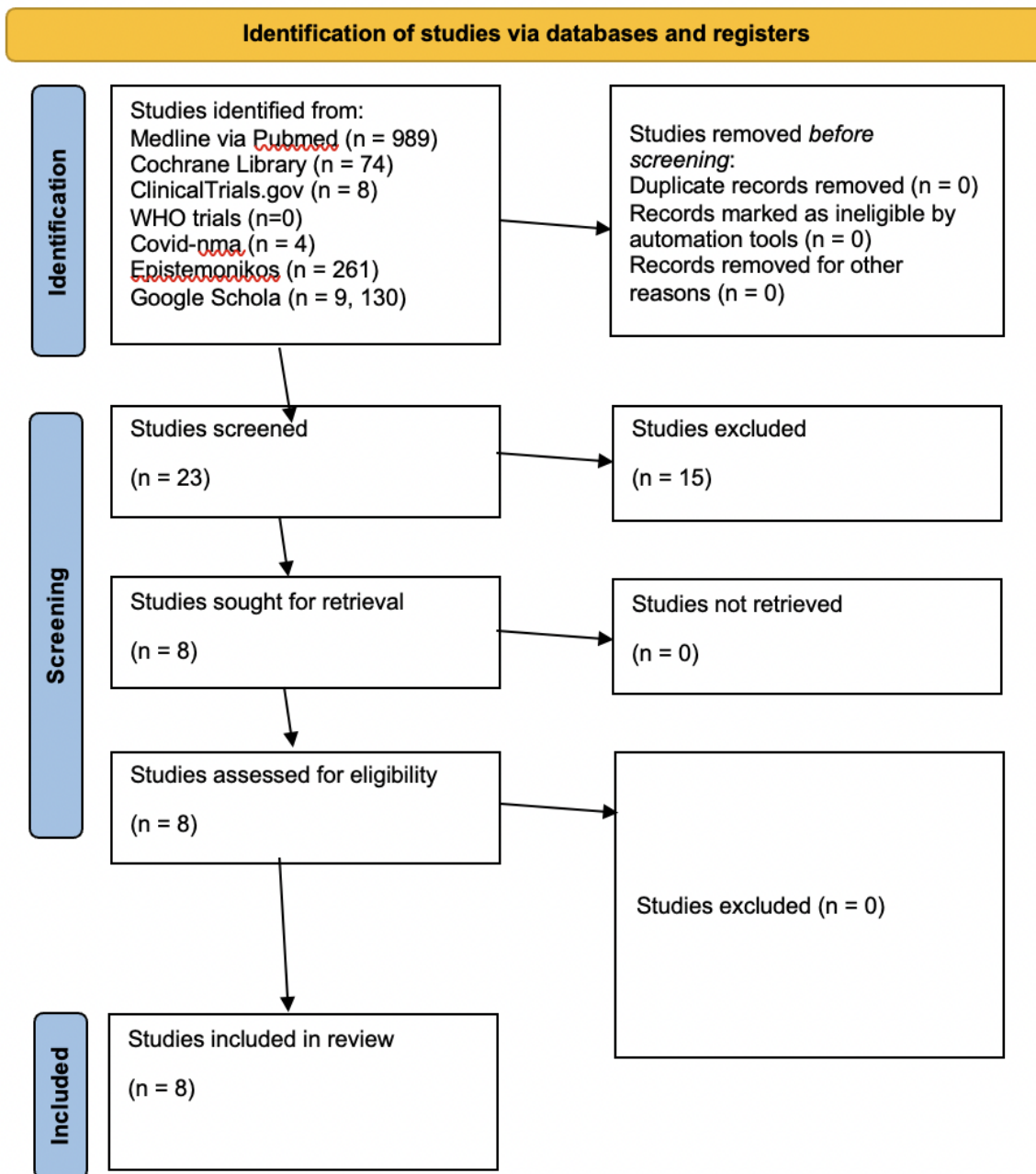
Utilizing the PICO for key search terms of each clinical question, search was conducted through PubMed with the following study type filters: meta-analysis, RCTs, systematic review

DATABASE	SEARCH STRATEGY / SEARCH TERMS	DATE AND TIME OF SEARCH	RESULTS	
			Yield	Eligible
Medline	("diabetes mellitus, type 2"[MeSH] OR "Type 2 diabetes"[Title/Abstract] OR "DMT2"[Title/Abstract]) AND ("self-monitoring of blood glucose"[Title/Abstract] OR "SMBG"[Title/Abstract] OR "blood glucose monitoring"[Title/Abstract] OR "self-blood glucose monitoring"[MeSH]) AND ("glycemic control"[MeSH] OR "glycemic management"[Title/Abstract] OR "glucose control"[Title/Abstract]) Filters: Meta-Analysis, Randomized Controlled Trial, Systematic Review, in the last 10 years	March 3, 2023 8PM	28	5

Q13.3. PRISMA flow diagram

PRISMA FLOW DIAGRAM FOR UPDATING SYSTEMATIC REVIEWS

PRISMA Flow Diagram



Q13.4. Results

Adapted from Diabetes Canada Self-monitoring blood glucose frequency and pattern tool
<http://guidelines.diabetes.ca/self-management/smbg-tool>

I. Pattern Recommendation: Basal Insulin Only

NPH or long-acting insulin analog (Detemir, Glargine, Degludec) is typically given at bedtime.

Perform SMBG at least as often as insulin is being given.

Upon basal insulin initiation, before breakfast SMBG reading is used for basal insulin adjustment. If before breakfast SMBG is already at target but HbA1c is still not, check before dinner or bedtime SMBG.

	breakfast		lunch		supper		bedtime	night
	before	after	before	after	before	after		
Insulin							NPH / long (basal)	
SMBG pattern	SMBG test							
Adjustment	Basal insulin ↑ if BG hi ↓ if BG low							

II. Pattern Recommendation: Premixed Insulin twice daily

Insulin doses are typically given pre-breakfast and pre-dinner.

Perform SMBG at least as often as insulin is being given.

SMBG BID before breakfast and before dinner for insulin dose adjustment.

SMBG BID (alternating times) once glycemic targets are met.

	breakfast		lunch		supper		bedtime	night
	before	after	before	after	before	after		
Insulin	Premix/ NPH				Premix/ NPH			
SMBG pattern 1: Starting	SMBG test		SMBG test		SMBG test		SMBG test	
SMBG pattern 2: Stable Alternating daily	SMBG test				SMBG test			
			SMBG test				SMBG test	
Adjustment	Pre-supper insulin ↑ if BG hi ↓ if BG low		Pre-breakfast insulin ↑ if BG hi ↓ if BG low		Pre-breakfast insulin ↑ if BG hi ↓ if BG low		Pre-supper insulin ↑ if BG hi ↓ if BG low	

III. Pattern Recommendation: DMT2 not on insulin

Recently diagnosed DMT2 in the past 6 months or not meeting glycemic targets.

A. SMBG pattern to learn the effects of various meals

	breakfast		lunch		supper		bedtime	night
	before	after	before	after	before	after		
Sunday								
Monday								
Tuesday								
Wednesday								
Thursday								
Friday								
Saturday								

B. SMBG pattern to learn the effects of exercise

	breakfast		lunch		supper		bedtime	night
	before	after	before	after	before	after		
Sunday				Exercise	TEST			
Monday								
Tuesday								
Wednesday								
Thursday						Exercise	TEST	
Friday								
Saturday								

IV. Pattern Recommendations: acutely-ill or starting steroids

	breakfast		lunch		supper		bedtime	night
	before	after	before	after	before	after		
Sunday								
Monday								
Tuesday								
Wednesday								
Thursday								
Friday								
Saturday								

Q13.5. GRADE Evidence Profile

Table 1: Evidence summary table of SMBG versus no SMBG among non-insulin treated patients with Type 2DM.

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	SMBG	no SMBG	Relative (95% CI)	Absolute (95% CI)		

mean HbA1c difference (assessed with: mean HbA1c difference)*

14	randomised trials	serious	not serious	not serious	not serious	none	1752	1524	-	MD 0.3% lower (0.4 lower to 0.2 lower)	⊕⊕ ⊕○ Moderate	CRITICAL
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severe hypoglycemia (assessed with: incidence rate)#

3	randomised trials	serious	not serious	not serious	not serious	none	1/853 (0.1%)	4/727 (0.6%)	RR 0.35 (0.07 to 1.77)	4 fewer per 1,000 (from 5 fewer to 4 more)	⊕⊕ ⊕○ Moderate	CRITICAL
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CI: confidence interval; **MD:** mean difference; **RR:** risk ratio

*Chircop J, Sheffield D, Kotera Y. Systematic Review of Self-Monitoring of Blood Glucose in Patients With Type 2 Diabetes. Nurs Res. 2021 Nov-Dec 01;70(6):487-497.

#NICE Grade tables and metaanalysis results. Accessed <https://www.nice.org.uk/guidance/ng28/evidence/appendix-d-grade-tables-and-metaanalysis-results-pdf-78671532573>

Table 2: Evidence Summary of SMBG: 8-14 measurements /week compared to SMBG: none or less than one measurement /week for glycemic control among patients with non-insulin treated T2DM

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	SMBG: 8-14 measurements /week	SMBG: none or less than one measurement /week	Relative (95% CI)	Absolute (95% CI)		

HBA1c reduction (follow-up: mean 12 months; assessed with: mean difference)

3	randomised trials	not serious	not serious	not serious	not serious	none	619	629	-	MD 0.2 % lower (0.29 lower to 0.11 lower)	⊕⊕⊕ ⊕ ○ Moderate	CRITICAL
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HBA1c reduction (follow-up: mean 6 months; assessed with: mean difference)

7	randomised trials	not serious	not serious	not serious	not serious	none	707	697	-	MD 0.46 % lower (0.54 lower to 0.39 lower)	⊕⊕⊕ ⊕ ○ Moderate	CRITICAL
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Citation: Xu Y, Tan DHY, Lee JY. Evaluating the impact of self-monitoring of blood glucose frequencies on glucose control in patients with type 2 diabetes who do not use insulin: A systematic review and meta-analysis. Int J Clin Pract. 2019 Jul;73(7):e13357.

Table 3: Evidence Summary of SMBG: upto 7 measurements /week compared to SMBG: none or less than one measurement /week for glycemic control among patients with non-insulin treated T2DM

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	SMBG: upto 7 measurements/week	SMBG: one or less than one measurement/week	Relative (95% CI)	Absolute (95% CI)		

HBA1c reduction (follow-up: mean 12 months; assessed with: mean difference)

4	randomised trials	not serious	not serious	not serious	not serious	none	462	466	-	MD 0.07 % higher (0.05 lower to 0.2 higher)	⊕⊕ ⊕ ○ Moderate	CRITICAL
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HBA1c reduction (follow-up: mean 6 months; assessed with: mean difference)

3	randomised trials	not serious	not serious	not serious	not serious	none	402	404	-	MD 0.15 % lower (0.4 lower to 0.1 higher)	⊕⊕ ⊕ ○ Moderate	CRITICAL
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Citation: Xu Y, Tan DHY, Lee JY. Evaluating the impact of self-monitoring of blood glucose frequencies on glucose control in patients with type 2 diabetes who do not use insulin: A systematic review and meta-analysis. Int J Clin Pract. 2019 Jul;73(7):e13357.

Table 4: Evidence Summary of SMBG results applied to adjust diabetes medication compared to SMBG results not applied for glycemic control among non-insulin treated T2DM

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	SMBG results applied to adjust diabetes medication	SMBG results not applied	Relative (95% CI)	Absolute (95% CI)		

HBA1c reduction (follow-up: mean 12 months; assessed with: mean difference)

4	randomised trials	not serious	not serious	not serious	not serious	none	836	862	-	MD 0.23% lower (0.31 lower to 0.15 lower)	⊕⊕ ⊕○ Moderate	CRITICAL
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Citation: Xu Y, Tan DHY, Lee JY. Evaluating the impact of self-monitoring of blood glucose frequencies on glucose control in patients with type 2 diabetes who do not use insulin: A systematic review and meta-analysis. Int J Clin Pract. 2019 Jul;73(7):e13357.

Q13.6. Recommendations from other groups

Group or Agency	Recommendation	Strength of Recommendation/ Certainty/Quality of Evidence
ADA 2023	Yes No mention regarding frequency	Grade A Grade B Grade B
NICE 2022	"Do not routinely offer" except if on insulin, hypoglycemia-prone No mention regarding frequency	Not mentioned
CDA	Yes (Grade A level 1 T1DM) (Grade C, level 3 T2DM) Individualized frequency (Grade D consensus)	Not mentioned
IDF	Yes Mandatory for patients on insulin Useful for treatment adjustment and acute illness No mention regarding frequency	Not mentioned
SIGN	Yes for those on insulin and prone to hypoglycemia or planning to become pregnant	Not mentioned
Malaysian	Yes, recommended If on insulin, desired if on OAD - Individualized frequency	Grade C

Guideline Question 14.

Should we use a continuous glucose monitor (CGM) to monitor glycemic control among nonpregnant adults with type 2 DM?

Q1.1. Methods

The selected guidelines (refer to the main methodology) were carefully reviewed to ascertain their appropriateness in addressing the clinical question. Additionally, the scientific evidence forming the basis for the recommendations, along with their adaptability to the local context, was scrutinized.

To supplement guideline adaptation, a systematic search of the literature was conducted to evaluate the effectiveness of continuous glucose monitoring (CGM) for glycemic monitoring among nonpregnant adults with type 2 diabetes mellitus (T2DM). Searches were performed in MEDLINE (via PubMed), the Cochrane Library (CENTRAL), and HERDIN Plus from database inception to March 31, 2024, and were subsequently updated to include newly published studies through September 2025. Both Medical Subject Headings (MeSH) and free-text terms were used, including: “continuous glucose monitoring,” “CGM,” “real-time CGM,” “intermittently scanned CGM,” “flash glucose monitoring,” “glycemic monitoring,” and “type 2 diabetes.”

Eligible study designs included randomized controlled trials, prospective or retrospective cohort studies, and systematic reviews comparing CGM with usual care, self-monitoring of blood glucose (SMBG), or other glycemic monitoring strategies. Studies were required to report at least one critical or important outcome based on the PICO framework, including HbA1c reduction, time in range (TIR), time below range (TBR), hypoglycemia, glycemic variability, treatment intensification, patient satisfaction, adherence, and quality of life. Studies exclusively involving type 1 diabetes, gestational diabetes, pregnant individuals with T2DM, or pediatric populations were excluded. Only full-text publications written in English were considered.

Two reviewers independently screened titles, abstracts, and full texts; extracted relevant data using a standardized data collection form; and examined reference lists of included articles for additional eligible studies. Risk of bias was assessed using validated tools appropriate for each study design: the Cochrane Risk of Bias 2 (RoB 2) tool for randomized trials, the ROBINS-I tool for nonrandomized studies, and AMSTAR 2 for systematic reviews. Certainty of evidence was evaluated using the GRADE approach.

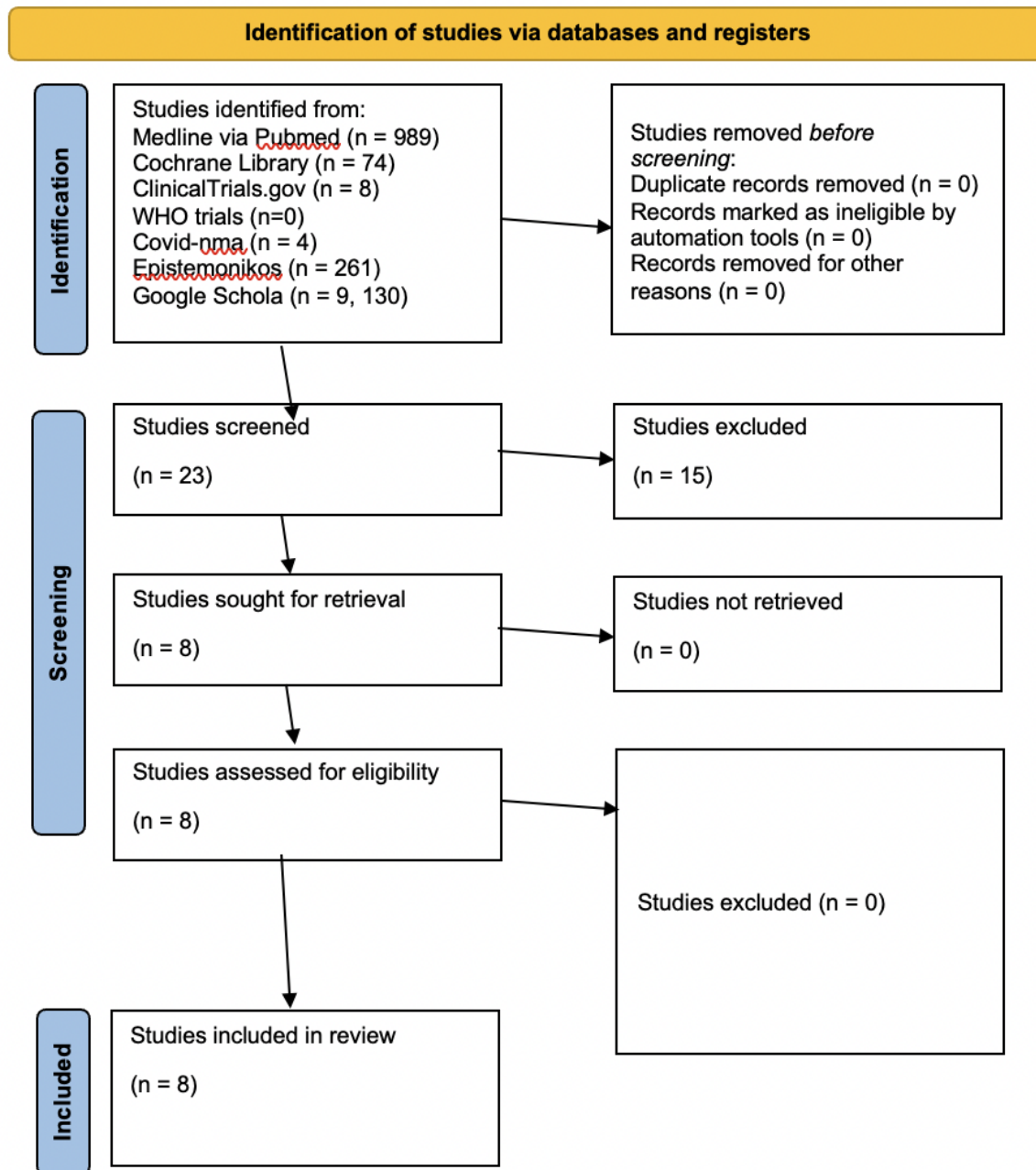
Q14.2. Search strategy

Utilizing the PICO for key search terms of each clinical question, search was conducted through PubMed with the following study type filters: meta-analysis, RCTs, systematic review

DATABASE	SEARCH STRATEGY / SEARCH TERMS	DATE AND TIME OF SEARCH	RESULTS	
			Yield	Eligible
Medline	("Continuous Glucose Monitoring"[MeSH] OR "Continuous Glucose Monitoring" OR "CGM") AND ("Self-Monitoring of Blood Glucose"[MeSH] OR "Self-Monitoring Blood Glucose" OR "SMBG") Filters: in the last 5 years, Meta-Analysis (("search"[All Fields] OR "searched"[All Fields] OR "searches"[All Fields] OR "searching"[All Fields] OR "searchs"[All Fields]) AND ("Continuous Glucose Monitoring"[MeSH Terms] OR "Continuous Glucose Monitoring"[All Fields] OR "CGM"[All Fields]) AND ("Self-Monitoring Blood Glucose"[All Fields] OR "SMBG"[All Fields])) AND ((y_5[Filter]) AND (meta-analysis[Filter]))	March 3, 2024 8PM	7	2

Q14.3. PRISMA flow diagram

PRISMA Flow Diagram



Q14.4. GRADE Evidence Profile

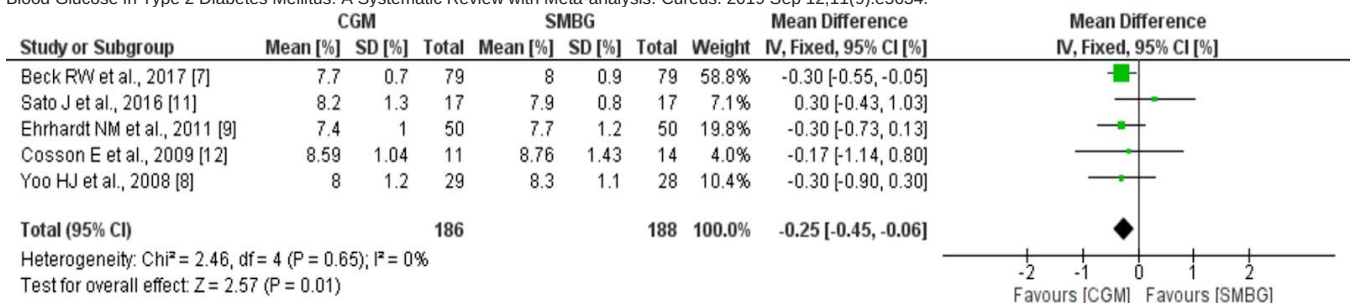
Evidence Summary: CGM (both RTCGM and FGM) compared to SMBG for monitoring glycemic control among individuals with DMT2

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	CGM (both RTCGM and FGM)	SMBG	Relative (95% CI)	Absolute (95% CI)		

HbA1c reduction (follow-up: range 3 months to 8 months)

5	randomised trials	not serious	not serious	not serious	not serious	none	186	188	-	MD 0.25 % lower (0.45 lower to 0.06 lower)	⊕⊕⊕ ○ Low	CRITICAL
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Citation: Janapala RN, Jayaraj JS, Fathima N, Kashif T, Usman N, Dasari A, Jahan N, Sachmechi I. Continuous Glucose Monitoring Versus Self-monitoring of Blood Glucose in Type 2 Diabetes Mellitus: A Systematic Review with Meta-analysis. Cureus. 2019 Sep 12;11(9):e5634.



Citation: Janapala RN, Jayaraj JS, Fathima N, Kashif T, Usman N, Dasari A, Jahan N, Sachmechi I. Continuous Glucose Monitoring Versus Self-monitoring of Blood Glucose in Type 2 Diabetes Mellitus: A Systematic Review with Meta-analysis. Cureus. 2019 Sep 12;11(9):e5634.

Evidence Summary: Flash CGM compared to SMBG for glycemic control among patients with type 2 DM on insulin

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Flash CGM	SMBG	Relative (95% CI)	Absolute (95% CI)		

Insulin-treated T2DM: HBA1c reduction (follow-up: mean 17 weeks; assessed with: mean difference in HBA1c)

2	randomised trials	not serious	not serious	not serious	not serious	none	202	123	-	MD 0.22 % lower (0.72 lower to 0.28 higher)	⊕⊕⊕ ○ Low	CRITICAL
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Citation: Castellana M, Parisi C, Di Molfetta S, Di Gioia L, Natalicchio A, Perrini S, Cignarelli A, Laviola L, Giorgino F. Efficacy and safety of flash glucose monitoring in patients with type 1 and type 2 diabetes: a systematic review and meta-analysis. BMJ Open Diabetes Res Care. 2020 Jun;8(1):e001092.

Q14.5. Recommendations from other groups

Group or Agency	Recommendation	Strength of Recommendation/Certainty/Quality of Evidence
ADA 2023	Continuous glucose monitoring is recommended for individuals with type 1 diabetes and for certain individuals with type 2 diabetes to improve glycemic control and reduce the risk of hypoglycemia.	Grade A
AACE 2023	CGM is recommended for individuals with type 1 diabetes and for those with type 2 diabetes who are on insulin therapy.	Grade A
NICE 2022	Continuous glucose monitoring should be offered to individuals with type 1 diabetes, particularly those who experience frequent episodes of hypoglycemia or have difficulty managing their blood glucose levels.	Grade A
Canadian Diabetes Association 2023	Continuous glucose monitoring (CGM) is recommended for individuals with diabetes, particularly those with type 1 diabetes and some with type 2 diabetes who are on intensive insulin therapy.	Grade A

Guideline Question 15.

Should we use aspirin among patients with type 2 DM for primary prevention of cardiovascular events?

Q15.1. Methods

A systematic search of the literature was conducted to evaluate the effectiveness and safety of aspirin or clopidogrel for the primary prevention of cardiovascular events among adults with type 2 diabetes mellitus (T2DM). Searches were performed in MEDLINE (via PubMed) and HERDIN Plus from database inception to March 31, 2024, and were subsequently updated to include newly published studies through September 2025. A combined Medical Subject Headings (MeSH) and free-text strategy was used, including the terms: “aspirin,” “acetylsalicylic acid,” “clopidogrel,” “antiplatelet therapy,” “primary prevention,” “cardiovascular disease,” and “type 2 diabetes.” Filters for randomized controlled trials were applied when available.

Eligible study designs included randomized controlled trials comparing aspirin or clopidogrel with placebo, usual care, or no antiplatelet therapy among nonpregnant adults with T2DM without established atherosclerotic cardiovascular disease. To be included, studies were required to report at least one critical or important outcome based on the PICO framework, such as major adverse cardiovascular events (MACE), myocardial infarction, stroke, all-cause mortality, cardiovascular mortality, major bleeding, and other clinically relevant harms. Studies involving secondary-prevention populations, mixed populations without diabetes subgroup data, noncomparative designs, or publications not written in English were excluded.

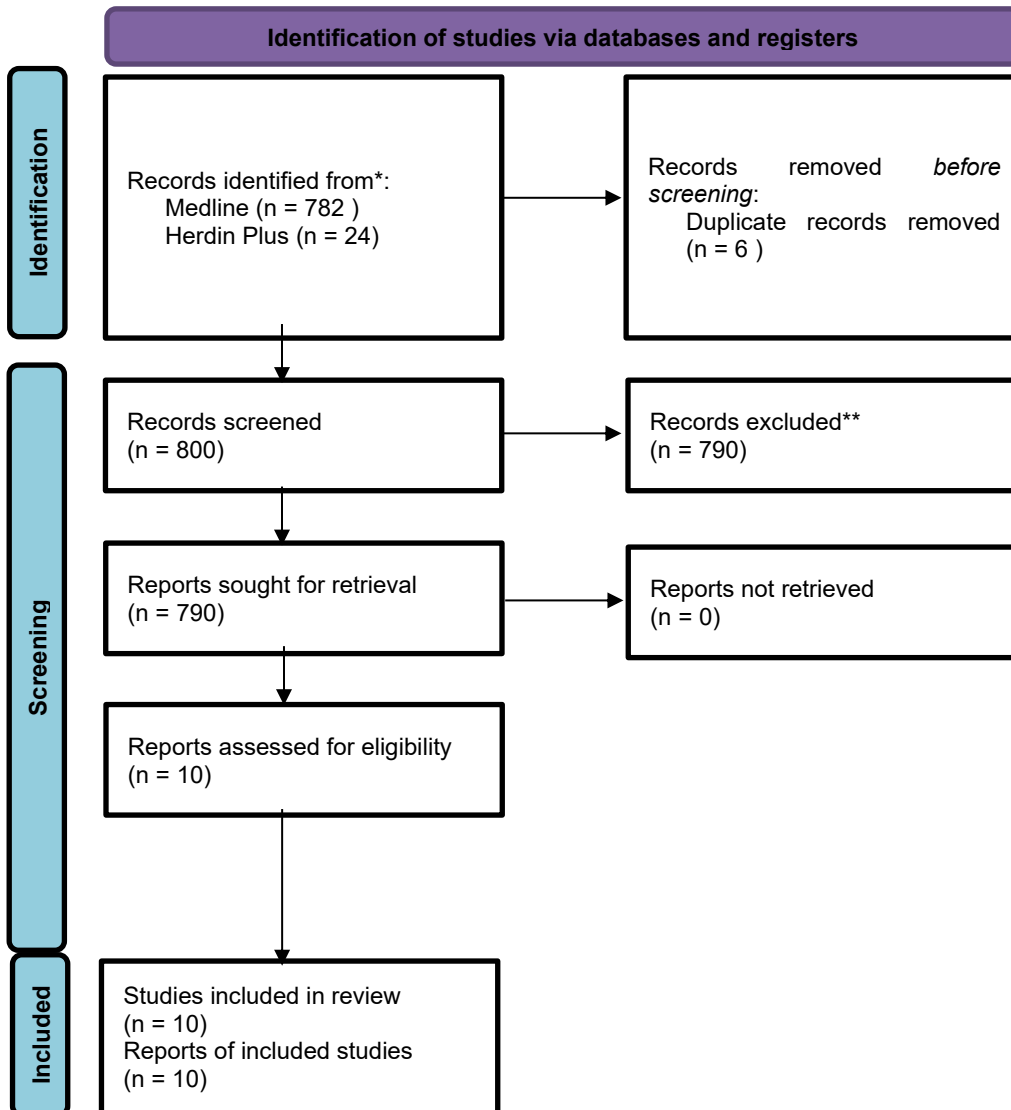
Two reviewers independently screened titles, abstracts, and full texts; extracted data using a standardized form; and examined reference lists of included studies for additional eligible articles. Risk of bias was assessed using validated tools appropriate for each study design—primarily the Cochrane Risk of Bias tool for randomized controlled trials. Certainty of evidence was evaluated using the GRADE approach.

Q15.2. Search strategy

DATABASE	SEARCH STRATEGY / SEARCH TERMS	DATE AND TIME OF SEARCH	RESULTS	
			Yield	Eligible
Medline	("acetylsalicylic acid"[Title/Abstract] OR "Aspirin"[Title/Abstract] OR ("Aspirin"[MeSH Terms] OR "Aspirin"[Title/Abstract])) AND ("Primary Prevention"[MeSH Terms] OR "Diabetes Mellitus"[MeSH Terms]) AND (("randomized controlled trial"[Publication Type] OR "controlled clinical trial"[Publication Type] OR "randomized"[Title/Abstract] OR "placebo"[Title/Abstract] OR "Drug Therapy"[MeSH Terms:noexp] OR "randomly"[Title/Abstract] OR "trial"[Title/Abstract]) NOT ("Animals"[MeSH Terms] NOT "Humans"[MeSH Terms]))	MARCH 31, 2024	782	10
Herdin	Aspirin OR Clopidogrel	MARCH 31, 2024	24	0

Q15.3. PRISMA flow diagram

Appendix: <PRISMA Flow Diagram for de novo/updating SRs>



Q15.4. Characteristics of the Included Studies

Trials, Year published	Design, Location	Total number of patients (% of patients with DM in the trial)	Nr DM	Mean age (or % above certain age) Male %	Intervention (Aspirin dose and posology)	Control	Follow-up (mean)	Funding from pharmaceutical company?
PPP 2003	RCT (open label) Italy	4495 (23%)	1031	64.2 years	Aspirin (100 mg/day, oral)	No aspirin	3.7 years	Yes
JPPP 2008	RCT (open label), Japan	2539 (100%)	2539	48.2% 65 years 55%	Aspirin (81 or 100 mg/day, oral)	No aspirin	4.37 years	Yes
JPAD 2014	RCT (open label), Japan	14,464 (33.9%)	4903	70.6 55.2% ≥ 70 y 42.3	Aspirin (100 mg/day, oral)	No aspirin	6.5 years	No
ASPREE 2018	RCT (double-blinded), Australia and USA	19,114 (11%)	2057	74 years 44%	Aspirin (100 mg/day, oral)	Placebo	4.7 years	No
WHHS 2005	RCT (double-blinded) USA	39,876 (2.6%)	1027	54.6 years 0%	Aspirin (100 mg on alternate days, oral)	Placebo	10.1 years	No
ETDRS 1992	RCT (single-blinded), USA	3711 (100%)	3711	51.7% ≥ 50 years 56%	Aspirin (650 mg/day, oral, given as two tablets of 325 mg per day)	Placebo	5 years	No
Physicians' Health Study 1989	RCT (double-blinded) USA	22,071 (2.4%)	533	59% ≥ 50 years 100%	Aspirin + beta carotene Aspirin + placebo (325 mg, alternate days)	Placebo	5 years	No
POPADAD 2008	RCT (double-blinded) Scotland	1276 (100%)	1276	60.5 years 47%	Aspirin + antioxidant Aspirin + placebo (100 mg/day, oral)	Placebo	6.7 years	No
ASCEND 2018	RCT (double-blinded) UK	15,480 (100%)	15,480	63.3 years 61.1%	Aspirin (100 mg/day)	Placebo	7.4 years	No
HOT 1998	RCT (double-blinded) International	18,790 (8%)	1501	61.5 years 53%	Aspirin (75 mg/day)	Placebo	3.8 years	Yes

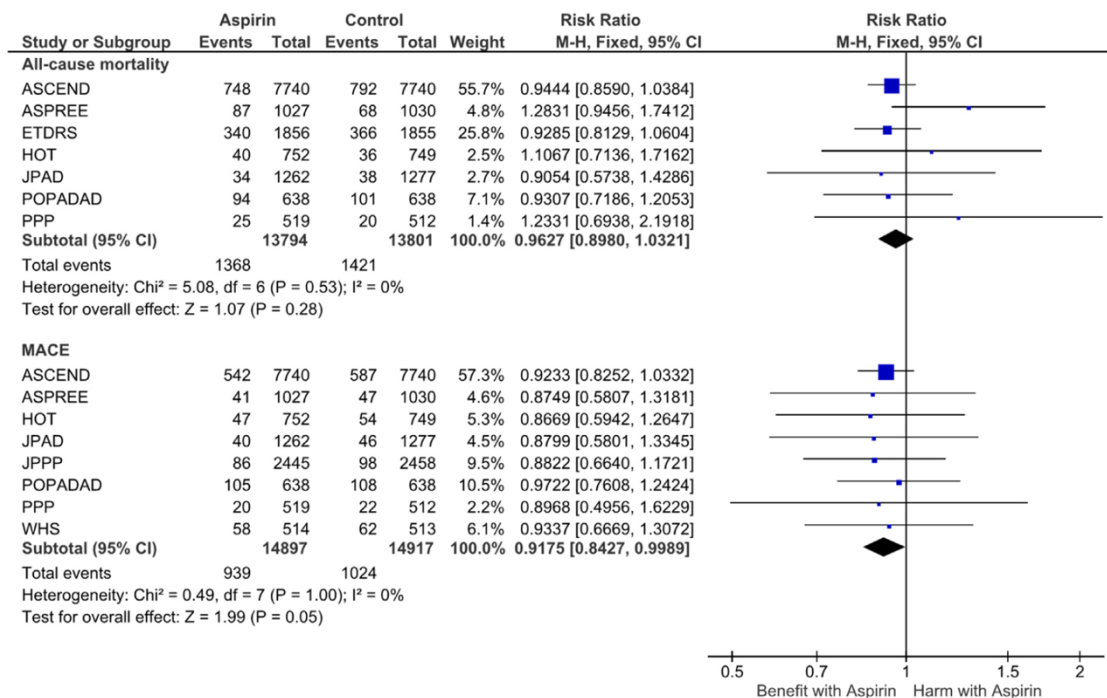
D. Cordero et al. / Primary Care Diabetes 14 (2020) 213–221

Q15.5. Assessment of Risk of Bias

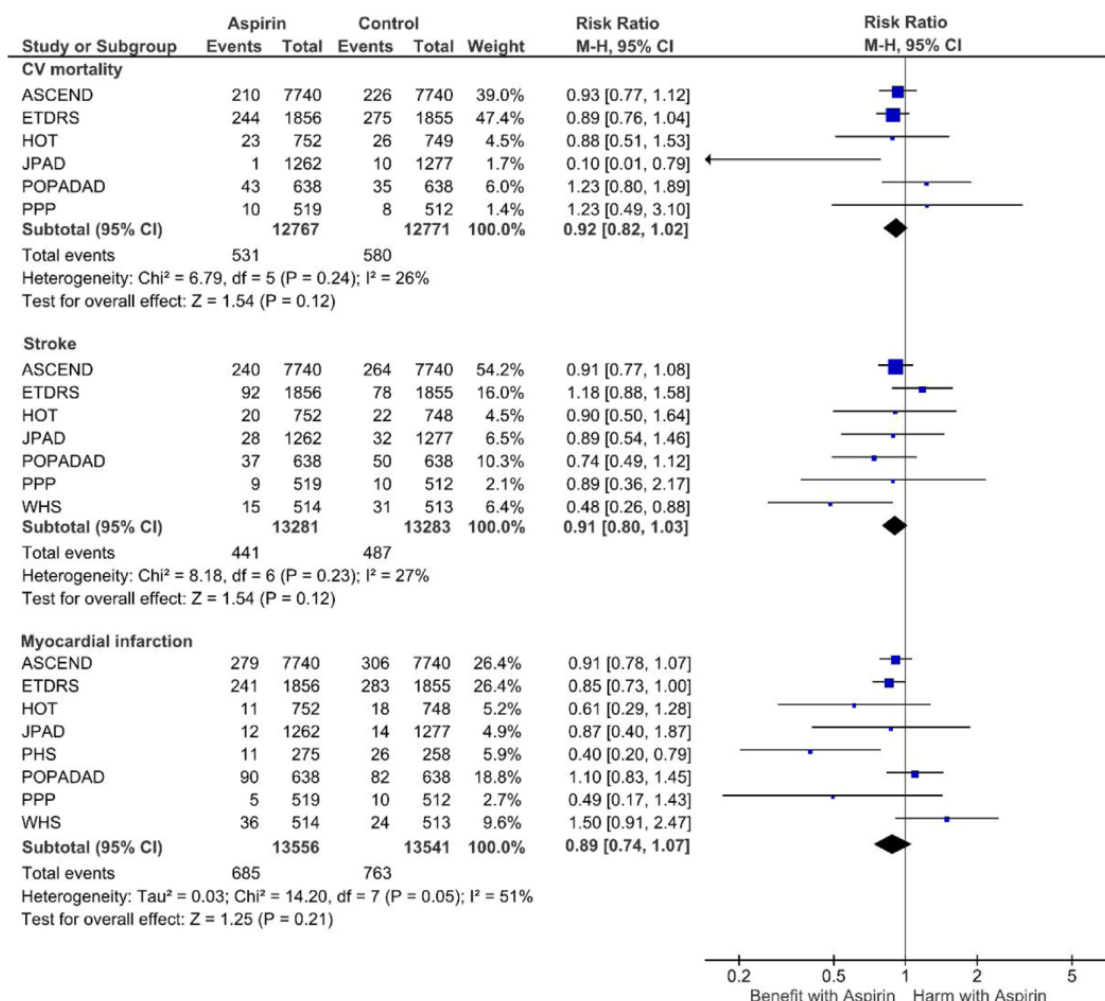
	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
ASCEND	+	+	+	+	+	+	+
ASPRE	+	+	+	+	+	-	+
ETDRS	+	?	+	+	+	+	+
HOT	+	+	+	+	+	-	+
JPAD	+	?	-	+	+	+	+
JPPP	+	+	-	+	+	-	+
PHS	+	?	+	+	-	-	+
POPADAD	+	+	+	+	+	+	+
PPP	+	+	-	+	+	+	-
WHS	+	?	+	+	+	-	+

Q15.6. Results

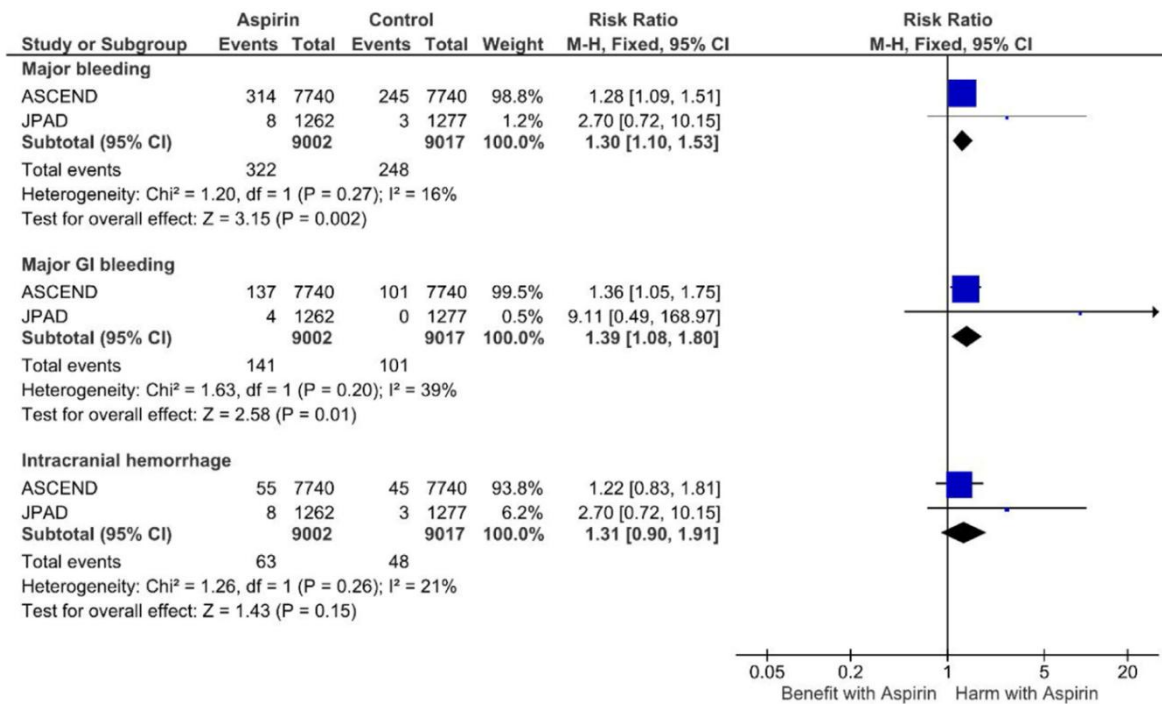
Forrest Plot for All-Cause Mortality and MACE



Appendix B.2. Forrest Plot for CV mortality, Stroke and Myocardial Infarction



Appendix B.3. Forrest Plot for Major Bleeding, Major GI Bleeding and Intracranial Hemorrhage



Q15.7. GRADE Evidence Profile

Certainty assessment							N _e of patients		Effect		Certainty	Importance
N _e of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	aspirin	placebo	Relative (95% CI)	Absolute (95% CI)		
All Cause Mortality												
7	randomised trials	not serious	not serious	not serious	serious ^a	none	1368/13794 (9.9%)	1421/13801 (10.3%)	RR 0.96 (0.90 to 1.03)	4 fewer per 1,000 (from 10 fewer to 3 more)	⊕⊕⊕ ○ Moderate	CRITICAL
MACE												
8	randomised trials	serious ^b	not serious	not serious	serious ^a	none	939/14987 (6.3%)	1024/14917 (6.9%)	RR 0.920 (0.840 to 0.999)	5 fewer per 1,000 (from 11 fewer to 0 fewer)	⊕⊕○ ○ Low	CRITICAL
CV Mortality												
6	randomised trials	not serious	not serious	not serious	serious ^a	none	531/12767 (4.2%)	580/12771 (4.5%)	RR 0.92 (0.82 to 1.02)	4 fewer per 1,000 (from 8 fewer to 1 more)	⊕⊕⊕ ○ Moderate	CRITICAL
Myocardial Infarction												
8	randomised trials	serious ^c	serious ^d	not serious	serious ^a	none	685/13556 (5.1%)	763/13541 (5.6%)	RR 0.89 (0.74 to 1.07)	6 fewer per 1,000 (from 15 fewer to 4 more)	⊕○○ ○ Very low	CRITICAL
Stroke												
7	randomised trials	serious ^c	not serious	not serious	serious ^a	none	441/13281 (3.3%)	487/13283 (3.7%)	RR 0.91 (0.80 to 1.03)	3 fewer per 1,000 (from 7 fewer to 1 more)	⊕⊕○ ○ Low	CRITICAL
Major Bleeding												
2	randomised trials	not serious	not serious	not serious	serious ^a	none	322/9002 (3.6%)	248/9017 (2.8%)	RR 1.30 (1.10 to 1.53)	8 more per 1,000 (from 3 more to 15 more)	⊕⊕⊕ ○ Moderate	CRITICAL
Major Gastrointestinal Bleeding												
2	randomised trials	not serious	not serious	not serious	serious ^a	none	141/9002 (1.6%)	101/9017 (1.1%)	RR 1.39 (1.08 to 1.80)	4 more per 1,000 (from 1 more to 9 more)	⊕⊕ ⊕○ Moderate	CRITICAL
Intracranial Hemorrhage												
2	randomised trials	not serious	not serious	not serious	serious ^a	none	63/9002 (0.7%)	48/9017 (0.5%)	RR 1.31 (0.90 to 1.91)	2 more per 1,000 (from 1 fewer to 5 more)	⊕⊕ ⊕○ Moderate	CRITICAL

CI: confidence interval; RR: risk ratio

Explanations

a. 95% CI overlaps no effect (i.e. CI includes RR of 1.0)

b. Involve three studies with analysis of diabetes subgroup that is not prespecified.

- c. involve two studies with analysis of diabetes subgroup that is not prespecified.
- d. $I^2 = 51\%$
- e. wide confidence interval

APPENDIX C.2 GRADE Summary of Findings Table

Patient or population:

primary prevention of CV events among T2DM patients

Setting:

Intervention:

aspirin

Comparison:

placebo

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	No of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with placebo	Risk with aspirin				
All Cause Mortality	103 per 1,000	99 per 1,000 (93 to 106)	RR 0.96 (0.90 to 1.03)	27595 (7 RCTs)	⊕⊕⊕○ Moderate ^a	
MACE	69 per 1,000	63 per 1,000 (58 to 69)	RR 0.920 (0.840 to 0.999)	29904 (8 RCTs)	⊕⊕○○ Low ^{a,b}	
CV Mortality	45 per 1,000	42 per 1,000 (37 to 46)	RR 0.92 (0.82 to 1.02)	25538 (6 RCTs)	⊕⊕⊕○ Moderate ^a	
Myocardial Infarction	56 per 1,000	50 per 1,000 (42 to 60)	RR 0.89 (0.74 to 1.07)	27097 (8 RCTs)	⊕○○○ Very low ^{a,c,d}	
Stroke	37 per 1,000	33 per 1,000 (29 to 38)	RR 0.91 (0.80 to 1.03)	26564 (7 RCTs)	⊕⊕○○ Low ^{a,c}	
Major Bleeding	28 per 1,000	36 per 1,000 (30 to 42)	RR 1.30 (1.10 to 1.53)	18019 (2 RCTs)	⊕⊕⊕○ Moderate ^e	
Major Gastrointestinal Bleeding	11 per 1,000	16 per 1,000 (12 to 20)	RR 1.39 (1.08 to 1.80)	18019 (2 RCTs)	⊕⊕⊕○ Moderate ^e	
Intracranial Hemorrhage	5 per 1,000	7 per 1,000 (5 to 10)	RR 1.31 (0.90 to 1.91)	18019 (2 RCTs)	⊕⊕⊕○ Moderate ^a	

*The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI).

CI: confidence interval; RR: risk ratio

GRADE	Working Group	grades of evidence
High certainty:	we are very confident that the true effect lies close to that of the estimate of the effect.	
Moderate certainty:	we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.	
Low certainty:	our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.	
Very low certainty:	we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.	

Explanations

a. 95% CI overlaps no effect (i.e. CI includes RR of 1.0)

b. involve three studies with analysis of diabetes subgroup that is not prespecified.

c. involve two studies with analysis of diabetes subgroup that is not prespecified.

d. I² = 51%

e. wide confidence interval

Q15.8. Recommendations from other groups

Recommendations From Other Groups

The 2023 American Diabetes Association (ADA) standards of care recommend that Aspirin therapy (75–162 mg/day) may be considered as a primary prevention strategy in those with diabetes who are at increased cardiovascular risk, after a comprehensive discussion with the patient on the benefits versus the comparable increased risk of bleeding.² This was also echoed by the Scottish Intercollegiate Guidelines Network (2010)³, 6th Edition of the Malaysian CPG for the Management of Type 2 Diabetes Mellitus (2021)⁴ and the KDIGO (2020) Diabetes Management in CKD Guideline.⁵

The 2020 NICE Type 2 Diabetes in Adults Management update focuses on the new evidence and safety issues of the off-label use of aspirin and clopidogrel in the primary prevention of cardiovascular disease.⁶ It states that we should not offer antiplatelet therapy for adults with type 2 diabetes without cardiovascular disease. There are no recommendations for using Aspirin as primary prevention for cardiovascular disease for the 2017 IDF CPG for managing Type 2 Diabetes in Primary care.⁷ The 2018 Diabetes Canada Clinical Practice Guidelines for the Prevention and Management of Diabetes in Canada discusses the uncertainties of the use of Aspirin in the prevention of CVD and potential benefits but does not recommend its use.⁸

The rest of the guidelines in this study do not mention or recommend the use of Aspirin in the primary prevention of cardiovascular disease among patients with type 2 diabetes mellitus.

Group/ Guidelines	Recommendation	Strength of recommendation and certainty of evidence
American Diabetes Association (2022)	Aspirin therapy (75–162 mg/day) may be considered as a primary prevention strategy in those with diabetes who are at increased cardiovascular risk, after a comprehensive discussion with the patient on the benefits versus the comparable increased risk of bleeding.	Grade A
Scottish Intercollegiate Guidelines Network (2010)	Low-dose aspirin is not recommended for primary prevention of vascular disease in patients with diabetes.	Grade A
National Institute for Health and Care Excellence (2015)	Do not offer antiplatelet therapy (aspirin or clopidogrel) for adults with type 2 diabetes without cardiovascular disease.	Strength of Recommendation not specified.
Diabetes Canada (2018)	ASA should not be used routinely for the primary prevention of CVD events in people with diabetes [Grade A, Level 1A (62–64)]. ASA may be used in the presence of additional CV risk factors [Grade D, Consensus].	Grade A
Malaysia Clinical Practice Guidelines for the Management of Type 2 Diabetes Mellitus (6 th Edition)	Primary prevention of ASCVD with low dose aspirin (100 mg) is only recommended in patients at increased cardiovascular risk after discussion with the patient on benefits vs. risk of bleeding.	Grade A

Kidney Improving Outcomes (2020)	Disease: Global (KDIGO)	Aspirin may be considered for primary prevention among high-risk individuals, ¹ but it should be balanced against increased risk for bleeding including thrombocytopathy with low GFR.	Strength of Recommendation not specified.
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Guideline Question 16.

Should we use aspirin among patients with type 2 DM for secondary prevention of cardiovascular events?

Q16.1. Methods

A systematic search of the literature was conducted to evaluate the effectiveness and safety of aspirin for the secondary prevention of cardiovascular events among adults with type 2 diabetes mellitus (T2DM). Searches were performed in MEDLINE (via PubMed) and the Cochrane Library (CENTRAL) from database inception to March 31, 2024, and were subsequently updated to include newly published studies through September 2025. A combined Medical Subject Headings (MeSH) and free-text strategy was used, including the terms: “aspirin,” “acetylsalicylic acid,” “antiplatelet therapy,” “secondary prevention,” “cardiovascular disease,” “myocardial infarction,” “stroke,” and “type 2 diabetes.” Filters for randomized controlled trials were applied when possible.

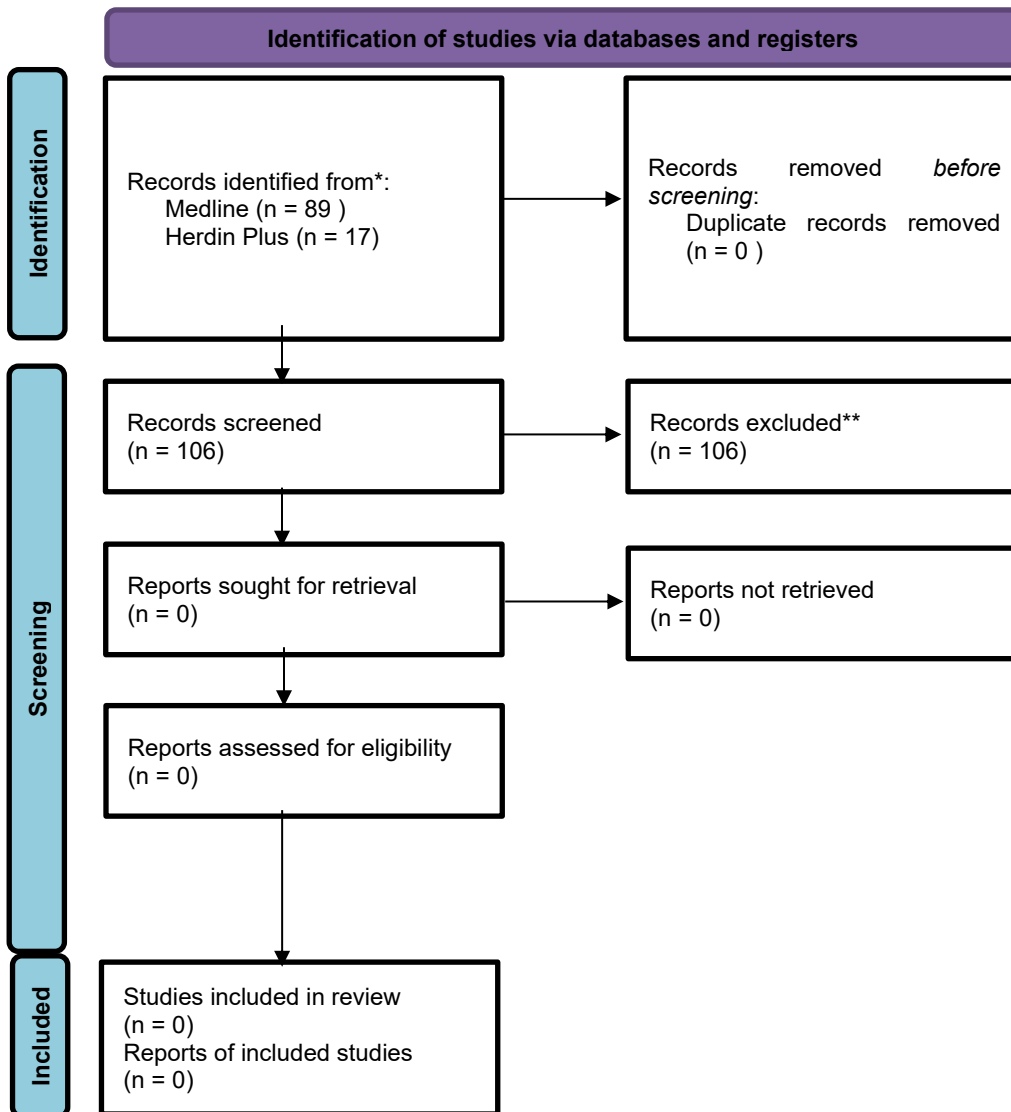
Eligible study designs included randomized controlled trials comparing aspirin with placebo, usual care, or no antiplatelet therapy among nonpregnant adults with T2DM who had established atherosclerotic cardiovascular disease (ASCVD), such as prior myocardial infarction, ischemic stroke, or peripheral arterial disease. Studies were required to report at least one critical or important outcome consistent with the PICO framework, such as recurrent major adverse cardiovascular events (MACE), myocardial infarction, stroke, cardiovascular mortality, all-cause mortality, major bleeding, or other significant harms. Studies involving primary prevention populations, mixed cohorts that did not report subgroup data for secondary prevention, noncomparative study designs, and articles not written in English were excluded.

Two reviewers independently screened all titles, abstracts, and full texts; extracted relevant data using a standardized data extraction form; and reviewed reference lists of included studies to identify additional eligible sources. Risk of bias was assessed using validated instruments appropriate for each study design—primarily the Cochrane Risk of Bias tool for randomized controlled trials. Certainty of evidence was evaluated using the GRADE approach.

Q16.2. Search strategy

DATABASE	SEARCH STRATEGY / SEARCH TERMS	DATE AND TIME OF SEARCH	RESULTS	
			Yield	Eligible
Medline	("Aspirin"[MeSH Terms] OR "Aspirin"[Title/Abstract] OR "acetylsalicylic acid"[Title/Abstract] OR "ASA"[Title/Abstract] OR "Ecotrin"[Title/Abstract] OR "Bufferin"[Title/Abstract]) AND ("diabetes mellitus, type 2"[MeSH Terms] OR "type 2 diabetes"[Title/Abstract] OR "T2DM"[Title/Abstract] OR "adult-onset diabetes"[Title/Abstract] OR "non-insulin-dependent diabetes"[Title/Abstract] OR "NIDDM"[Title/Abstract]) AND ("Secondary Prevention"[MeSH Terms] OR "Secondary Prevention"[Title/Abstract] OR "cardiovascular prevention"[Title/Abstract] OR "CVD prevention"[Title/Abstract] OR "cardiovascular events"[Title/Abstract] OR "myocardial infarction"[Title/Abstract] OR "stroke"[Title/Abstract] OR "ischemic heart disease"[Title/Abstract] OR "coronary artery disease"[Title/Abstract] OR "thrombosis"[Title/Abstract]) AND ("Randomized Controlled Trial"[Publication Type] OR "Randomized Controlled Trial"[Title/Abstract] OR "RCT"[Title/Abstract] OR "clinical trial"[Title/Abstract])	March 31, 2024	89	0
Herdin	aspirin AND diabetes	March 31, 2024	17	0

Q16.3. PRISMA flow diagram



Annex B.4. AGREE II Guideline Evaluation

Source Guideline	Scope and Purpose	Stakeholder Involvement	Rigor of Development	Clarity of Presentation	Applicability	Editorial Independence	Overall Assessment
Management of the Axilla in Early-Stage Breast Cancer: Ontario Health (Cancer Care Ontario) and ASCO Guideline	91.7	80.6	77.1	91.7	50.0	66.7	75.0
Early breast cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up	52.8	50.0	60.4	91.7	10.4	83.3	50.0
Pan-Asian adapted ESMO Clinical Practice Guidelines for the management of patients with early breast cancer: a KSMO-ESMO initiative endorsed by CSCO, ISMPO, JSMO, MOS, SSO and TOS	88.9	55.6	44.8	100.0	29.2	83.3	66.7

Annex B.5. Source Guidelines Characteristics

Title	Code	Publisher	Country, language	Publication date	Search Duration	Recommendation Standards	AGREE II Score (Rigor)
Breast Cancer Screening and Diagnosis, Version 1.2021	Bca-NCCN2021S CR	NCCN	USA, English	2021	07/21 to 05/22	NCCN Categories of Evidence and Consensus	86.5
Early and locally advanced breast cancer: diagnosis and management	Bca-NICE2018	NICE	UK, English	2018	07/21 to 05/22	n/a	100
Early breast cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up	Bca-ESMO2019	Oxford University Press	Europe, English	2019	07/21 to 05/22	IDSA Levels of evidence and grades of recommendation	60.4
Pan-Asian adapted ESMO Clinical Practice Guidelines for the management of patients with early breast cancer: a KSMO-ESMO initiative endorsed by CSCO, ISMPO, JSMO, MOS, SSO and TOS	Bca-ESMO2020	Elsevier Ltd.	Europe, English	2020	07/21 to 05/22	IDSA Levels of evidence and grades of recommendation	44.8

Q16.4. Characteristics of the Included Studies

Appendix A: Table of characteristics of included studies

Web Table 1: Baseline characteristics of the primary and secondary prevention trials

	Number of participants	Male	Age, years	Blood pressure (SBP/DBP), mmHg	Total cholesterol, mmol/L	Current smokers	Body mass index, kg/m ²	Diabetes mellitus	Hypertension	Any vascular disease
Primary prevention trials										
British Doctors Study	5139	100%	61 (7)	136 (17) / 83 (10)	-	31%	24.4 (2.5)	2%	10%	8%
US Physicians Health Study	22071	100%	53 (10)	126 (12) / 79 (8)	5.5 (1.2)	11%	24.9 (3.0)	2%	24%	1%
Thrombosis Prevention Trial	5085	100%	57 (7)	139 (18) / 83 (10)	6.4 (1.0)	41%	27.4 (3.6)	2%	16%	<1%
Hypertension Optimal Treatment Trial	18790	53%	61 (7)	170 (14) / 105 (3)	6.0 (1.1)	16%	28.4 (4.7)	8%	100%	3%
Primary Prevention Project	4495	43%	64 (8)	145 (16) / 85 (8)	6.1 (1.2)	15%	27.6 (4.7)	17%	68%	4%
Women's Health Study	39876	0%	54 (7)	124 (13) / 77 (8)*	5.2 (1.0)*	13%	26.0 (5.1)	3%	26%	<1%
Subtotal: 6 trials	95456	46%	56 (9)	136 (22) / 84 (13)	5.6 (1.1)	16%	26.3 (4.6)	4%	41%	2%
Secondary prevention post-MI trials										
Cardiff-I	1239	100%	55 (8)	-	-	-	-	-	-	100%
Cardiff-II	1725	85%	56 (10)	143 (29) / 90 (18)	-	-	-	5%	-	100%
PARIS-I	1216	87%	56 (8)	132 (18) / 83 (9)	-	-	-	10%	-	100%
AMIS	4524	89%	55 (8)	128 (16) / 80 (9)	-	-	-	11%	-	100%
CDP-A	1529	100%	56 (7)	132 (18) / 81 (10)	-	-	-	14%	-	100%
Gamis	626	78%	59 (7)	-	-	-	-	20%	19%	100%
Subtotal: 6 trials	10859	90%	56 (8)	132 (21) / 82 (12)	-	-	-	11%	19%	100%
Secondary prevention post TIA / stroke trials										
AITIA	319	70%	58 (14)	-	-	-	-	-	-	100%
UK-TIA	2435	73%	60 (9)	151 (25) / 88 (12)	-	-	-	4%	27%	100%
Reuther	60	65%	58 (10)	-	-	-	-	17%	50%	100%
CA Co-op	283	67%	61 (9)	146 (23) / 85 (11)	-	-	-	8%	37%	100%
Toulouse TIA	303	86%	63 (9)	-	-	-	-	-	-	100%
AICLA	402	68%	64 (10)	150 (21) / 90 (12)	-	-	-	23%	64%	100%
Danish Co-op	203	73%	59 (9)	138 (22) / 84 (12)	-	-	-	6%	-	100%
Britton	505	62%	68 (10)	-	-	-	-	17%	46%	100%
Danish Low Dose	301	65%	59 (8)	149 (23) / 85 (12)	-	-	-	7%	-	100%
SALT	1359	66%	67 (7)	-	-	-	-	13%	47%	100%
Subtotal: 10 trials	6170	70%	62 (10)	149 (24) / 87 (12)	-	-	-	9%	38%	100%

- = Not available, MI = myocardial infarction, TIA = transient ischaemic attack. * In the Women's Health Study, individual blood pressure and cholesterol levels were imputed based on categories provided by the investigators (in 10 mmHg ranges for SBP, 5 mmHg ranges for DBP and 10 mmol/L, 0.25 mmol/L ranges for cholesterol). Continuous data are presented as mean (SD). Percentages are based on the proportions among those participants with data available. Some patients in the primary prevention trials were found, after randomisation, to have had vascular disease (i.e. prior history of myocardial infarction, cerebrovascular disease, angina pectoris, peripheral arterial disease or heart failure).

Page 1 of Aspirin in the primary and secondary prevention of vascular disease web appendix (Lancet 2009)

Web Table 2: Number of events in the primary and secondary prevention trials

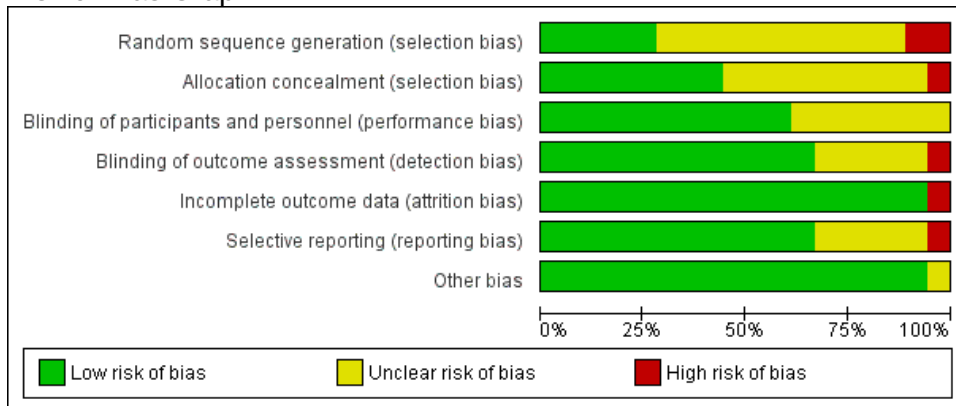
Trial	Serious vascular event	Major coronary events	Non fatal myocardial infarction	Any stroke	Stroke of unknown cause	Mortality							Major extracranial bleed	Fatal bleed
						CHD	Stroke	Other vascular	Any known vascular	Non vascular	Unknown cause	All causes		
Primary prevention trials														
British Doctors Study (2:1†)	434	267	149	133	87	136	42	46	224	194	3	421	30	4
US Physicians Health Study	686	459	342	219	10	127	22	28	177	205	62	444	78	2
Thrombosis Prevention Trial	468	353	233	100	22	141	25	28	194	197	48	439	33	5
Hypertension Optimal Treatment Trial	712	345	182	317	291	170‡	51	63	284	305	0	589	176	10
Primary Prevention Project	112	46	36	39	6	10	7	35	52	76	12	140	9	4
Women's Health Study	999	493	365	487	4	134	58	55	247	850	154	1251	218	1
Subtotal: 6 trials	3411	1963	1307	1295	420	718	205	255	1178	1827	279	3284	544	26
Secondary prevention post-MI trials														
Cardiff-I	133	129	25	1	0	104	1	2	107	6	1	114	0	0
Cardiff-II	316	306	96	10	7	206	10	4	220	10	0	230	0	0
PARIS-I (2:1†)	212	195	84	20	18	110	5	3	118	19	0	137	0	0
AMIS	795	707	317	101	97	388	10	10	408	52	5	465	0	0
CDP-A	178	146	59	25	24	85	5	14	104	6	0	110	0	0
Gamris	78	61	26	2	0	35	2	7	44	7	8	59	0	0
Subtotal: 6 trials	1712	1544	607	159	146	928	33	40	1001	100	14	1115	0	0
Secondary prevention post TIA / stroke trials														
AITIA	61	17	6	40	39	7	6	10	23	3	0	26	3	0
UK-TIA (2:1†)	558	245	77	320	224	163	55	35	253	77	13	343	15	2
Reuther	7	1	0	6	4	1	2	0	3	0	0	3	1	0
CA Co-op	63	16	4	43	43	12	9	5	26	5	0	31	0	0
Toulouse TIA	27	7	2	16	16	5	5	4	14	11	0	25	0	0
AICLA	79	15	11	53	49	4	6	5	15	12	10	37	0	0
Danish Co-op	50	20	10	32	29	10	4	1	15	5	1	21	0	0
Britton	114	42	21	63	15	15	20	22	57	14	0	71	5	0
Danish Low Dose	42	16	2	21	20	14	1	3	18	4	2	24	0	0
SALT	307	119	53	180	11	54	32	34	120	35	5	160	4	0
Subtotal: 10 trials	1308	498	186	774	450	285	140	119	544	166	31	741	28	2

† Allocation ratio 2:1; in tables or figures where adjusted numbers of controls are given, the number of events in the control group of this study is doubled. ‡ Includes 149 sudden deaths.

Page 2 of Aspirin in the primary and secondary prevention of vascular disease web appendix (Lancet 2009)

Q16.5. Assessment of Risk of Bias

Risk of Bias Graph



D.2 Risk of Bias Summary

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Aims 1980	?	?	?	?	+	?	+
Bousser AICLA 1983	?	+	?	?	+	?	+
Boysen 1988	?	+	+	+	+	+	+
Breddinuberla 1980	?	+	?	+	+	+	+
Britton Swedish Cooperative 1987	+	?	?	+	+	+	+
Canadian Cooperative 1978	+	+	+	+	+	?	+
Candellise 1982	?	?	+	?	+	+	+
Coronary Drug Project 1976	?	+	+	+	+	+	+
Elwood 1974	+	+	+	+	+	+	+
Elwood 1979	+	+	+	+	+	+	+
Fields 1977	?	+	+	+	+	+	+
Fields 2 1978	?	?	+	?	+	+	+
Paris Trial 1980	+	?	+	+	+	+	+
Paris Trial 2 1986	+	?	+	+	+	+	+
SALT 1991	?	+	?	+	+	?	+
Sorensen 1983	+	?	+	?	+	+	+
UK-TIA Final 1991	?	?	?	+	+	+	?
UK-TIA Interim 1988	?	?	?	+	+	?	+

Q16.6. Results

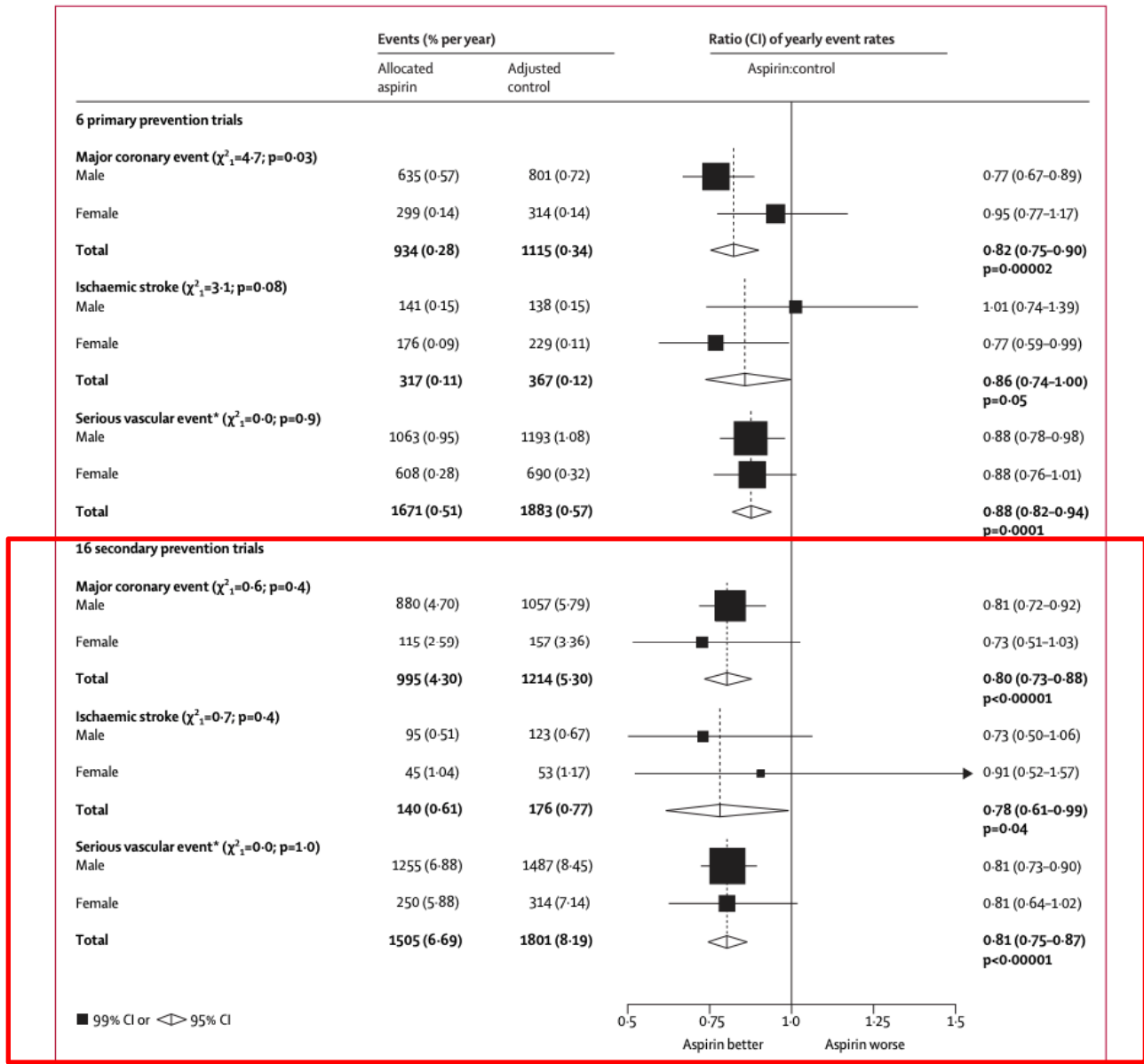


Figure 3: Selected outcomes in primary and secondary prevention trials of aspirin, by sex

Actual numbers for aspirin-allocated trial participants, and adjusted numbers for control-allocated trial participants, are presented together with the corresponding mean yearly event rate (in parentheses). Rate ratios (RRs) for all trials are indicated by squares and their 99% CIs by horizontal lines. Subtotals and their 95% CIs are represented by diamonds. Squares or diamonds to the left of the solid line indicate benefit. *Myocardial infarction, stroke (haemorrhagic or other), or vascular death.

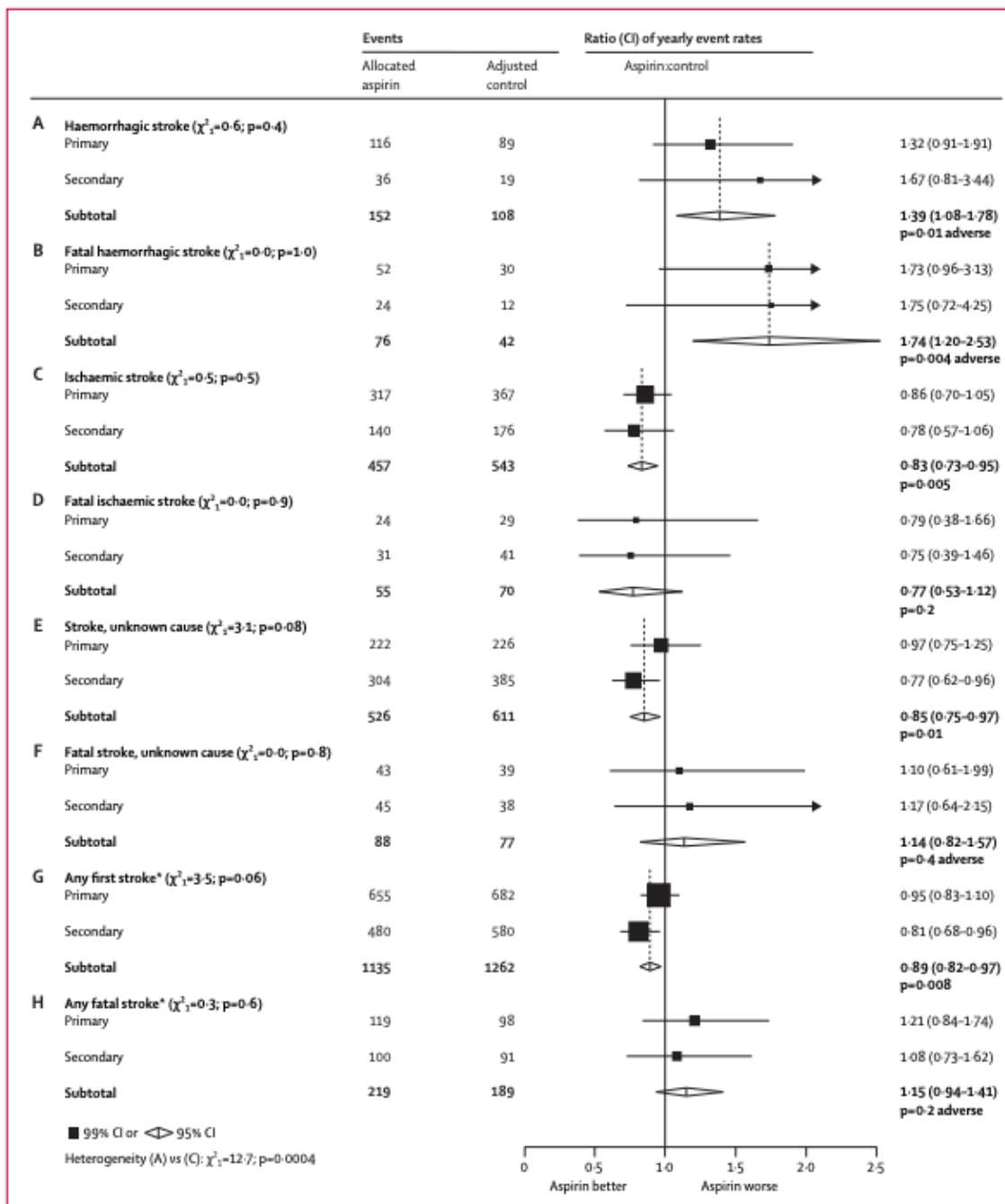


Figure 4: Stroke subtypes in primary and secondary prevention trials

Actual numbers for aspirin-allocated trial participants, and adjusted numbers for control-allocated trial participants, are presented. Rate ratios (RRs) for all trials are indicated by squares and their 99% CIs by horizontal lines. Subtotals and their 95% CIs are represented by diamonds. Squares or diamonds to the left of the solid line indicate benefit. *Haemorrhagic, ischaemic, or unknown cause.

Q16.7. GRADE Evidence Profile

Certainty assessment							No of patients		Effect	Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	aspirin	no aspirin	Relative (95% CI)		
Non Fatal Myocardial Infarction											
16	randomised trials	not serious	not serious	serious ^a	not serious	none	357/43000	505/43000	Rate ratio 0.69 (0.60 to 0.80)	⊕⊕⊕○ Moderate	CRITICAL
Coronary Heart Disease Mortality											
16	randomised trials	not serious	not serious	serious ^a	not serious	none	614/43000	696/43000	Rate ratio 0.87 (0.78 to 0.98)	⊕⊕⊕○ Moderate	CRITICAL
Stroke											
16	randomised trials	not serious	not serious	serious ^a	not serious	none	480/43000	580/43000	Rate ratio 0.81 (0.71 to 0.92)	⊕⊕⊕○ Moderate	CRITICAL
Vascular Death											
16	randomised trials	not serious	not serious	serious ^a	not serious	none	825/43000	896/43000	Rate ratio 0.91 (0.82 to 1.00)	⊕⊕⊕○ Moderate	CRITICAL
Any serious vascular event											
16	randomised trials	not serious	not serious	serious ^a	not serious	none	1505/43000	1801/43000	Rate ratio 0.81 (0.75 to 0.87)	⊕⊕⊕○ Moderate	CRITICAL
Major extracranial bleeding											
5	randomised trials	not serious	not serious	serious ^a	serious ^a	strong association	23/43000	6/43000	Rate ratio 2.61 (1.25 to 5.76)	⊕⊕○○ Low  (Ctrl)	CRITICAL
Hemorrhagic Stroke											
16	randomised trials	not serious	not serious	serious ^a	serious ^a	none	36/43000	19/43000	Rate ratio 1.67 (0.97 to 2.90)	⊕⊕○○ Low	CRITICAL

CI: confidence interval

Explanations

- a. not specifically for diabetic patients
- b. wide confidence interval
- c. RR include 1.0

Q16.8. Recommendations from other groups

The 2023 American Diabetes Association recommends the use of aspirin therapy (75–162 mg/day) as a secondary prevention strategy in those with diabetes and a history of atherosclerotic cardiovascular disease.³ Earlier guidelines made similar recommendations from the Scottish Intercollegiate Guidelines Network (2010)⁴, Diabetes Canada (2018)⁵, KDIGO (2020)⁶, Malaysian CPG for the Management of type 2 diabetes mellitus (2020)⁷ and IDF (2017)⁸.

Group/ Guidelines	Recommendation	Strength of recommendation and certainty of evidence
American Diabetes Association (2022)	Use aspirin therapy (75–162 mg/day) as a secondary prevention strategy in those with diabetes and a history of atherosclerotic cardiovascular disease.	Grade A
Scottish Intercollegiate Guidelines Network (2010)	Aspirin (<i>75 mg per day</i>) should be given routinely and continued long term in patients with diabetes and coronary heart disease.	Grade A
Diabetes Canada (2018)	In people with established CVD, low-dose ASA therapy (81–162mg) should be used to prevent CV events	Grade A
Kidney Disease: Improving Global Outcomes (KDIGO) (2020)	Aspirin should generally be used lifelong for secondary prevention among those with established cardiovascular disease and may be considered for primary prevention among high-risk individuals, with dual antiplatelet therapy used in patients after acute coronary syndrome or percutaneous coronary intervention.	Strength of Recommendation not specified.
Malaysia Clinical Practice Guidelines for the Management of Type 2 Diabetes Mellitus (6 th Edition)	Aspirin should be used for secondary prevention in T2DM patients. For patients with ASCVD and documented aspirin allergy, clopidogrel (75 mg/day) should be used.	Grade A
IDF Clinical Practice Recommendations for managing Type 2 Diabetes in Primary Care - 2017	Low-dose aspirin (75-350 mg/d) should be started and maintained lifelong in people with T2D and established CVD (secondary prevention).	Strength of Recommendation not specified.

Guideline Question 17.

Should we use clopidogrel for the secondary prevention of cardiovascular events among patients with type 2 DM?

Q17.1. Methods and search strategies

Evidence for the use of clopidogrel for secondary prevention was primarily drawn from a recent meta-analysis by Aggarwal et al. (March 2022), which synthesized the comparative efficacy and safety of P2Y12 inhibitor monotherapy versus aspirin. In this meta-analysis, MEDLINE, Embase, and the Cochrane Central Register of Controlled Trials (CENTRAL) were searched from database inception until June 2021 to identify eligible randomized controlled trials. For the purposes of this guideline, the evidence base was extended through a supplementary search in MEDLINE (via PubMed) and the Cochrane Library from database inception to March 31, 2024, and was subsequently updated to include newly published studies through September 2025. Search terms included: “clopidogrel,” “P2Y12 inhibitor,” “ticagrelor,” “antiplatelet therapy,” “secondary prevention,” “cardiovascular disease,” “myocardial infarction,” “stroke,” and “type 2 diabetes mellitus.” Filters for randomized controlled trials were applied where appropriate.

The Aggarwal meta-analysis included nine randomized controlled trials (total N = 61,623 participants) comparing P2Y12 inhibitor monotherapy (clopidogrel or ticagrelor) with aspirin monotherapy for the secondary prevention of cardiovascular events in patients with established atherosclerotic cardiovascular disease, including coronary artery disease, cerebrovascular disease, and peripheral arterial disease. The primary efficacy outcome was major adverse cardiovascular events (MACE), defined as a composite of stroke, myocardial infarction (MI), or death. The primary safety outcome was major bleeding.

For the guideline, studies were eligible if they:

- involved nonpregnant adults with T2DM and established ASCVD;
- compared clopidogrel (or other P2Y12 inhibitor) monotherapy with aspirin;
- reported at least one critical or important outcome (MACE, MI, stroke, cardiovascular mortality, all-cause mortality, major bleeding).

Studies focused solely on primary prevention, mixed cohorts without extractable diabetes or secondary-prevention subgroup data, noncomparative designs, and non-English publications were excluded.

Two reviewers independently screened titles, abstracts, and full texts; extracted relevant clinical and methodological data using a standardized template; and reviewed reference lists of included systematic reviews and trials to identify additional eligible evidence. Risk of bias for randomized controlled trials was assessed using the Cochrane Risk of Bias (RoB 2) tool. Certainty of evidence was evaluated according to the GRADE approach.

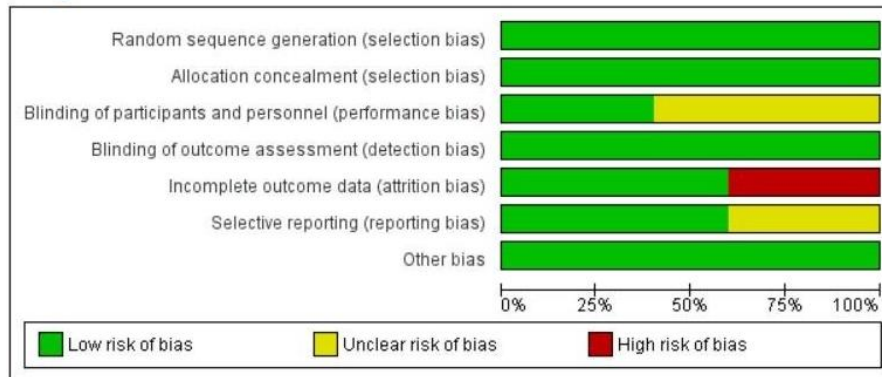
Q17.2. Characteristics of the Included Studies

Characteristics of included studies

Study	Study design	Number of patients (diabetic)	Population (qualifying event)	Intervention/Treatment	Control/Comparison	Outcomes	Follow-up (months)
CAPRIE 1996 multinational	RCT multicentre double blind	19185 (20.0%)	Patients with prior ischemic stroke, peripheral artery disease, myocardial infarction	Clopidogrel 75mg daily	Aspirin 325mg daily	MI, ischemic stroke, vascular death	36
ASCET 2012 Norway	RCT single centre double blind	1001 (19.9%)	Patients with coronary artery disease	Clopidogrel 75mg daily	Aspirin 160 mg daily	death, MI, ischemic stroke, unstable angina	24
HOST EXAM 2021 South Korea	RCT multicentre open label	5438 (34.2%)	Patients with CAD post-PCI	Clopidogrel 75mg daily	Aspirin 100mg daily	All cause mortality, MI, stroke, readmission for ACS	24
CADET 2004 UK	RCT multicentre double blind	184 (<i>not specified</i>)	Patients with recent MI	Clopidogrel 75mg daily	Aspirin 75 mg daily	Thrombotic variables and CRP	6
CHANCE 2013 China	RCT multicentre double blind	5170 (21.2%)	Patients with stroke or high risk TIA	Clopidogrel 75mg daily	Aspirin 75 mg daily	Thrombotic variables and CRP	3

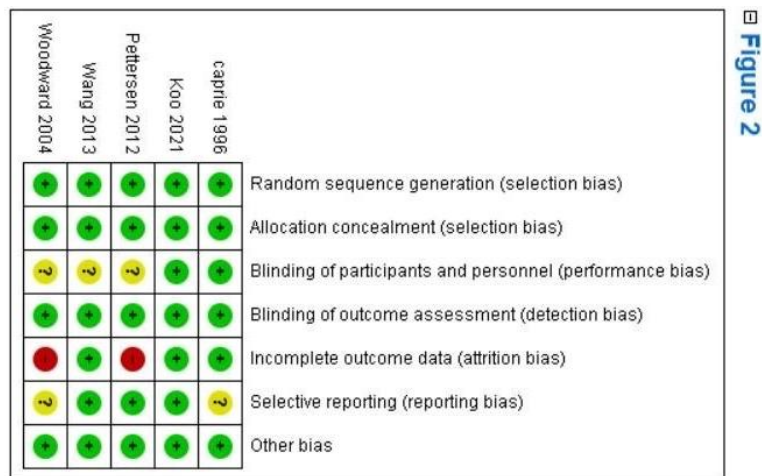
Q17.5. Assessment of Risk of Bias

Assessment of Risk of Bias of Included Studies



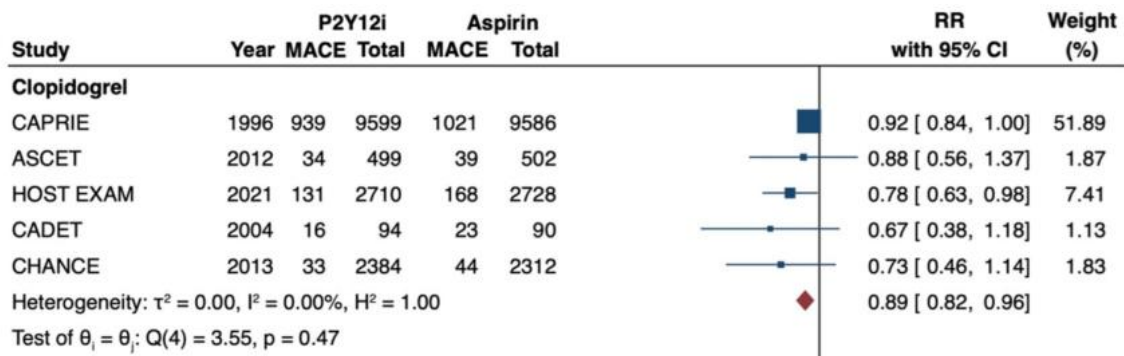
Caption

Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

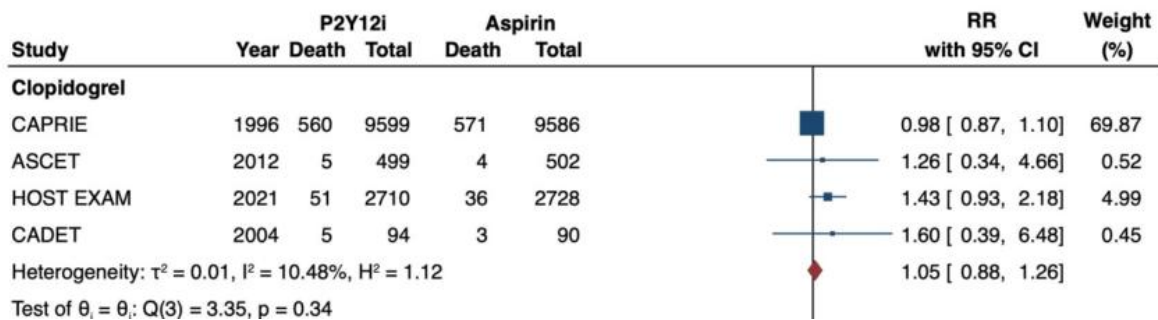


Q17.6. Results Forest Plots

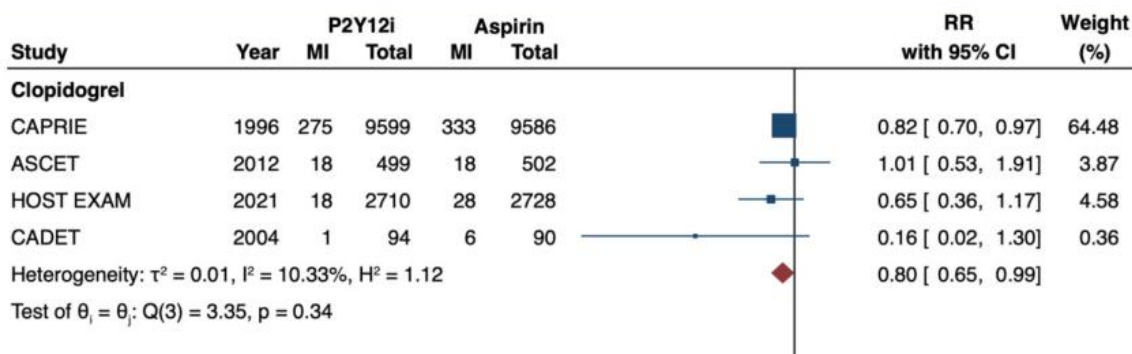
Annex 3.1 MACE



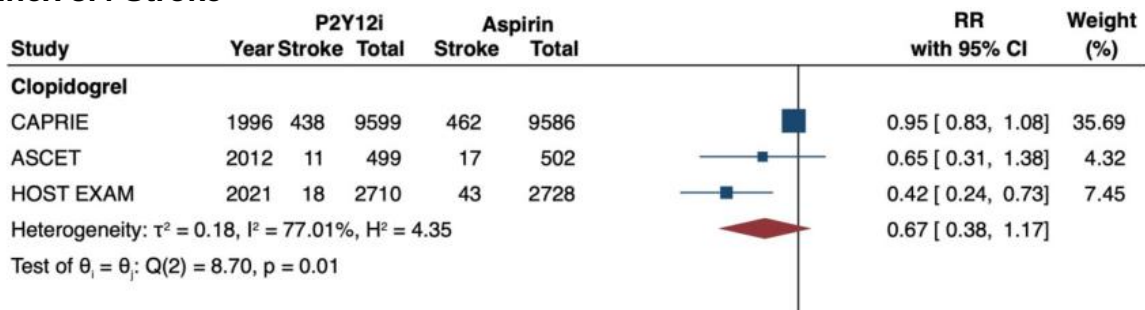
Annex 3.2 All-cause mortality



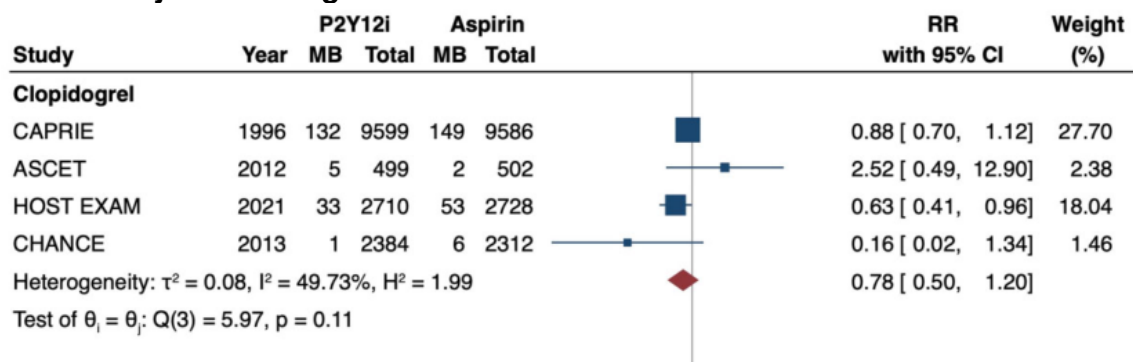
Annex 3.3 Myocardial Infarction



Annex 3.4 Stroke



Annex 3.5 Major Bleeding



Q17.7. GRADE Evidence Profile

Author(s):
Question: Clopidogrel compared to aspirin for secondary prevention of CV events among persons with T2DM with established ASCVD
Setting:
Bibliography:

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	clopidogrel	aspirin	Relative (95% CI)	Absolute (95% CI)		
Major adverse cardiovascular events (follow-up: range 3 months to 36 months)												
5	randomised trials	not serious	not serious	serious ^a	serious ^b	none	1153/15286 (7.5%)	1295/15218 (8.5%)	OR 0.89 (0.82 to 0.96)	9 fewer per 1,000 (from 14 fewer to 3 fewer)	⊕⊕○○ Low	CRITICAL
All-cause death (Total Mortality) (follow-up: range 6 months to 36 months)												
4	randomised trials	not serious	not serious	serious ^a	not serious	none	621/12902 (4.8%)	614/12906 (4.8%)	OR 1.05 (0.88 to 1.26)	2 more per 1,000 (from 5 fewer to 12 more)	⊕⊕⊕○ Moderate	CRITICAL
Vascular death (Cardiovascular Death) (follow-up: range 12 months to 36 months)												
1	randomised trials	not serious	not serious	serious ^a	not serious	none	373/9599 (3.9%)	405/9586 (4.2%)	OR 0.92 (0.79 to 1.06)	3 fewer per 1,000 (from 9 fewer to 2 more)	⊕⊕⊕○ Moderate	CRITICAL
Stroke (Cerebrovascular Disease) (follow-up: range 6 months to 36 months)												
3	randomised trials	not serious	serious ^c	serious ^a	not serious	none	467/12808 (3.6%)	522/12816 (4.1%)	OR 0.67 (0.38 to 1.17)	13 fewer per 1,000 (from 25 fewer to 7 more)	⊕⊕○○ Low	CRITICAL
Myocardial infarction (follow-up: range 6 months to 36 months)												
4	randomised trials	not serious	not serious	serious ^a	not serious	none	312/12902 (2.4%)	385/12906 (3.0%)	OR 0.80 (0.65 to 0.99)	6 fewer per 1,000 (from 10 fewer to 0 fewer)	⊕⊕⊕○ Moderate	CRITICAL
Major bleeding (follow-up: range 3 months to 36 months)												
4	randomised trials	not serious	serious ^d	serious ^a	not serious	none	171/15192 (1.1%)	210/15128 (1.4%)	OR 0.78 (0.50 to 1.20)	3 fewer per 1,000 (from 7 fewer to 3 more)	⊕⊕○○ Low	

CI: confidence interval; OR: odds ratio

Explanations

- a. less than 50% of study population involves diabetic subjects
b. CI falls within the range of ignorable benefit
c. I2 = 77%
d. I2 = 49.73%

Q17.8. Recommendations from other groups

Group/ Guidelines	Recommendation	Strength of Recommendation and Certainty of Evidence
ADA 2022	For patients with atherosclerotic cardiovascular disease and documented aspirin allergy, clopidogrel (75 mg/day) should be used. Dual antiplatelet therapy (with low-dose aspirin and a P2Y12 inhibitor) is reasonable for a year after an acute coronary syndrome and may have benefits beyond this period.	Grade A Grade A
Diabetes Canada CPG 2022	Clopidogrel 75 mg may be used in people unable to tolerate ASA.	Grade D, consensus
Malaysia CPG 2020	For patients with ASCVD and documented aspirin allergy, clopidogrel (75 mg/day) should be used.	Grade A

Guideline Question 18.

Should we recommend low dose combination drugs for the control of hypertension among persons with type 2 DM and hypertension?

Q1.1. Methods and search strategy

The selected clinical practice guidelines (refer to the main methodology) were carefully reviewed to determine their appropriateness in addressing the use of low-dose combination antihypertensive therapy among adults with type 2 diabetes mellitus (T2DM) and hypertension. The scientific evidence supporting their recommendations, including the strength and consistency of evidence and their adaptability to the Philippine context, was critically evaluated using the ADAPTE framework.

To supplement the guideline adaptation process, a focused search of the literature was conducted to identify additional evidence relevant to low-dose combination antihypertensive regimens in this population. Searches were performed in MEDLINE (via PubMed) and the Cochrane Library from database inception to March 31, 2024, and were subsequently updated to include newly published studies through September 2025. Medical Subject Headings (MeSH) and free-text terms included: “low-dose combination therapy,” “fixed-dose combination,” “hypertension,” “type 2 diabetes,” “antihypertensive agents,” “combination antihypertensive,” and “blood pressure control.”

Eligible evidence included clinical practice guidelines, randomized controlled trials, meta-analyses, and high-quality observational studies that assessed low-dose or fixed-dose combination antihypertensive therapy compared with monotherapy or standard stepwise titration. Studies were required to report at least one critical or important outcome based on the PICO framework, including systolic or diastolic blood pressure reduction, treatment adherence, adverse events, cardiovascular outcomes, renal outcomes, or treatment discontinuation. Studies involving populations without diabetes, pediatric cohorts, or noncomparative designs were excluded. Only full-text English-language publications were considered.

Two reviewers independently screened the identified guidelines and retrieved citations, extracted relevant data, and assessed the methodological quality of the included evidence. When applicable, risk of bias in primary studies was evaluated using validated instruments appropriate for each design (e.g., RoB 2 for randomized trials, ROBINS-I for nonrandomized studies). Certainty of evidence was appraised using the GRADE approach.

Q18.2. Assessment of Risk of Bias

Table S1 – Description of quality evaluation of included studies

Author	Trial	Concealment of randomization	RCT Stopped early	Patients blinded	Health care providers blinded	Data collectors blinded	Outcome assessors blinded	Intention to treat
Bakris	INVEST	Yes	No	No	No	No	Yes	Yes
Barnett	DETAIL	Yes	No	Yes	Yes	Yes	Yes	unclear
Berle Lewis	IDNT	Yes	No	Yes	Yes	Yes	Yes	Yes
Brenner	RENAAL	Yes	No	Yes	Yes	Yes	Yes	Yes
Curb	SHEP	Yes	No	Yes	Yes	Yes	Yes	Yes
Estacio	ABCD	unclear	Yes	Yes	Yes	Yes	Yes	Yes
Fogari		unclear	No	No	No	No	Yes	No
Hansson	NORDIL	Yes	No	No	No	No	Yes	Yes
Imai E	ORIENT	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Lindholm	LIFE	unclear	No	Yes	Yes	Yes	Yes	Yes
Lindholm	STOP-2	unclear	No	No	No	No	Yes	Yes
Mancia	INSIGHT	Yes	No	Yes	Yes	Yes	Yes	Yes
Marre	NESTOR	Yes	No	Yes	Yes	Yes	Yes	Yes
Muramatsu	NAGOYA HEART	Yes	No	No	No	No	Yes	Yes
Nakao	CASE-J	Yes	No	No	No	No	Yes	Yes
Nielsen		unclear	No	Yes	No	unclear	unclear	Yes
Niskanen	CAPPP	unclear	No	No	No	No	Yes	Yes
Ostergren	ASCOT	unclear	No	No	No	No	Yes	Yes
Parving	IRMA	unclear	No	Yes	Yes	Yes	Yes	Yes
Rugeneri Remuzzi	BENEDICT-A	Yes	No	Yes	Yes	Yes	Yes	Yes
Rugeneri	BENEDICT-B	Yes	No	Yes	Yes	Yes	Yes	Yes
Safare Tuomilehto	SYST-EUR	Yes	No	Yes	Yes	Yes	Yes	Yes
Tatti	FACET	Yes	No	No	No	No	Yes	Yes

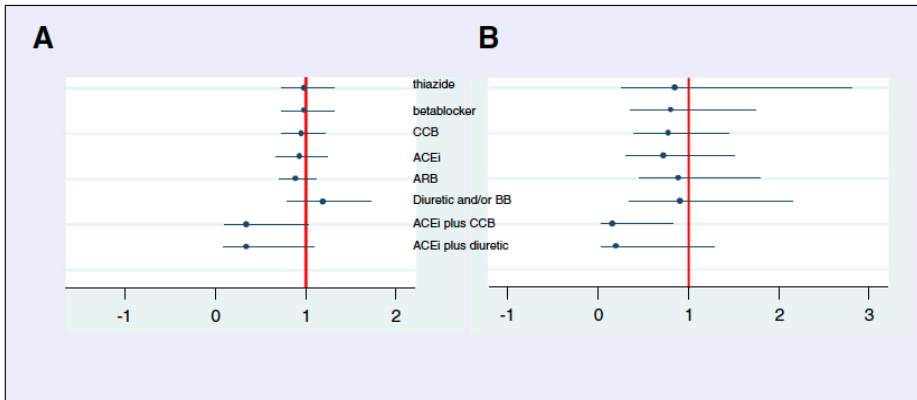
Weber	ACCOMPLISH	unclear	No	Yes	Yes	Yes	Yes	Yes
Whelton	ALLHAT	unclear	No	Yes	Yes	unclear	unclear	Yes
Yui	JMIC-B	Yes	No	No	No	No	Yes	Yes
	UKPDS 39	Yes	No	No	No	No	Yes	Yes

Source: Supplementary materials from <https://doi.org/10.1016/j.jdiacomp.2016.04.020> from Luciana R. Remonti, Sofia Dias, Cristiane B. Leitão, Caroline K. Kramer, Lucas P. Klassman, Nicky J. Welton, A.E. Ades, Jorge L. Gross, Classes of antihypertensive agents and mortality in hypertensive patients with type 2 diabetes—Network meta-analysis of randomized trials, Journal of Diabetes and its Complications, Volume 30, Issue 6, 2016. Pages 1192-1200, ISSN 1056-8727.

All studies were randomized, however in ten they could not define the method used for randomization and, therefore, its concealment. Eleven trials were not double blinded; however in all but 2 trials the outcome evaluators were blinded. From the 21 studies included in cardiovascular mortality analysis, 16 had all events adjudicated by an independent committee. The other 5 trials do not describe if outcomes were adjudicated and, in 3 of these, the events of death were described in adverse event section. In 9 trials, the study describes clearly a standardized method for [blood pressure measurement](#). Six trials describes that clinical assessments including blood pressure were conducted according to the study protocol. Only in one case, there is no information regarding blood pressure measurement technique.

Q18.3. Results

Annex 4: SR/Meta-analysis Forest Plot (if applicable)



Hazard ratio for total mortality (A) and cardiovascular mortality (B) considering placebo as reference treatment. CCB = calcium channel blocker, ACEi = angiotensin converting enzyme inhibitor, ARB = angiotensin receptor blocker, BB = betablocker. Vertical line represents the no effect line. X-axis represents the Hazard ratio.

Reference: L.R. Remonti et al. / Journal of Diabetes and Its Complications 30 (2016) 1192–1200

Q18.4. GRADE Evidence Profile

Table S2: GRADE assessment

Criteria	Points
Study design (randomized trials only)	+ 4
Risk of bias	0
Inconsistency	-1
Indirectness	0
Imprecision	0
Large effect	0
Dose response	0
All plausible residual confounding	0
TOTAL	+ 3 (moderate)

According to GRADE system, the quality of the evidence was considered moderate (Supplemental Table S2).

Source: Supplementary materials from <https://doi.org/10.1016/j.jdiacomp.2016.04.020> from Luciana R. Remonti, Sofia Dias, Cristiane B. Leitão, Caroline K. Kramer, Lucas P. Klassman, Nicky J. Welton, A.E. Ades, Jorge L. Gross, Classes of antihypertensive agents and mortality in hypertensive patients with type 2 diabetes—Network meta-analysis of randomized trials, Journal of Diabetes and its Complications, Volume 30, Issue 6, 2016. Pages 1192-1200, ISSN 1056-8727.

Table 2. Comparisons of the effects of antihypertensive drug classes in total and cardiovascular (CV) mortality [median hazard ratio (95% CrI)].

Placebo	0.85 (0.24–2.79)	0.81 (0.35–1.74)	0.78 (0.37–1.44)	0.72 (0.29–1.51)	0.89 (0.45–1.79)	0.90 (0.33–2.14)	<u>0.16</u> (0.01–0.82)	0.19 (0.01–1.28)
0.98 (0.72–1.32)	Thiazide	0.94 (0.30–2.95)	0.91 (0.32–2.48)	0.85 (0.26–2.43)	1.04 (0.33–3.47)	1.06 (0.30–3.4)	0.19 (0.02–1.18)	0.23 (0.01–1.79)
0.98 (0.72–1.31)	1.0 (0.74–1.34)	BB	0.97 (0.55–1.58)	0.89 (0.45–1.56)	1.10 (0.58–2.21)	1.12 (0.48–2.38)	<u>0.20</u> (0.02–0.98)	0.24 (0.02–1.53)
0.95 (0.72–1.20)	0.97 (0.75–1.20)	0.97 (0.78–1.17)	CCB	0.93 (0.53–1.51)	1.14 (0.67–2.20)	1.16 (0.59–2.22)	<u>0.21</u> (0.02–0.97)	0.25 (0.02–1.54)
0.93 (0.66–1.23)	0.95 (0.70–1.20)	0.95 (0.71–1.20)	0.97 (0.79–1.18)	ACEi	1.23 (0.64–2.78)	1.24 (0.65–2.48)	0.23 (0.02–1.03)	0.27 (0.028–1.65)
0.89 (0.70–1.11)	0.90 (0.67–1.22)	0.90 (0.69–1.18)	0.93 (0.75–1.18)	0.95 (0.73–1.30)	ARB	1.02 (0.39–2.25)	<u>0.18</u> (0.02–0.91)	0.21 (0.02–1.41)
1.18 (0.78–1.72)	1.20 (0.81–1.71)	1.20 (0.82–1.70)	1.24 (0.90–1.69)	1.26 (0.93–1.74)	1.32 (0.89–1.91)	Diuretic ± BB	<u>0.18</u> (0.02–0.91)	0.21 (0.02–1.44)
0.34 (0.08–1.03)	0.35 (0.09–1.04)	0.35 (0.09–1.05)	0.36 (0.09–1.06)	0.37 (0.09–1.08)	0.38 (0.09–1.15)	<u>0.29</u> (0.07–0.89)	ACEi + CCB	1.20 (0.44–3.24)
0.34 (0.08–1.09)	0.34 (0.08–1.09)	0.34 (0.08–1.1)	0.35 (0.08–1.12)	0.36 (0.09–1.14)	0.38 (0.09–1.21)	<u>0.28</u> (0.07–0.94)	0.98 (0.67–1.46)	ACEi + diuretic

□ HR for total mortality (95% CrI)

▤ HR for CV mortality (95% CrI)

Numbers express the HR for the treatments in the lower line compared to the treatment in the upper line. In total mortality section, HR b1 favors the line-defining treatment.

In CV mortality section, HR b1 favors the row-defining treatment. Results with evidence of benefit are in bold and underlined.

BB = betablocker, CCB = calcium channel blocker, ACEi = angiotensin-converting enzyme inhibitor, ARB = angiotensin receptor blocker.

Q18.5. Recommendations from other groups

A review of the 13 guidelines show that they unanimously have set the target blood pressure to be < 130/80 mm Hg. Ten out of the 13 guidelines recommend this target, while the rest recommend a slightly higher number. Two of the guidelines set a different target for the elderly; the UK NICE sets a target of below 135/85 mmHg for adults aged under 80, and below 145/85 mmHg for adults aged 80 and over, while the ACC/AHA states that in older people aged $\geq$ 65 years old, the target SBP range of 130-139 mm Hg. Data from ACCORD has also been the basis for the recommendation not to lower the BP to less than 120/70 mm Hg due to increased risk of adverse outcomes including increased creatinine and hyperkalemia.

Guideline Question 19.

Should we recommend statins for primary prevention of cardiovascular (CV) events among adults with type 2 diabetes mellitus?

Q1.1. Methods

The selected clinical practice guidelines (refer to the main methodology) were carefully reviewed to determine their appropriateness in addressing the question of statin use for the primary prevention of cardiovascular (CV) events among adults with type 2 diabetes mellitus (T2DM). The scientific evidence underpinning their recommendations, including the strength and consistency of findings and their adaptability to the Philippine context, was critically appraised using the ADAPTE framework.

To supplement guideline adaptation, a focused search of the literature was conducted to identify additional evidence on the benefits and harms of statins for primary prevention in adults with T2DM without established atherosclerotic cardiovascular disease. Searches were performed in MEDLINE (via PubMed), the Cochrane Library (CENTRAL), and HERDIN Plus from database inception to March 31, 2024, and were subsequently updated to include newly published studies through September 2025. A combination of Medical Subject Headings (MeSH) and free-text terms was used, including: “statins,” “HMG-CoA reductase inhibitors,” “atorvastatin,” “simvastatin,” “rosuvastatin,” “pravastatin,” “primary prevention,” “cardiovascular disease,” and “type 2 diabetes mellitus.”

Eligible evidence included clinical practice guidelines, randomized controlled trials, meta-analyses, and large prospective cohort studies evaluating statin therapy versus placebo, usual care, or no statin therapy for primary prevention in nonpregnant adults with T2DM. Studies were required to report at least one critical or important outcome consistent with the PICO framework, such as major adverse cardiovascular events (MACE), myocardial infarction, stroke, cardiovascular mortality, all-cause mortality, or key safety outcomes (e.g., myopathy, liver enzyme elevations, new-onset or worsening diabetes, treatment discontinuation). Studies exclusively involving secondary prevention populations, mixed populations without extractable data for primary prevention in T2DM, noncomparative designs, or non-English-language publications were excluded.

Two reviewers independently screened titles, abstracts, and full texts; extracted data using a standardized data collection form; and examined reference lists of included guidelines, trials, and meta-analyses to identify additional relevant studies. Where applicable, risk of bias in randomized controlled trials was assessed using the Cochrane Risk of Bias (RoB 2) tool, and observational studies were appraised using appropriate validated instruments (e.g., ROBINS-I). Certainty of evidence for each critical and important outcome was evaluated using the GRADE approach.

Q19.2. Characteristics of the Included Studies

Characteristics of Statin Primary Prevention Trials Among Individuals with Diabetes

Study/year	Population	Baseline lipid values Mean (SD) mg/dL	Intervention	Duration of follow up	Outcome Measures
AFCAPS/TexCAPS (1998)	Subgroup of 155 (out of 6605) diabetic individuals (45-73 years old) without clinically evident CVD	TC – 221 (21) LDL – 150 (17) HDL – 36 (5) (men) & 40 (5) (women)	Lovastatin 20-40 mg OD	5.2 years	First acute major coronary event defined as fatal or nonfatal myocardial infarction, unstable angina, or sudden cardiac death
ALL-HAT LLA (2002) (DM Sub-study)	Subgroup of 3,638 diabetics from a total of 10,355 hypertensives with at least 1 other CHD risk factor aged 55 years or older	TC = 224 LDL-C = 146 HDL-C = 48 TG = 152	Pravastatin 40 mg OD	8 years	All cause mortality; nonfatal myocardial infarction or fatal CHD (CHD events) combined, cause-specific mortality, and cancer
ASCOT-LLA (2003) (DM Sub-study)	Subgroup of 2,532 diabetics out of 19,342 hypertensive individuals (40-79 years old) with at least 3 CV risk factors: LVH, PAD, smoking, proteinuria, ECG abnormalities	TC = 212.4 (40.4) LDL = 130 (27) TG = 150 (88) HDL = 50.2 (11.4) Inclusion is TC < 6.5 mmol/L	Atorvastatin 10 mg OD	3.3 years (median), prematurely terminated	Total CV events and procedures, Non-fatal MI & Fatal CHD, Fatal and non-fatal stroke
ASPEN (2006) Primary & secondary prevention	2,410 subjects with type 2 diabetes, 40-75 years (Europe, USA, South Africa) For primary prevention: LDL_C ≤ 160 mg/dl. TG required to be ≤ 600 mg/dl at all visits.	TC = 195 (31) LDL-C = 114 (26) TG = 145 (99-205) HDL = 48 (14)	Atorvastatin 10 mg OD	4 years	Time to the first occurrence of a composite end point of CV death (fatal MI, fatal stroke, sudden cardiac death, heart failure, or arrhythmic non-sudden CV death), nonfatal or silent MI, nonfatal stroke, CABG, resuscitated cardiac arrest, or worsening unstable angina requiring hosp

CARDS (2004)	2,838 individuals with DM (40-75 years old) [UK, Ireland] No previous history of CV disease, LDL-C $\leq$ 4.14 mmol/L, TG $\leq$ 6.78 mmol/L & at least one of the ff: retinopathy, albuminuria, current smoking, or hypertension.	TC= 203 (31) LDL-C = 115 (26.6) TG = 150 (106- 211) HDL-C = 54 (13)	Atorvastatin 10 mg OD	2 years, early termination	Time to first occurrence of the following: acute coronary heart disease events, coronary revascularisation, or stroke
HPS (2003)	5,963 individuals with diabetes (40-80 yrs old) & 14,573 with occlusive artery disease but no DM, with non-fasting TC of at least 3.5 mmol/L (135 mg/dL)	TC = 220 (38) LDL-C = 129 (31) TG = 202 (140) HDL-C = 40 (13.7)	Simvastatin 40 mg OD	5 years	First major coronary event (ie, non-fatal MI or coronary death) and of first major vascular event (ie, major coronary event, stroke or revascularisation)
MEGA (2008) (DM Sub-study)	1746 patients with DM & 464 patients with IFG; men 40-70 yrs & menopausal women up to 80 years	TC = 220-270 mg/dL TC = 243 LDL-C = 157, HDL-C = 58 Trig= 127	Pravastatin 10-20 mg OD	5.3 years	First occurrence of CHD (fatal & nonfatal MI, angina, cardiac or sudden death, cardiac or vascular intervention)
PROSPER (2002) (DM Sub-study)	623 pts with diabetes among 5,804 individuals (70-82 yrs old) with increased risk of vascular disease due to smoking, diabetes or hypertension in Scotland, Ireland, Netherlands	TC = 227 (33.8) LDL-C = 150 (31) TG= 139 (58) HDL-C = 52.44 (14)	Pravastatin 40 mg OD	3.2 years	

Q19.3. Assessment of Risk of Bias

Across all outcomes evaluated—including total mortality, fatal cardiovascular death, fatal and non-fatal myocardial infarction, unstable angina, CVD/stroke, acute major CVD events, and interventional procedures—the body of evidence consistently demonstrated a low risk of bias. All

included studies were randomized controlled trials, and for every outcome, the GRADE assessment rated the risk of bias as “not serious.” This indicates that the trials were generally well-conducted, with adequate randomization, allocation concealment, and blinding procedures where applicable, and that there were no notable concerns related to deviations from intended interventions, missing outcome data, selective reporting, or measurement of outcomes. As a result, no downgrading for risk of bias was applied for any outcome.

Q19.4. GRADE Evidence Profile

GRADE Summary of Findings on Statins and CV Outcomes Among Persons with Diabetes.

Quality assessment							N _e of patients		Effect		Quality	Importance
N _e of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Statins	Placebo	Relative (95% CI)	Absolute (95% CI)		
Total Mortality												
1	randomised trials	not serious	not serious	serious $\downarrow$	not serious	none	61/1428 (4.3%)	82/1409 (5.8%)	RR 0.73 (0.53 to 1.01)	NA	⊕⊕⊕ ○ MODERATE	CRITICAL
Fatal CHD/Cardiovascular death												
3	randomised trials	not serious	not serious	serious $\downarrow$	not serious	none	60/3645 (1.6%)	58/3629 (1.6%)	RR 0.98 (0.68 to 1.41)	NA	⊕⊕⊕ ○ MODERATE	CRITICAL
Fatal and Non-fatal MI												
4	randomised trials	not serious	not serious	serious $\downarrow$	not serious	none	378/13914 (2.7%)	518/13896 (3.7%)	RR 0.73 (0.64 to 0.83)	10 fewer per 1000 (from 6 fewer to 13 fewer)	⊕⊕⊕ ○ MODERATE	CRITICAL
Unstable angina												
3	randomised trials	not serious	not serious	serious $\downarrow$	not serious	none	51/3645 (1.4%)	60/3629 (1.7%)	RR 0.77 (0.53 to 1.01)	NA	⊕⊕⊕ ○ MODERATE	IMPORTANT
CVD/Stroke												
4	randomised trials	not serious	not serious	serious $\downarrow$	not serious	none	224/13914 (1.6%)	298/13896 (2.1%)	RR 0.75 (0.63 to 0.89)	5 fewer per 1000 (from 2 fewer to 8 fewer)	⊕⊕⊕ ○ MODERATE	CRITICAL
Acute Major CVD Events (composite)												
8	randomised trials	not serious	not serious	serious $\downarrow$	not serious	strong association	597/8083 (7.4%)	766/8012 (9.6%)	RR 0.78 (0.7 to 0.86)	21 fewer per 1000 (from 13 fewer to 29 fewer)	⊕⊕⊕⊕ HIGH	IMPORTANT
Interventional Procedures												
3	randomised trials	not serious	not serious	serious $\downarrow$	not serious	none	328/12908 (2.5%)	390/12875 (3.0%)	RR 0.84 (0.73 to 0.97)	5 fewer per 1000 (from 1 fewer to 8 fewer)	⊕⊕⊕ ○ MODERATE	IMPORTANT

Q19.5. Recommendations from other groups

Other guidelines have similar recommendations but add on a layer of risk on top of diabetes mellitus. For example, the Canadian Diabetes Association guidelines recommend statin therapy for diabetic individuals with an indication for lipid-lowering therapy. The American Diabetes Association on the other hand recommends high-intensity statin for patients of all ages with diabetes and overt CVD, or for those who are at least 40 years old and with additional CV risk factors (total of 3 risk factors: $\geq$ 40 years old, diabetes and another CV risk factor). Those who have diabetes and are aged 40-75 years old should consider using moderate-intensity statins. It is silent though for diabetic individuals who are less than age 40.

Guideline Question 20.

Should we use sodium-glucose cotransporter-2 (SGLT2) inhibitors among patients with type 2 diabetes mellitus with established atherosclerotic cardiovascular disease (ASCVD)?

Q20.1. Methods and search strategy

Evidence for this clinical question was primarily derived from a systematic review and meta-analysis conducted by Zhao et al., published in April 2022. Their review synthesized data from four major randomized controlled trials (RCTs) enrolling a total of 42,568 adult participants with type 2 diabetes mellitus (T2DM) who were either diagnosed with established cardiovascular disease (CVD) or identified as having elevated cardiovascular risk. Eligible participants were individuals aged 18 years or older with T2DM and baseline HbA1c levels between 7.0% and 10.5%. The populations studied included those with documented ASCVD as well as those with significant atherosclerotic cardiovascular risk factors.

To ensure that no relevant studies were missed beyond the Zhao et al. review, a supplementary targeted search was conducted in MEDLINE (via PubMed) and the Cochrane Library from database inception to March 31, 2024, and was subsequently updated to include newly published evidence through September 2025. Search terms included: “type 2 diabetes,” “glucose-lowering therapy,” “cardiovascular outcomes trial,” “major adverse cardiovascular events,” “randomized controlled trial,” and the names of relevant pharmacologic agents evaluated in the included RCTs. Filters for randomized controlled trials were applied when appropriate. Reference lists of the included trials and the Zhao et al. meta-analysis were also screened to identify additional eligible studies.

Eligible evidence included RCTs that:

- enrolled nonpregnant adults with T2DM and either established CVD or elevated CV risk;
- compared a glucose-lowering intervention with placebo or standard of care;
- reported at least one critical cardiovascular outcome such as major adverse cardiovascular events (MACE), nonfatal myocardial infarction, nonfatal stroke, cardiovascular mortality, or all-cause mortality;
- followed participants for sufficient duration to allow assessment of cardiovascular risk reduction.

Trials not reporting cardiovascular outcomes, nonrandomized studies, studies in type 1 diabetes or gestational diabetes, pediatric populations, and non-English publications were excluded.

Two reviewers independently screened the RCTs included in the Zhao et al. meta-analysis and any additional studies identified in the supplementary search. Data were extracted using a standardized form capturing study design, population characteristics, interventions, comparators, follow-up duration, and cardiovascular outcomes. Risk of bias for each included RCT was assessed using the Cochrane Risk of Bias 2 (RoB 2) tool. Certainty of evidence for each outcome was appraised using the GRADE approach.

Q20.2 Characteristics of the Included Studies

Characteristics of included studies

Study	Design	Number of diabetic patients/total number of patients (%)	Intervention/Treatment	Control/Comparator	Outcomes	Median Follow-up Duration
VERTIS CV (2020)	RCT	8,246 (100%)	Ertugliflozin 5/15mg OD	Placebo	MACE (composite of CV death, nonfatal MI, or stroke)	6.1 years
DECLARE-TIMI 58 (2019)	RCT	17,160 (100%)	Dapagliflozin 10mg OD	Placebo	MACE, composite of CV death or HHF	5.2 years
CANVAS Program (2018)	RCT	10,142 (100%)	Canagliflozin 100/300mg OD	Placebo	Composite of CV death, nonfatal MI or stroke	3.6 years
EMPA-REG OUTCOME (2017)	RCT	7,020 (100%)	Empagliflozin 10/25mg OD	Placebo	Composite of CV death, nonfatal MI, or stroke	4.6 years

Q20.3. Assessment of Risk of Bias

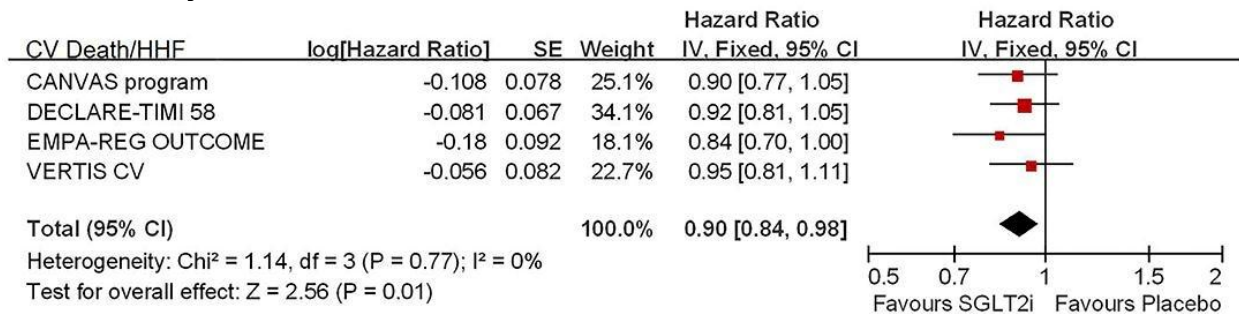
Risk of bias

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
CANVAS Program	+	+	+	+	+	+	?
DECLARE-TIMI	+	+	+	+	+	+	?
EMPA-REG OUTCOME	+	+	+	+	+	+	?
VERTIS CV	+	+	+	+	+	+	?

Q20.4. Results

Forest plots

3.1 Meta-analysis of SGLT2i on CVD death or HHF



3.2 Meta-analysis of SGLT2i on HHF

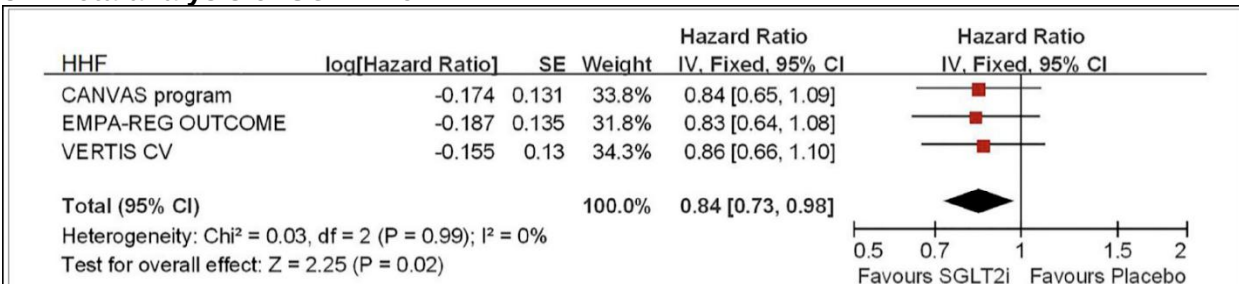


FIGURE 2 | Meta-analysis of SGLT2i on HHF.

3.3 Renal composite outcomes

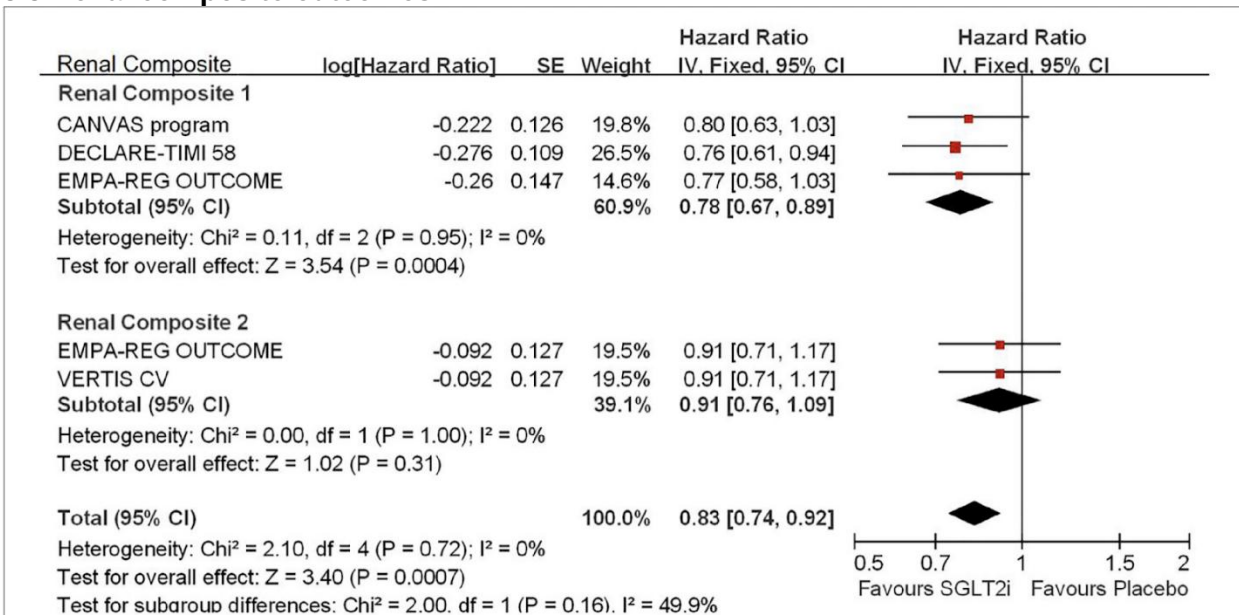
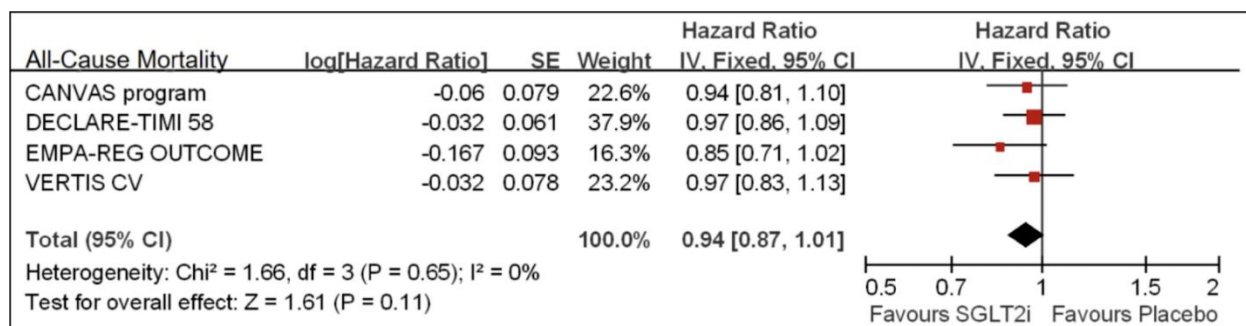


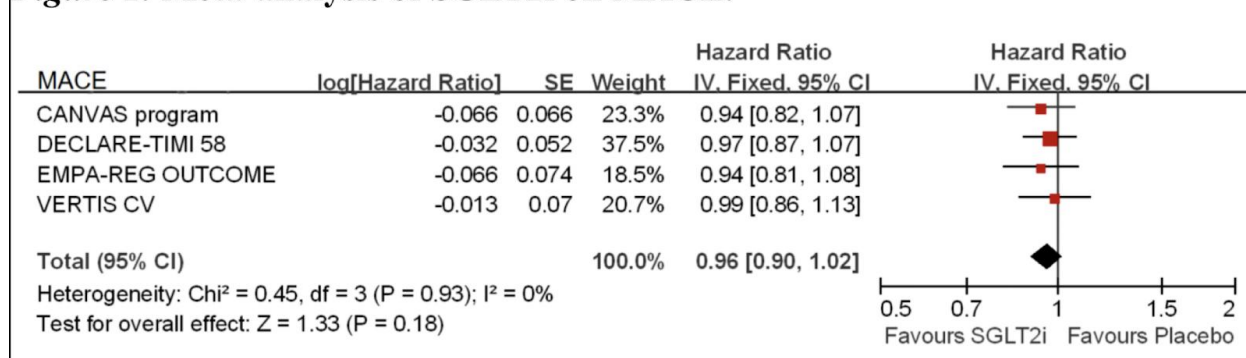
FIGURE 3 | Meta-analysis of SGLT2i on renal composite outcomes. Renal Composite 1 concluded 40% decrease in eGFR, end-stage kidney disease, or renal death; renal Composite 2 included renal death, renal dialysis/transplant, or doubling of serum creatinine from the baseline.

3.4 All-cause Death

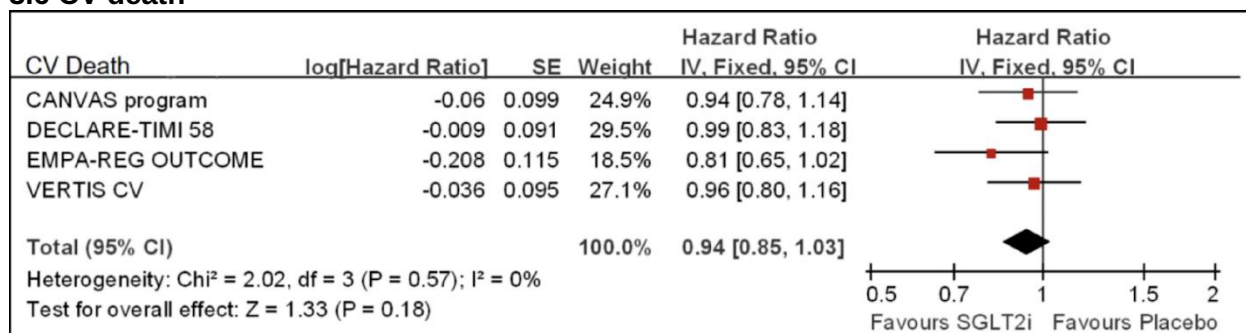


3.5 MACE

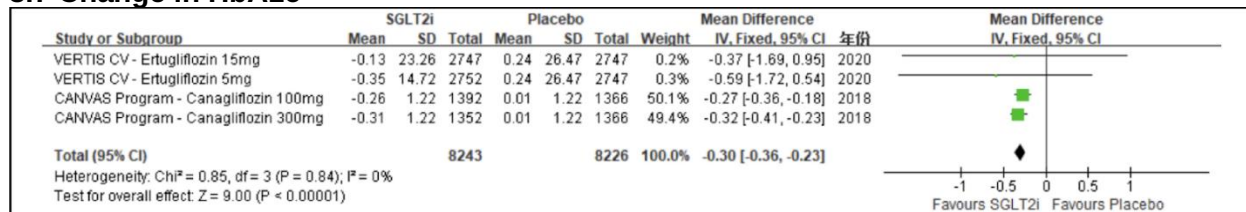
Figure 2. Meta-analysis of SGLT2i on MACE.



3.6 CV death



3.7 Change in HbA1c



Q20.5. GRADE Evidence Profile

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	SGLT2i	Placebo	Relative (95% CI)	Absolute (95% CI)		
Major Adverse Cardiovascular Events (follow-up: range 3.6 years to 6.1 years; assessed with: Hazard ratio)												
4	randomised trials	not serious	not serious	not serious	serious ^a	none			HR 0.96 (0.90 to 1.02)	1 fewer per 1,000 (from 1 fewer to 1 fewer)	⊕⊕⊕○ Moderate	CRITICAL
Hospitalization for Heart Failure												
3	randomised trials	not serious	not serious	not serious	not serious	none			HR 0.84 (0.73 to 0.98)	1 fewer per 1,000 (from 1 fewer to 1 fewer)	⊕⊕⊕⊕ High	CRITICAL
Cardiovascular Death/Hospitalization for Heart Failure (follow-up: range 3.6 years to 6.1 years)												
4	randomised trials	not serious	not serious	not serious	not serious	none			HR 0.90 (0.84 to 0.98)	1 fewer per 1,000 (from 1 fewer to 1 fewer)	⊕⊕⊕⊕ High	CRITICAL
Glycemic Control (Change in HbA1c from Baseline)												
2	randomised trials	not serious	not serious	not serious	not serious	none			-- (-0.36 to -0.23)	-- per 1,000 (from -- to --)	⊕⊕⊕⊕ High	IMPORTANT
All-cause Death/Mortality												
4	randomised trials	not serious	not serious	not serious	serious ^a	none			HR 0.94 (0.87 to 1.01)	1 fewer per 1,000 (from 1 fewer to 1 fewer)	⊕⊕⊕○ Moderate	CRITICAL
Renal composite outcome 1 (40% decrease in eGFR, end-stage kidney disease, or renal death)												
3	randomised trials	not serious	not serious	not serious	not serious	none			HR 0.78 (0.67 to 0.89)	1 fewer per 1,000 (from 1 fewer to 1 fewer)	⊕⊕⊕⊕ High	IMPORTANT
Renal composite outcome 2 (renal death, renal dialysis/transplant, or doubling of serum creatinine from the baseline)												
2	randomised trials	not serious	not serious	not serious	serious ^a	none			HR 0.91 (0.76 to 1.09)	1 fewer per 1,000 (from 1 fewer to 1 fewer)	⊕⊕⊕○ Moderate	IMPORTANT

CI: confidence interval; HR: hazard Ratio

Explanations

a. Crossed the line of no effect

Q20.6. Recommendations from other groups

Group/ Guidelines	Recommendation	Strength of Recommendation and Certainty of Evidence
ADA 2023	In people with type 2 diabetes and established ASCVD, multiple risk factors for ASCVD, or diabetic kidney disease, a sodium-glucose cotransporter 2 inhibitor with demonstrated cardiovascular benefit is recommended to reduce the risk of major adverse cardiovascular events and/or heart failure hospitalization.	Grade A
Diabetes Canada CPG 2020 Update	In adults with type 2 diabetes with clinical CVD in whom glycemic targets are not achieved with existing antihyperglycemic medication(s) and with an eGFR >30ml/min/1.73m ² , an antihyperglycemic agent with demonstrated CV outcome benefit should be added to reduce the risk of major CV events.	Grade A, Level 1A for Empagliflozin Grade C, Level 2 for Canagliflozin
	In adults with type 2 diabetes with ASCVD, HF, and/or CKD, treatment should include agents from the following classes with demonstrated CV or renal benefits: a. In adults with type 2 diabetes and ASCVD, a GLP1-RA or SGLT2i with CV or renal benefit should be used. b. In adults with type 2 diabetes and a history of HF (reduced ejection fraction <40%), an SGLT2i should be used to reduce the risk of HHF or CV death, if the eGFR is >30 mL/min/1.73m² c. In adults with type 2 diabetes and CKD and an eGFR >30 mL/min/1.73m², an SGLT2i should be used to reduce the risk of progression of nephropathy, HHF and MACE.	Nephropathy Grade A, Level 1A for canagliflozin Grade A, Level 1 for empagliflozin and dapagliflozin HHF Grade A, Level 1 for canagliflozin, dapagliflozin, and empagliflozin MACE Grade B, Level 2 for canagliflozin Grade C, Level 3 for empagliflozin
Malaysia CPG 2020	In patients with T2DM with established atherosclerotic cardiovascular disease, consider adding SGLT2i or GLP1-RA with demonstrated CV benefit as part of the glucose lowering regimen.	Grade A

	Among patients with ASCVD with pre-existing or at high risk of heart failure (HFrEF), SGLT2i should be considered, even if HbA1c is at target.	Grade A
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Guideline Question 21.

Should we use GLP-1 Receptor agonists (GLP-1 RA) among adults with Type 2 DM and established atherosclerotic cardiovascular disease (ASCVD)?

Q21.1. Methods

The evidence review was conducted from July 2021 to November 2023 and subsequently updated to include relevant guidelines, systematic reviews, and individual studies addressing the clinical questions, covering publications from database inception through October 2024...

Search, Retrieval, and Selection of Guidelines

From September 2021 to February 2022, a systematic review of the literature was undertaken by independent reviewers. A targeted search strategy was used to identify relevant clinical practice guidelines (CPGs) addressing pharmacologic management of type 2 diabetes mellitus (T2DM) with cardiovascular comorbidities. Searches were conducted in MEDLINE/PubMed, the National Guidelines Clearinghouse, the Guideline International Network, and the National Institute for Clinical Evidence. Additional searches of professional society websites and other medical databases were conducted by the Technical Working Group (TWG) to locate CPGs not captured in the primary search (see Appendix A and B). The guideline recommendations, their evidentiary bases, and applicability to the Philippine context were reviewed in accordance with the ADAPTE framework.

Supplementary Systematic Review of Primary Studies

Because guideline recommendations relied heavily on cardiovascular outcomes trials, and to ensure that all relevant evidence was evaluated, a supplementary systematic search was performed to identify individual randomized controlled trials (RCTs) and meta-analyses evaluating GLP-1 receptor agonists in adults with T2DM and established ASCVD. Searches were performed in MEDLINE (via PubMed) and the Cochrane Library (CENTRAL) from database inception to March 31, 2024, **and were subsequently updated to include newly published studies through September 2025.** Both Medical Subject Headings (MeSH) and free-text terms were used,

including:
“GLP-1 receptor agonist,” “liraglutide,” “semaglutide,” “dulaglutide,”
“cardiovascular outcomes trial,” “major adverse cardiovascular events,”
“ASCVD,” “type 2 diabetes mellitus.”

Eligibility Criteria

The following criteria were applied for inclusion:

- **Population:** Nonpregnant adults with T2DM and *established* ASCVD (e.g., previous myocardial infarction, stroke, PAD).
- **Intervention:** GLP-1 receptor agonists at therapeutic doses approved for T2DM.
- **Comparator:** Placebo or standard of care in addition to optimized background therapy.
- **Outcome:** At least one critical or important cardiovascular outcome, including major adverse cardiovascular events (MACE), cardiovascular mortality, all-cause mortality, myocardial infarction, stroke, hospitalization for heart failure, and key adverse events.
- **Study design:** RCTs, prespecified subgroup analyses of CVOTs, and meta-analyses of RCTs. Studies focusing solely on glycemic outcomes without cardiovascular endpoints, nonrandomized trials, studies in type 1 diabetes, gestational diabetes, pediatric populations, and non-English publications were excluded.

Screening, Data Extraction, and Appraisal

Two reviewers independently screened titles, abstracts, and full texts; extracted study characteristics and outcome data using a standardized data collection form; and reviewed reference lists of included CPGs and CVOTs to identify additional eligible studies.

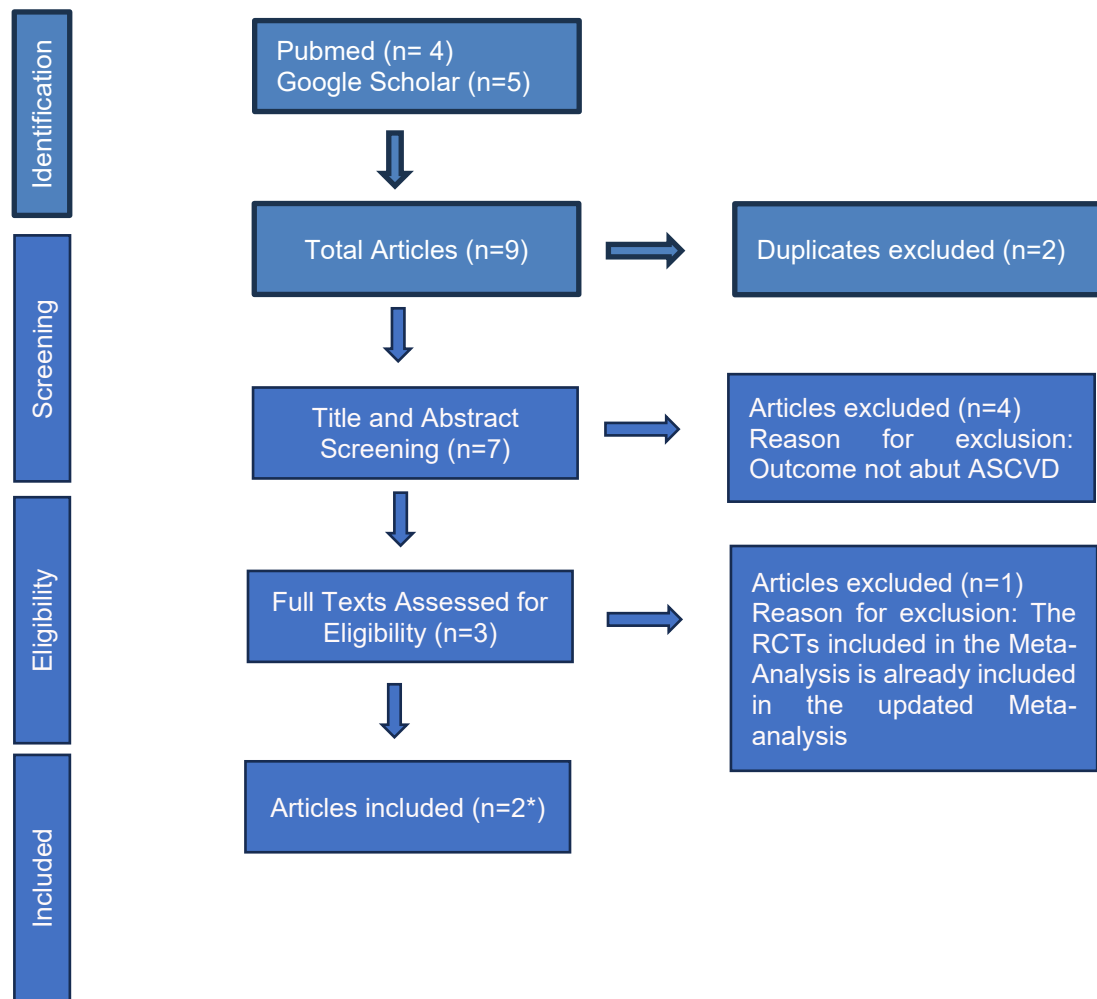
Risk of bias for randomized trials was assessed using the **Cochrane Risk of Bias 2 (RoB 2)** tool. Certainty of evidence for each outcome was graded using the **GRADE** approach.

Q21.2. Search strategy

Utilizing PICO for key search terms of this clinical question, search strategy was conducted with the following study type filters used, database, and filter for dates of publication.

Database	Search Terms	Date and Time of Search	Results	
Google Scholar	Diabetes Mellitus and Glucagon-like Peptide-1 Receptor Agonists and Cardiovascular Outcomes and Meta-Analysis	February 22, 2022 12PM	Yield 5	Eligible 2
PubMed	"Diabetes Mellitus, Type 2"[Mesh] AND "Glucagon-like Peptide-1 Receptor Agonists" [Mesh] Filters applied: Meta-Analysis, Systematic Review, in the last 5 years	February 21, 2022 7:25PM	0	
	"Diabetes Mellitus"[Mesh] AND "Glucagon-Like Peptide-1 Receptor Agonists" [Mesh] Filters applied: Meta-Analysis, Systematic Review, in the last 5 years			
PubMed	"Glucagon-Like Peptide-1 Receptor Agonists"[Majr] Filters applied: Meta-Analysis, Systematic Review, in the last 5 years	February 21, 2022 7PM	4	0

Q21.3. PRISMA flow diagram
Appendix B.3. PRISMA Flow



***2 Meta-analysis**

Q21.4. Characteristics of the Included Studies

Characteristics of Included Studies

Title/Author	Study design	Country	Number of patients	Population	Intervention Group(s)	Outcomes
Dario Giugliano, Lorenzo Scappaticcio, Miriam Longo, Paola Caruso, Maria Ida Maiorino, Giuseppe Bellastella, Antonio Ceriello, Paolo Chiodini, Katherine Esposito. GLP-1 receptor agonists and cardiorenal outcomes in type 2 diabetes: an updated meta-analysis of eight CVOTs. Cardiovascular Diabetology (2021) 20:189.	Meta-Analysis	Italy	60,080	Type 2 DM patients	GLP-1 RA	MACE
Tsapas A, Karagiannis T, Avgerinos I, Liakos A, Bekiari E. GLP-1 receptor agonists for cardiovascular outcomes with and without metformin. A systematic review and meta-analysis of cardiovascular outcomes trials. Diabetes Res Clin Pract. 2021 Jul;177:108921. doi: 10.1016/j.diabres.2021.108921. Epub 2021 Jun 15. PMID: 34144086.	Meta-Analysis	Greece	43,456	Type 2 DM patients	GLP-1 RA	MACE

Q1.5. Assessment of Risk of Bias

Summary of Risk of Bias

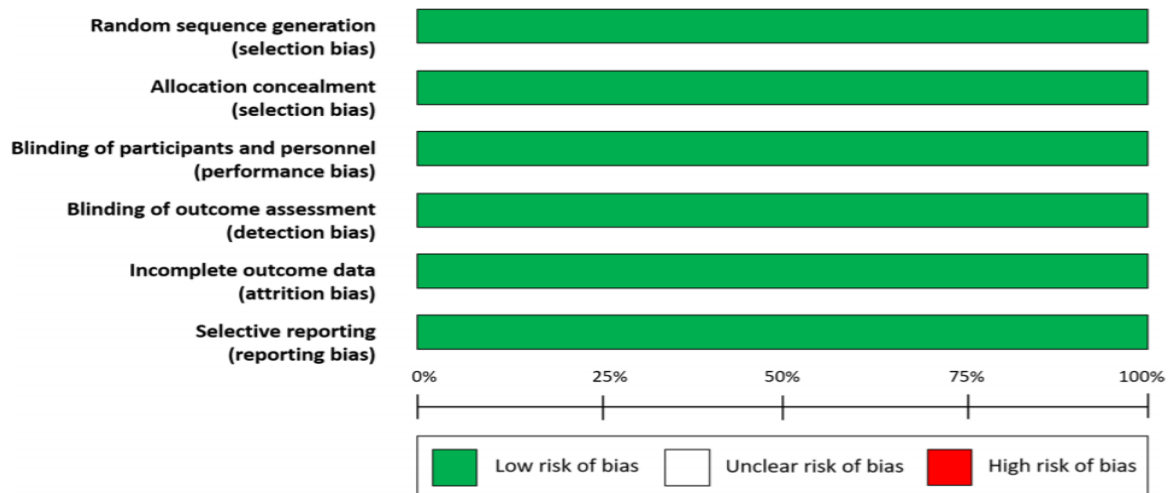


Figure 1. Cochrane Risk of Bias for Meta-Analysis done by Giugliano et al.

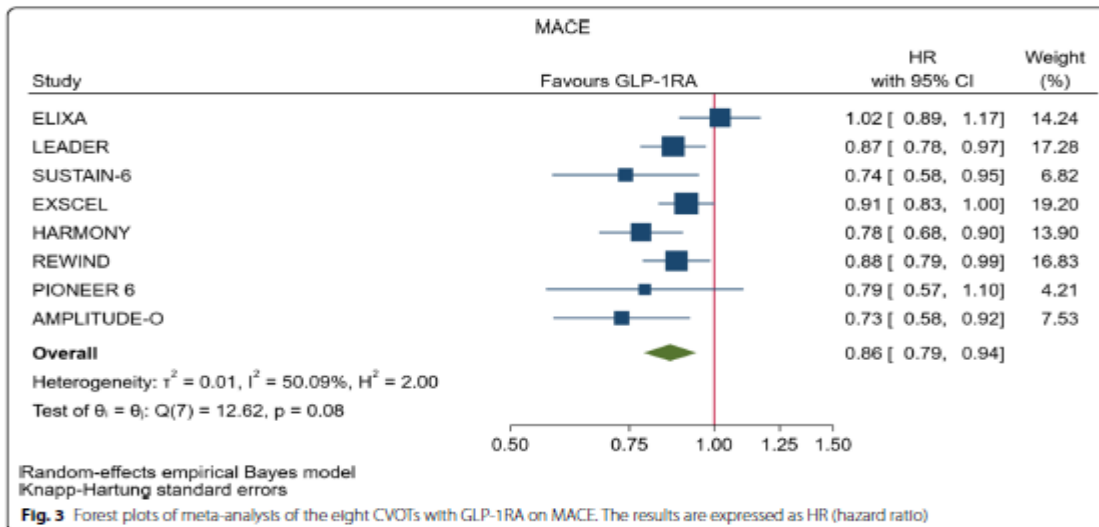
Table 1: Risk of Bias for Meta-Analysis by Tsapas et al.

I	APPRAISING DIRECTNESS	
	Does the study provide a direct enough answer to your clinical question in terms of type of patients (P), exposure/ intervention (E) and outcome (O)?	Yes
II.	APPRAISING VALIDITY	
1.	Were the criteria for inclusion of studies appropriate?	Yes
2.	Was the search for eligible studies thorough?	Yes
3.	Was the validity of the included studies assessed?	Yes
4.	Were the assessments of the studies reproducible?	Yes
III.	APPRAISING RESULTS	
1.	What are the overall results of the review?	See forest plot below
2.	How precise were the results?	See forest plot below

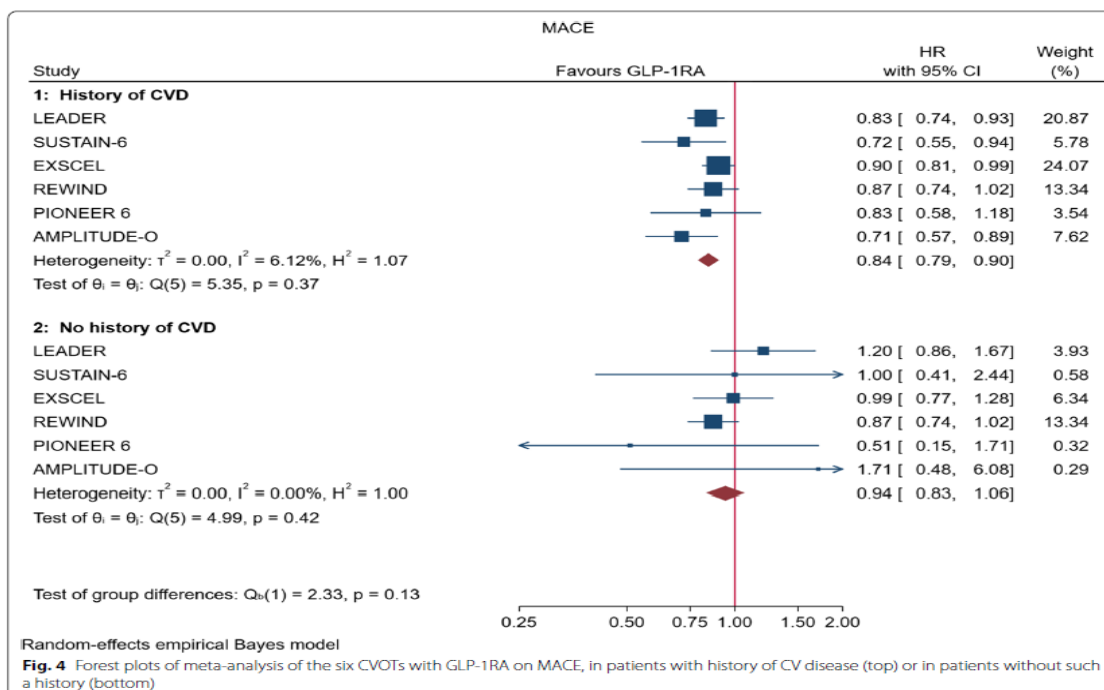
3.	Were the results similar from study to study?	See forest plot below
IV.	ASSESSING APPLICABILITY	
1.	Are there biologic issues affecting applicability? (Consider the influence of sex, co-morbidity, race, age and pathology)	None
2.	Are there socio-economic issues affecting applicability?	Yes
3.	If the overall results of the review are not directly applicable to your patient, are there credible subgroup analyses that you could use?	Yes
V.	INDIVIDUALIZING THE RESULTS	
1.	What is the implication of study findings on your individual patient? (Estimate the individualized NNTs for your patient)	77
2.	Would you offer the treatment to your patients?	Yes

Q21.6. Results

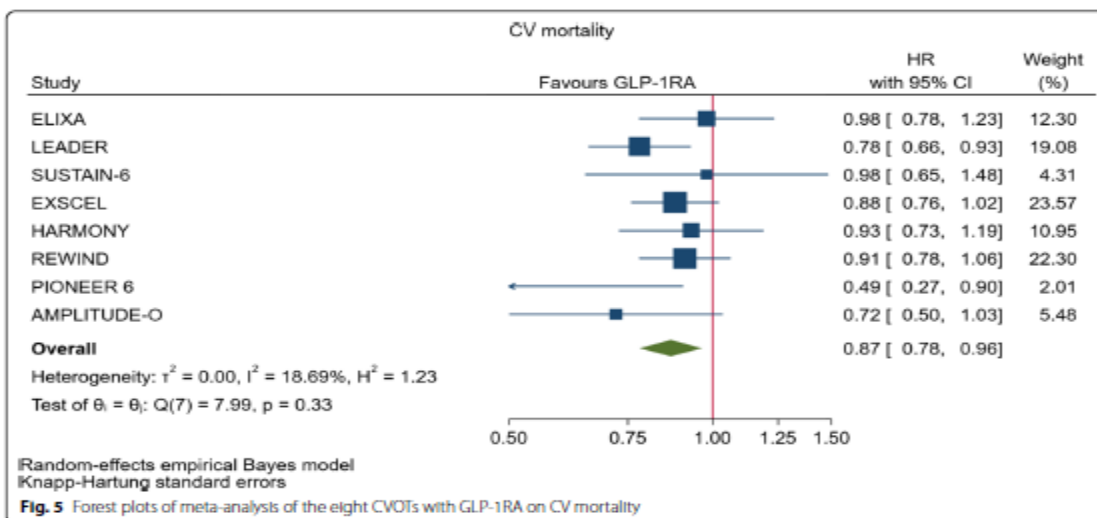
Forest Plots



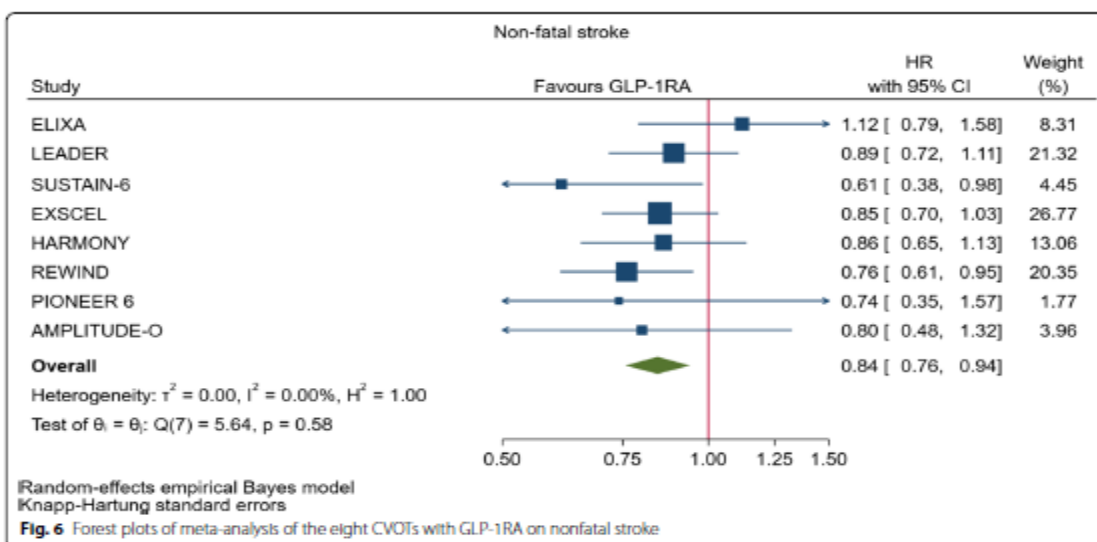
Forest Plot 1.1. Meta-analysis by Giugliano et al. (Major Cardiovascular Events)



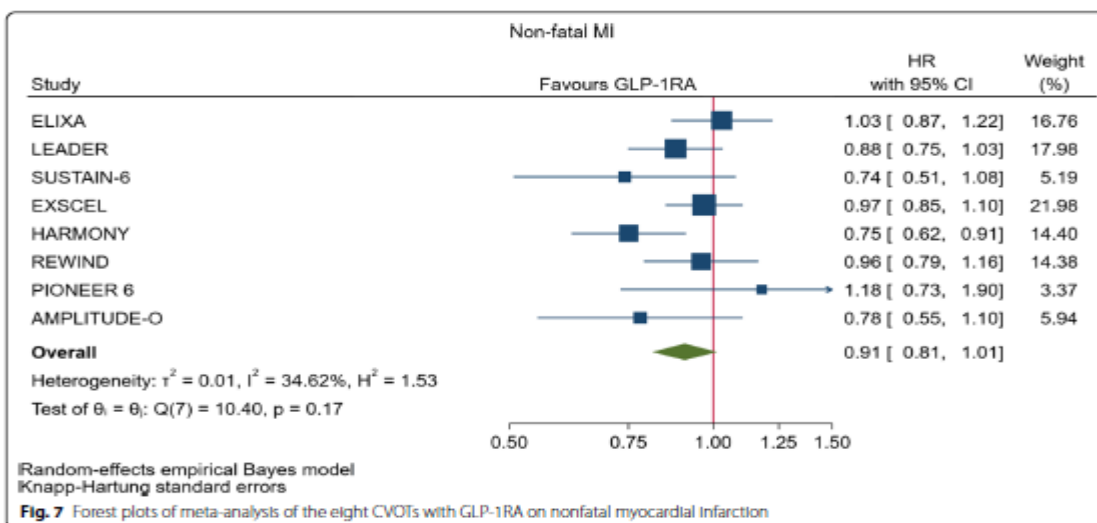
Forest Plot 1.2. Meta-analysis by Giugliano et al. (Major Cardiovascular Events-Subgroup Analysis)



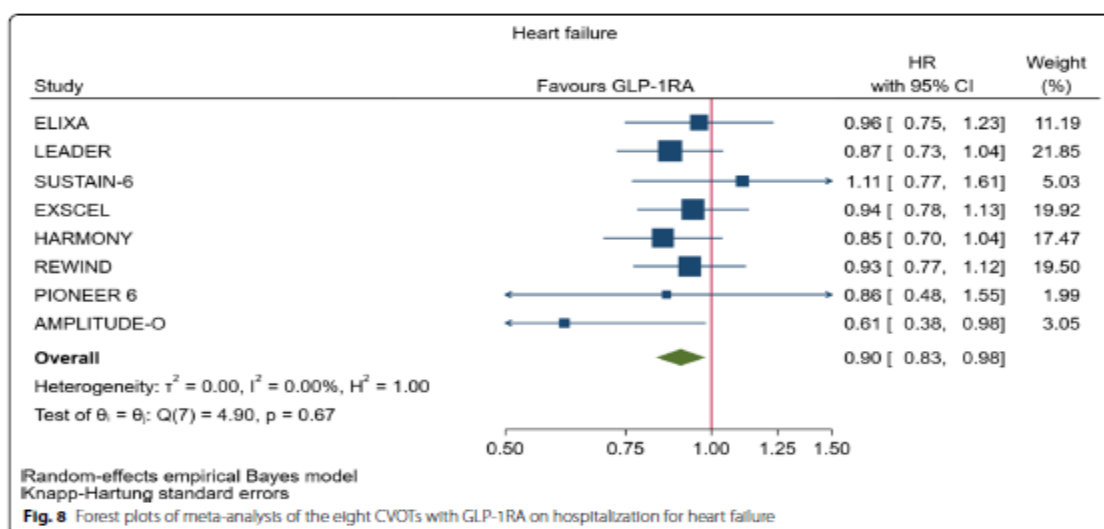
Forest Plot 1.3. Meta-analysis by Giugliano et al. (Cardiovascular Mortality)



Forest Plot 1.4. Meta-analysis by Giugliano et al. (Non-fatal Stroke)



Forest Plot 1.5. Meta-analysis by Giugliano et al. (Non-fatal Myocardial Infarction)



Forest Plot 1.6. Meta-analysis by Giugliano et al. (Heart failure)

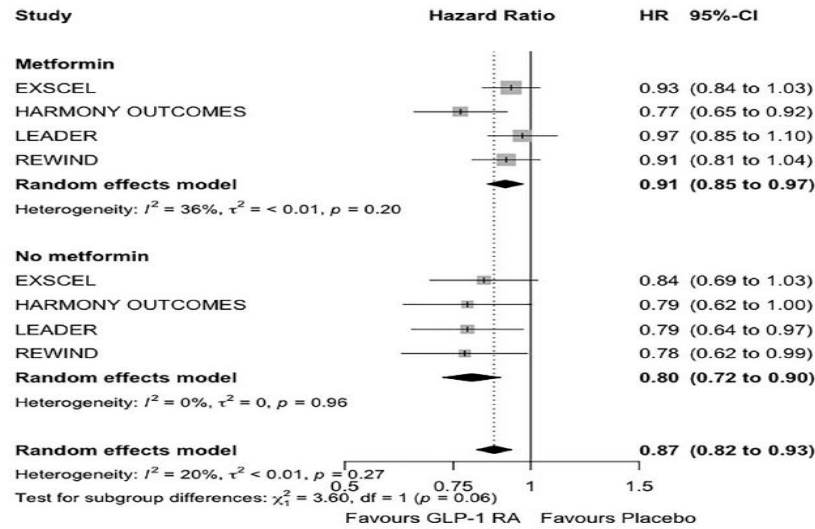


Fig. 1 – Effect of GLP-1 RAs on MACE by baseline metformin use. CI, confidence interval; GLP-1 RAs, glucagon-like peptide-1 receptor agonists; MACE, major adverse cardiovascular events. MACE were defined as non-fatal myocardial infarction, non-fatal stroke or cardiovascular death.

Forest Plot 2.1. Meta-Analysis by Tsapas et al. (Major Cardiovascular Events)

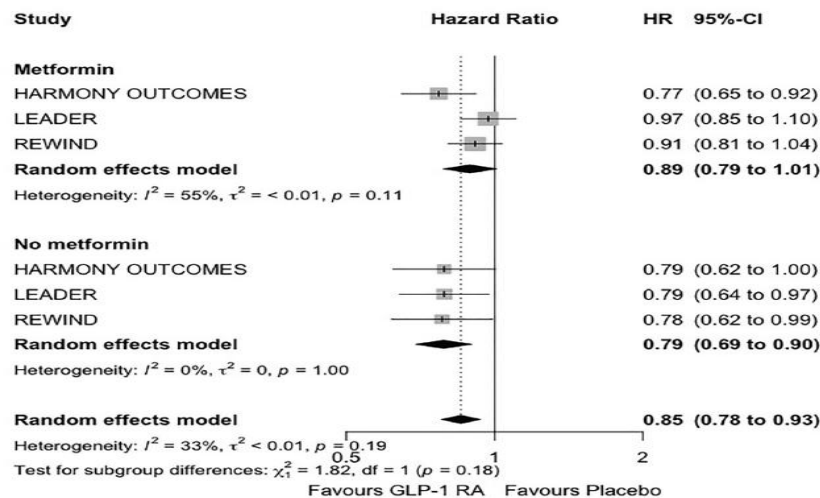


Fig. 2 – Sensitivity analysis for effect of GLP-1 RAs on MACE by baseline metformin use. CI, confidence interval; GLP-1 RAs, glucagon-like peptide-1 receptor agonists.

Forest Plot 2.2. Meta-Analysis by Tsapas et al. (Major Cardiovascular Events- Sensitivity Analysis)

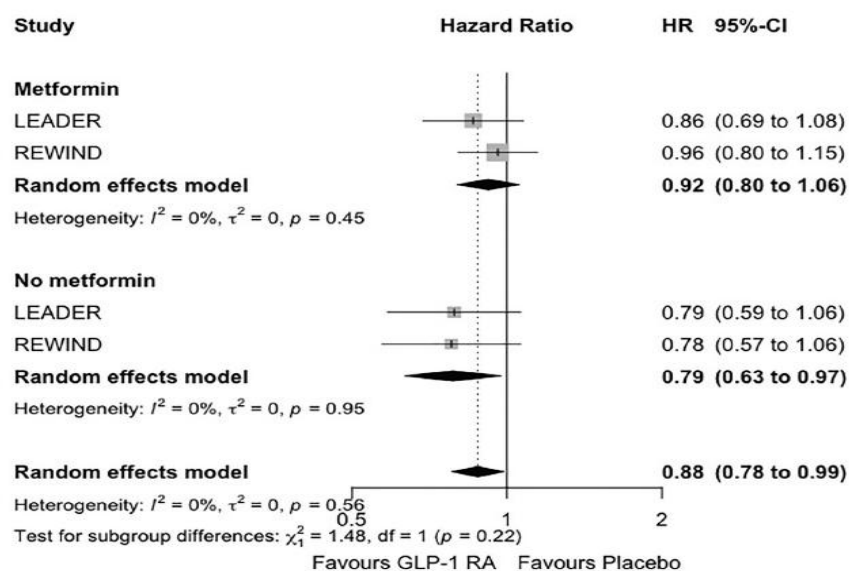


Fig. 3 – Effect of GLP-1 RAs on cardiovascular mortality by baseline metformin use. CI, confidence interval; GLP-1 RAs, glucagon-like peptide-1 receptor agonists.

Forest Plot 2.3. Meta-Analysis by Tsapas et al. (Cardiovascular Mortality)

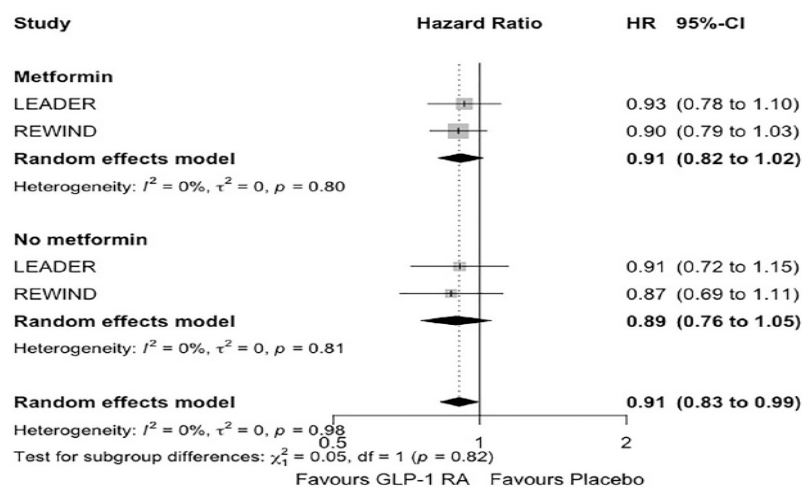


Fig. 4 – Effect of GLP-1 RAs on all-cause mortality by baseline metformin use. CI, confidence interval; GLP-1 RAs, glucagon-like peptide-1 receptor agonists.

Forest Plot 2.4. Meta-Analysis by Tsapas et al. (All-Cause Mortality)

Q21.7. GRADE Evidence Profile

GRADE Evidence Profile Table

Author(s): Dario Giugliano, Lorenzo Scappaticcio, Miriam Longo, Paola Caruso, Maria Ida Maiorino, Giuseppe Bellastella, Antonio Ceriello, Paolo Chiodini, Katherine Esposito.

Bibliography: GLP-1 receptor agonists and cardiorenal outcomes in type 2 diabetes: an updated meta-analysis of eight CVOTs. Cardiovascular Diabetology (2021) 20:189.

Certainty assessment							N ₂ of patients		Effect		Certainty	Importance
N ₂ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	GLP-1 RA	Placebo	Relative (95% CI)	Absolute (95% CI)		
MACE												
8	randomised trials	not serious	not serious	not serious	not serious	none	3137/30694 (10.2%)	3429/29386 (11.7%)	RR 0.86 (0.79 to 0.94)	16 fewer per 1,000 (from 25 fewer to 7 fewer)	⊕⊕⊕ ⊕ High	CRITICAL
Cardiovascular Mortality												
8	randomised trials	not serious	not serious	not serious	not serious	none	1352/30694 (4.4%)	1521/29386 (5.2%)	HR 0.87 (0.78 to 0.96)	7 fewer per 1,000 (from 11 fewer to 2 fewer)	⊕⊕⊕ ⊕ High	CRITICAL
Fatal or Non-Fatal Stroke												
8	randomised trials	not serious	not serious	not serious	not serious	none	809/30694 (2.6%)	908/29386 (3.1%)	HR 0.84 (0.76 to 0.94)	5 fewer per 1,000 (from 7 fewer to 2 fewer)	⊕⊕⊕ ⊕ High	CRITICAL
Fatal or Non-Fatal Myocardial Infarction												
8	randomised trials	not serious	not serious	not serious	not serious	none	1716/30694 (5.6%)	1773/29386 (6.0%)	HR 0.91 (0.81 to 1.01)	5 fewer per 1,000 (from 11 fewer to 1 more)	⊕⊕⊕ ⊕ High	CRITICAL
Heart Failure												
8	randomised trials	not serious	not serious	not serious	not serious	none	976/30694 (3.2%)	1047/29386 (3.6%)	HR 0.90 (0.83 to 0.98)	4 fewer per 1,000 (from 6 fewer to 1 fewer)	⊕⊕⊕ ⊕ High	CRITICAL

Author(s): Apostolos Tsapas, Thomas Karagiannis, Ioannis Avgerinos, Aris Liakos a,1, Eleni Bekiari

Bibliography: GLP-1 receptor agonists for cardiovascular outcomes with and without metformin. A systematic review and meta-analysis of cardiovascular outcomes trials. Diabetes Res Clin Pract. 2021 Jul;177:108921. doi: 10.1016/j.diabres.2021.108921. Epub 2021 Jun 15. PMID: 34144086.

Certainty assessment							No of patients		Effect		Certainty	Importance
No of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	GLP-1 RA with or without Metformin	Placebo	Relative (95% CI)	Absolute (95% CI)		

MACE

4	randomised trials	not serious	not serious	not serious	not serious	none	2379/21704 (11.0%)	2690/21752 (12.4%)	HR 0.87 (0.82 to 0.93)	15 fewer per 1,000 (from 21 fewer to 8 fewer)	⊕⊕⊕⊕ ⊕ High	CRITICAL
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Cardiovascular Mortality

2	randomised trials	not serious	not serious	not serious	not serious	none	536/9617 (5.6%)	622/9624 (6.5%)	HR 0.88 (0.78 to 0.99)	8 fewer per 1,000 (from 14 fewer to 1 fewer)	⊕⊕⊕⊕ ⊕ High	CRITICAL
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All Cause Mortality

2	randomised trials	not serious	not serious	not serious	not serious	none	917/9617 (9.5%)	1039/9624 (10.8%)	HR 0.91 (0.83 to 0.99)	9 fewer per 1,000 (from 17 fewer to 1 fewer)	⊕⊕⊕⊕ ⊕ High	CRITICAL
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Q21.8. Recommendations from other groups

(based from the guidelines adapted in the formulation of this CPG)

American Diabetes Association	2022	Glucagon-like peptide 1 [GLP-1] receptor agonists, sodium–glucose cotransporter 2 [SGLT2] inhibitors), with or without metformin based on glycemic needs, are appropriate initial therapy for individuals with type 2 diabetes with or at high risk for atherosclerotic cardiovascular disease (ASCVD), HF, and/or chronic kidney disease (CKD). (Grade A)
American Association of Clinical Endocrinologists	2022	In persons with Type 2 Diabetes and established ASCVD or at high risk for ASCVD, use GLP-1 RAs with proven CV benefits to reduce the risk of myocardial infarction, stroke, or CV death regardless of other glucose-lowering or CV therapies and independent of A1C. (Grade A)
Philippine Consensus Statement	2021	For patients with established ASCVD, either SGLT2 inhibitors or GLP-1 RA, added on top of metformin, is recommended to reduce the risk of major cardiovascular events. If either SGLT2-Inhibitors or GLP-1 RA have been given and target blood sugar has not been reached, a combination of these two agents or other regimen may be considered.
Malaysian CPG on Management of Type 2 DM	2020	GLP1-RA have been shown to demonstrate CV benefit in those with ASCVD or who are at high CV risk irrespective of HbA1c achieved. (Level 1)
Canadian Journal of Diabetes	2020	Our recommendations have been updated to include Dulaglutide and subcutaneous Semaglutide as options for patients with ASCVD. This has led to a recommendation that a GLP1-RA with proven CV outcome benefit can be considered in patients aged 60 years or older with at least 2 CV risk factors, with the strongest evidence for Dulaglutide followed by Liraglutide and subcutaneous Semaglutide. In adults with type 2 diabetes and ASCVD, a GLP1-RA or SGLT2i with CV or renal benefit should be used to reduce the risk of MACE [Grade A, Level 1A, for Liraglutide and Dulaglutide; Grade B, Level 2 for subcutaneous Semaglutide.
European Association for the Study of Diabetes	2019	Add SGLT-2i or GLP-1 RA in patients with T2DM who have established cardiovascular disease. If HF or CKD predominates, prefer SGLT-2i. If SGLT-2i not tolerated or contraindicated, use GLP-1 RA.
American College of Cardiology/American Heart Association Primary Prevention of Cardiovascular Guidelines	2019	For patients with T2DM who have additional cardiovascular risk factors, it is reasonable to initiate SGLT-2i or GLP-1 RA.
Canadian Diabetes Association	2018	GLP-1 RA or SGLT2-I may be considered as initial add-on therapy in patients with T2DM who have ASCVD who have not achieved target A1C.
Scottish Intercollegiate Guidelines Network	2017	For individuals with type 2 diabetes and established cardiovascular disease, GLP-1 receptor agonist therapies with proven cardiovascular benefit (currently liraglutide) should be considered.